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CUI et al.(10) **Pub. No.: US 2025/0277211 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **LPA INHIBITORS AND USES THEREOF**(71) Applicant: **KYLONOVA (XIAMEN)**
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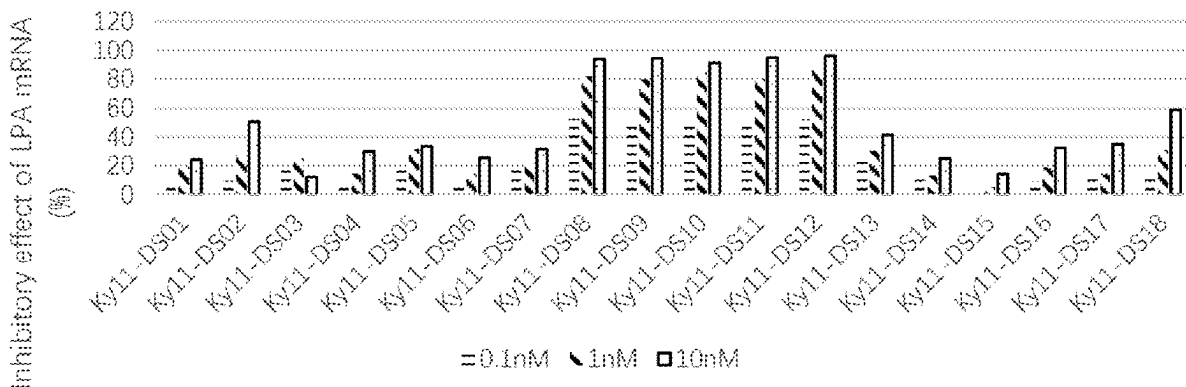
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(2013.01); **C12N 2310/351** (2013.01); **C12N**
2320/32 (2013.01)(57) **ABSTRACT**

The present application relates to an RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting the expression of an LPA gene comprising a complementary region of an antisense strand, the complementary region comprises at least 15 consecutive nucleotides, wherein the antisense strand comprises the following nucleotide sequence: any sequence of SEQ ID NO. 414-418, a sequence having at least 15 consecutive nucleotides identical to that in SEQ ID NO. 414-418, or a sequence having no more than three different nucleotides from the above sequences. The RNA inhibitor of the present application directly degrades LPA mRNA, continuously and efficiently inhibits LPA gene expression, and can be used to treat or prevent cardio-cerebrovascular diseases mediated by LPA genes.

Specification includes a Sequence Listing.

Inhibitory effect of LPA mRNA (%) in HepG2 cells



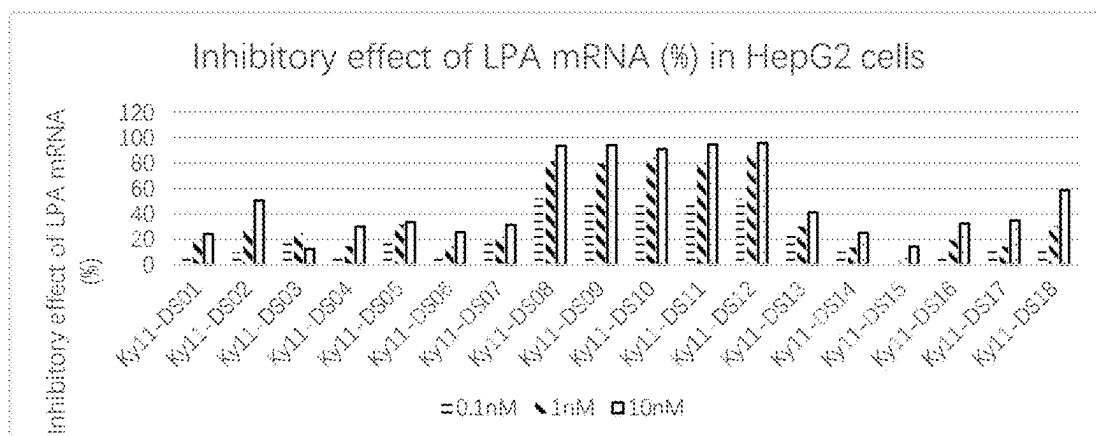


Figure 1

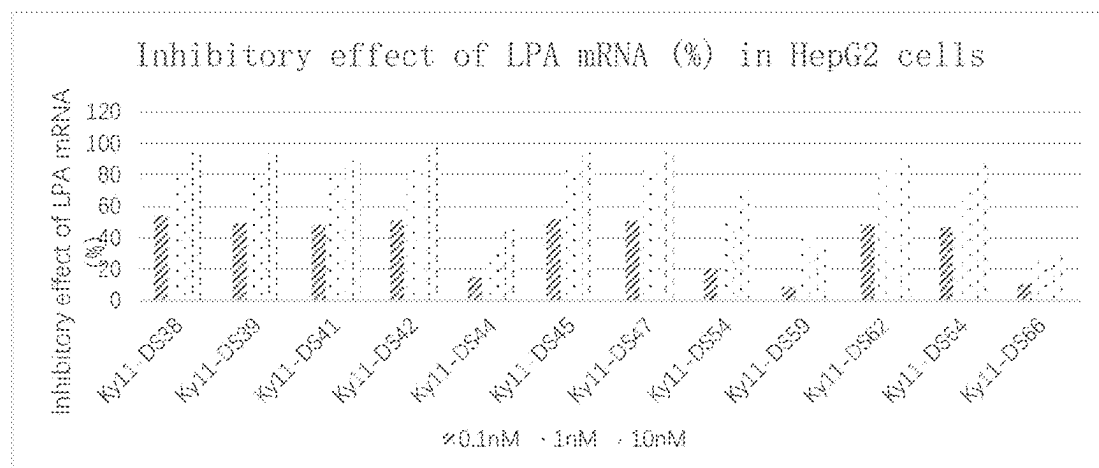


Figure 2

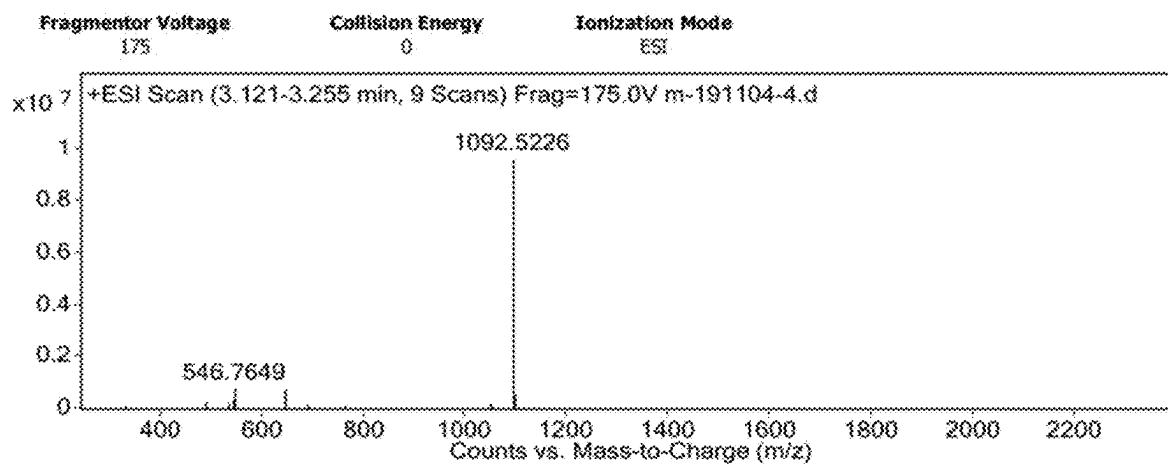


Figure 3

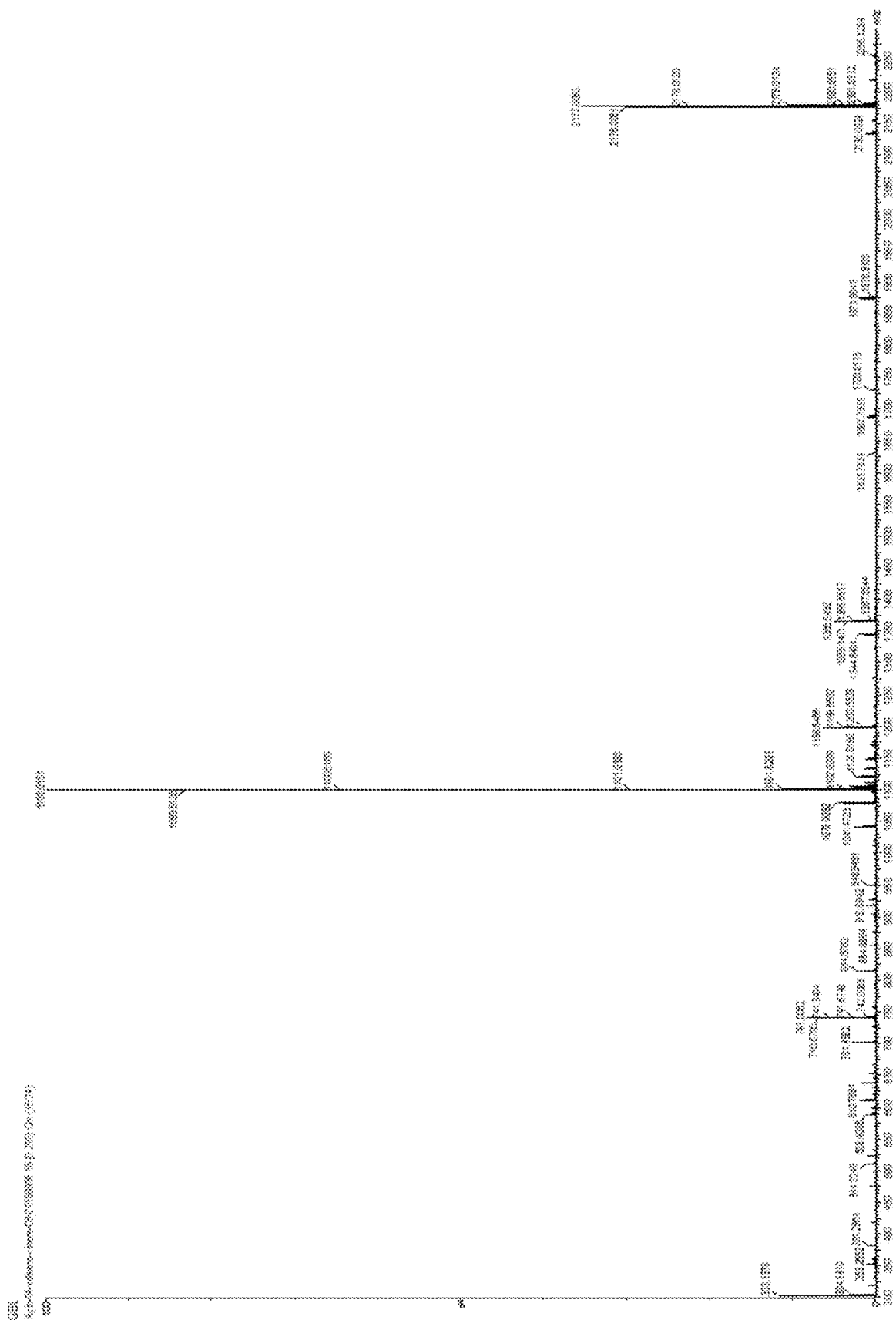


Figure 4

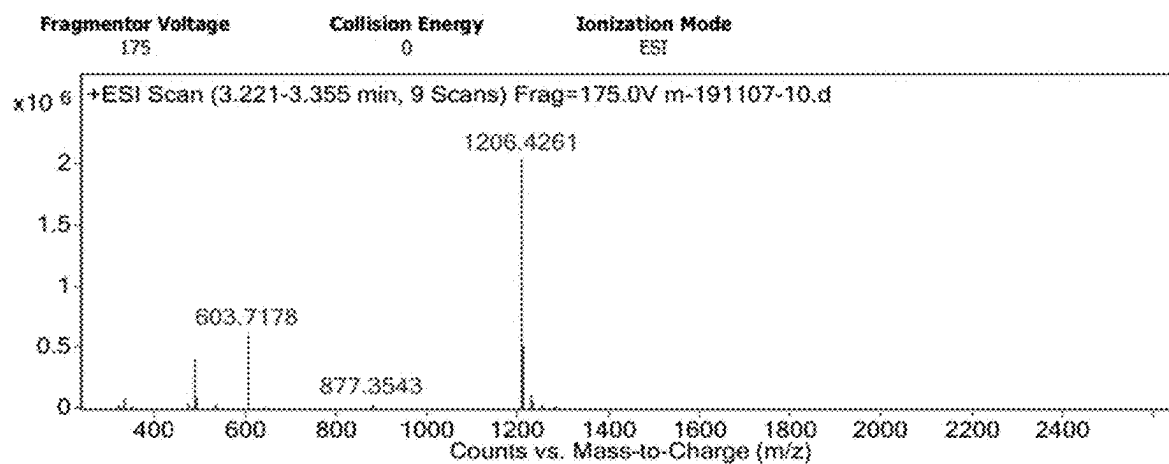


Figure 5

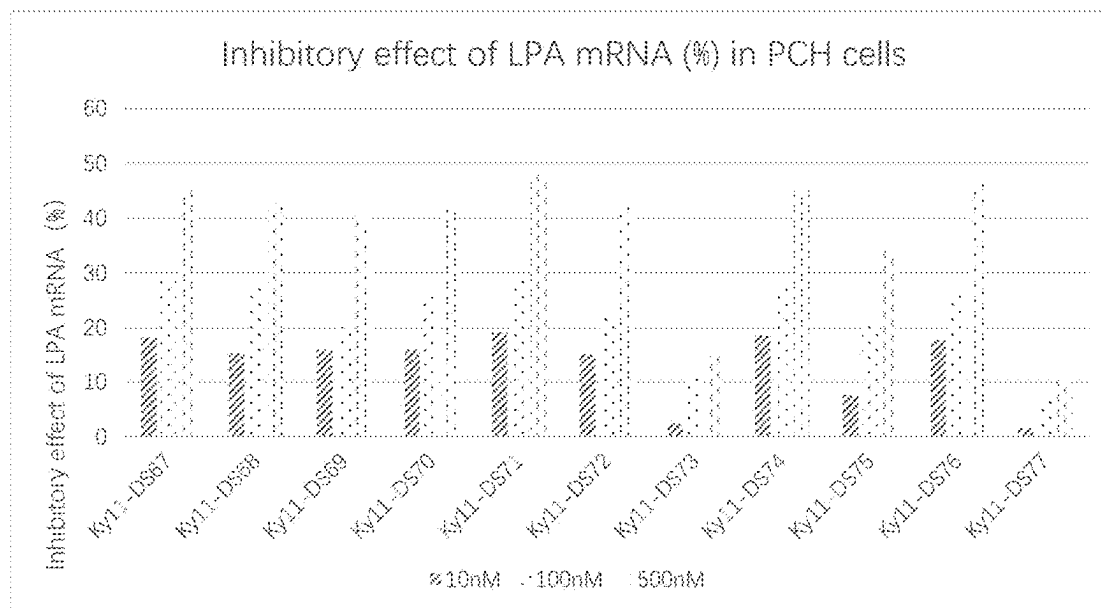


Figure 6

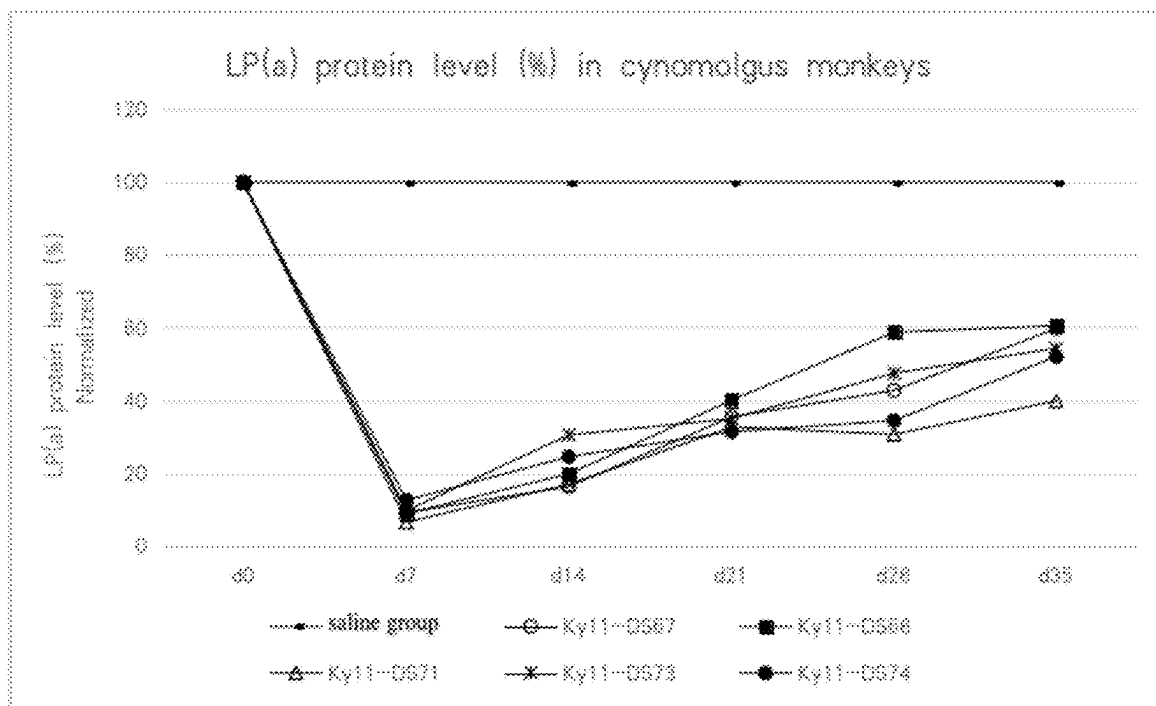


Figure 7

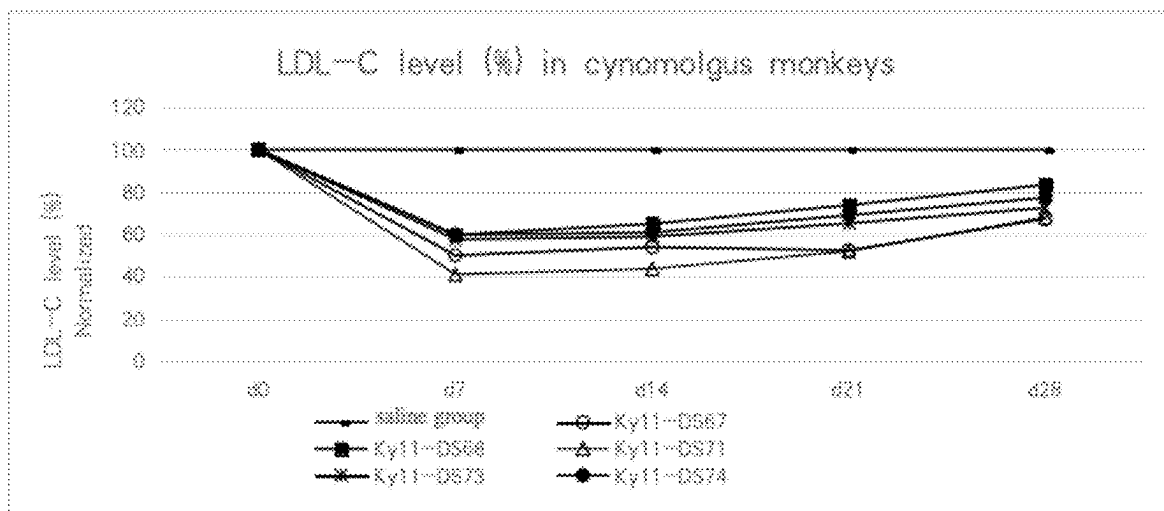


Figure 8

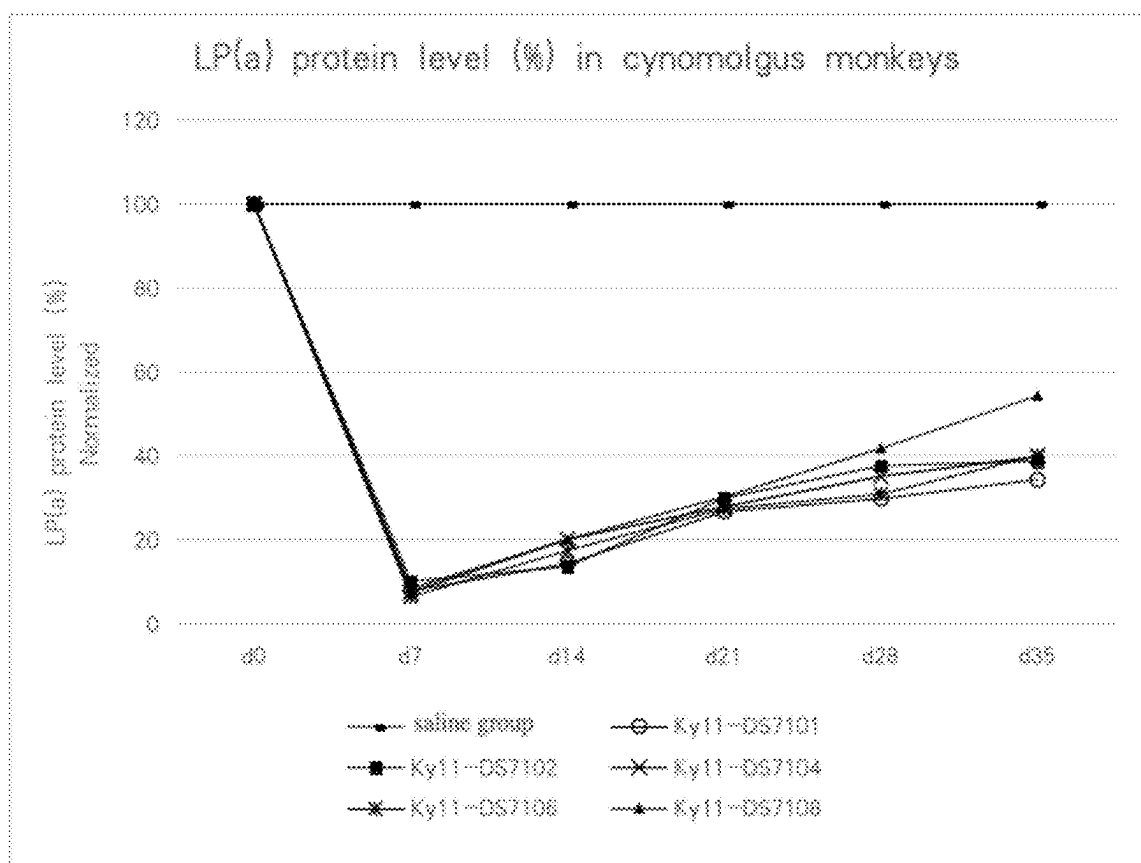


Figure 9

LPA INHIBITORS AND USES THEREOF

CROSS-REFERENCE OF THE RELEVANT APPLICATION

[0001] The present application is a 371 of international Patent Cooperation Treaty (PCT) patent application serial number PCT/CN2023/080708, filed Mar. 10, 2023, which claims priority to the Chinese patent application 202210241706.1 with the name of “LPA inhibitor and uses thereof” filed on Mar. 11, 2022, the entire content of which are incorporated in the present application by reference.

TECHNICAL FIELD

[0002] The present application relates to the biomedical field, specifically involving an RNA inhibitor that suppresses LPA gene expression and its application.

BACKGROUND ART

RNAi

[0003] RNAi (RNA interference) was discovered by Andrew Z. Fire et al. during antisense RNA inhibition experiments in *Caenorhabditis elegans* in 1998, and this process was called RNAi. This discovery was rated as one of the top ten scientific advances in 2001 by Science magazine, and ranked first among the top ten scientific advances in 2002. Since then, siRNA with RNAi as its mechanism of action has received widespread attention as a potential gene therapy drug. In 2006, Andrew Z. Fire and Craig C. Mello won the Nobel Prize in physiology or medicine for their contributions in the study of RNAi mechanism. RNAi is triggered by double stranded RNA (dsRNA) in many organisms, including animals, plants and fungi. In the process of RNAi, an endonuclease called “Dicer” cleaves or “dices” long strand of dsRNA into small segments of 21-25 nucleotides long. These small segments are known as small interfering RNAs (siRNAs), whose antisense strands are loaded onto Argonaute protein (AGO2). AGO2 loading occurs in the RISC-loading complex, a ternary complex consisting of Argonaute protein, Dicer, and dsRNA binding protein (TRBP for short). During loading, the sense strand is cleaved by AGO2 and discharged. Then, AGO2 uses the antisense strand to bind to mRNA containing fully complementary sequences, and then catalyzes the cleavage of these mRNA, resulting in mRNA cleavage to loss the role as a translation template, thereby preventing the synthesis of related proteins. After cleavage, the cleaved mRNA is released, and the RISC-loading complex loaded with the antisense strand is recycled for another round of cleavage.

[0004] According to statistics, among the disease-related proteins in the human body, about more than 80% of them cannot be targeted by current conventional small molecule drugs and biological macromolecule preparations, and are considered undruggable proteins. Gene therapy, which aims to treat diseases through gene expression, silencing and other functions, is considered by the industry to be the third generation of therapeutic drugs after chemical small molecule drugs and biological macromolecular drugs. This therapy treats diseases at the genetic level and is not restricted by undruggable proteins. As the most mainstream type of gene therapy, RNAi technology treats diseases at the mRNA level and has higher efficiency as compared with

chemical small molecule drugs and biological macromolecular drugs at the protein level. RNAi technology can be used to design the sense strand and antisense strand sequences of siRNAs with high specificity and good inhibitory effect according to the specific gene sequence. These single strand sequences are synthesized through solid-phase synthesis, and then the sense strand and antisense strand are paired into siRNA in a specific annealing buffer according to the principle of base pairing. Finally, it is delivered to the corresponding target site in the body through the vehicle system, degrading the target mRNA, and destroying the function of the target mRNA as a translation template, thus preventing the synthesis of related proteins.

Delivery System of siRNA

[0005] siRNA is unstable in blood and tissues and easily degraded by nucleases. In order to improve the stability of siRNA, the sense strand and antisense strand of siRNA can be modified, but these chemical modifications only provide limited protection from nuclease degradation and may ultimately affect the activity of siRNA. Therefore, a corresponding delivery system is needed to ensure that siRNA can safely and efficiently pass through cell membrane. Due to its large molecular weight, large amount of negative charge, and high water solubility, siRNA itself cannot smoothly pass through cell membrane to enter the cell.

[0006] Liposome is basically composed of a hydrophilic core and a phospholipid bilayer. It has a phospholipid bilayer similar to a biological membrane and has high biocompatibility, therefore liposome once became the most popular and widely used siRNA vehicle. Liposome-mediated siRNA delivery mainly encapsulates siRNA into liposomes to protect siRNA from degradation by nucleases, improves the efficiency of siRNA passing through cell membrane barriers, and thereby promotes cellular absorption. Examples of liposomes include, for example, anionic liposomes, pH sensitive liposomes, immunoliposomes, fusogenic liposomes, and cationic lipids. Despite some progress, liposomes themselves are prone to trigger inflammatory reactions, a variety of antihistamines and hormones such as cetirizine and dexamethasone must be used before administration to reduce possible acute inflammatory reactions. Therefore, they are not suitable for all treatment fields in actual clinical application, especially for diseases with long treatment cycles such as chronic hepatitis B. Possible accumulated toxicity resulting from long-term use is a potential safety hazard. Therefore, a safer and more effective vehicle system is needed to deliver siRNA.

[0007] Asialoglycoprotein receptor (ASGPR) in the liver is a receptor specifically expressed on hepatocytes and is a highly efficient endocytic receptor. Due to the fact that under physiological conditions in the body, after acid and enzymatic hydrolysis of sialic acids, the exposed secondary ends of various glycoproteins are galactose residues, the sugar specifically bound to ASGPR is galactosyl, so ASGPR is also called galactose-specific receptor. Monosaccharide and polysaccharide molecules such as galactose, galactosamine, and N-acetylgalactosamine all have high affinity for ASGPR. ASGPR mainly has the physiological function of mediating the clearance of asialoglycoprotein, lipoprotein and other substances in the blood, and it is closely related to the occurrence and development of liver diseases such as viral hepatitis, liver cirrhosis, liver cancer and so on. The discovery of this characteristic of ASGPR plays an important role in the diagnosis and treatment of hepatogenic

diseases (Ashwell G and Harford J, Carbohydrate specific Receptors of the Liver, Ann Rev Biochem 1982 51:531-554). The therapeutic drug for hepatogenic diseases containing galactose or galactosamine and their derivatives in the structure can have specific affinity with ASGPR, so it actively targets liver and does not need other vehicle systems to deliver.

LPA and Lp (a)

[0008] LPA is the name of the gene encoding apolipoprotein (a) (apo (a)), which is mainly expressed in the liver, and its expression is restricted to humans and non-primates. Apolipoprotein (a) is attached to apo (B)-100 via a disulfide bond and combined with the lipid core to form lipoprotein (a) (Lp (a)) particles. Lp(a) particles are special cholesterol-rich macromolecular lipoproteins with surface wrapped by cholesterol and phospholipids and embedded with hydrophilic apolipoprotein fractions, apo (a) and apo (B)-100. Lp (a) can enter and deposit on the blood vessel wall, promoting atherosclerosis. Lp (a) is structurally homologous to plasminogen (PLG) and can compete with fibrozygens for binding to the fibrin site, thus inhibiting fibrinogen hydrolysis and promoting thrombosis. Therefore, Lp (a) is closely associated with atherosclerosis and thrombosis. Studies have shown that blood Lp (a) level is an independent risk factor for cardiovascular disease, stroke, and atherosclerotic stenosis.

[0009] High Lp (a) levels are mainly related to genetics and do not change significantly with diet, exercise and other lifestyle changes. Lp (a) levels greater than 300 mg per liter in humans are considered high Lp (a). High Lp (a) level-soften indicate a significantly increased risk of atherosclerosis and thrombosis. Detection of Lp (a) is relatively important for early identification of the risk of atherosclerosis. In China, there are about 330 million people suffering from cardiovascular diseases, but the awareness, treatment and control rates of dyslipidemia among the general public is generally at a low level, and the awareness of the risk of Lp (a) is even lower. Most hospitals do not include it in their routine lipid examination, and there are no targeted therapeutic drugs in the clinic at home and abroad. Therefore, an effective therapeutic drug is urgently needed in this field.

Invention Content

[0010] The purpose of the present application is to provide an LPA RNA inhibitor that prevents the formation of Lp (a) particles by interfering with LPA mRNA and destroying its function as a translation template. The inhibitor can continuously and efficiently inhibit LPA gene expression and effectively reduce Lp (a) level, and is an efficient drug for treatment of diseases caused by LPA gene expression.

[0011] In one aspect, the present application provides an RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting the expression of an LPA gene comprising a complementary region of an antisense strand, the complementary region comprises at least 15 consecutive nucleotides, wherein the antisense strand comprises the following nucleotide sequence: any sequence of SEQ ID NO. 414-418, a sequence having at least 15 consecutive nucleotides identical to that in SEQ ID NO. 414-418, or a sequence having no more than three different nucleotides from the above sequences. It is to be noted that the scope of protection of the present application is not limited to any of the sequences in

SEQ ID NO. 414-418, but also includes: sequences having 15 or more consecutive nucleotides identical to those in SEQ ID NO. 414-418, or sequences having less than or equal to three different nucleotides from the above sequences. The above sequence refers to any one of SEQ ID NO. 414-418 and the sequences having 15 or more consecutive nucleotides identical to those in SEQ ID NO. 414-418.

[0012] The aforementioned RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene, wherein the sense strand comprises the following nucleotide sequence: any one of SEQ ID NO. 396-400, or a sequence having at least 15 consecutive nucleotides identical to that in SEQ ID NO. 396-400, or a sequence having no more than three different nucleotides from the above sequences.

[0013] The aforementioned RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene, wherein the sense and antisense strands are selected from one or more of the following combinations:

- 1) The sense strand: (SEQ ID NO. 396)
ugguaauggacagaguuaauu,
the antisense strand: (SEQ ID NO. 414)
auaacucuguccauuaccau;
- 2) The sense strand: (SEQ ID NO. 397)
acagaguuaucgaggcucatt,
the antisense strand: (SEQ ID NO. 415)
ugagccucgaaacucuguuu;
- 3) The sense strand: (SEQ ID NO. 398)
guaauggacagaguuaucgau,
the antisense strand: (SEQ ID NO. 416)
cgauaacucuguccauuacca;
- 4) The sense strand: (SEQ ID NO. 399)
gccccuugguguauuacaa,
the antisense strand: (SEQ ID NO. 417)
uuguauaacaccaaggggcug;
and
- 5) The sense strand: (SEQ ID NO. 400)
gcuccuuauuguauuacga,
the antisense strand: (SEQ ID NO. 418)
ucguuaacaauaaggagcug.

[0014] The aforementioned RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene, wherein one or more nucleotides on the sense strand and/or antisense strand are modified to form a modified nucleotide sequence.

[0015] The aforementioned RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene, wherein the sense and antisense strands are selected from one or more of the following combinations:

1) The sense strand:
(SEQ ID NO. 456)
UsGsGUfAAfUfGfGACAGAUUAUsUsU,
the antisense strand:
(SEQ ID NO. 492)
AsUsAACUfCUGUCfCAfUUACCAsUsU;
2) The sense strand:
(SEQ ID NO. 457)
AsCsAGfAGfUfUfAUCGAGGCUCAsTsT,
the antisense strand:
(SEQ ID NO. 493)
UsGsAGCCfUCGAUfAAfCUCUGUsUsU;
3) The sense strand:
(SEQ ID NO. 460)
GsUsAAfUGfGfAfCAGAGUUAUCGsAsU,
the antisense strand:
(SEQ ID NO. 496)
CsGsAUAAfCUCUGfUCfCAUUAACsCsA;
4) The sense strand:
(SEQ ID NO. 463)
GsCsfcCCfCUfUfGfGfGUfUfUfUAcSAsA,
the antisense strand:
(SEQ ID NO. 499)
UsfUsGfUAfUAfACACCAfAGGGGCSUsG;
and
5) The sense strand:
(SEQ ID NO. 465)
GsCsfcUCfCUfUfAfUUfGUfUfUAcSGsA,
the antisense strand:
(SEQ ID NO. 501)
UsfCsGFUAfUAfACAUAfAGfGAGCsUsG:

wherein, G = 2'-methylguanylate, A = 2'-O-methyladenylate, U = 2'-O-methyluridylylate, C = 2'-O-methylcytidylate, Gs = 2'-O-methyl-3'-thioguanylate, As = 2'-O-methyl-3'-thioadenylate, Us = 2'-O-methyl-3'-thiouridylylate, Cs = 2'-O-methyl-3'-thiocytidylate, fG = 2'-fluoroguanylate, fA = 2'-fluoroadenylate, fU = 2'-fluorouridylylate, fC = 2'-fluorocytidylate, fGs = 2'-fluoro-3'-thioguanylate, fAs = 2'-fluoro-3'-thioadenylate, fUs = 2'-fluoro-3'-thiouridylylate, fCs = 2'-fluoro-3'-thiocytidylate.

[0016] The aforementioned RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene, wherein the RNA inhibitors are selected from one or more of Ky 11-DS38, Ky 11-DS39, Ky 11-DS42, Ky 11-DS45, and Ky 11-DS47.

[0017] The aforementioned RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene, which further contains ligands, wherein the ligand is conjugated to the sense and/or antisense strands.

[0018] The aforementioned RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene, wherein the RNA inhibitors are selected from one or more of Ky 11-DS71 and Ky 11-DS7101 to Ky 11-DS7109.

[0019] In another aspect, the present application also provides a pharmaceutical composition containing the aforementioned RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene, a delivery vehicle, and/or physiologically acceptable excipients and/or carriers and/or diluent.

[0020] In yet another aspect, the present application further provides the following technical solution: use of the aforementioned RNA inhibitor or its pharmaceutically acceptable salt for inhibiting LPA gene expression, as well as the aforementioned pharmaceutical composition in the

preparation of drugs for the prevention or treatment of diseases or conditions or for reducing the risk of diseases or conditions.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The specific features of the invention involved in the present application are shown in the appended claims. The features and advantages of the invention covered by the present application can be better understood by referring to the exemplary embodiments and drawings described in detail below. A brief description of the attached drawings is as follows:

[0022] FIG. 1 shows the inhibitory effect (%) of LPA mRNA in HepG 2 cells by the RNAi inhibitor in Example 2 of the present application;

[0023] FIG. 2 shows the inhibitory effect (%) of LPA mRNA in HepG 2 cells by the RNAi inhibitor in Example 3 of the present application;

[0024] FIG. 3 shows the high resolution mass spectrum of ERCd-01-c2 synthesized in Example 4 1.1.5 of the present application;

[0025] FIG. 4 shows the high resolution mass spectrum of 3'MVIP17-c1 synthesized in Example 4 1.2.6 of the present application;

[0026] FIG. 5 shows the high resolution mass spectrum of 5'MVIP09-ERCd-PFP-c2 synthesized in Example 4 2.1.2 of the present application;

[0027] FIG. 6 shows the inhibitory effect (%) of LPA mRNA in PCH cells by the RNAi inhibitor in Example 6 of the present application;

[0028] FIG. 7 shows the LP (a) protein level (%) in cynomolgus monkeys in Example 7 of the present application;

[0029] FIG. 8 shows the LDL-C protein level (%) in cynomolgus monkeys in Example 7 of the present application;

[0030] FIG. 9 shows the LP (a) protein level (%) in cynomolgus monkeys with different 5'MVIP/3'MVIP combinations of RNA inhibitors in Example 8 of the present application.

EMBODIMENTS

[0031] The implementation of the present invention is described by specific examples below. People familiar with the technology can easily understand other advantages and effects of the present invention from the content disclosed in the specification.

DEFINITION OF TERMS

[0032] In the present application, the terms “iRNA”, “RNAi agent”, “iRNA agent” and “RNA interfering agent”, “RNA inhibitor” can be used interchangeably, and generally refer to agents containing RNA as the terms are defined herein, which can mediate the targeted cleavage of RNA transcripts through RNA-induced silencing complex (RISC) pathway. The iRNA guides the sequence-specific degradation of mRNA via a process called RNA interference (RNAi). The iRNA regulates (e.g., inhibits) the expression of LPA genes in cells (e.g., cells in a subject such as a mammalian subject).

[0033] In some embodiments, the RNAi agent may be single stranded siRNA (ssRNAi) introduced into cell or organism to inhibit target mRNA. The single stranded RNAi

agent binds to the RISC endonuclease Argonaute 2, which then cleaves the target mRNA. The single stranded siRNAs are generally 15 to 30 nucleotides in length and chemically modified. The design and test of single stranded siRNA are described in U.S. Pat. No. 8,101,348 and Lima et al. (2012) Cell 150:883-894, and their complete contents are incorporated herein by reference.

[0034] In some embodiments, the “iRNA” used in the present application is a double stranded RNA, and is referred to herein as “double stranded RNAi agent”, “double stranded RNA (dsRNA) molecule”, “dsRNA agent” or “dsRNA”. The term “dsRNA” refers to a complex of ribonucleic acid molecules with a duplex structure containing two antiparallel and essentially complementary nucleic acid strands, known as having a “sense” and “antisense” orientations relative to the target RNA (i.e., LPA gene). In some embodiments of the present application, double stranded RNA (dsRNA) triggers the degradation of target RNA (e.g., mRNA) through a post transcriptional gene silencing mechanism (herein referred to as RNA interference or RNAi).

[0035] The duplex structure can be any length that allows the required target RNA to be specifically degraded through the RISC pathway, and can have a length in range of about 19 to 36 base pairs, for example, the length of about 19-30 base pairs, for example, the length of about 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35 or 36 base pairs. The ranges and lengths within the above ranges and lengths are also included as part of the present application. In some embodiments, the iRNA agent of the present application is a dsRNA containing 15-23 nucleotides in each strand, which interacts with the target RNA sequence (e.g., LPA gene) to guide the cleavage of the target RNA. In some embodiments, the iRNA of the present application is a dsRNA containing 24-30 nucleotides, which interacts with the target RNA sequence (e.g., LPA target mRNA sequence) to guide the cleavage of the target RNA.

[0036] Generally, most nucleotides of each strand of dsRNA molecule are ribonucleotides, but as described in detail herein, each strand or both strands can also include one or more non-ribonucleotides, such as deoxyribonucleotides or modified nucleotides. In addition, the “iRNA” used in this specification may include ribonucleotides with chemical modifications; the iRNA can include substantial modifications at multiple nucleotides. The term “modified nucleotide” used herein refers to a nucleotide that independently has a modified sugar moiety, a modified internucleotide linkage or a modified nucleobase, or any combination thereof. Therefore, the term “modified nucleotide” covers the substitution, addition, or removal of functional groups or atoms of internucleotide linkages, sugar moieties, or nucleobases. The modifications applicable to the agent of the present invention include all types of modifications disclosed herein or known in the art.

[0037] In the present application, the terms “nucleic acid” and “polynucleotide” can be used interchangeably and refer to the polymeric form of nucleotides (deoxyribonucleotides or ribonucleotides or analogues thereof) of any length. Polynucleotides can have any three-dimensional structure and can perform any function. The following are non-limiting examples of polynucleotides: genes or gene segments (such as probes, primers, EST or SAGE tags), exons, introns, messenger RNA (mRNA), transfer RNA, ribosomal RNA, ribozymes, cDNA, recombinant polynucleotides,

branched polynucleotides, plasmids, vectors, isolated DNA of any sequence, isolated RNA of any sequence, nucleic acid probes, siRNA, miRNA, shRNA, RNAi reagents and primers. Polynucleotides may be modified or substituted at one or more bases, sugars, and/or phosphates with any of the various modifications or substitutions described in the present application or known in the art. Polynucleotides can contain modified nucleotides, such as methylated nucleotides and nucleotide analogues. If present, modifications to the nucleotide structure can be made before or after polymer assembly. Nucleotide sequences can be blocked by non-nucleotide components. Polynucleotides can be modified after polymerization, for example by conjugating with labelling components. The terms can refer to double stranded and single stranded molecules. Unless otherwise stated or required, any embodiment herein as a polynucleotide includes the double stranded form and each of the two complementary single stranded forms known or predicted to constitute the double stranded form.

[0038] In the present application, the term “target nucleic acid” or “target sequence” generally refers to the consecutive part of the nucleotide sequence of the mRNA molecule formed during the transcription of the LPA gene, including mRNA as the main transcription product of RNA processing. The target part of the sequence should be at least long enough to serve as a substrate for iRNA-guided cleavage at or near that portion of the nucleotide sequence of mRNA molecule formed during LPA gene transcription. In one embodiment, the target sequence is within the protein coding region of LPA. The target sequence can be about 19-36 nucleotides in length, for example, it is preferred to be about 19-30 nucleotides in length. The ranges and lengths within the above ranges and lengths are also included as part of the present application.

[0039] In the present application, the term “nucleotide sequence” usually refers to a series or a certain sequence of nucleobases, nucleotides and/or nucleosides, whether modified or unmodified, described by a series of letters using standard nucleotide nomenclature and a list of symbols for modified nucleotides described in the present application.

[0040] In the present application, the term “oligonucleotide” generally refers to a polymer composed of multiple nucleotide residues (deoxyribonucleotide or ribonucleotide, or related structural variant or synthetic analog thereof) connected by phosphodiester bond (or related structural variant or synthetic analog thereof). Therefore, although the term “oligonucleotide” generally refers to the nucleotide polymer in which the nucleotide residues and the linkage between them are naturally occurred, it should be understood that the scope of the term also includes various analogues, including but not limited to: peptide nucleic acid (PNA), aminophosphate, thiophosphate, methyl phosphonate, 2-O-methyl ribonucleic acid, etc. The exact size of the molecule may depend on the specific application. Oligonucleotides are generally short in length, usually about 10-30 nucleotide residues, but the term can also refer to molecules of any length, although the term “polynucleotide” or “nucleic acid” is generally used for larger oligonucleotides.

[0041] In some embodiments, the oligonucleotides comprise one or more unmodified ribonucleosides (RNA) and/or unmodified deoxyribonucleosides (DNA) and/or one or more modified nucleosides. The term “modified oligonucle-

otide” generally refers to an oligonucleotide containing at least one modified nucleoside and/or at least one modified internucleoside linkage.

[0042] In the present application, the term “modified nucleoside” generally means a nucleoside containing at least one chemical modification compared with a naturally occurring RNA or DNA nucleoside. Modified nucleosides contain modified sugar moieties and/or modified nucleobases.

[0043] In the present application, the term “nucleobase” generally refers to a heterocyclic pyrimidine or purine compound, which is a component of all nucleic acids and includes adenine (A), guanine (G), cytosine (C), thymine (T) and uracil (U). Nucleotides may include modified nucleotides or nucleotide mimics, abasic sites (Ab or X), or the part substituted by the substituent. As used in the present application, “nucleobase sequence” generally means the sequence of consecutive nucleobases that does not depend on any sugar, linkage or nucleobase modification. The term “unmodified nucleobase” or “naturally occurring nucleobase” generally means naturally occurring heterocyclic nucleobases of RNA or DNA: purine bases adenine (A) and guanine (G); and the pyrimidine bases thymine (T), cytosine (C) (including 5-methyl C) and uracil (U). “Modified nucleobase” usually means any nucleobase that is not a naturally occurring nucleobase.

[0044] In the present application, the term “sugar moiety” generally refers to the naturally occurring sugar moiety or modified sugar moiety of a nucleoside. The term “naturally occurring sugar moiety” generally refers to a ribofuranosyl as found in naturally occurring RNA or a deoxyribofuranosyl as found in naturally occurring DNA. “Modified sugar moiety” means a substituted sugar moiety or sugar substitute.

[0045] In the present application, the term “internucleoside linkage” generally refers to the covalent linkage between adjacent nucleosides in oligonucleotides. “Naturally occurring internucleoside linkage” means 3' to 5' phosphodiester linkage. “Modified internucleoside linkage” means any internucleoside linkage other than the naturally occurring internucleoside linkage.

[0046] In the present application, the term “antisense oligonucleotide” refers to a single stranded oligonucleotide molecule with a nucleobase sequence complementary to the corresponding segment of the target nucleic acid (e.g., the target genome sequence, mRNA precursor, or mRNA molecule). In some embodiments, the antisense oligonucleotide is 12 to 30 nucleobases in length. In some embodiments, the antisense oligonucleotide is an unmodified or modified nucleic acid with a nucleotide sequence complementary to the sequence of target nucleic acid (such as LPA polynucleotide).

[0047] In the present application, the term “antisense strand” generally refers to the strand of RNAi agent (such as dsRNA) that includes a region that are substantially complementary to the target sequence. When used herein, the term “complementary region” generally refers to a region on the antisense strand that is substantially complementary to the sequence (e.g., target sequence) defined in the present application. When the complementary region is not fully complementary to the target sequence, the mismatch can be in the inner or terminal region of the molecule. In general, the most tolerated mismatches are in the terminal region, for example, within 5, 4, 3, or 2 nucleotides at the 5' and/or 3' ends.

[0048] In the present application, the term “sense strand” (s) generally refers to such a strand of RNAi agents, which includes a region that is substantially complementary to a region of the antisense strand as the term defined herein. The “sense” strand is sometimes referred to as the “passenger” strand or “anti-guide” strand. With their sequences, the antisense strand targets the desired mRNA, while the sense strand targets different targets. Therefore, if the antisense strand is incorporated into RISC, the correct target is targeted. The incorporation of sense strand can lead to off-target effects. These off-target effects can be limited by the use of modifications or 5' end caps on the sense strand.

[0049] In the present application, the term “complementary” when used to describe the first nucleotide sequence (such as RNAi agent sense strand or LPA mRNA) in terms of the second nucleotide sequence (such as RNAi agent antisense strand) refers to the ability of oligonucleotides or polynucleotides containing the first nucleotide sequence to hybridize (form base pair hydrogen bonds) with oligonucleotides or polynucleotides containing the second nucleotide sequence and form duplex or double helix structures under certain conditions. Complementary sequences include Watson-Crick base pairs or non-Watson-Crick base pairs, and include natural or modified nucleotides or nucleotide mimics, as long as the above requirements for their hybridization ability are realized. “Complementation” does not require nucleobase complementarity on each nucleoside. On the contrary, some mismatches can be tolerated.

[0050] In the present application, the term “fully complementary” generally means that all (100%) bases in the consecutive sequence of the first polynucleotide will hybridize with the same number of bases in the consecutive sequence of the second polynucleotide. The consecutive sequence may comprise all or part of the first or second nucleotide sequence. As used in the present application, “partially complementary” generally means that in the hybridized nucleobase sequence pairs, at least about 70% of the bases in the consecutive sequence of the first polynucleotide will hybridize with the same number of bases in the consecutive sequence of the second polynucleotide. As used in the present application, “substantially complementary” generally means that in the hybridized nucleobase sequence pairs, at least about 90% of the bases in the consecutive sequence of the first polynucleotide will hybridize with the same number of bases in the consecutive sequence of the second polynucleotide. The terms “complementary”, “fully complementary” and “substantially complementary” as used in the present application can be used in terms of base matching between the sense strand and antisense strand of RNAi agent or between the antisense strand of RNAi agent and the sequence of LPA mRNA. Sequence identity or complementarity does not depend on modification. For the purpose of determining identity or complementarity, for example, a and Af are complementary to U (or T) and identical to A.

[0051] In the present application, the term “homologous” or “homology” generally refers to the number of nucleotides of the subject nucleic acid sequence that have matched the same nucleotide of the reference nucleic acid sequence, which is usually determined by sequence analysis programs (for example, Karlin and Altschul, 1990, PNAS 87:2264-2268; Karlin and Altschul, 1993, PNAS 90:5873-5877), or by visual inspection. As used herein, the term “complete homology” or “fully homologous” generally refers to com-

plete (100%) homology or “identity” between the reference sequence and the subject nucleic acid sequence. As used herein, the term “substantially homologous” or “substantial homology” generally refers to the fact that nucleotides at the same nucleotide positions in the subject sequence and the reference sequence are at least 50% (e.g., at least 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99%) homologous.

[0052] In the present application, the term “ligand” generally refers to any compound or molecule that can be covalently or otherwise chemically bound to bioactive substances (such as oligonucleotides). In some embodiments, the ligand is able to interact directly or indirectly with another compound such as a receptor, the receptor interacting with the ligand may exist on the cell surface, or alternatively may be an intracellular and/or intercellular receptor, the interaction of the ligand with the receptor may lead to a biochemical reaction, or may simply be a physical interaction or binding.

[0053] In the present application, the terms “induce”, “inhibit”, “strengthen”, “elevate”, “increase”, “decrease”, “reduce” and the like usually indicate quantitative differences between two states. For example, “the amount that effectively inhibits the activity or expression of LPA” means that the level of LPA activity or expression in the treated samples will be lower than that in the untreated samples. The terms described apply, for example, to expression levels and activity levels. The terms “decrease” and “reduce” are used interchangeably and usually indicate any changes less than the original. “Decrease” and “reduce” are relative terms that need to be compared before and after measurement. “Decrease” and “reduce” include complete depletion.

[0054] In some embodiments, the term “reduce” may refer to an overall reduction of about 5%, 10%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95%, or 100% in the expression level/amount of a gene/gene product such as protein or biomarker in a first sample compared to the expression level/amount of the corresponding gene/gene product such as protein or biomarker in a second sample, as detected by standard methods known in the art, such as those described in the present application. In some embodiments, the term “reduce” refers to a reduction in the expression level/amount of a gene or biomarker in the first sample, wherein the reduction is at least about 0.9 times, 0.8 times, 0.7 times, 0.6 times, 0.5 times, 0.4 times, 0.3 times, 0.2 times, 0.1 times, 0.05 times, or 0.01 times the expression level/amount of the corresponding gene or biomarker in the second sample. In some embodiments, the first sample is a sample obtained from a subject, while the second sample is a reference sample.

[0055] In the present application, the term “expression” generally refers to the process by which a gene eventually produces a protein. The expression includes, but is not limited to, transcription, post transcriptional modifications (e.g., splicing, polyadenylation, addition of 5'-caps), and translation.

[0056] In the present application, the term “pharmaceutically acceptable” generally refers to one or more non-toxic substances that do not interfere with the biological activity and effectiveness of the active ingredient. Such preparations may generally contain salts, excipients, buffers, preservatives, compatible carriers, and optionally other therapeutic agents. Such pharmaceutically acceptable preparations may also generally contain compatible solid or liquid fillers,

diluents, or encapsulation materials suitable for administration to humans. When used in medicine, salts should be pharmaceutically acceptable salts, but non-pharmaceutically acceptable salts can be conveniently used to prepare pharmaceutically acceptable salts, which cannot be excluded from the scope of the present application. Such pharmacologically and pharmaceutically acceptable salts include but are not limited to those prepared from the following acids: hydrochloric acid, hydrobromic acid, sulfuric acid, nitric acid, phosphoric acid, maleic acid, acetic acid, salicylic acid, citric acid, boric acid, formic acid, malonic acid, succinic acid, etc. The pharmaceutically acceptable salts can also be prepared into alkali metal salts or alkaline earth metal salts, such as sodium salt, potassium salt or calcium salt.

[0057] In the present application, the term “lipid nanoparticles” or “LNP” generally refers to nanoparticles containing encapsulated pharmacologically active molecules, such as nucleic acid molecules, for example, iRNA or plasmid from which iRNA is transcribed. LNP is described in, for example, Chinese patent No. CN103189057B, the complete contents of which are incorporated herein by reference.

[0058] The term “cynomolgus monkey” is also known as: crab-eating macaque (scientific name: *Macaca fascicularis*); also called the long-tailed macaque.

[0059] In the present application, the term “prevention and/or treatment” not only includes the prevention and/or treatment of disease, but also generally includes the prevention of the onset of disease, slowing or reversing the progression of disease, preventing or slowing the onset of one or more symptoms related to disease, reducing and/or alleviating one or more symptoms related to disease, reducing the severity and/or duration of the disease and/or any symptom associated with it and/or preventing further increases in the severity of the disease and/or any symptom associated with it, preventing, reducing or reversing any physiological damage caused by the disease, and any pharmacological effects that are generally beneficial to the patient being treated. The RNAi agent or pharmaceutical composition of the present application does not need to achieve complete cure or eradication of any symptoms or manifestations of the disease to form a feasible therapeutic agent. As recognized in relevant fields, drugs used as therapeutic agents can reduce the severity of a given disease state, but do not need to eliminate every manifestation of the disease to be considered as useful therapeutic agents. Similarly, treatments administered prophylactically to constitute viable prophylactic agents do not need to be completely effective in preventing the onset of the condition. Simply reducing the impact of disease in subjects (for example, by reducing the numbers or severity of their symptoms, or by improving the effectiveness of another treatment, or by producing another beneficial effect), or reducing the likelihood of disease occurrence or exacerbation is sufficient.

[0060] In the present application, the term “disease” or “condition” can be used interchangeably, and generally refers to any deviation from the normal state of the subject, such as any change in the state of the body or some organs, which hinders or disturbs the performance of the function, and/or causes symptoms such as discomfort, dysfunction, pain or even death in the person who is ill or in contact with him. Disease or condition can also be called distemper, ailing, ailment, malady, disorder, sicknesses, illnesses, complaints, inderdispositions or affection.

[0061] In the present application, the term “administration” generally refers to the introduction of the pharmaceutical preparation of the present application into the subject’s body through any route of introduction or delivery. Any method known to those skilled in the art for contacting a cell, organ or tissue with the drug can be adopted. The administration may include, but is not limited to, intravenous, intra-arterial, intranasal, intra-abdominal, intramuscular, subcutaneous transdermal, aerosol inhalation, or oral administration. The daily dose can be divided into one, two or more suitable forms of dose to be administered at one, two or more times during a certain time period.

[0062] In the present application, the term “contact” generally refers to the contact of two or more different types of substances in any order, in any way and for any duration. Contact can occur in vivo, ex vivo, or in vitro. In some embodiments, it may mean that the RNAi agent or composition of the present application directly contacts a cell or tissue. In other embodiments, the term refers to making the RNAi agent or composition of the present application contact a cell or tissue indirectly. For example, the method of the present application includes a method in which the subject is exposed to the RNAi agent or composition of the present application, and then the RNAi agent or composition contacts the cell or tissue through diffusion or any other active or passive transport process known in the art through which the compound circulates in the body.

[0063] In the present application, the term “effective amount” or “effective dose” generally refers to an amount sufficient to achieve or at least partially achieve the desired effect. The “therapeutically effective amount” or “therapeutically effective dose” of a drug or therapeutic agent is generally any amount of drug that, when used alone or in combination with another therapeutic agent, promotes disease regression (as evidenced by a decrease in the severity of disease symptoms, an increase in the frequency and duration of asymptomatic periods of disease, or prevention of impairment or disability due to the disease). The “preventive effective amount” or “preventive effective dose” of a drug usually refers to the amount of drug that inhibits the development or recurrence of disease when administered alone or in combination with another therapeutic agent to subjects at risk of disease development or disease recurrence. A variety of methods known to those skilled in the art can be used to evaluate the ability of therapeutic or preventive agents to promote disease regression or inhibit disease development or recurrence, such as predicting efficacy in humans in human subjects during clinical trials and in animal model systems, or determining the activity of the agent in in vitro assays. In some embodiments, “effective amount” refers to the amount of RNAi agent that produces the expected pharmacological, therapeutic or preventive results.

[0064] In the present application, the term “subject” generally refers to humans or non-human animals (including mammals) that need diagnosis, prognosis, improvement, prevention and/or treatment of diseases, such as humans, non-human primates (apes, gibbons, gorillas, chimpanzees, orangutans, macaques), domestic animals (dogs and cats), farm animals (poultry such as chickens and ducks, horses, cattle, goats, sheep, pigs) and experimental animals (mice, rats, rabbits, guinea pigs). Human subjects include fetal, newborn, infant, adolescent and adult subjects. Subjects include animal disease models.

[0065] In the present application, the terms “include”, “comprise”, “have”, “may/can”, “contain” and their variants are generally intended to be open-ended transitional phrases, terms or words, which do not exclude the possibility of additional actions or structures. The term “consisting of/composed of” usually means that no other component (or similarly, feature, integer, step, etc.) can exist. Unless otherwise specified in the context, the singular forms such as “a”, “an”, “the” in English generally include the plural form of the thing referred to.

[0066] In the present application, the term “about” usually means approximately, in the region of, roughly, or around. When the term “about” is used to refer to the range of values, the cut-off value or specific value is used to indicate that the stated value may differ by up to 10% from the enumerated value. Therefore, the term “about” can be used to cover variation from a specific value of $\pm 10\%$ or less, $\pm 5\%$ or less, $\pm 1\%$ or less, $\pm 0.5\%$ or less, or $\pm 0.1\%$ or less.

[0067] It should be understood that the term “at least” before a number or series of numbers includes the number adjacent to the term “at least”, and all subsequent numbers or integers logically included, as defined from the context. For example, the number of nucleotides in a nucleic acid molecule must be an integer. For example, “at least 15 nucleotides in a nucleic acid molecule of 21 nucleotides” means that 15, 16, 17, 18, 19, 20, or 21 nucleotides have the indicated properties. When “at least” appears before a series of numbers or ranges, it should be understood that “at least” can modify each number in the series or ranges.

[0068] It should be understood that “no more than” used herein can be exchanged with “not more than” or “less than or equal to”, and refers to a value or integer that is adjacent to the phrase and logically lower. For example, duplexes with protruding end of “no more than 3 nucleotides” have protruding end of 3, 2, 1, or 0 nucleotides. When “no more than” appears before a series of numbers or ranges, it should be understood that “no more than” can modify each number in the series or ranges. The scope used herein includes both upper and lower limits.

[0069] The term “ligand” refers to a ligand that is introduced into a target tissue receptor to alter the distribution, targeting, or stability of RNAi agents. For example, a specific ligand can provide an enhanced affinity for a selected target (e.g., molecule, cell, or cell type, compartment (e.g., cell or organ compartment, body tissue, organ, or region) compared to species where no ligand is present.

[0070] Ligands can include naturally occurring substances, such as proteins (such as human serum albumin (HSA), low-density lipoprotein (LDL), or globulin); carbohydrates (such as glucan, pullulan, chitin, chitosan, inulin, cyclodextrin, N-acetylglucosamine, N-acetylgalactosamine, or hyaluronic acid); or lipids. Ligands can also be recombinant or synthetic molecules, such as synthetic polymers, such as synthetic polyamino acids. Ligands may also include targeting groups, such as cell or tissue targeting agents, such as lectins, glycoproteins, lipids, or proteins, such as antibodies, that bind to a specified cell type such as kidney cells. The targeting groups can be thyrotropin, melanotropin, lectin, glycoprotein, surfactant protein A, mucin carbohydrate, multivalent lactose, multivalent galactose, N-acetylgalactosamine, N-acetyl-glucosamine multivalent mannose, multivalent fucose, glycosylated polyamino acids, multivalent galactose, transferrin, bisphosphonate, polyglutamic acid, polyaspartic acid, lipid, cholesterol, steroid, cholic

acid, folic acid, vitamin B12, vitamin A, biotin, or RGD peptide or RGD peptide mimetics. In some embodiments, the ligand is a multivalent galactose, for example, N-acetyl-galactosamine.

Detail of Invention

Antisense Strand and Sense Strand

[0071] In one aspect, the present application provides an RNAi agent inhibiting LPA gene expression.

[0072] In some embodiments, RNAi agents include single stranded oligonucleotides or double stranded ribonucleic acid (dsRNA) molecules used to inhibit the expression of the LPA gene in cells, such as cells of a subject (e.g., a mammal). The dsRNA includes an antisense strand with a complementary region complementary to at least part of the mRNA formed in the expression of LPA gene. The complementary region is about 12-30 nucleotides in length (e.g., about 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, or 12 nucleotides in length).

[0073] The dsRNA includes two RNA strands, which can be complementary and hybridize to form a duplex structure (complementary region) under the conditions in which the dsRNA is used. One strand (antisense strand) of the dsRNA includes a complementary region that is substantially, and often completely complementary to the target sequence. The target sequence can be derived from the sequence of mRNA formed during the expression of the LPA gene. The other strand (sense strand) includes a region complementary to the antisense strand, so that when combined under appropriate conditions, the two strands can hybridize and form a duplex structure. Typically, duplex structures are 12 to 30 base pairs in length. Similarly, the length of the region complementary to the target sequence is 12 to 30 nucleotides.

[0074] In some embodiments, the dsRNA is about 19 to about 23 nucleotides in length, or about 24 to about 30 nucleotides in length. In general, the dsRNA is long enough to be used as a substrate for Dicer. For example, it is known in the art that the dsRNA with a length greater than about 21-23 nucleotides can be used as a substrate for Dicer. Those skilled in the art also understand that the region of RNA targeted for cleavage is usually part of larger RNA molecules (usually mRNA molecules). A “part” of the target is the consecutive nucleotides of the mRNA target, which is long enough to allow it to be a substrate for RNAi-guided cleavage (i.e., cleavage via RISC pathway).

[0075] Those skilled in the art should also understand that the duplex region is the main functional part of dsRNA, for example, the duplex region of about 19 to about 30 base pairs, such as, about 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20 base pairs. Therefore, in one embodiment, in order to achieve the extent of becoming a functional duplex (for example, 15-30 base pairs) that targets the desired RNA for cleavage, RNA molecules or complexes of RNA molecules with duplex regions of more than 30 base pairs are dsRNA.

[0076] In some embodiments, the sense strand of the RNA inhibitor is selected from the sequences in Table 1 below or sequences that differ by one, two, or three nucleotides from each of the sequences in Table 1.

TABLE 1

Sense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Sense strand sequence 5'→3'
1	S1	accacatggctttgtcaggt
2	S2	attatgttgatgtggtgcat
3	S3	tcctattatgttgatgtggg
4	S4	tttctgtggctctattatgtt
5	S5	aataaggggctgccacaggat
6	S6	atagttggaggcgacggca
7	S7	aatagttggaggcgacggc
8	S8	taatagtggaggcgacgg
9	S9	taatgaagtatgtgccttgat
10	S10	acagtaaatgaagtatgtgcct
11	S11	tcttctgtgacagtaaatgaa
12	S12	aggatctggatttcggcagta
13	S13	aacaccaagggctgccacag
14	S14	ataacaccaagggctgccac
15	S15	tgcatctgagcatcgtgtcag
16	S16	tctgcatctgagcatcgtgtc
17	S17	attctgcatctgagcatcgtg
18	S18	agtccattctgcatctgagca
19	S19	acattcggagggacgaaggca
20	S20	aacattcggagggacgaaggc
21	S21	aataacattcggagggacgaa
22	S22	aaagcctctaggcttgagcc
23	S23	aaaaagcctctaggcttgag
24	S24	aaaaaagcctctaggcttgga
25	S25	tcaaaaaagcctctaggctt
26	S26	ttcaaaaaagcctctaggct
27	S27	tgcttggtcaaaaaagcctc
28	S28	agtgttggtcaaaaaagcc
29	S29	tttcctcagtcagtgctgtt
30	S30	tagcagtcctgtacccgggg
31	S31	agtagcagtcctgtacccgg
32	S32	tagtagcagtcctgtacccg
33	S33	aatggtagtagcagtcctgta
34	S34	ataatggtagtagcagtcctg
35	S35	tccataatggtagtagcagtc
36	S36	tgtccataatggtagtagcag

TABLE 1-continued

Sense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Sense strand sequence 5'→3'
37	S37	actctgtccataaatggtagta
38	S38	tctgtcaccaggcattgtgtc
39	S39	attctgtcaccaggcattgtg
40	S40	acttgattctgtcaccaggca
41	S41	agagttgcaaggacacttgat
42	S42	tggatctgggaccaccgtgag
43	S43	tcttcagaagaagcctctgtg
44	S44	tgcttcttcagaagaagcctc
45	S45	agcaatcctggaccccgggc
46	S46	tagcaatcctggaccccgggg
47	S47	tgttgctcctctgatgccagt
48	S48	tctgttgctcctctgatgccag
49	S49	ttctgttgctcctctgatgcca
50	S50	attctgttgctcctctgatgcc
51	S51	aatattctgttgctcctctgat
52	S52	ataatattctgttgctcctctg
53	S53	tggataaatattctgttgctcct
54	S54	tttgataaatattctgttgctc
55	S55	aaggactaatctcagcatctg
56	S56	aacaccaaggactaatctcag
57	S57	ataacaccaaggactaatctc
58	S58	atggataaacaccaaggacta
59	S59	aaacagccgtggacgtcgcaa
60	S60	agaaacagccgtggacgtcgc
61	S61	tcagaaacagccgtggacgtc
62	S62	ttcagaaacagccgtggacgt
63	S63	tggtcagaaacagccgtggac
64	S64	ttgttcagaaacagccgtgga
65	S65	tgcttggttcagaaacagccgt
66	S66	tagcagtcctggactgtgggg
67	S67	tgtggttctctgatgccagt
68	S68	attctgtggttctctgatgcc
69	S69	agtattctgtggttctctgat
70	S70	tccatcactggacattgcgtc
71	S71	ttccatcactggacattgcgt
72	S72	attccatcactggacattgcg

TABLE 1-continued

Sense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Sense strand sequence 5'→3'
73	S73	agagttgattccatcactgga
74	S74	ttgtgaggagagttgattcca
75	S75	aaggaagctctgtgcttgga
76	S76	tcagaaggaagctctgtgctt
77	S77	ttcagaaggaagctctgtgct
78	S78	tcttcagaaggaagctctgtg
79	S79	tgcttcttcagaaggaagctc
80	S80	tttcagttgggtgcttcttcag
81	S81	tgttttcagttgggtgcttctt
82	S82	tgctgttttcagttgggtgctt
83	S83	tagcagtcctggaccccgatg
84	S84	atcacctcggtagcagtcctg
85	S85	tccatcacctcggtagcagtc
86	S86	tgccatcacctcggtagcag
87	S87	actctgtccatcacctcggt
88	S88	aactctgtccatcacctcggt
89	S89	taactctgtccatcacctcggt
90	S90	atagtggtggagagtggtgctt
91	S91	aatgggatcctccgatgcca
92	S92	taatgggatcctccgatgcca
93	S93	ataatgggatcctccgatgcc
94	S94	agtataatgggatcctccgat
95	S95	tagtataatgggatcctccga
96	S96	atagataatgggatcctccg
97	S97	tggatagataatgggatcct
98	S98	atttgatagataatgggat
99	S99	ttgtgaggacactcgattctg
100	S100	tgtgcttgaacccggggccac
101	S101	ttttctcagtggtgcttgtt
102	S102	atatgcctcgataaactccgtc
103	S103	agttctcggtcaggccagcat
104	S104	tagttctcggtcaggccagca
105	S105	agtagttctcggtcaggccag
106	S106	tgcagtagttctcggtcaggc
107	S107	attctcagtagttctcggt
108	S108	tctaggacacctgattctggt

TABLE 1-continued

Sense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Sense strand sequence 5'→3'
109	S109	tctctaggacacctgattctg
110	S110	tcagaaggaggccctaggctt
111	S111	ttcagaaggaggccctaggct
112	S112	tcttggtcagaaggaggccct
113	S113	agtcttggtcagaaggaggcc
114	S114	tacagtcttggtcagaaggag
115	S115	atacagtcttggtcagaagga
116	S116	aacatacagtcttggtcagaa
117	S117	tttgccctggaatgaacgtg
118	S118	atttgccctggaatgaacgt
119	S119	tttatttgccctggaatgaa
120	S120	atgtcaccatcagggttacgg
121	S121	attgatgtcaccatcagggtt
122	S122	attcattgtgtaccagggg
123	S123	tggattcattgtgtaccacca
124	S124	tcttgattcattgtgtagca
125	S125	tttcttgattcattgtgtga
126	S126	aaaaagtttcttggttcat
127	S127	agtcaaaaagtttcttggtat
128	S128	tagtcaaaaagtttcttgga
129	S129	tatcacagtagtcaaaaagtt
130	S130	acagagagggatatacacagta
131	S131	aatgaagaggatgcacagag
132	S132	aatgcttccaggacatttctt
133	S133	ttcaagcagtgagcagcagtc
134	S134	tcttcaagcagtgagcagcag
135	S135	ttcttcaagcagtgagcagca
136	S136	acttcttcaagcagtgagcag
137	S137	aggacttcttcaagcagtgag
138	S138	aacatgagattcgaggttcac
139	S139	ttcctgaacatgagattcgag
140	S140	tttctgaacatgagattcga
141	S141	tctatttctgaacatgagat
142	S142	tgcttggtgtgggtccaagaa
143	S143	atctgcttggtgtgggtccaa
144	S144	tatctgcttggtgtgggtcca

TABLE 1-continued

Sense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Sense strand sequence 5'→3'
145	S145	aatatctgcttggtgtgggtctc
146	S146	aaggcaatatctgcttggtgtg
147	S147	ttagcaaggcaatatctgctt
148	S148	tagtctggggatggcagacaa
149	S149	tgtagtctggggatggcagac
150	S150	accatgtagtctggggatggc
151	S151	ttcagtcctggcggtgacctat
152	S152	attcagtcctggcggtgacca
153	S153	acattcagtcctggcggtgac
154	S154	taacattcagtcctggcggtg
155	S155	tgtaacattcagtcctggcggtg
156	S156	atgtaacattcagtcctggcg
157	S157	taacaaggagctgggtcttctt
158	S158	tcaataacaaggagctgggtct
159	S159	tctcaataacaaggagctgggt
160	S160	attctcaataacaaggagctgt
161	S161	ttcattctcaataacaaggag
162	S162	acacttcattctcaataacaa
163	S163	tatagtgtgtgcacacttcat
164	S164	acttatagtgtgtgcacactt
165	S165	atacttatagtgtgtgcacac
166	S166	atatacttatagtgtgtgcac
167	S167	aatatacttatagtgtgtgca
168	S168	agcacaaataacttatagtgt
169	S169	atgctcagcacaaataactt
170	S170	tgatgcttcactctgtctccc
171	S171	aggttgatgcttcactctgtct
172	S172	aagttaggtgatgcttcactct
173	S173	tcagcttctcaagttaggtgat
174	S174	ttacccacggttctcagcttcta
175	S175	tccttaccacggttctcagctt
176	S176	atccttaccacggttctcagct
177	S177	aaatccttaccacggttctcag
178	S178	attgctgtctattatttccag
179	S179	tcttcggttgattgtgtctcta
180	S180	atgccaaaggttggcatagct

TABLE 1-continued

Sense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Sense strand sequence 5'→3'
181	S181	aaaatgccaggtttggcat
182	S182	aatacaaaaatgccaggtt
183	S183	aaatacaaaaatgccaggt
184	S184	acaaaaatacaaaaatgcc
185	S185	agcttatcacaaaaatacca
186	S186	accttaaaagcttatcacaa
187	S187	agaccttaaaagcttatcac
188	S188	tcagtcagaccttaaaagctt
189	S189	agaatttgtcagtcagacctt
190	S190	tacagaatttgtcagtcagac
191	S191	aatacagaatttgtcagtcag
192	S192	ttaatacagaatttgtcagtc
193	S193	agctatgacaccttaatacag
194	S194	aatcaaaataagtgcagagtt

Where, g = guanylate, a = adenylate, t = thymidine, c = cytidylate.

[0077] In some embodiments, the antisense strand of the RNA inhibitor is selected from the sequences in Table 2 below or sequences that differ by one, two, or three nucleotides from each of the sequences in Table 2.

TABLE 2

Antisense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Antisense strand sequence 5'→3'
195	AS1	cugagcaaagccauguggucn
196	AS2	gacaccacaucaacauaauan
197	AS3	ccacaucaacauauaggacn
198	AS4	cauaauaggaccacagaaaan
199	AS5	ccuguggcagcccccuaauugn
200	AS6	ccgucgcgcuccaacuauun
201	AS7	cgucgcgcuccaacuauuan
202	AS8	gucgcgcuccaacuauuacn
203	AS9	caaggcacauacuauuacn
204	AS10	gcacauacuauuacugucn
205	AS11	cauuacugucacaggaagaan
206	AS12	cugccgaaauccagaucucgn
207	AS13	guggcagcccccugguguuan
208	AS14	ggcagcccccugguguuauan

TABLE 2-continued

Antisense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Antisense strand sequence 5'→3'
209	AS15	gacacgaugcucagaugcagn
210	AS16	cacgaugcucagaugcagaan
211	AS17	cgaugcucagaugcagaaugn
212	AS18	cucagaugcagaauaggacugn
213	AS19	ccuucgucuccuccgauguun
214	AS20	cuucgucuccuccgauguuan
215	AS21	cguccuccgauguuuuucn
216	AS22	cuccaagccuagaggcuuuun
217	AS23	ccaagccuagaggcuuuuun
218	AS24	caagccuagaggcuuuuun
219	AS25	gccuagaggcuuuuuugaan
220	AS26	ccuagaggcuuuuuugaacn
221	AS27	ggcuuuuuugaacaagcacn
222	AS28	cuuuuuugaacaagcacugn
223	AS29	caagcacugacugaggaaacn
224	AS30	ccggguacaggacugcuacn
225	AS31	ggguacaggacugcuacuacn
226	AS32	ggguacaggacugcuacuacn
227	AS33	caggacugcuacuaccuauan
228	AS34	ggacugcuacuaccuauaun
229	AS35	cugcuacuaccuauauggacn
230	AS36	gcuacuaccuauauggacagn
231	AS37	cuaccuauauggacagaguun
232	AS38	cacaaugccuggugacagaan
233	AS39	caaugccuggugacagaaucn
234	AS40	ccuggugacagaaucaagugn
235	AS41	caaguguccuugcaacucucn
236	AS42	cacggguguccagauccaan
237	AS43	cagaggcuucucugaagaan
238	AS44	ggcuucucugaagaagcacn
239	AS45	cccgggguccaggauugcuan
240	AS46	ccgggguccaggauugcuacn
241	AS47	cuggcaucagaggacaacagn
242	AS48	ggcaucagaggacaacagaan
243	AS49	gcaucagaggacaacagaun
244	AS50	caucagaggacaacagaauan

TABLE 2-continued

Antisense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Antisense strand sequence 5'→3'
245	AS51	cagaggacaacagaauuuuan
246	AS52	gaggacaacagaauuuuauucn
247	AS53	gacaacagaauuuuuuuccaan
248	AS54	caacagaauuuuuuuccaaaun
249	AS55	gaugcugagauuaguccuugn
250	AS56	gagauuaguccuugguguuan
251	AS57	gauuaguccuugguguuauan
252	AS58	guccuugguguuauuaccaugn
253	AS59	gcgacguccacggcuguuuucn
254	AS60	gacguccacggcuguuucugn
255	AS61	cguccacggcuguuucugaan
256	AS62	guccacggcuguuucugaacn
257	AS63	ccacggcuguuucugaacaan
258	AS64	cacggcuguuucugaacaagn
259	AS65	ggcuguuucugaacaagcacn
260	AS66	ccacaguccaggacugcuacn
261	AS67	cuggcaucagagaaccacagn
262	AS68	caucagagaaccacagaauan
263	AS69	cagagaaccacagaauacuau
264	AS70	cgcaauguccagugauggaan
265	AS71	gcaauguccagugauggaaun
266	AS72	caauguccagugauggaaucn
267	AS73	cagugauggaaucaacucucn
268	AS74	gaaucaacucuccucacaacn
269	AS75	ccaagcacagagcuuccuucn
270	AS76	gcacagagcuuccuucugaan
271	AS77	cacagagcuuccuucugaagn
272	AS78	cagagcuuccuucugaagaan
273	AS79	gcuuccuucugaagaagcacn
274	AS80	gaagaagcaccaacugaaaan
275	AS81	gaagcaccaacugaaaacagn
276	AS82	gcaccaacugaaaacagcacn
277	AS83	cugggguccaggacugcuacn
278	AS84	ggacugcuaccgaggugaugn
279	AS85	cugcuaccgaggugauggacn
280	AS86	gcuaccgaggugauggacagn

TABLE 2-continued

Antisense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Antisense strand sequence 5'→3'
281	AS87	ccgaggugauggacagaguun
282	AS88	cgaggugauggacagaguuan
283	AS89	gaggugauggacagaguuaun
284	AS90	gcacacucuccaccacuacn
285	AS91	ggcaucggaggaucccauuan
286	AS92	gcaucggaggaucccauuuau
287	AS93	caucggaggaucccauuuauan
288	AS94	cggaggaucccauuuauacuau
289	AS95	ggaggaucccauuuauacuauun
290	AS96	gaggaucccauuuauacuacn
291	AS97	gaucccauuuauacuauccaan
292	AS98	cccauuuauacuauccaaaun
293	AS99	gaaucgaguguccucacaacn
294	AS100	ggccccggguccaagcacagn
295	AS101	caagcaccaccugagaaaagn
296	AS102	cggaguuauccgaggcauacn
297	AS103	gcuggccugaccgagaacuau
298	AS104	cuggccugaccgagaacuacn
299	AS105	ggccugaccgagaacuacuagn
300	AS106	cugaccgagaacuacugcagn
301	AS107	cgagaacuacugcaggaaucn
302	AS108	cagaauccagguguccuagagn
303	AS109	gaauccagguguccuagagacn
304	AS110	gccuaggggccuccuucugaan
305	AS111	ccuaggggccuccuucugaacn
306	AS112	ggccuccuucugaacaagacn
307	AS113	ccuccuucugaacaagacugn
308	AS114	ccuucugaacaagacuguaun
309	AS115	cuucugaacaagacuguaugn
310	AS116	cugaacaagacuguauguuun
311	AS117	cguucauuccagggacaaaun
312	AS118	guucauuccagggacaaaun
313	AS119	cauuccagggacaaaauun
314	AS120	guaaccucgaggugacauacn
315	AS121	cccugauggugacaucauagn
316	AS122	cuggugcuacacaugaauacn

TABLE 2-continued

Antisense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Antisense strand sequence 5'→3'
317	AS123	gugcuacacaaugaauccaan
318	AS124	cuacacaaugaauccaagaan
319	AS125	cacaaugaauccaagaaaacn
320	AS126	gaauccaagaaaacuuuuugn
321	AS127	ccaagaaaacuuuuugacuan
322	AS128	caagaaaacuuuuugacuacn
323	AS129	cuuuuugacuacugugauaun
324	AS130	cugugauaucccucucugugn
325	AS131	cugugcauccucuucauuugn
326	AS132	gaaauguccuggaagcauugn
327	AS133	cugcugcucacugcuugaagn
328	AS134	gcugcucacugcuugaagaan
329	AS135	cugcucacugcuugaagaagn
330	AS136	gcucacugcuugaagaagucn
331	AS137	cacugcuugaagaaguccucn
332	AS138	gaaccucgaaucucauguuucn
333	AS139	cgaaucucauguucaggaaan
334	AS140	gaaucucauguucaggaaun
335	AS141	cucauguucaggaaauagaan
336	AS142	cuuggagcccacacaagcagn
337	AS143	ggagcccacacaagcagauan
338	AS144	gagcccacacaagcagauaun
339	AS145	gcccacacaagcagauuuugn
340	AS146	cacaagcagauuuugccuugn
341	AS147	gcagauuuugccuugcuaaan
342	AS148	gucugccauccccagacuacn
343	AS149	cugccauccccagacuacaun
344	AS150	cauccccagacuacaugguucn
345	AS151	ggucaccgccaggacugaaun
346	AS152	gucaccgccaggacugaaugn
347	AS153	caccgccaggacugaauguun
348	AS154	ccgccaggacugaauguuacn
349	AS155	gccaggacugaauguuacaun
350	AS156	ccaggacugaauguuacaucn
351	AS157	gaagcccagcuccuuguuaun
352	AS158	cccagcuccuuguuaugagn

TABLE 2-continued

Antisense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Antisense strand sequence 5'→3'
353	AS159	cagcuccuuguuaugagaaan
354	AS160	gcuccuuguuaugagaaugn
355	AS161	ccuuguuaugagaaugaagn
356	AS162	guuaugagaaugaagugugn
357	AS163	gaagugugcaaucaacuuaan
358	AS164	gugugcaaucaacuuaaguan
359	AS165	gugcaaucaacuuaaguauan
360	AS166	gcaaucaacuuaaguauuuun
361	AS167	caaucaacuuaaguauuuun
362	AS168	cuauaaguauuuugugcugn
363	AS169	guauuuuugugcugagcauun
364	AS170	gagacagagugaagcaucaan
365	AS171	cagagugaagcaucaaccuan
366	AS172	gugaagcaucaaccuacuuan
367	AS173	caaccuacuuagaagcugaan
368	AS174	gaagcugaaacgugguuaagn
369	AS175	gcugaaacguggguuaaggaun
370	AS176	cugaaacguggguuaaggauun
371	AS177	gaaacguggguuaaggauuuun
372	AS178	ggaaaauaauagacagcaaucn
373	AS179	gacagcaaucaaacgaagacn
374	AS180	cuauGCCAAACCUUGGCAUUN
375	AS181	gCCAAACCUUGGCAUUUUGN
376	AS182	CCUUGGCAUUUUGGUUUUUN
377	AS183	CUUGGCAUUUUGGUUUUUUN
378	AS184	GCAUUUUGGUUUUUUGUGN
379	AS185	GUUUUUUGUGUUAUAGCUUN
380	AS186	GUGUUAUAGCUUUUAAGGUCN
381	AS187	GUUAUAGCUUUUAAGGUCGN
382	AS188	GCUUUUAAGGUCUGACUGACN
383	AS189	GGUCUGACUGACAAAUUCUGN
384	AS190	CUGACUGACAAAUUCUGUAUN
385	AS191	GACUGACAAAUUCUGUAUUN
386	AS192	CUGACAAAUUCUGUAUUAAGN

TABLE 2-continued

Antisense strand of the RNAi agent		
SEQ ID NO.	Single strand code	Antisense strand sequence 5'→3'
387	AS193	guauuaaggugucauagcuan
388	AS194	cucugcacuuuuuugauuun

Where, g = guanylate, a = adenylate, u = uridylylate, c = cytidylate, n = guanylate, adenylate, uridylylate, or cytidylate.

[0078] In some embodiments, the RNA inhibitor also comprises a sense strand, wherein at least about 80% of the bases are complementary between the sense strand and the antisense strand.

[0079] In some embodiments, the sense and antisense strands are each independently 15-30 nucleotides.

[0080] In some embodiments, the sense and antisense strands are each independently 17-25 nucleotides.

[0081] In some embodiments, the sense and antisense strands are each independently 19-23 nucleotides.

[0082] In some screened embodiments, the sense and antisense strands of the RNA inhibitor are selected from the sequences in Table 3 below or sequences that differ by one, two, or three nucleotides from each of the sequences in Table 3.

TABLE 3

Screened RNA inhibitor sequences						
SEQ ID NO.	Single strand code	Sense strand sequence 5'→3'	SEQ ID NO.	Single strand code	Antisense strand sequence 5'→3'	RNA inhibitor code
389	S195	gacauuuguuaaaaauaa a	407	AS195	uuuuuuuuuaacaauguc au	Ky11-DS01
390	S196	gaaugaugagaaaaauu a	408	AS196	uaauuuuuucucauauuc cc	Ky11-DS02
391	S197	gggcaggucuggaaaaaa a	409	AS197	uuuuuuuccagaccugccc au	Ky11-DS03
392	S198	gguuuuuuuacuuuuu ua	410	AS198	uaaaagaaguagaagaacc ac	Ky11-DS04
393	S199	ggacaaaacauuuuaca a	411	AS199	uuguaaaaugauuuuguc cuu	Ky11-DS05
394	S200	ggaggaucccauuuacu a	412	AS200	uaguauaugggauccucc ga	Ky11-DS06
395	S201	ccuuguuuuugagaaug aa	413	AS201	uucauucucaauaacaagg ag	Ky11-DS07
396	S202	ugguauggacagaguu auuu	414	AS202	auaacucuguccauuacca uu	Ky11-DS08
397	S203	acagaguuaucgaggcuc att	415	AS203	ugagccucgauaacucugu uu	Ky11-DS09
398	S204	guaauggacagaguauuc gau	416	AS204	cgauaacucuguccauuac ca	Ky11-DS10
399	S205	gccccuugguguuuuaca a	417	AS205	uuguauaacaccaaggggc ug	Ky11-DS11
400	S206	gcuccuuuuguuuuac ga	418	AS206	ucguauaacaauaaggagc ug	Ky11-DS12
401	S207	caugguauggacagagu uau	419	AS207	aacucuguccauuaccaug uu	Ky11-DS13
402	S208	ggacagaguuaucgaggc uca	420	AS208	agccucgauaacucugucc uu	Ky11-DS14
403	S209	agaguuaucgaggcucau uau	421	AS209	aaugagccucgauaacucu gu	Ky11-DS15
404	S210	gaaggacaaaacauuuu a	422	AS210	uaaaauguauuuuguccuuc uc	Ky11-DS16
405	S211	cguucauuccagggacaa a	423	AS211	uuugucccuggaugaacg ug	Ky11-DS17
406	S212	cguguuuaagguuugu ua	424	AS212	uaacaaaccuugaaacacg ag	Ky11-DS18

Where, g = guanylate, a = adenylate, u = uridylylate, c = cytidylate.

[0083] For example, the sense strand could be SEQ ID NO. 396-400 or a sequence that differs by one, two or three nucleotides from each of them; and the antisense strand could be SEQ ID NO. 414-418 or a sequence that differs by one, two or three nucleotides from each of them:

1) The sense strand:	(SEQ ID NO. 396)
ugguaauggacagaguauuu,	
the antisense strand:	(SEQ ID NO. 414)
auaacucuguccauuaccau;	
2) The sense strand:	(SEQ ID NO. 397)
acagaguauaucgaggcucatt,	
the antisense strand:	(SEQ ID NO. 415)
ugagccucgauaacucuguuu;	
3) The sense strand:	(SEQ ID NO. 398)
guaauggacagaguauaucgau,	
the antisense strand:	(SEQ ID NO. 416)
cgauaacucuguccauuacca;	
4) The sense strand:	(SEQ ID NO. 399)
gccccuugguguuauacaa,	
the antisense strand:	(SEQ ID NO. 417)
uuguauaacaccaaggggcug;	
and	
5) The sense strand:	(SEQ ID NO. 400)
gcuccuuauuguuauacga,	
the antisense strand:	(SEQ ID NO. 418)
ucguauaacauaaggagcug.	

Chemical Modification of the Nucleotides

[0084] In order to enhance the stability of the above RNAi inhibitors in vivo, the sense strand and antisense strand of the RNAi inhibitors can be modified without affecting their activity or even enhancing their activity, and the nucleotide therein can have modifying group, and the whole strand or part of it can be modified. In some embodiments, one or more nucleotides on the sense strand and/or antisense strand are modified to form modified nucleotides.

[0085] The sense and antisense strands in the RNA inhibitor structure provided by the present application are 15-30 long, preferably 19-23, with at least 85% of the bases complementary to each other. In order to enhance the stability of the sense and antisense strands in vivo, the sense and antisense strands of RNA inhibitors can be modified without affecting or even enhancing activity. The nucleotides therein can have modifying groups, the strand can be wholly or partially modified, preferably wholly modified. The modification is a technique easily understood by researchers in the art, and any one or more of the following can be selected in the glycosyl portion: deoxyribonucleotides, nucleotide mimics, abasic nucleotides, 2'-modified

nucleotides, 3'-3'-linkage (inverted) nucleotides, nucleotides containing unnatural bases, bridged nucleotides, peptide nucleic acids (PNA), unlocked nucleobase analogues, locked nucleotides, 3'-O-methoxy (2' inter-nucleoside linkage) nucleotides, 2'-F-arabinose nucleotides, 5'-Me/2'-Fluoro-nucleotides, morpholino nucleotides, vinyl phosphonate deoxyribonucleotides, nucleotides containing vinyl phosphonate and nucleotides containing cyclopropyl phosphonate. Among them, the 2'-modified nucleotides include but are not limited to: 2'-O-methyl nucleotide, 2'-deoxy-2'-fluoronucleotide, 2'-deoxynucleotide, 2'-methoxyethyl nucleotide, 2'-amino nucleotide and 2'-alkyl nucleotide. In the RNA inhibitor provided by the present application, neither the sense strand nor the antisense strand of the RNA inhibitor requires uniform modification and they can be incorporated with one or more modifications in their single nucleotide. The modifications can also occur in the base moiety, modified nuclear bases include both synthetic and natural nuclear bases, such as 5-substituted pyrimidine, 6-azopyrimidine and N-2/N-6 and O-6-substituted purine, 5-methylcytosine, 5-hydroxymethylcytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-alkyl derivatives of adenine and guanine, 2-alkyl and other alkyl derivatives of adenine and guanine, 2-thiouracil, 2-thiothymine, 2-thiocytosine, 5-halouracil, cytosine, 5-propynyl uracil, 5-propynylcytosine, 6-azouracil, 6-azocytosine, 6-azothymine, 5-uracil, 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxy, and other 8-substituted adenine and guanine, 5-halo, 5-trifluoromethyl, and other 5-substituted uracil and cytosine, 7-methylguanine and 7-methyladenine, 8-azaguanine and 8-azaadenine, 7-deazaguanine, 7-deazaadenine, 3-deazaguanine and 3-deazaadenine.

[0086] Part or all of the sense strand and antisense strand of the RNA inhibitors of the present application are the 2'-O-methyl nucleotide and/or 2'-deoxy-2'-fluoronucleotide, and at least two consecutive phosphorothioate bonds exist between the 5' end of the sense strand and the 3' end of the antisense strand, preferably the phosphate bonds between the terminal three consecutive nucleotides are thioated.

[0087] In some embodiments, the modified nucleotides are selected from the group consisting of deoxyribonucleotides, nucleotide mimics, abasic nucleotides, 2'-modified nucleotides, 3'-3'-linkage (inverted) nucleotides, nucleotides containing unnatural bases, bridged nucleotides, peptide nucleic acids (PNA), unlocked nucleobase analogues, locked nucleotides, 3'-O-methoxy (2' inter-nucleoside linkage) nucleotides, 2'-F-arabinose nucleotides, 5'-Me/2'-Fluoro-nucleotides, morpholino nucleotides, vinyl phosphonate deoxyribonucleotides, nucleotides containing vinyl phosphonate and nucleotides containing cyclopropyl phosphonate.

[0088] In some embodiments, the 2'-modified nucleotide includes: 2'-O-methyl nucleotide, 2'-deoxy-2'-fluoronucleotide, 2'-deoxynucleotide, 2'-methoxyethyl nucleotide, 2'-amino nucleotide and/or 2'-alkyl nucleotide.

[0089] In some screened embodiments, the sense and antisense strands of the RNAi agents are selected from the sequences in Table 4 below or sequences that differ by one, two, or three nucleotides from each of the sequences in Table 4.

TABLE 4

Methoxylated RNA inhibitors						
SEQ ID NO.	Single strand code	SS5'→3'	SEQ ID NO.	Single strand code	AS5'→3'	RNA inhibitor code
425	S213	GAcAuuuGuuAAAAA uAAA	437	AS213	UUuAUUUUuAAcAAAu GUcAU	Ky11-DS19
426	S214	GAAuGAuGAGAAAu AAuuA	438	AS214	uAAUuAUUUCUcAUcA UUCCC	Ky11-DS20
427	S215	GGGcAGGucuGGAAA AAAA	439	AS215	UUUUUUUCcAGACCuG CCcAU	Ky11-DS21
428	S216	GGuucuucuAcuucuuuu A	440	AS216	uAAAAGAAGuAGAAG AACcAC	Ky11-DS22
429	S217	GGAcAAuAcAuuuu AcAA	441	AS217	UuGuAAAAuGuAUUuG UCCUU	Ky11-DS23
430	S218	GGAGGAucccAuuAuA cuA	442	AS218	uAGuAuAAuGGGAUCC UCCGA	Ky11-DS24
431	S219	ccuuGuuAuuGAGAAu GAA	443	AS219	UUcAUUCUcAAuAACA AGGAG	Ky11-DS25
432	S220	GAAGGAcAAuAcAu uuuA	444	AS220	uAAAAuGuAUUuGUCC UUCUC	Ky11-DS26
433	S221	GccccuuGGuGuuAuAc AA	445	AS221	UuGuAuAAcACcAAGGG GCuG	Ky11-DS27
434	S222	cGuucAuuccAGGGAc AAA	446	AS222	UUuGUCCCuGGAAUGA ACGuG	Ky11-DS28
435	S223	GcuuccuAuuGuuAuAc GA	447	AS223	UCGuAuAAcAAuAAGG AGCUG	Ky11-DS29
436	S224	cGuGuuucAAGGuuuG uuA	448	AS224	uAAcAAACCUuGAAAc ACGAG	Ky11-DS30

Where, G = 2'-methoxyguanylate, A = 2'-methoxyadenylate, U = 2'-methoxyuridylate, C = 2'-methoxycytidylate; u = uridylate, and c = cytidylate.

[0090] In some embodiments, at least one of the —OH at the 2' positions of the nucleotide glycosyls at positions 2, 4, 6, 7, 8, 12, 14 and 16 starting from the 5' end of the antisense strand are fluorinated.

[0091] The —OH at the 2' position of the nucleotide glycosyls at positions 7, 12, and 14 starting from the 5' end of the antisense strand are fluorinated, or the —OH at the 2' position of the nucleotide glycosyls at positions 2, 4, 6, 8, 14, and 16 starting from the 5' end of the antisense strand are fluorinated.

[0092] Except for nucleotides at positions 7, 12, and 14 starting from the 5' end of the antisense strand, at least one of the —OH at the 2' position of the remaining nucleotide glycosyls are substituted with methoxy; or except for nucleotides at positions 2, 4, 6, 8, 14, 16 starting from the 5' end of the antisense strand, at least one of the —OH at the 2' position of the remaining nucleotide glycosyls are substituted with methoxy.

[0093] The —OH at the 2' position of the nucleotide glycosyls at positions 7, 12, 14 starting from the 5' end of the antisense strand are fluorinated, and the —OH at the 2' position of the remaining nucleotide glycosyls are substituted with methoxy; or the —OH at the 2' position of the nucleotide glycosyls at positions 2, 4, 6, 8, 14, 16 starting from the 5' end of the antisense strand are fluorinated, and

the —OH at the 2' position of the remaining nucleotide glycosyls are substituted with methoxy.

[0094] At least one of the —OH at the 2' positions of the nucleotide glycosyls at positions 3, 5, 7, 8, 9, 11, 13, 15 starting from the 5' end of the sense strand are fluorinated.

[0095] The —OH at the 2' positions of the nucleotide glycosyls at positions 5, 7, 8, and 9 starting from the 5' end of the sense strand are fluorinated, or the —OH at the 2' positions of the nucleotide glycosyls at positions 3, 5, 7, 8, 9, 11, 13, 15 starting from the 5' end of the sense strand are fluorinated.

[0096] Except for nucleotides at positions 5, 7, 8, 9 starting from the 5' end of the sense strand, at least one of the —OH at the 2' position of the remaining nucleotide glycosyls are substituted with methoxy; or except for nucleotides at positions 3, 5, 7, 8, 9, 11, 13, 15 starting from the 5' end of the sense strand, at least one of the —OH at the 2' position of the remaining nucleotide glycosyls are substituted with methoxy.

[0097] The —OH at the 2' position of the nucleotide glycosyls at positions 5, 7, 8, 9 starting from the 5' end of the sense strand are fluorinated, and the —OH at the 2' position of the remaining nucleotide glycosyls are substituted with methoxy; or the —OH at the 2' position of the nucleotide glycosyls at positions 3, 5, 7, 8, 9, 11, 13, 15 starting from the 5' end of the sense strand are fluorinated, and the —OH

[0099] The —OH at the 2' position of the nucleotide glycosyls at positions 7, 12, 14 starting from the 5' end of the antisense strand are fluorinated, and the —OH at the 2' position of the remaining nucleotide glycosyls of the antisense strand are substituted with methoxy, and there are at least two consecutive phosphorothioate bonds between the three consecutive nucleotides at the end of the antisense strand; the —OH at the 2' position of the nucleotide glycosyls at positions 5, 7, 8, 9 starting from the 5' end of the sense strand are fluorinated, and the —OH at the 2' position of the remaining nucleotide glycosyls of the sense strand are substituted with methoxy, and there are at least two consecutive phosphorothioate bonds between the three consecutive nucleotides at the end of the sense strand; or

[0100] The —OH at the 2' position of the nucleotide glycosyls at positions 2, 4, 6, 8, 14, 16 starting from the 5' end of the antisense strand are fluorinated, and the —OH at the 2' position of the remaining nucleotide glycosyls of the antisense strand are substituted with methoxy, and there are at least two consecutive phosphorothioate bonds between the three consecutive nucleotides at the end of the antisense strand; the —OH at the 2' position of the nucleotide glycosyls at positions 3, 5, 7, 8, 9, 11, 13, 15 starting from the 5' end of the sense strand are fluorinated, and the —OH at the 2' position of the remaining nucleotide glycosyls of the sense strand are substituted with methoxy, and there are at least two consecutive phosphorothioate bonds between the three consecutive nucleotides at the end of the sense strand.

[0101] The sense strand in the RNA inhibitor is selected from the sequences in Table 5 below or sequences differing by one, two or three nucleotides from each of the sequences in Table 5.

The modified sense strand

SEQ ID NO.	Single strand code	Sense strand sequence 5'→3'
449	S225	UsGsGUfAAfUfGfGACfAGfAGUUAUsUsU
450	S226	AsCsAGfAGfUfUfAUfCfGAGGCUCAsTsT
451	S227	UsGsGfGUfAAfUfGfGfGfCAfCAfGfGUUAUsUsU
452	S228	AsCsAGfAGfUfUfAUfCGfAGGfCUCAsTsT
453	S229	UsGsGUfAAfUfGfGfGfCAGAGfUAUsUsU
454	S230	AsCsAGfAGfUfUfAUfCAGAGGfCUCAsTsT
455	S231	UsGsGUfAAfUfGfGACAGAGfGUUAUsUsU
456	S232	UsGsGUfAAfUfGfGACAGAGUUAUsUsU
457	S233	AsCsAGfAGfUfUfAUfCAGAGGCUCAsTsT
458	S234	GsUsfAAfUGfGfAfCAfCAfGUfUAUCGsAsU
459	S235	GsUsAAUGfGfAfCAGAGfUUAUCGsAsU
460	S236	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
461	S237	GsCsCCfCUfUfGfGUGUUAUACsAsA

The modified sense strand

SEQ ID NO.	Single strand code	Sense strand sequence 5'→3'
462	S238	GsCsCCfCUfUGfGUGUUAfUACsAsA
463	S239	GsCs fCCfCUfUfGfGUfGUfUAfUACsAsA
464	S240	GsCsUCfCUfUfAfUUGUUAUCsGsA
465	S241	GsCsfUCfCUfUfAfUUfGUfUAfUACsGsA
466	S242	GsUsAAfUGfGfAfCAGAGUfUAUCGsAsU
467	S243	GsUsAAUGfGfAfCAGAGUUAUCGsAsU
468	S244	GsUsAAfUGfGfAfCAGAGUUAUCfGsAsU
469	S245	GsUsfAAfUGfGfAfCAGAGUUAUCGsAsU
470	S246	GsfUsAAfUGfGfAfCAGAGUUAUCGsAsU
471	S247	GsfUsAAfUGfGAfCAGAGUUAUCGsAsU
472	S248	GsUsAAfUGfGfAfCfAGAGUUAUCGsAsU
473	S249	GsUsAAUGfGAfCfAfGAGUUAUCGsAsU
474	S250	GsUsAAUGfGACfAfGfAGUUAUCGsAsU
475	S251	GsUsAAfUGfGAfCfAGAGUUAUCGsAsU
476	S252	GsUsAAUGfGfAfCAGAGUfUAUCGsAsU
477	S253	fGsUsfAAfUGfGfAfCAGfAGfUUfAUcGsAsU
478	S254	GsUsAAfUGfGfAfCAGAGfUUAUCGsAsU
479	S255	GsUsAAUGGfAfCAGAGfUUAUCGsAsU
480	S256	GsCsUCCUUFafUUGUUfAUAcGsA
481	S257	GsCsUCCUfUAUFufGfGUUAUCsGsA
482	S258	AsCsfAGfAGfUfUfAUfcGfAGfGCUCAsTsT
483	S259	GsUsAAfUGfGfAfCAGAGUUFfAUCGsAsU
484	S260	GsCsUCfCUfUfAfUfUGUUAUAcGsA

Where, G = 2'-O-methylguanylate, A = 2'-O-methyladenylylate, U = 2'-O-methyluridylylate, C = 2'-O-methylcytidylylate, fG = 2'-fluoroguanlylate, fA = 2'-fluoroadenylylate, fU = 2'-fluorouridylylate, fC = 2'-fluorocytidylylate, Gs = 2'-O-methyl-3'-thioguanlylate, As = 2'-O-methyl-3'-thioadenylylate, Us = 2'-O-methyl-3'-thiouridylylate, Cs = 2'-O-methyl-3'-thiocytidylylate.

[0102] In some embodiments, the antisense strand in the RNA inhibitor is selected from the sequences in Table 6 below or sequences differing by one, two, or three nucleotides from each of the sequences in Table 6.

The screened antisense strand

SEQ ID NO.	Single strand code	Antisense strand sequence 5'→3'
485	AS225	AsfUsAACfUCfUfGfUCCAfUfUfACCAUsUsU
486	AS226	UsfGsAGCfCUfCfGfAUAAfCfUfCUGUsUsU

TABLE 6-continued

The screened antisense strand		
SEQ ID NO.	Single strand code	Antisense strand sequence 5'→3'
487	AS227	AsfUsAfACfUCfUGUCCAfUUfACCAsUsU
488	AS228	UsGsAfGCCfUCGAUfAAfCUCUGUsUsU
489	AS229	AsUsAfACUfCUGUCfCAfUUACCAsUsU
490	AS230	UsGsAGfCCfUfCfGAUAACUfCUGUsUsU
491	AS231	AsUsAAfCUfCfUfGUCCAUFUACCAsUsU
492	AS232	AsUsAACUfCUGUCCAfUUACCAsUsU
493	AS233	UsGsAGCCfUCGAUfAAfCUCUGUsUsU
494	AS234	CsfGsAfUAfACfUCUGUCfCAfUUACsCsA
495	AS235	CsGsAUAAfCfUfCUGUCfCAUUAcsCsA
496	AS236	CsGsAUAAfCUGUCfUfCAUUAcsCsA
497	AS237	UsUsGUUfAACACfCAfAGGGCsUsG
498	AS238	UsUsGUfAUfAAfCACCAfAGGGCsUsG
499	AS239	UsfUsGUfUAfUAfACACCAfAGfGGCsUsG
500	AS240	UsCsGUUfAACAAfUAfAGGAGCsUsG
501	AS241	UsfCsGUfUAfUAfACAAUfAGfGAGCsUsG
502	AS242	CsGsAUfAAfCfUfCUGUCCfAUUAcsCsA
503	AS243	CsGsAUAAfCfUfCUGUCCAUUAcsCsA
504	AS244	CsGsAUfAAfCfUfCUGUCCAUUAfCsCsA
505	AS245	CsGsAfUfAAfCfUfCUGUCCAUUAcsCsA
506	AS246	CsfGsAfUfAAfCfUfCUGUCCAUUAcsCsA
507	AS247	CsfGsAfUfAAfCfUfCUGUCCAUUAcsCsA
508	AS248	CsfGsAUfAAfCfUfCfUGUCfCAUUAcsCsA
509	AS249	CsGsAfUAACUCUGfUCfCAUUAcsCsA
510	AS250	CsfGsAUfAAfCfUfCfUGUCfCAfUUACsCsA
511	AS251	CsfGsAUAAUCfUGUCfCAfUUACsCsA
512	AS252	CsfGsAUfAAfCfUfCfUGUCfCAfUUACsCsA
513	AS253	CsfGsAfUAACfUCUGUCfCAfUUfACsCsA
514	AS254	CsfGsAUfAAfCfUfCfUGUCfCAfUUACsCsA
515	AS255	CsfGsAUfAAfCfUfCfUGUCfCAfUUACsCsA
516	AS256	UsCsGUUfAAfCAAUfAGGAGCsUsG
517	AS257	UsfCsGUUAACAAUfAGfGAGCsUsG
518	AS258	UsfGsAfGCFUfCGAUAAfCUfCUGUsUsU
519	AS259	CsGsAUfAAfCfUfCUGUCCAfUUACsCsA
520	AS260	UsfCsGUUfAAfCAAUfAGGAGCsUsG

TABLE 6-continued

The screened antisense strand		
SEQ ID NO.	Single strand code	Antisense strand sequence 5'→3'
521	AS261	UsfUsGUUfAAfCAfAGGGCsUsG
522	AS262	UsUsGUUfAAACACfCAfAGGGCsUsG
523	AS263	UsfUsGUUfAAfCAfAGGGCsUsG
524	AS264	UsCsGUUfAAACAAUfAGGAGCsUsG
525	AS265	UsfCsGUUfAAfCAfAAUfAGfGAGCsUsG
526	AS266	UsfCsGUUfAAfCAfAAUfAGfGAGCsUsG
527	AS267	UsfCsGUUfAAACAAUfAGfGAGCsUsG
528	AS268	UsfCsGUUfAAACAAUfAGfGAGCsUsG
529	AS269	UsfCsGUUfAAfCAAUfAGfGAGCsUsG
530	AS270	UsfCsGUUfAAfCAAUfAGfGAGCsUsG
531	AS271	AsfUsAACfUCfUGfUCCAfUUfACCAsUsU
532	AS272	AsfUsAACUCUGUCCAfUUfACCAsUsU
533	AS273	AsfUsAACfUCUGUCCAfUUfACCAsUsU
534	AS274	AsfUsAfACUCfUGUCCAfUUfACfCAUsUsU
535	AS275	AsfUsAACfUCfUGUCCAfUUfACCAsUsU
536	AS276	UsfGsAGCfUCGUAUAAfCfCUGUsUsU
537	AS277	UsfGsAfGCCUfCGUAUAAfCfCUGUsUsU
538	AS278	UsfGsAGCfUCfCGfUAUAAfCfCUGUsUsU

Where, G = 2'-O-methylguanylate, A = 2'-O-methyladenylylate, U = 2'-O-methyluridylylate, C = 2'-O-methylcytidylylate, fG = 2'-fluoroguanlylate, fA = 2'-fluoroadenylylate, fU = 2'-fluorouridylylate, fC = 2'-fluorocytidylylate, Gs = 2'-O-methyl-3'-thioguanlylate, As = 2'-O-methyl-3'-thioadenylylate, Us = 2'-O-methyl-3'-thiouridylylate, Cs = 2'-O-methyl-3'-thiocytidylylate.

[0103] In some embodiments, the RNA inhibitor is selected from the sequences in Table 7 below.

TABLE 7

The modified RNA inhibitors		
Sense strand SEQ ID NO.	Antisense strand SEQ ID NO.	RNA inhibitor code
449	485	Ky11-DS31
450	486	Ky11-DS32
451	487	Ky11-DS33
452	488	Ky11-DS34
453	489	Ky11-DS35
454	490	Ky11-DS36
455	491	Ky11-DS37
456	492	Ky11-DS38
457	493	Ky11-DS39
458	494	Ky11-DS40
459	495	Ky11-DS41
460	496	Ky11-DS42
461	497	Ky11-DS43
462	498	Ky11-DS44
463	499	Ky11-DS45
464	500	Ky11-DS46
465	501	Ky11-DS47
466	502	Ky11-DS48
467	503	Ky11-DS49

TABLE 7-continued

The modified RNA inhibitors		
Sense strand SEQ ID NO.	Antisense strand SEQ ID NO.	RNA inhibitor code
468	504	Ky11-DS50
469	505	Ky11-DS51
470	506	Ky11-DS52
471	507	Ky11-DS53
472	508	Ky11-DS54
473	509	Ky11-DS55
474	510	Ky11-DS56
475	511	Ky11-DS57
476	512	Ky11-DS58
477	513	Ky11-DS59
478	514	Ky11-DS60
479	515	Ky11-DS61
449	533	Ky11-DS62
481	517	Ky11-DS63
482	518	Ky11-DS64
483	519	Ky11-DS65
484	520	Ky11-DS66

RNAi Inhibitors Conjugated to Ligand

[0104] Another aspect of the RNA inhibitor of the present application relates to a method to conjugate interfering nucleic acids to ligand to enhance the stability, activity, cellular distribution or cellular uptake of RNAi agents.

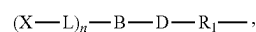
[0105] The sense strand and antisense strand contained in the RNA inhibitors of the present application can be conveniently and routinely prepared by the well-known technique of solid-phase synthesis. Additionally or alternatively, any other method known in the art for such synthesis, such as liquid-phase synthesis or fermentation, may be used. It is also known to use similar techniques to prepare other oligonucleotides, such as phosphorothioates and alkylated derivatives.

[0106] In some embodiments, in addition to standard nucleoside phosphoramidite monomers and non-standard nucleoside phosphoramidite monomers that are commercially available and routinely used in oligonucleotide synthesis, the oligonucleotides or linked nucleotides of the present application can be synthesized by an automated synthesizer using phosphoramidite method derived from the lignd-nucleoside phosphoramidite monomer.

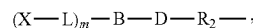
[0107] In some embodiments, the ligand conjugation method described in the present invention is to conjugated the ligand structure to the 5' and/or 3' ends of the antisense strand, and/or the 5' end and/or 3' end of the sense strand.

[0108] For example, the ligand structure can be conjugated to the 5' end and/or 3' end of the sense strand; or the ligand structure can be conjugated to the 5' end of the antisense strand and the ligand structure is coupled to the 3' end of the sense strand; or the ligand structure can be conjugated to the 3' end of the antisense strand, and the ligand is conjugated to the 5' end of the sense strand; or the ligand structure is conjugated to the 5' and 3' ends of the sense strand.

[0109] In some embodiments, the ligand structure includes 5'MVIP and 3'MVIP, wherein the 5'MVIP is conjugated at the 5' end of the sense strand and/or antisense strand, the 3'MVIP is conjugated at the 3' end of the antisense strand and/or sense strand, the structure of the 5'MVIP is shown in formula I, the structure of the 3'MVIP is shown in formula II,



I



II

[0110] wherein,

[0111] X is a targeting specific ligand;

[0112] L is a branched chain;

[0113] B is a linker;

[0114] D is a linking chain;

[0115] R_1 and R_2 are transition points;

[0116] The 5'MVIP is connected with the 5' end of the sense strand or the 5' end of the antisense strand through the transition point R_1 , and the 3'MVIP is connected with the 3' end of the sense strand or the 3' end of the antisense strand through the transition point R_2 . n and m are each independently any integer from 0 to 4.

[0117] In some embodiments, the X is tissue targeting specific ligand.

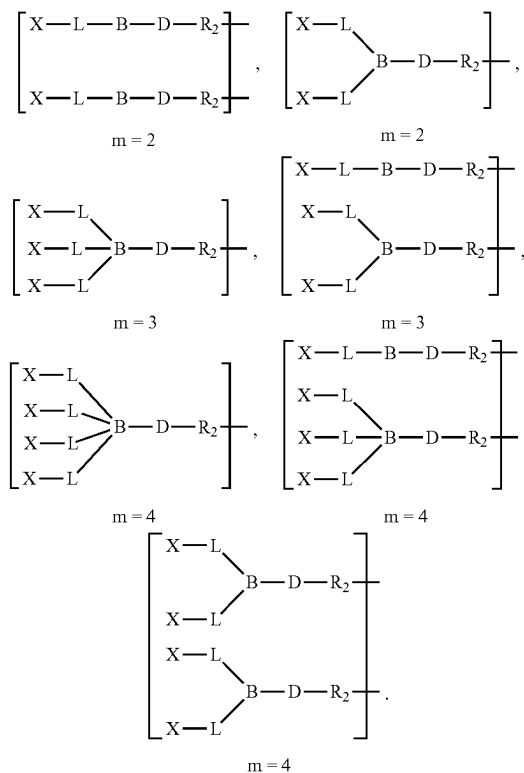
[0118] In some embodiments, the X is liver targeting specific ligand.

[0119] In some embodiments, the R_1 or R_2 is connected to the sense or antisense strand through a phosphate or modified phosphate, preferably through a phosphate or phosphorothioate.

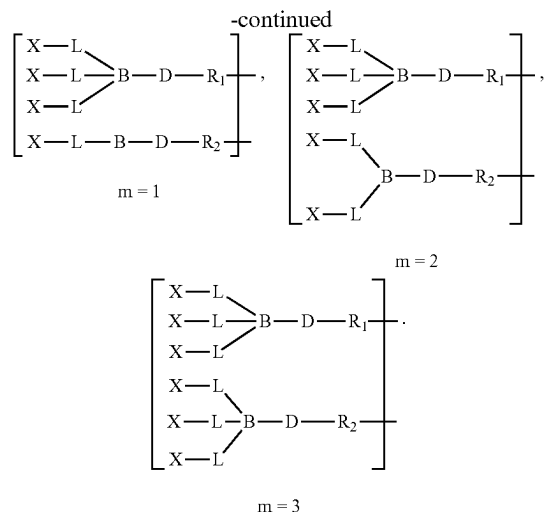
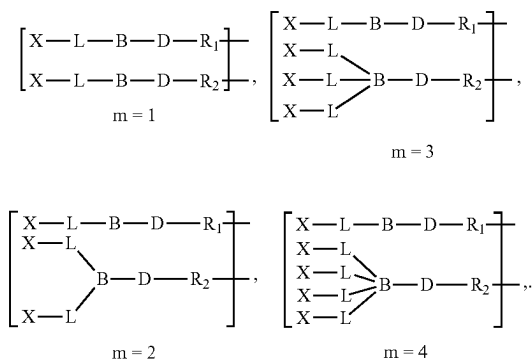
[0120] In some embodiments, $n+m$ is an integer of 2-6, preferably $n+m=2, 3$ or 4, more preferably 4.

[0121] In some embodiments, m or n can be 0, that is, there is no 3'MVIP or 5'MVIP.

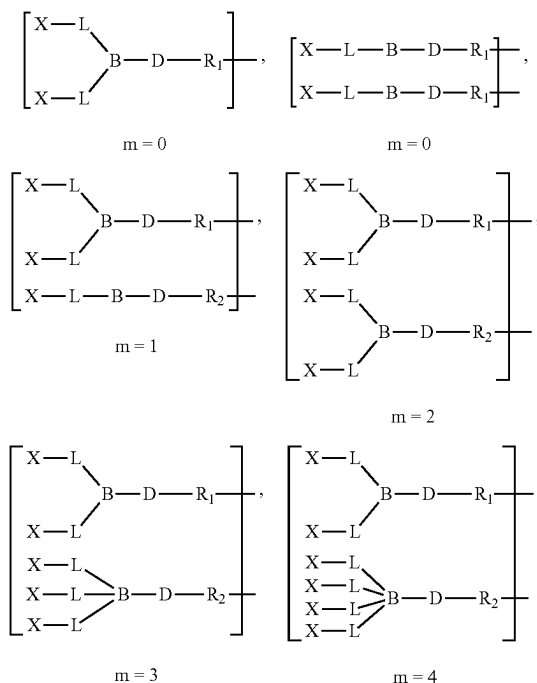
[0122] In some embodiments, when $n=0$ (that is, there is no 5'MVIP), the structure of the 3'MVIP can be:



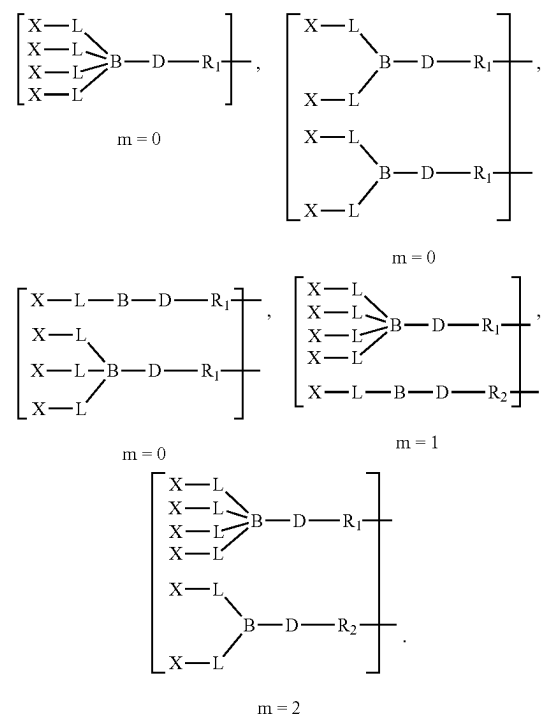
[0123] In some embodiments, when $n=1$, the structure of the 3' MVIP may be:



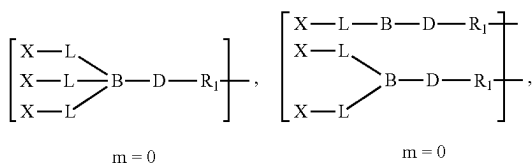
[0124] In some embodiments, when $n=2$, the structure of the 3' MVIP may be:



[0126] In some embodiments, when $n=4$, the structure of the 3' MVIP may be:



[0125] In some embodiments, when $n=3$, the structure of the 3' MVIP may be:



[0127] In some embodiments, the n refers to the sum of n in the 5'MVIP at 5' end of the sense strand and antisense strand of the RNAi agent at the same time, and the m refers to the sum of m in the 3'MVIP at 3' end of the sense strand and antisense strand of the RNAi agent at the same time.

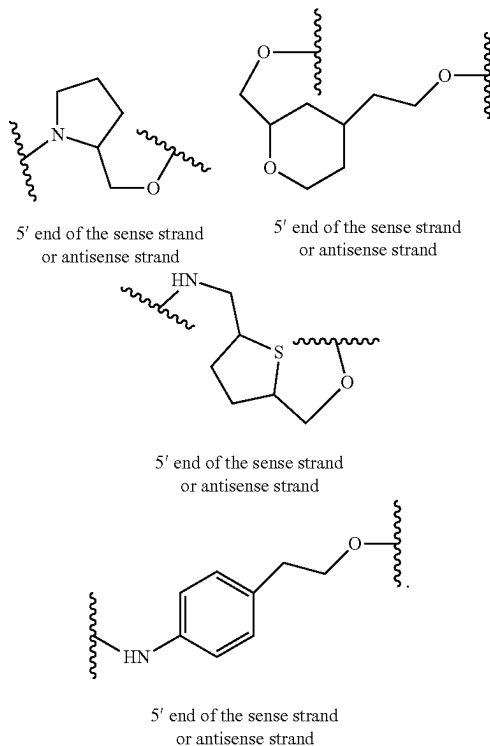
[0128] In some embodiments, the transition points R_1 and R_2 contain $-\text{NH}-$, $-\text{S}-$, and/or $-\text{O}-$ in their structures, R_1 and R_2 are respectively connected with the linking chain D and the 5' and 3' ends of the sense strand and/or antisense strand through $-\text{NH}-$, $-\text{S}-$, or $-\text{O}-$, and R_1 and R_2 are the same or different.

[0129] In some embodiments, the R_1 and R_2 are optionally straight chain, or straight chain or cyclic structure with amido, carboxyl, or alkyl branched chain, and the cyclic structure includes saturated or unsaturated aliphatic carbocyclyl, or five or six membered heterocyclyl or aromatic hydrocarbonyl containing sulfur, oxygen, or nitrogen atom.

[0130] In some embodiments, the R_1 and/or R_2 are $-E_1(CH_2)_xCH_2E_2-$, wherein x is any integer of 3-12, and the groups E_1 and E_2 can be $-NH-$, $-S-$, or $-O-$, respectively.

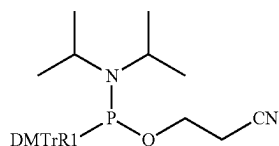
[0131] In some embodiments, the R_1 and/or R_2 are $-E_1(CH_2)_{x1}CH(OH)(CH_2)_{x2}E_2-$, wherein $X1$ or $X2$ are each independently any integer of 3-10, and E_1 and E_2 can be $-NH-$, $-S-$, or $-O-$, respectively.

[0132] In some embodiments, the R_1 is a heterocyclic or carbocyclic structure containing N, S or O:

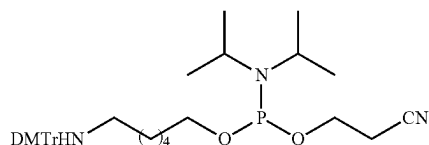


[0133] In some embodiments, the transition point R_1 is $-NH(CH_2)_xCH_2O-$, wherein x is any integer of 3-12, preferably any integer of 4-6, which can be introduced by the following two phosphoramidite monomers:

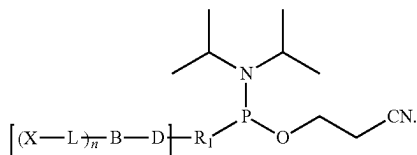
[0134] i. One of $-O-$ or $-S-$ is used for the synthesis of R_1 phosphoramidite monomer, and is connected to the 5' end of the sense strand or antisense strand of RNA inhibitor by solid-phase synthesis. In the structure, $-NH-$, $-S-$, or $-O-$ is used to connect with the linking chain D in the 5'MVIP, thereby introducing the liver targeting specific ligand X at the 5' end of the sense strand or antisense strand of the RNA inhibitor. An exemplary structure of a monomer introduced into the 5' end of the sense or antisense strand of the RNA inhibitor is as follows:



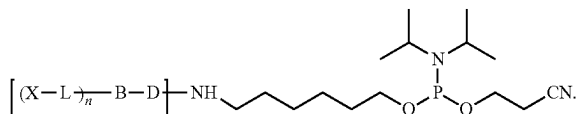
[0135] In some embodiments, the following structure is preferred:



[0136] ii. In R_1 structure, $-NH-$, $-S-$, or $-O-$ is first connected with the linking chain D, and another $-NH-$, $-S-$, or $-O-$ is used to form ester with phosphoramidite in the synthesis of 5'MVIP phosphoramidite monomer. The exemplary structure of sense strand or antisense strand 5'MVIP phosphoramidite monomer is as follows:



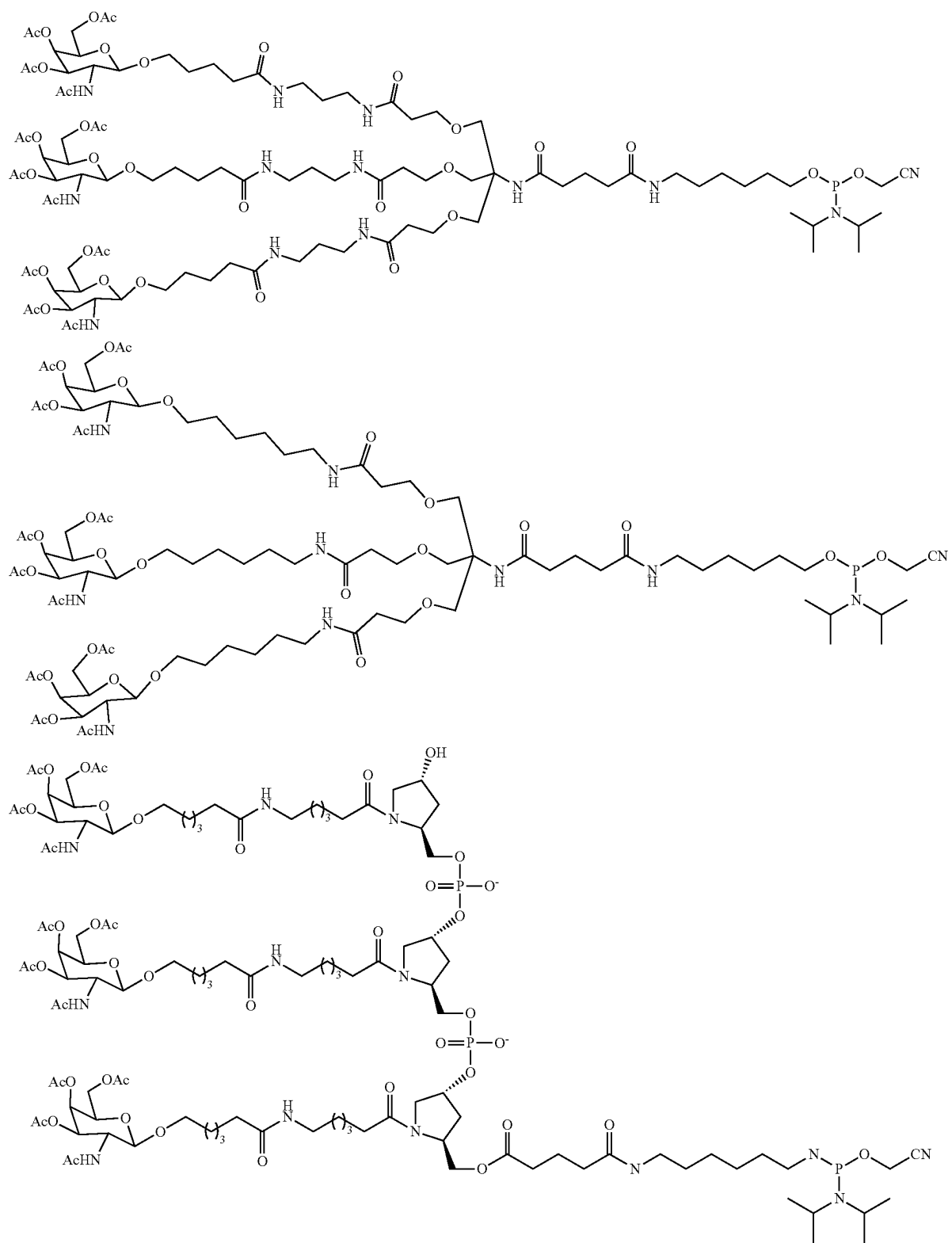
[0137] In some embodiments, the sense strand or antisense strand 5'MVIP phosphoramidite monomer preferably has the following structure:



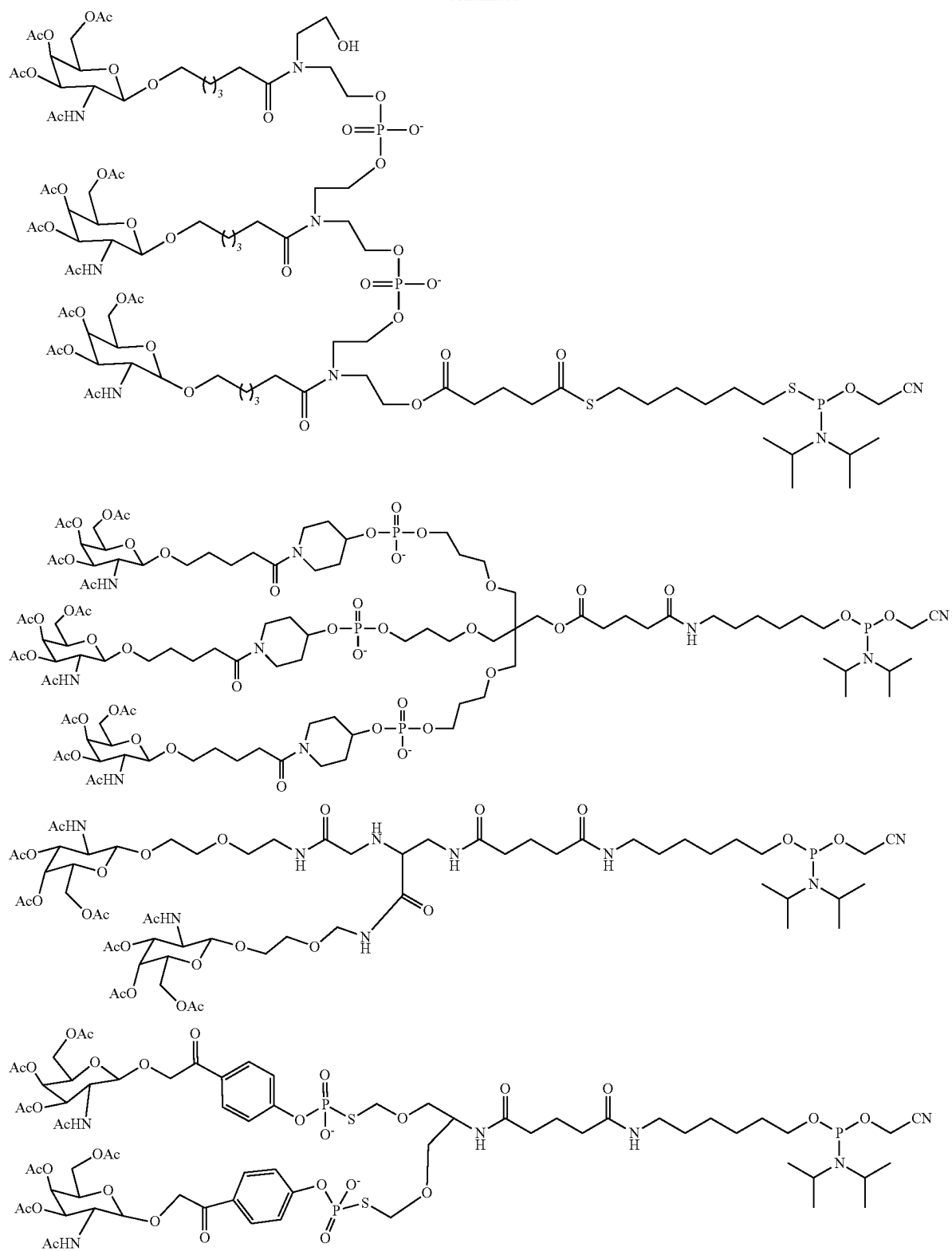
[0138] When n in the general formula is 1-4, the linker B part of the above monomer is branched for 1 to 4 times respectively to obtain the corresponding monomer compounds. With the help of the above monomer compounds, the liver targeting specific ligand X is introduced to the 5' end of the sense or antisense strand through solid-phase synthesis.

[0139] In some embodiments, the transition point R_1 is $-NH(CH_2)_xCH_2O-$, wherein x can be an integer of 3-12, preferably an integer of 4-6.

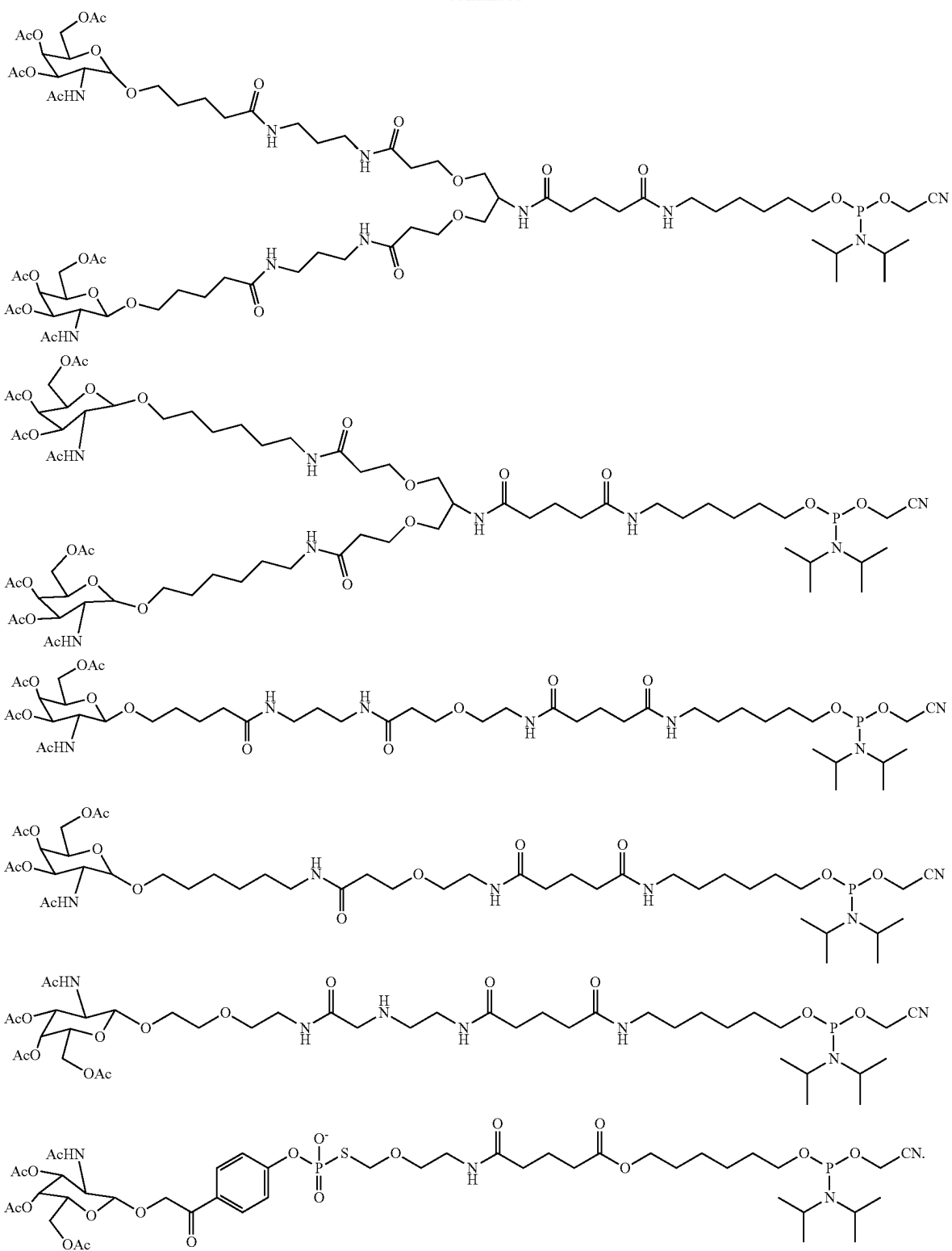
[0140] In some embodiments, the 5'MVIP phosphoramidite monomer has the structure selected from the following structures:



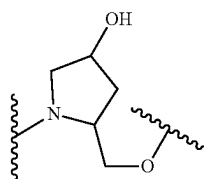
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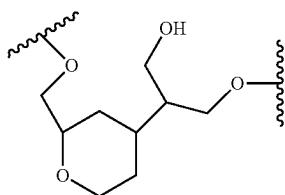
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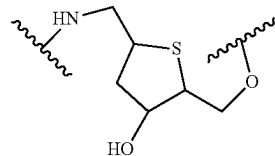
[0141] In some embodiments, the transition point R_2 is a heterocyclic or carbocyclic structure containing N, S or O:



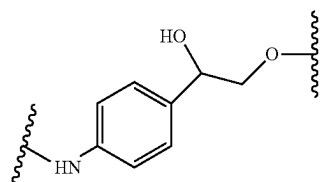
3' end of the sense strand
or antisense strand



3' end of the sense strand
or antisense strand



3' end of the sense strand
or antisense strand



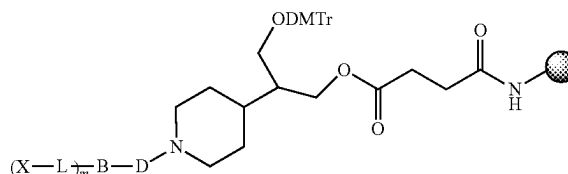
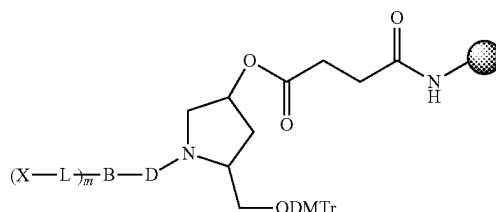
3' end of the sense strand
or antisense strand

[0142] In some embodiments, the R_2 is $\text{—NH(CH}_2\text{)}_{x1}\text{CH(OH)(CH}_2\text{)}_{x2}\text{CH}_2\text{O—}$, wherein $x1$ is any integer of 1-4 and $x2$ is any integer of 0-4.

[0143] The R_2 described in the present application forms ester or amide with —NH— , —S— , or —O— in the R_2 structure through succinic anhydride, and at the same time conjugates with —NH— in blank solid support to form 3'MVIP solid support, and then 3'MVIP is introduced into the 3' end of the sense strand or antisense strand by phosphoramidite solid-phase synthesis.

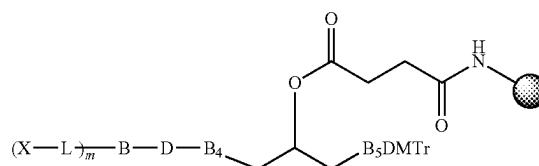
[0144] In some embodiments, the heterocycle in the R_2 structure is a pyrrole ring or a piperidine ring, which is connected with the linking chain D of 3'MVIP through the

nitrogen heteroatom in the ring. The exemplary structures of 3'MVIP solid support are as follows:



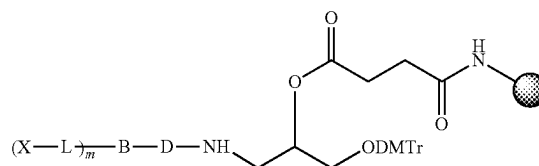
[0145] When m in the general formula is 1-4, the linker B part in the monomer above is branched for 1 to 4 times respectively to obtain the corresponding solid supports.

[0146] In some embodiments, R_2 is $\text{—B}_4(\text{CH}_2)_{x1}\text{CH(OH)(CH}_2\text{)}_{x2}\text{CH}_2\text{B}_5\text{—}$, wherein $x1$ is an integer of 1-4, $x2$ is an integer of 0-4, and B_4 and B_5 are —NH— , —S— , or —O— , respectively, the exemplary formula of the introduced 3' MVIP solid support is as follows:



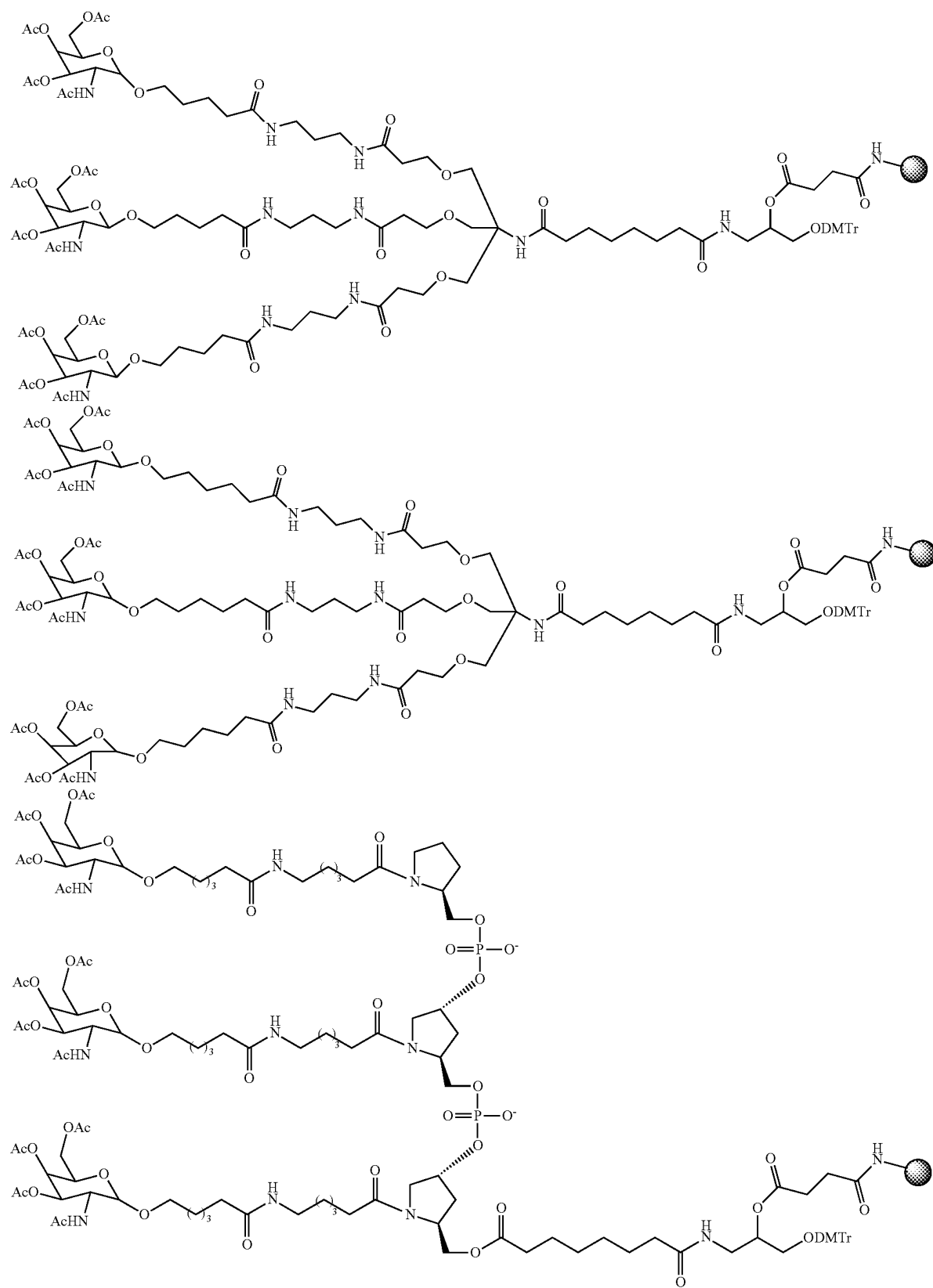
[0147] When m is 1-4, the linker B part in the above monomer is branched for 1 to 4 times to obtain the corresponding Solid Supports.

[0148] In some embodiments, the R_2 is $\text{—NHCH}_2\text{CH(OH)CH}_2\text{O—}$. The exemplary formula of the introduced 3' MVIP solid support is as follows:

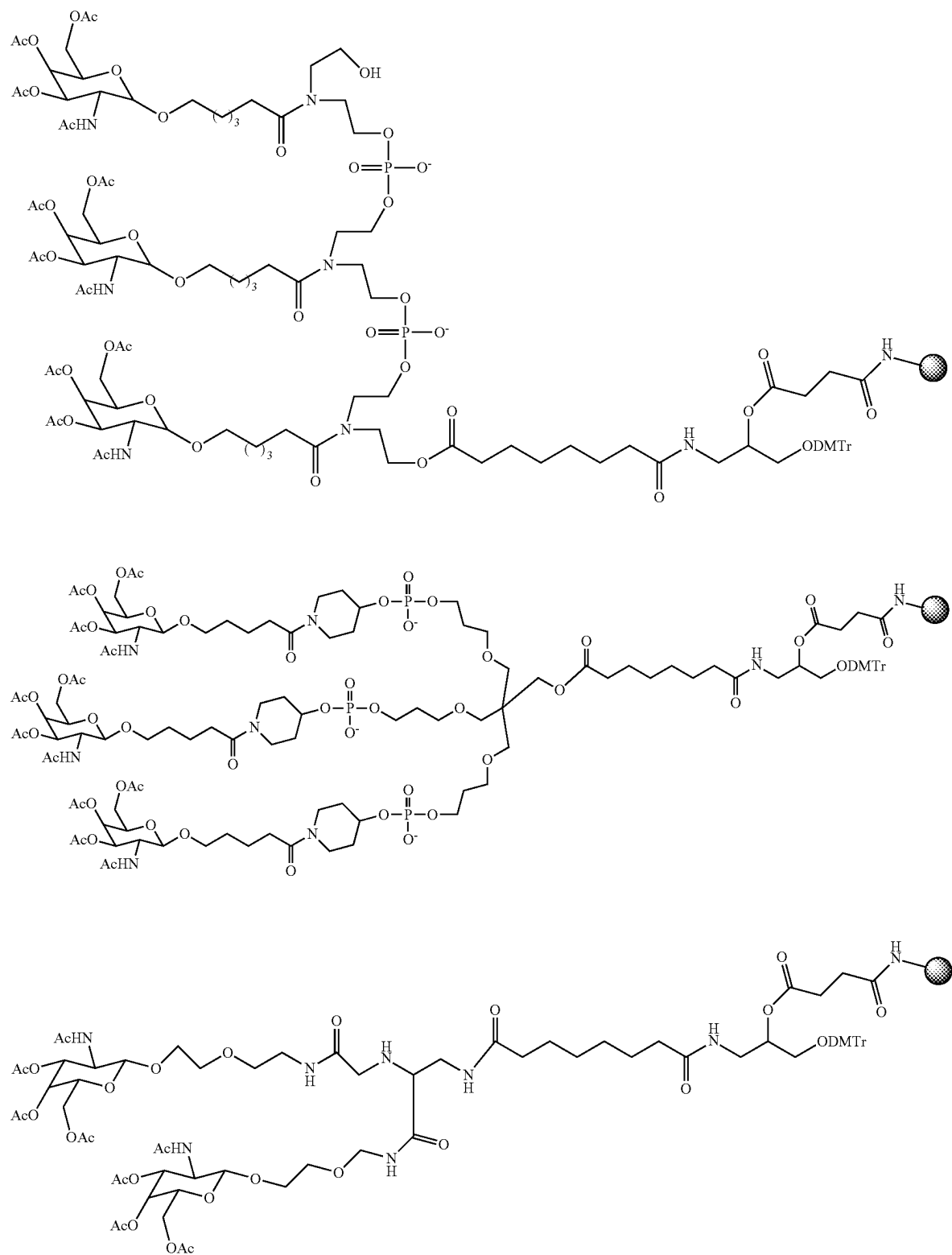


[0149] When m in the general formula is 1-4, the linker B part in the monomer above is branched for 1 to 4 times respectively to obtain the corresponding solid supports.

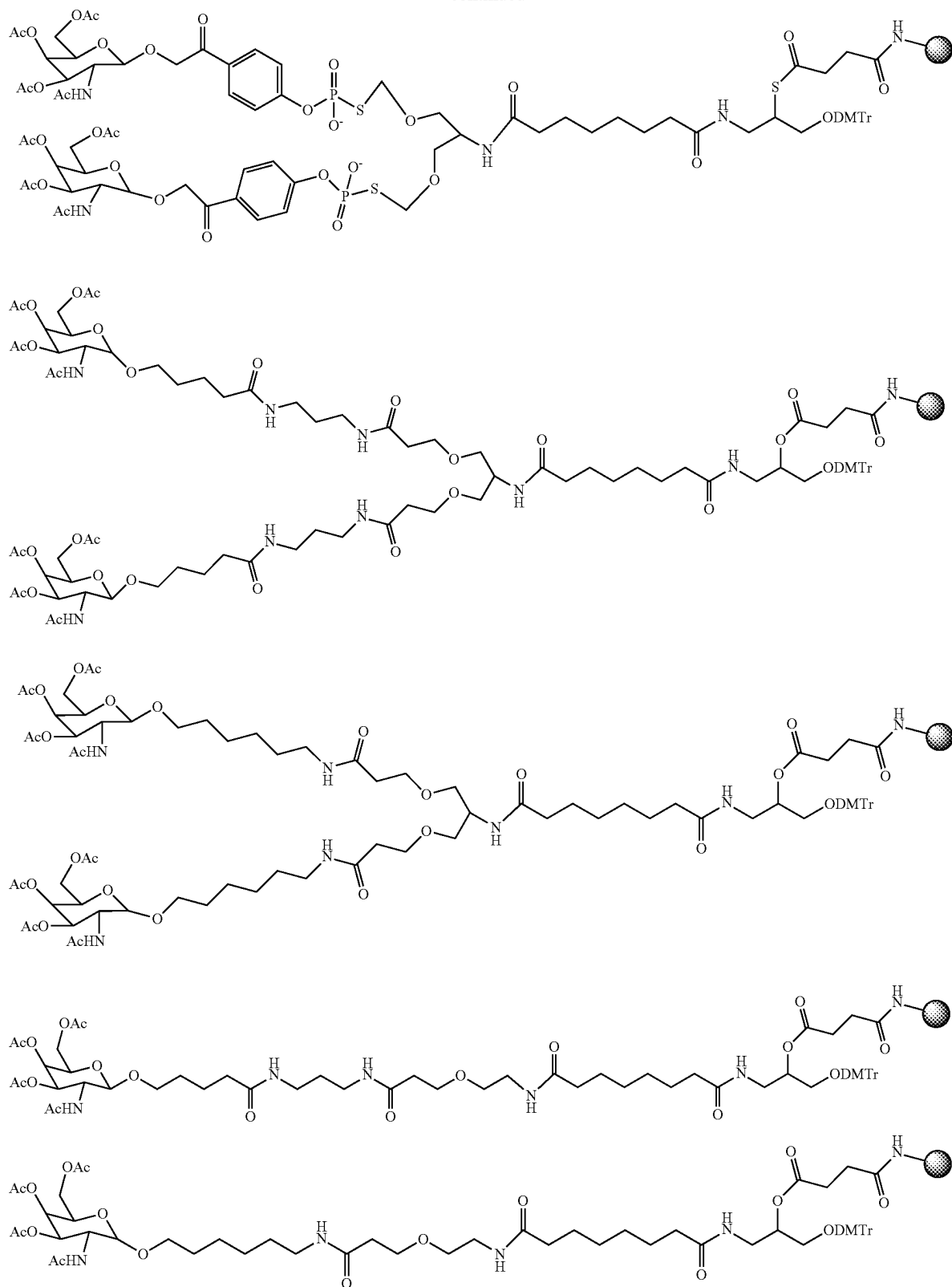
[0150] In some embodiments, the 3'MVIP solid support has a structure as follows:



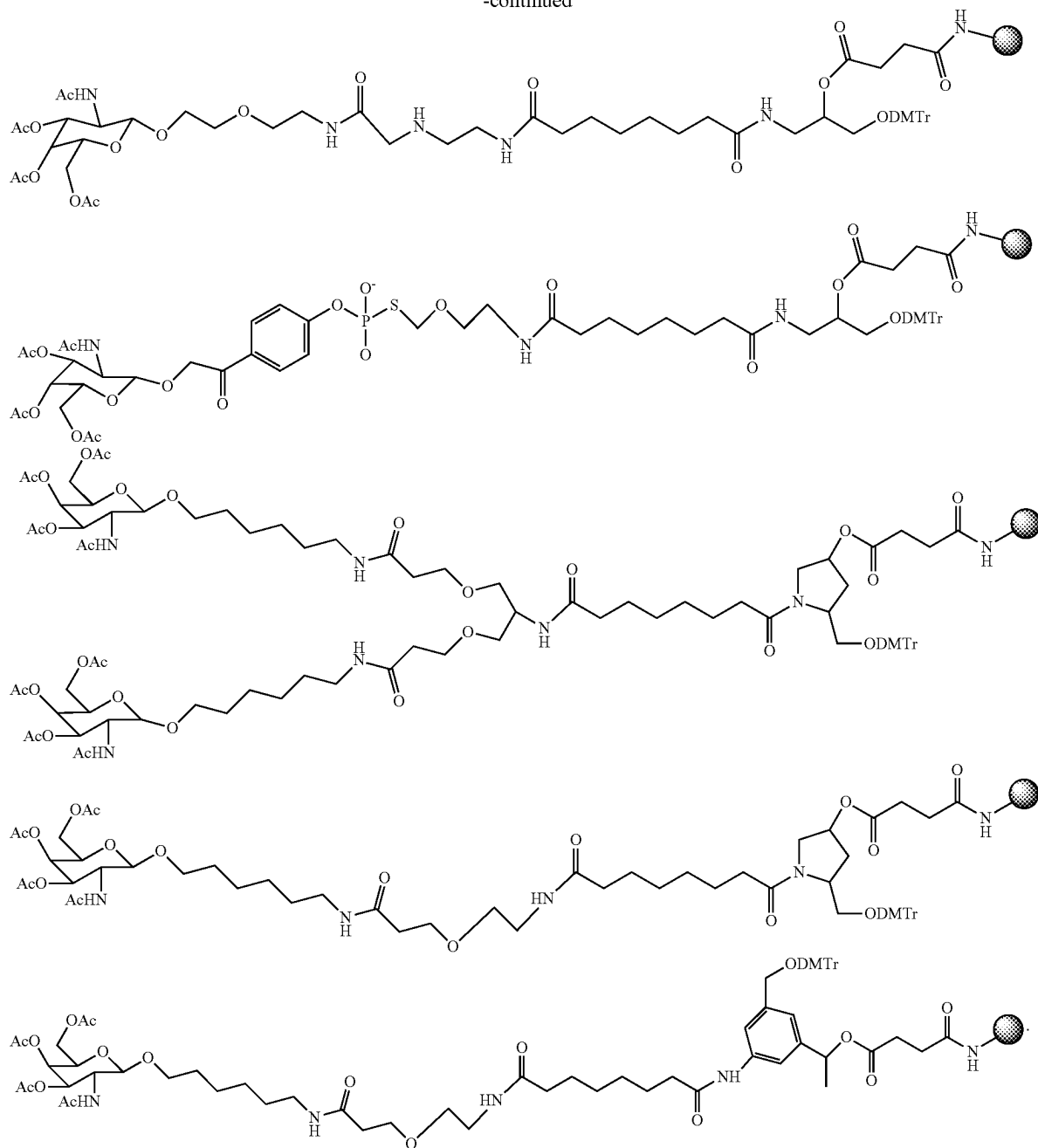
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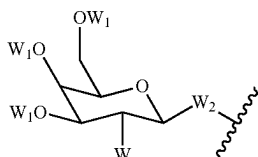
[0151] In some embodiments, the liver targeting specific ligand X is selected from structures used to enhance the uptake of RNAi agents by hepatocytes, which can be lipids, steroids, vitamins, sugars, proteins, peptides, polyamines and peptide mimetic structures. In the RNAi agent provided by the present application, the liver targeting specific ligand X introduced into the end of the sense strand or antisense strand of the RNAi agent can be the same or different. For example, in terms of characteristics, some can enhance liver targeting, some can be the structure that regulates the pharmacokinetics of the RNAi agent in vivo, and some can be the structure with dissolution activity in vivo. In some

embodiments, the liver targeting specific ligand X is selected from one or more monosaccharides and derivatives thereof in the following structures.

[0152] In some embodiments, the monosaccharide is selected from one or more of the following structures: mannose, galactose, D-arabinose, glucose, fructose, xylose, glucosamine, ribose. The monosaccharide derivative is selected from mannose derivatives, galactose derivatives, glucose derivatives, ribose derivatives, and other derivatives.

[0153] In some embodiments, the liver targeting specific ligand X is selected from galactose, galactosamine,

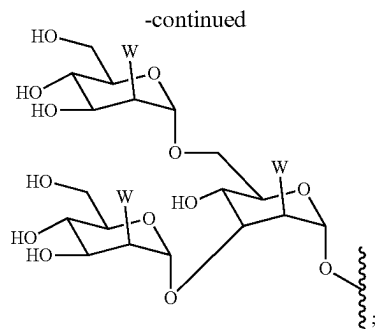
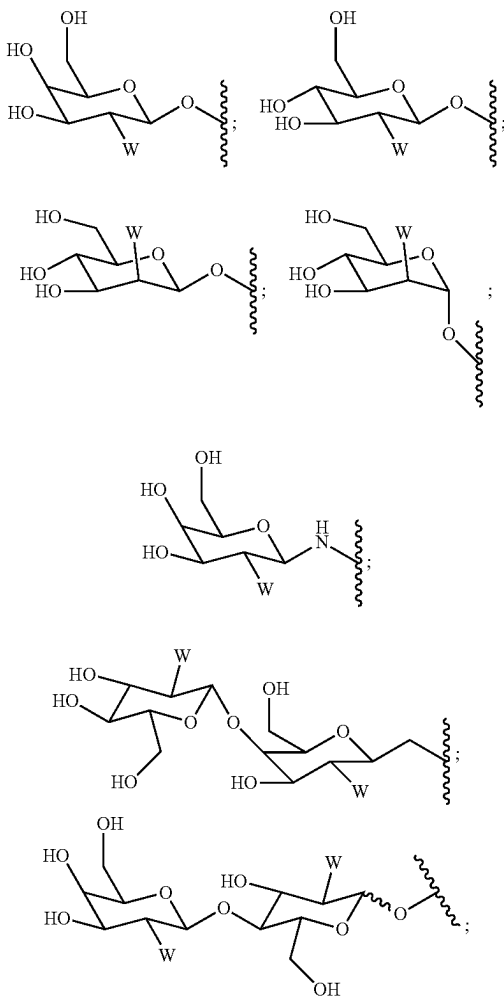
N-acetylgalactosamine and their derivatives, and its general structural formula is as follows:



[0154] wherein, W_1 is a hydrogen or hydroxyl protecting group, which can be the same or different; W is $-\text{OH}$, $-\text{NHCOOH}$ or $-\text{NHCO}(\text{CH}_2)_q\text{CH}_3$, wherein q is an integer of 0-4; W_2 is $-\text{NH}-$, O , S or C .

[0155] In some embodiments, the liver targeting specific ligand X is N-acetylgalactosamine and its derivatives.

[0156] In some embodiments, the liver targeting specific ligand X is selected from the following structures:



[0157] wherein W is selected from one or two of $-\text{OH}$, $-\text{NHCOOH}$ or $-\text{NHCO}(\text{CH}_2)_q\text{CH}_3$, wherein q is an integer of 0-4.

[0158] In some embodiments, the liver targeting specific ligand X may be the same or different in the same 5'MVIP or 3'MVIP structure.

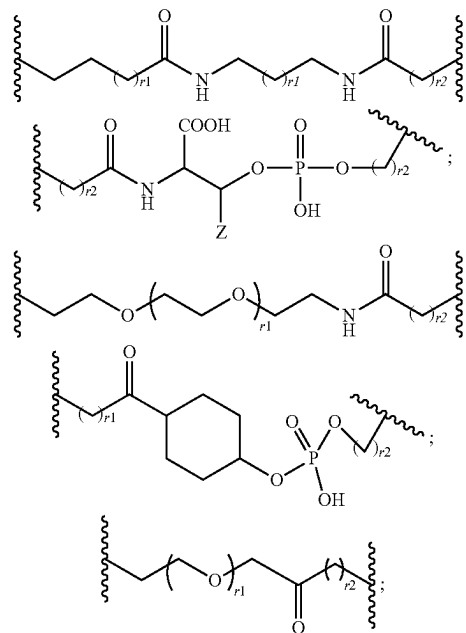
[0159] In some embodiments, the X between 5'MVIP and 3'MVIP can be the same or different.

[0160] In some embodiments, the branched chain L is a C_4 - C_{18} straight chain containing $-\text{NH}-$, $-\text{C}(=\text{O})-$, $-\text{O}-$, $-\text{S}-$, amido, phosphoryl, thiophosphoryl, C_4 - C_{10} aliphatic carbocyclyl, phenyl, or a combination of these groups.

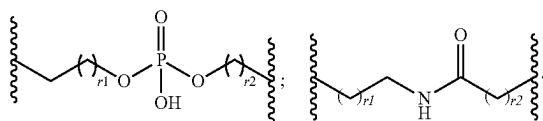
[0161] In some embodiments, the L also has a side chain of hydroxyethyl or carboxylic acids group.

[0162] In some embodiments, the L is C_7 - C_{18} straight chain containing amido group or six-membered aliphatic carbocyclyl group.

[0163] In some embodiments, the L is selected from one or more of the following structures:



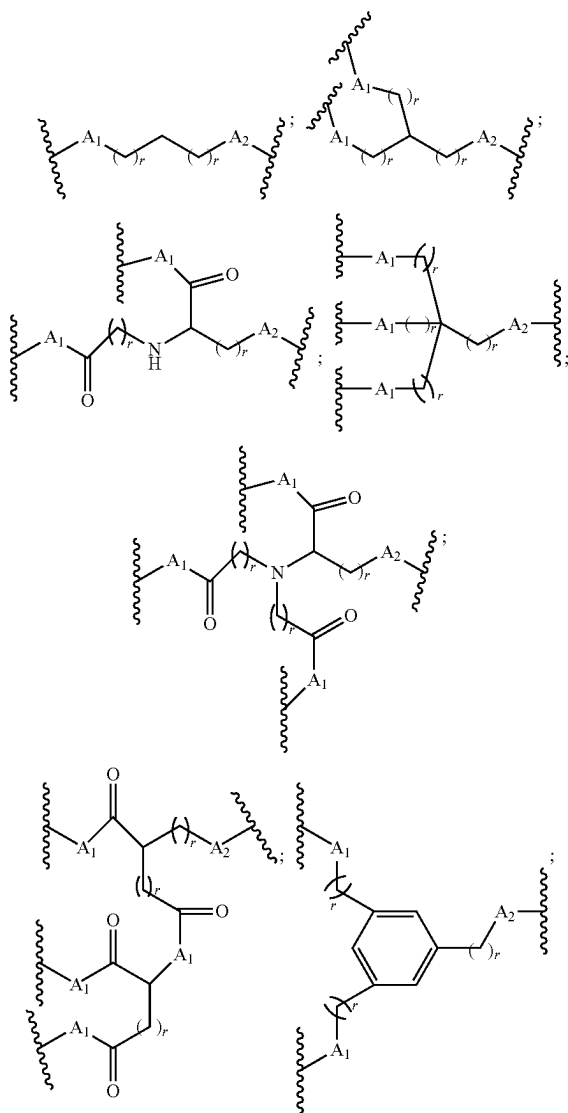
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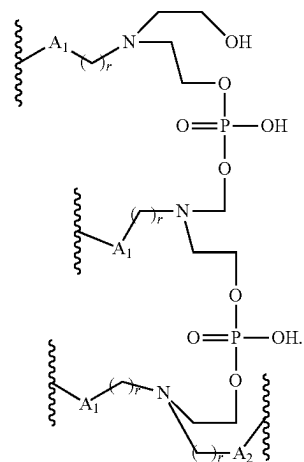
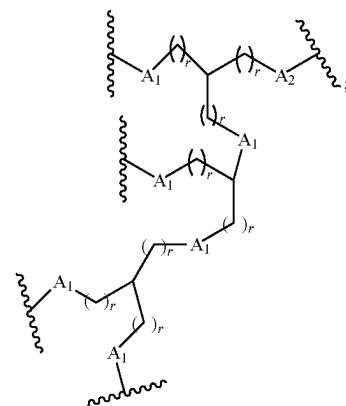
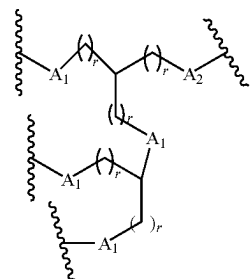
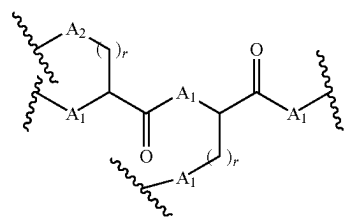
[0164] Wherein r_1 is any integer from 1 to 12, r_2 is any integer from 0 to 20, and Z is H, alkyl or amido group.

[0165] In some embodiments, the structure of the linker B is related to the number of X that can be introduced. The B contains —NH—, C, O, S, amido, phosphoryl, thiophosphoryl. When n or m is 1, it is a straight chain, and when n or m is 2, 3, or 4, the number of branches is 2, 3, or 4, respectively.

[0166] In some embodiments, the B is selected from the following structures:



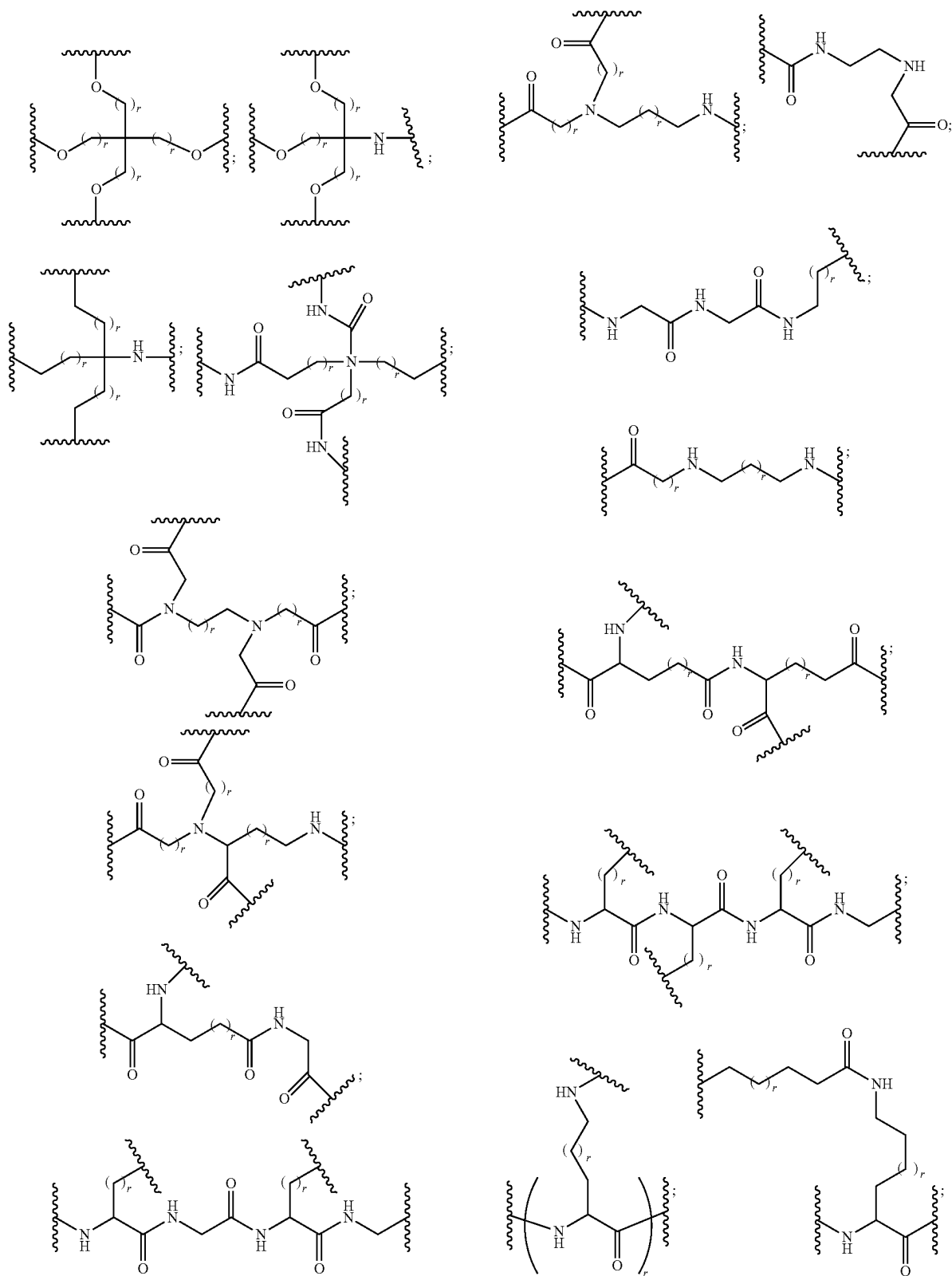
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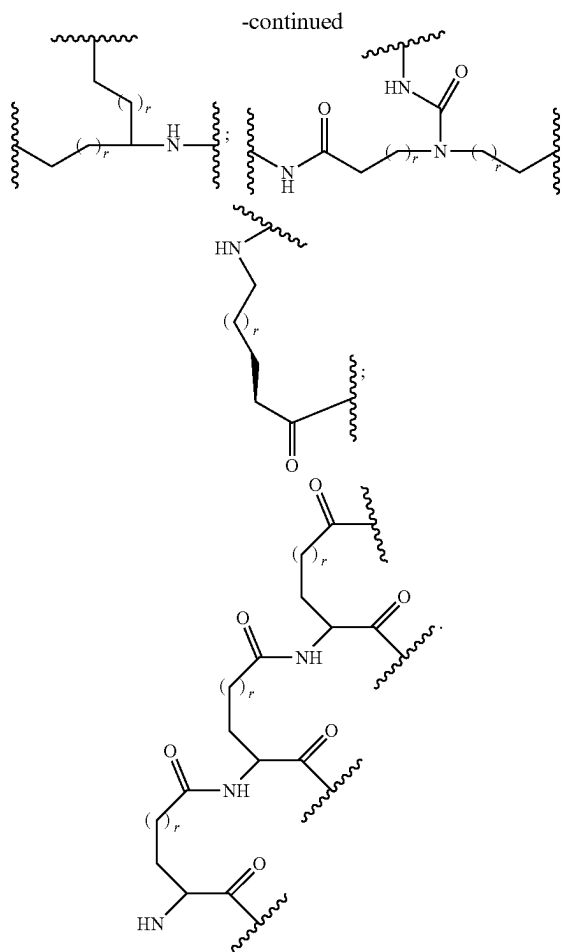
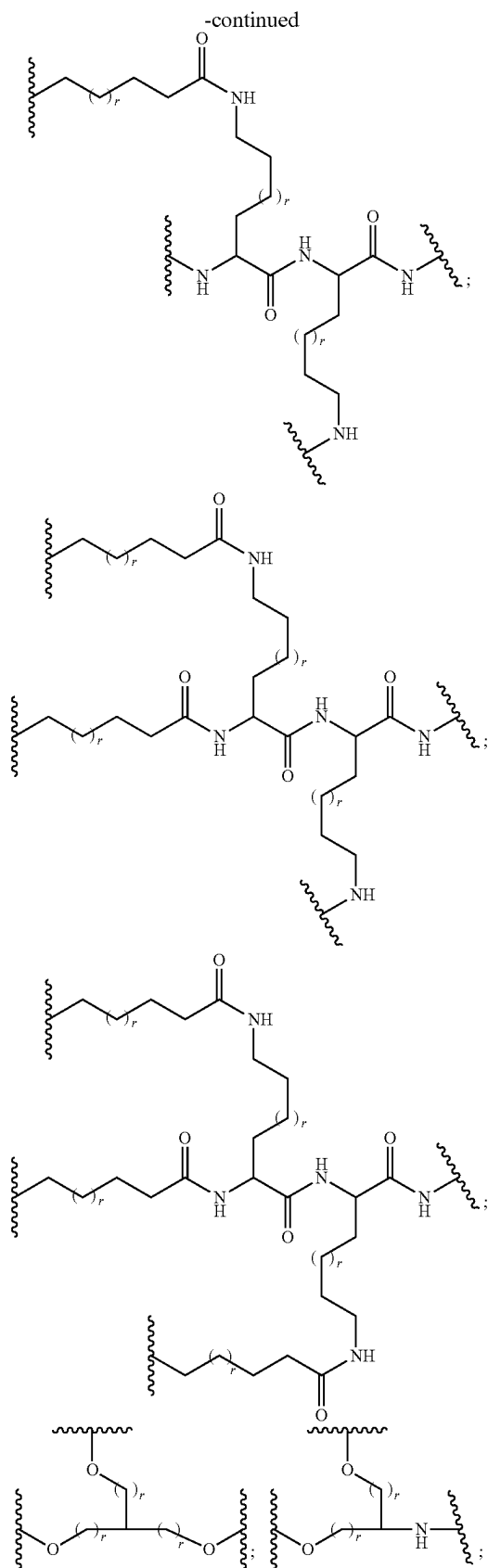


[0167] wherein A_1 and A_2 are each independently C, O, S, —NH—, carbonyl, amido, phosphoryl or thiophosphoryl, and r is an integer of 0-4.

[0168] In some embodiments, the B is selected from the following structures:

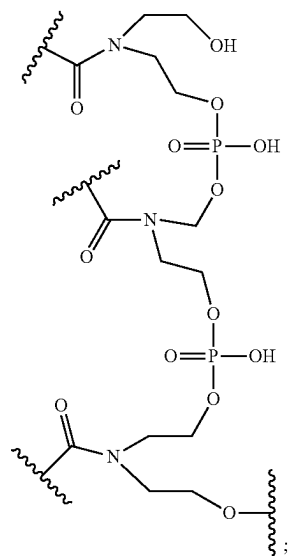
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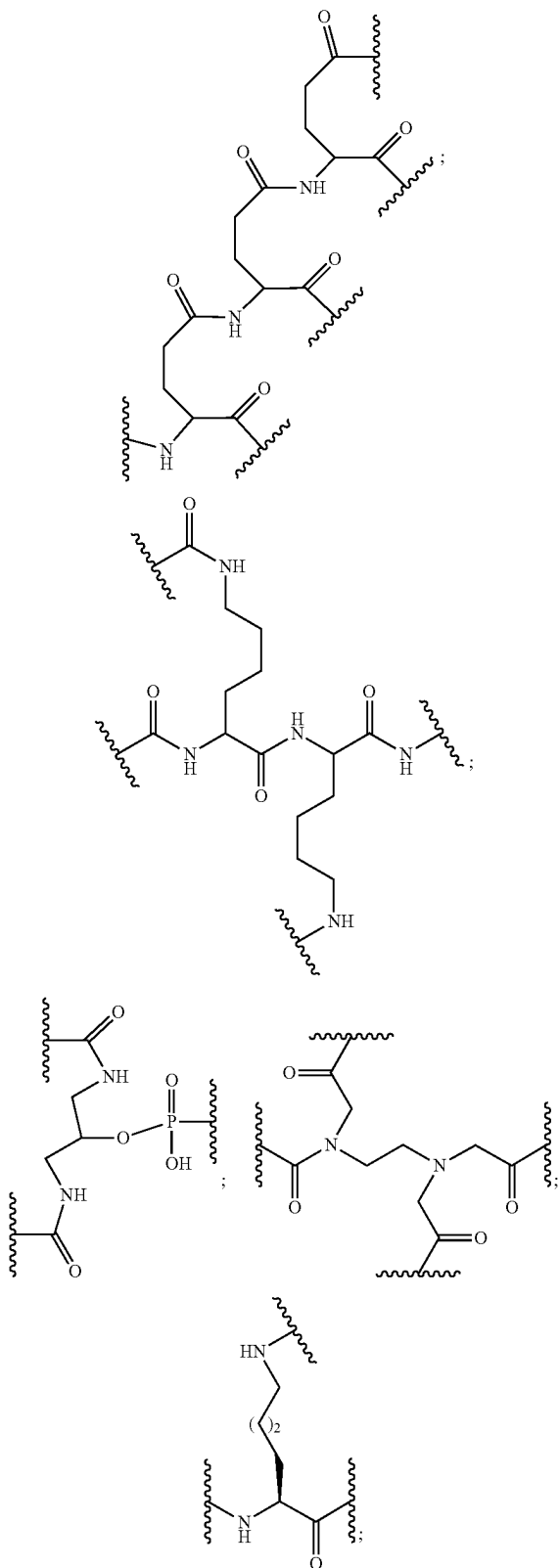


[0169] wherein r is any integer from 0 to 4.

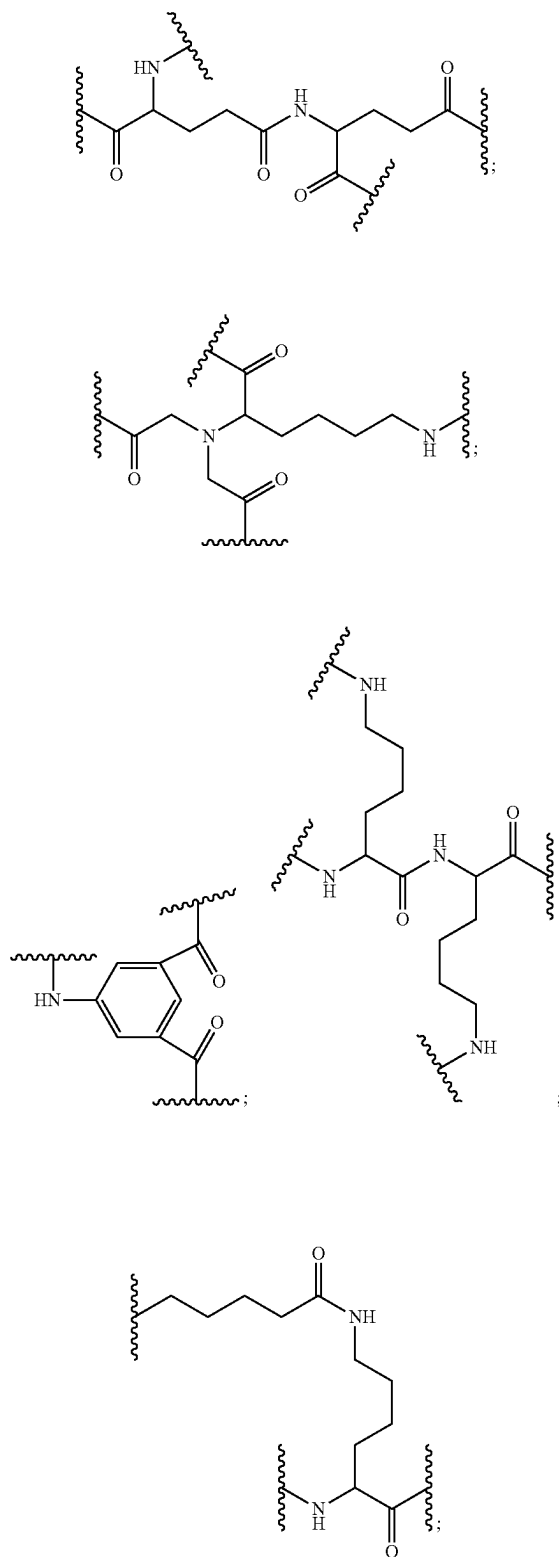
[0170] In some embodiments, the B is selected from the following structures:

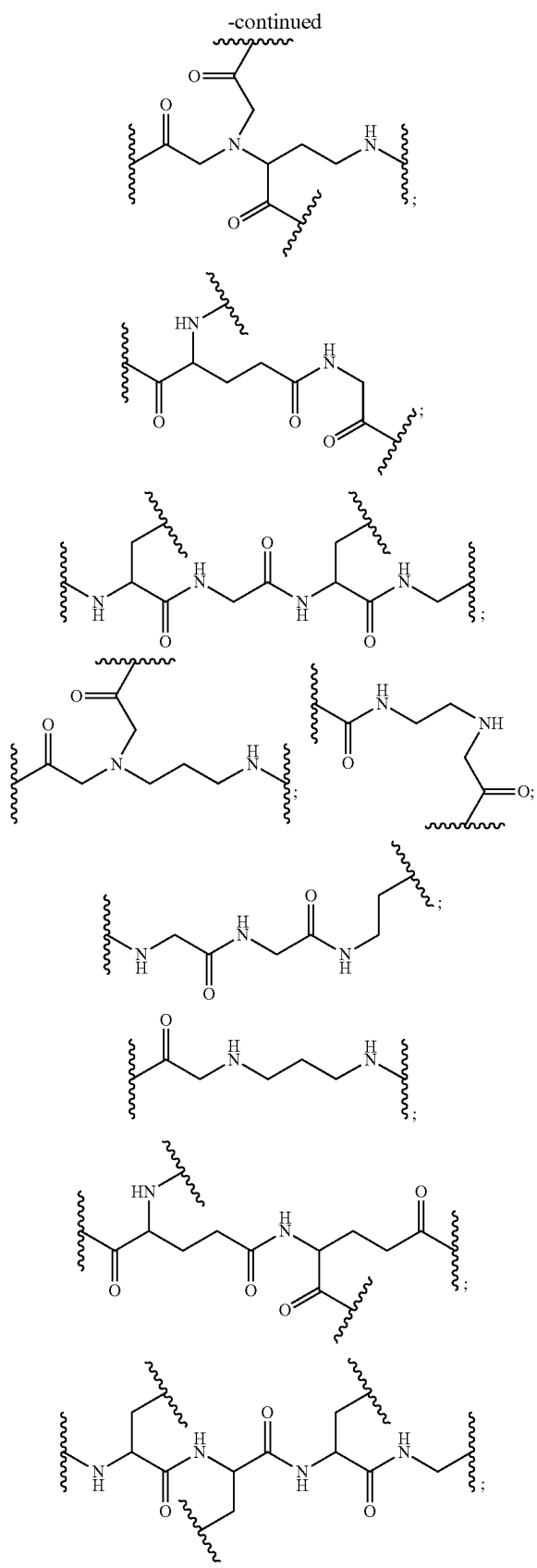
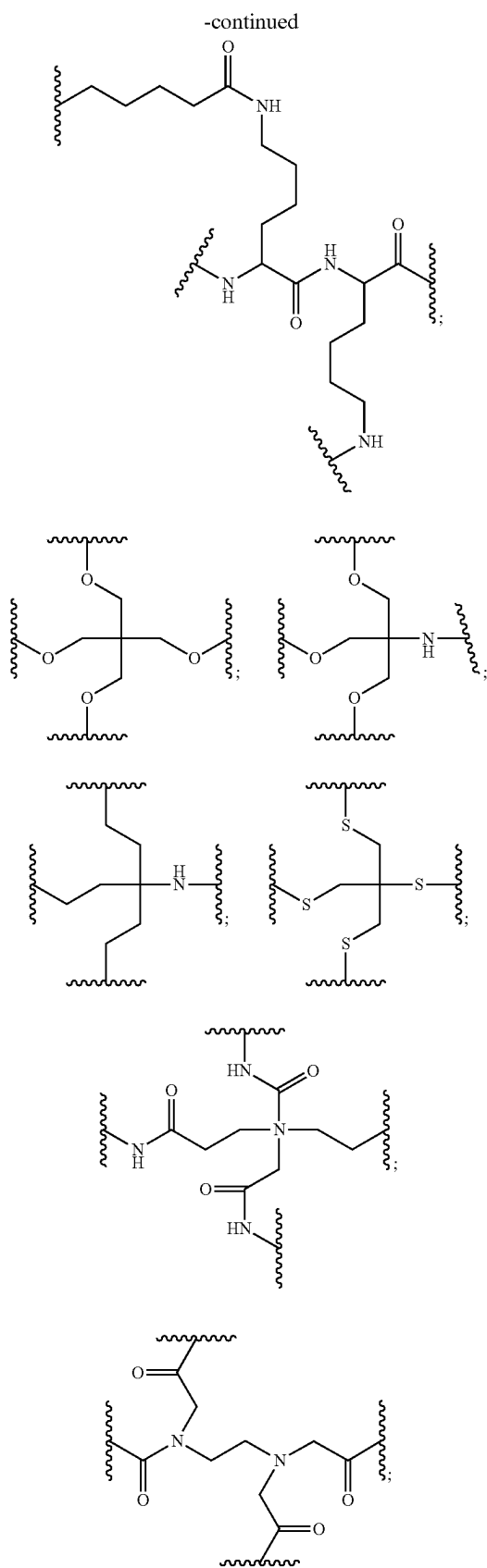


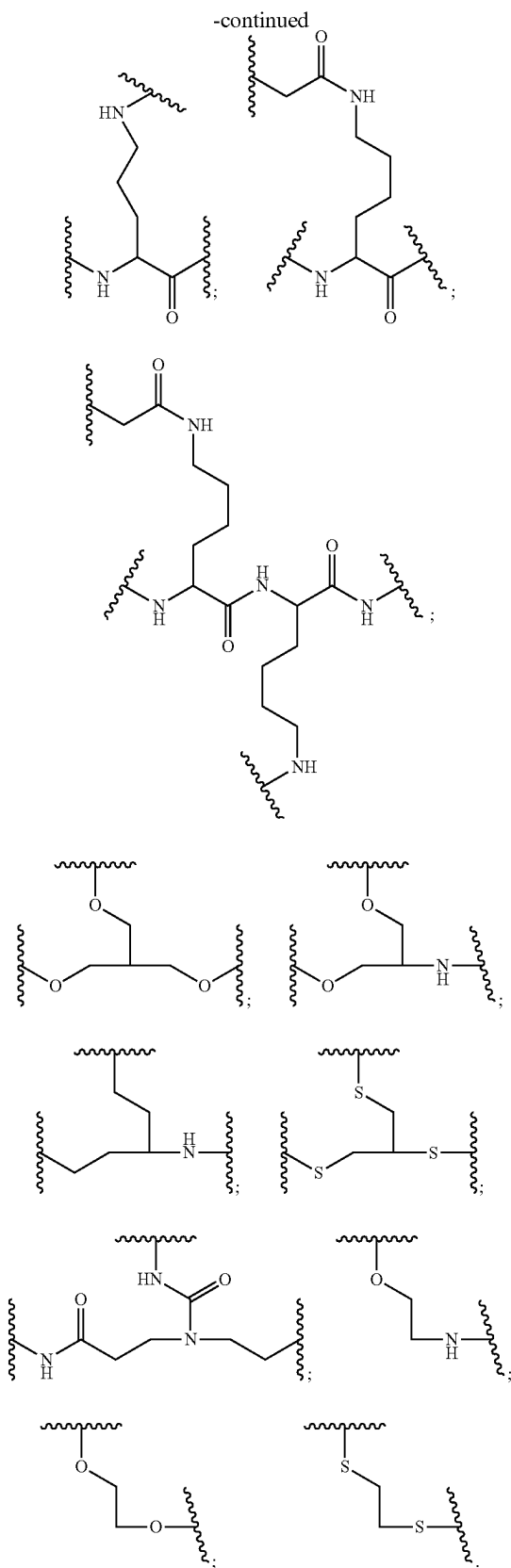
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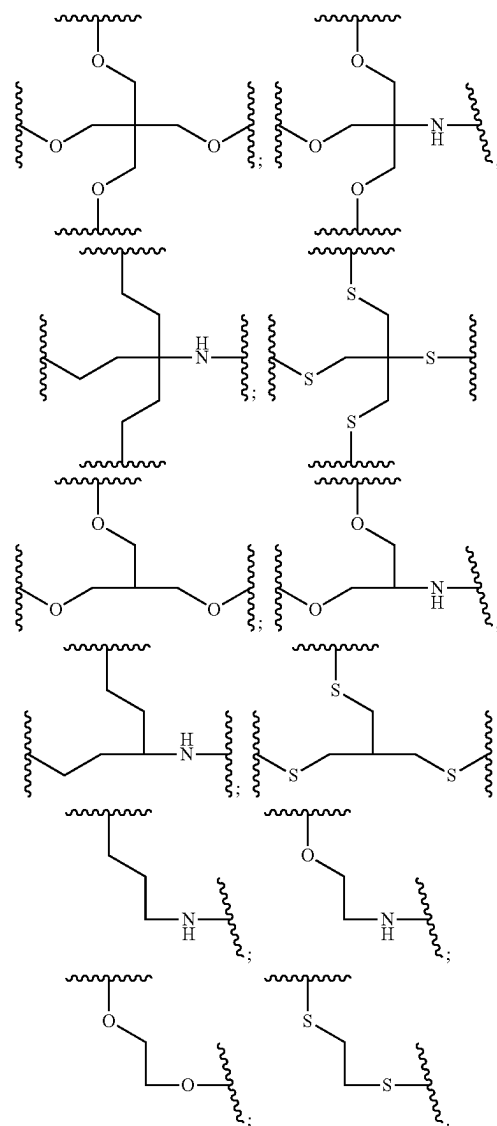
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[0171] In some embodiments, the B is selected from the following structures:



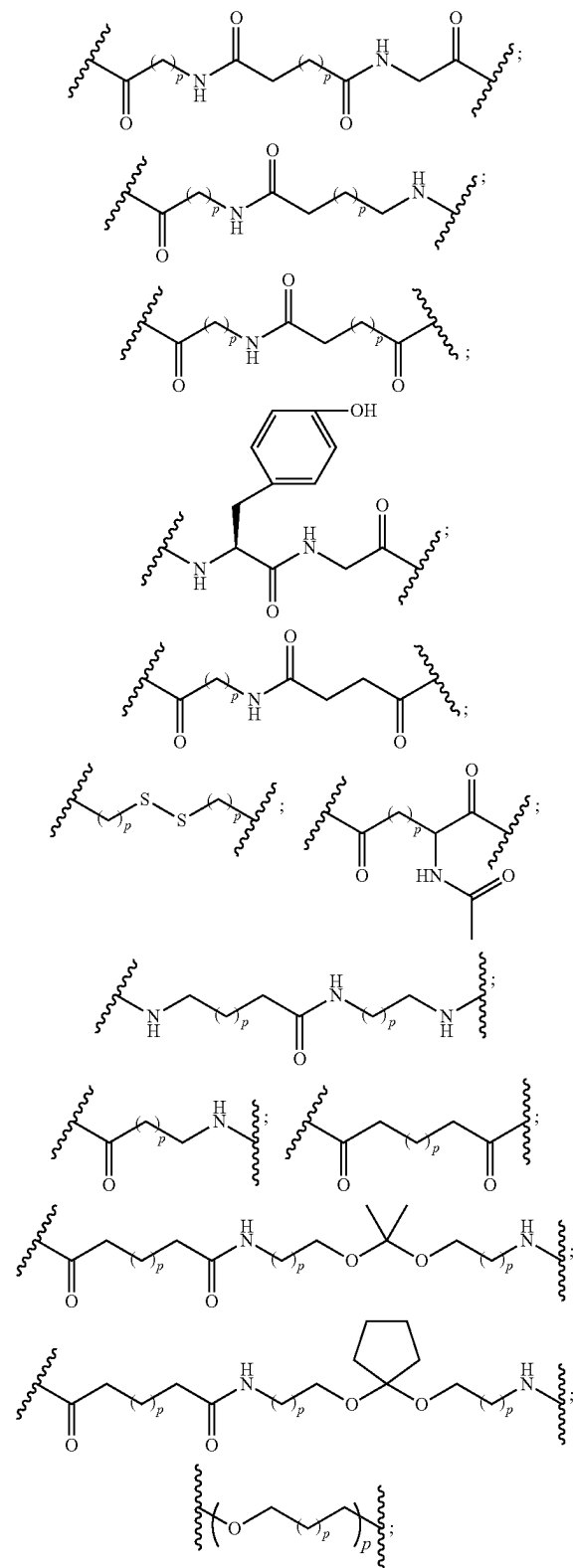
[0172] In some embodiments, the linking chain D is a C_3 - C_{18} straight chain containing $-NH-$, $C=O$, O , S , amido, phosphoryl, thiophosphoryl, aromatic hydrocarbyl, C_4 - C_{10} aliphatic carbocyclyl, five- or six-membered heterocyclyl containing 1-3 nitrogen, or a combination of these groups.

[0173] In some embodiments, the D also has side chains of hydroxymethyl, methyl tert-butyl, methylphenyl, and C_5 - C_6 aliphatic cyclic groups.

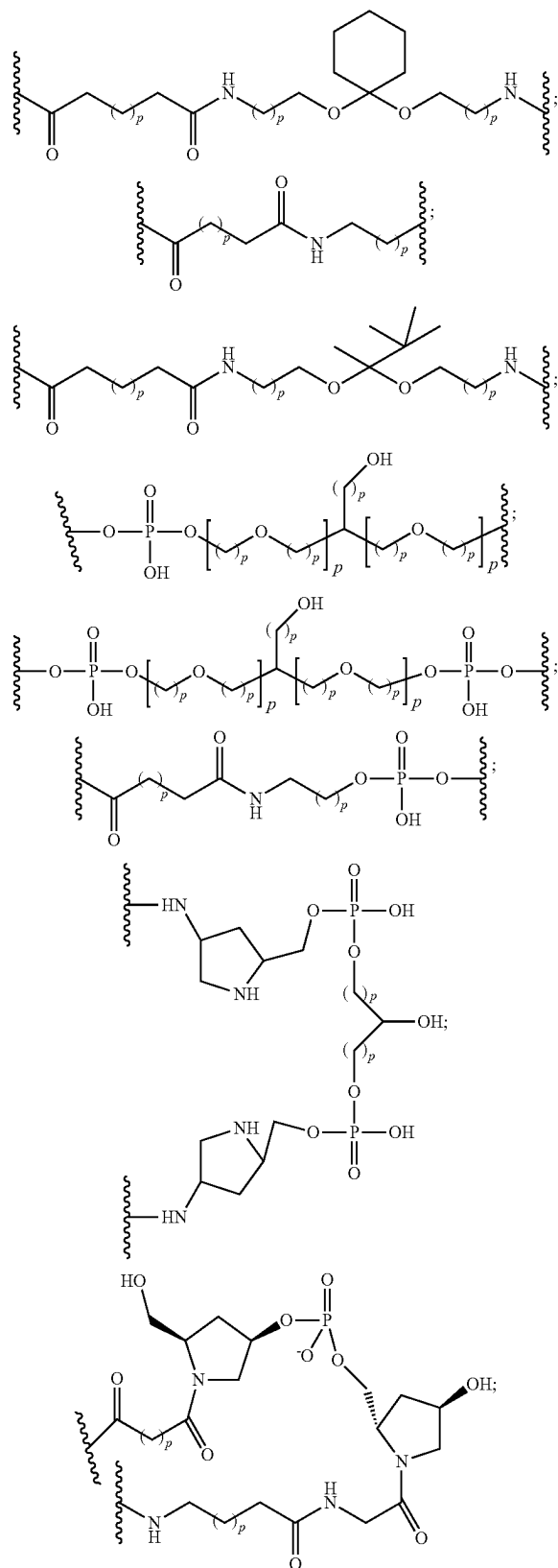
[0174] In some embodiments, the D is a C_3 - C_{10} straight chain containing two $C=O$, six-membered aliphatic carbocyclyl or phenyl groups.

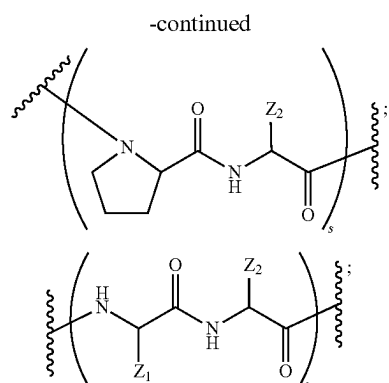
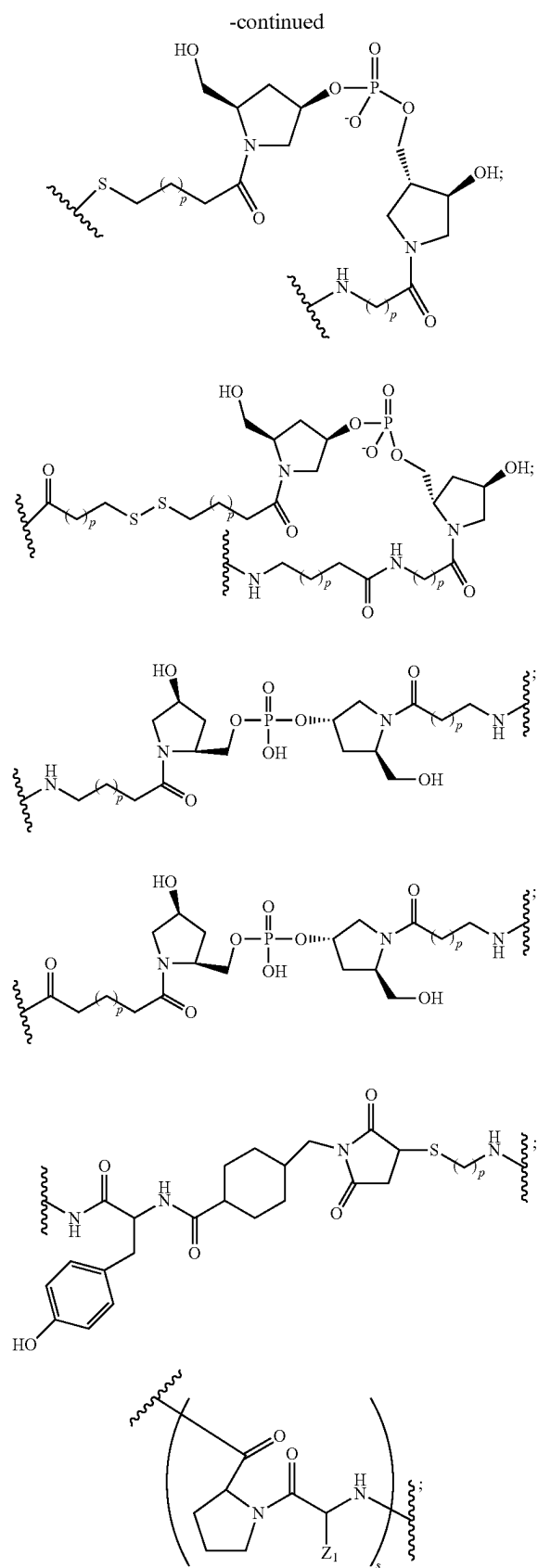
[0175] In some embodiments, the D is a C_3 - C_{10} straight chain containing two $C=O$.

[0176] In some embodiments, the D is selected from the following structures:



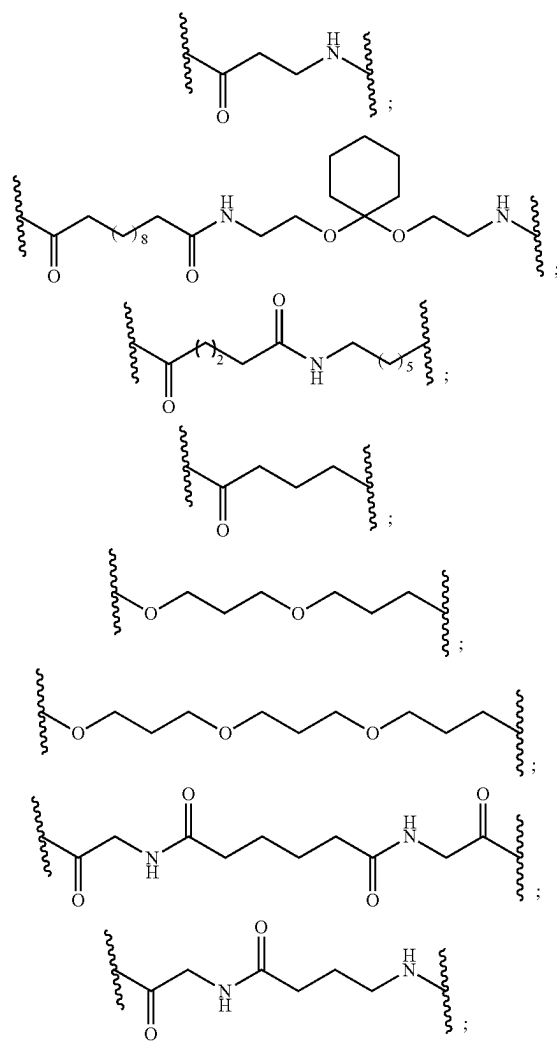
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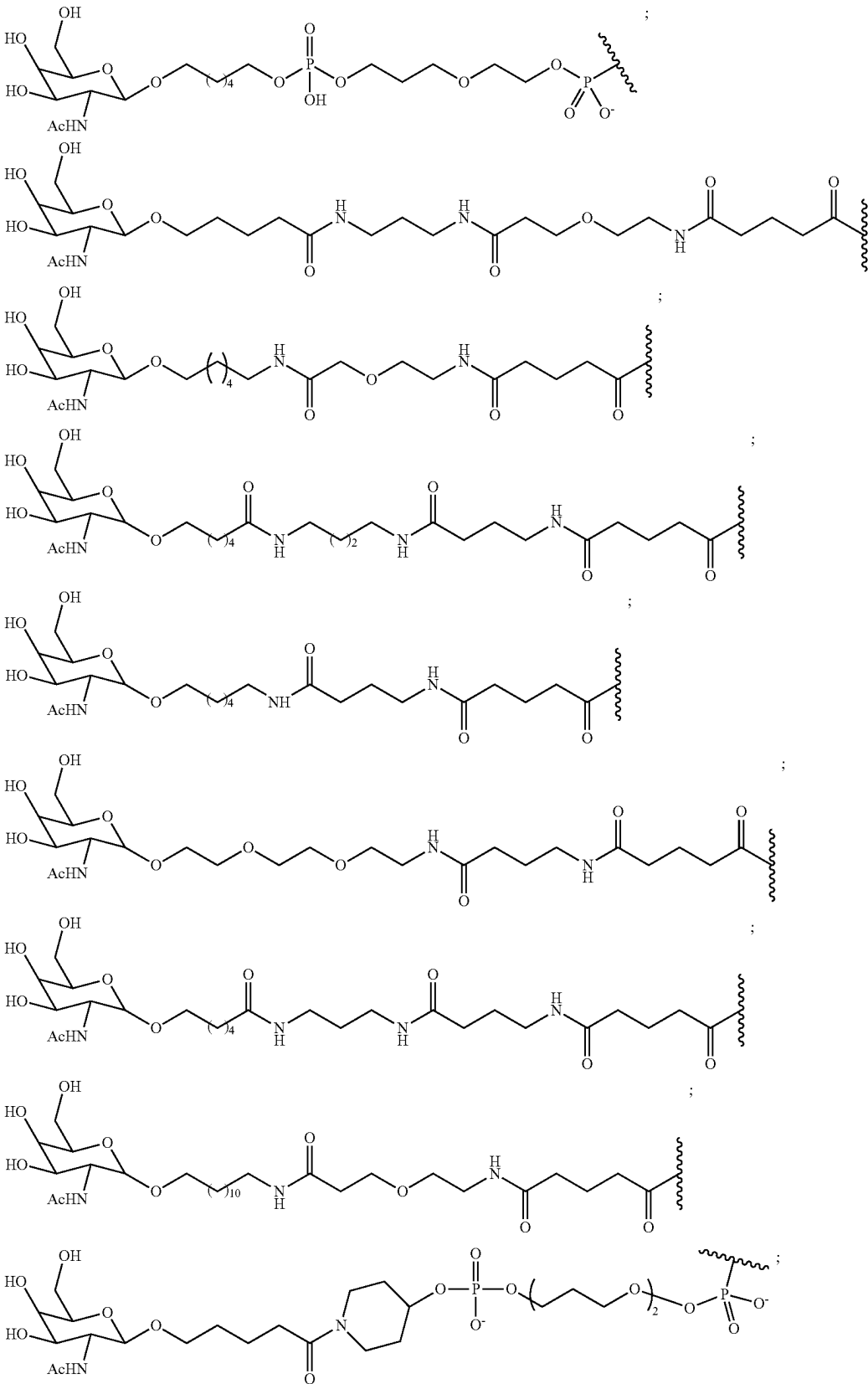




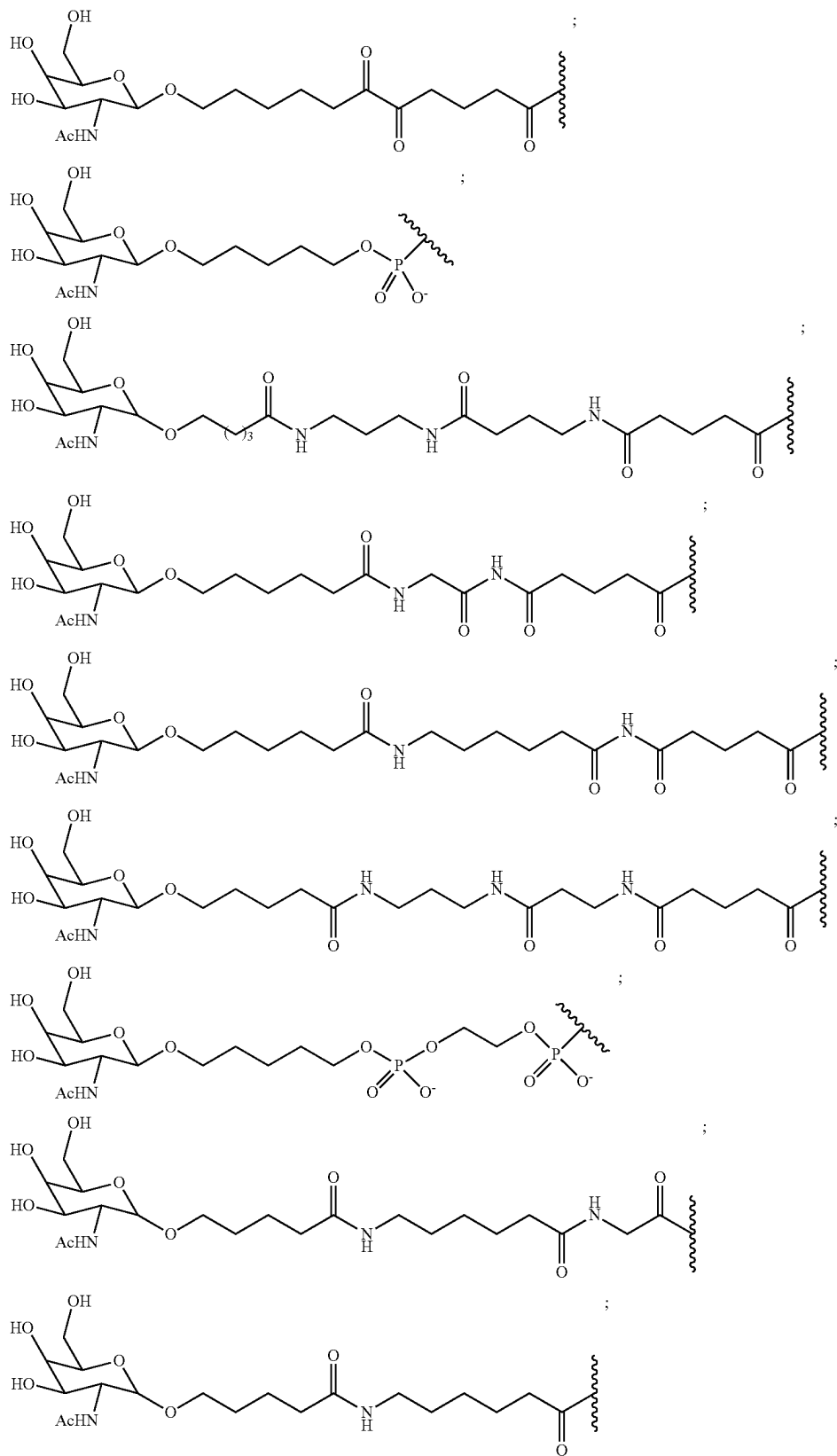
[0177] wherein, each p is independently any integer from 1 to 20; s is an integer of 2-13; Z_1 and Z_2 are the same or different substituents.

[0178] In some embodiments, the D is selected from the following structures:

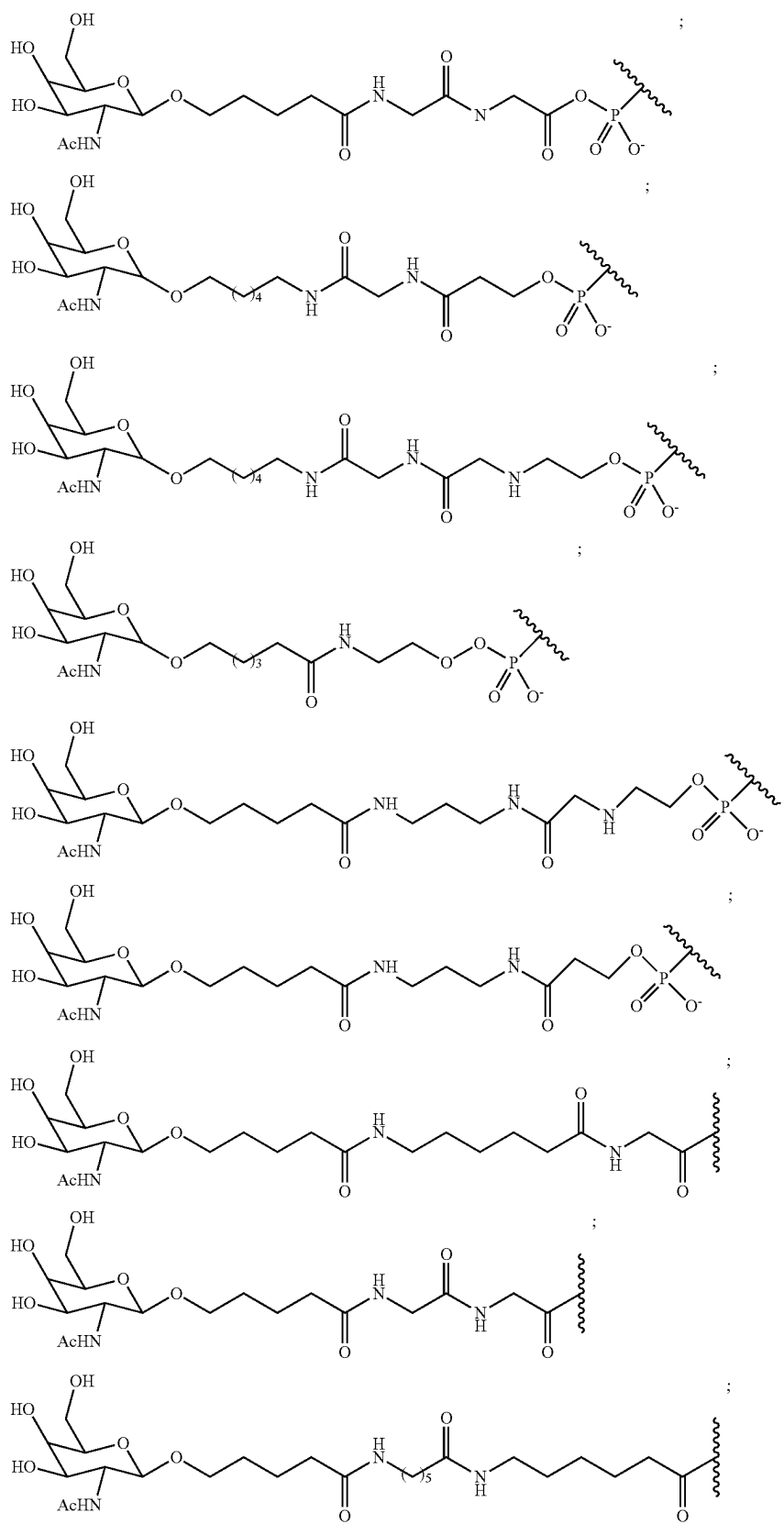




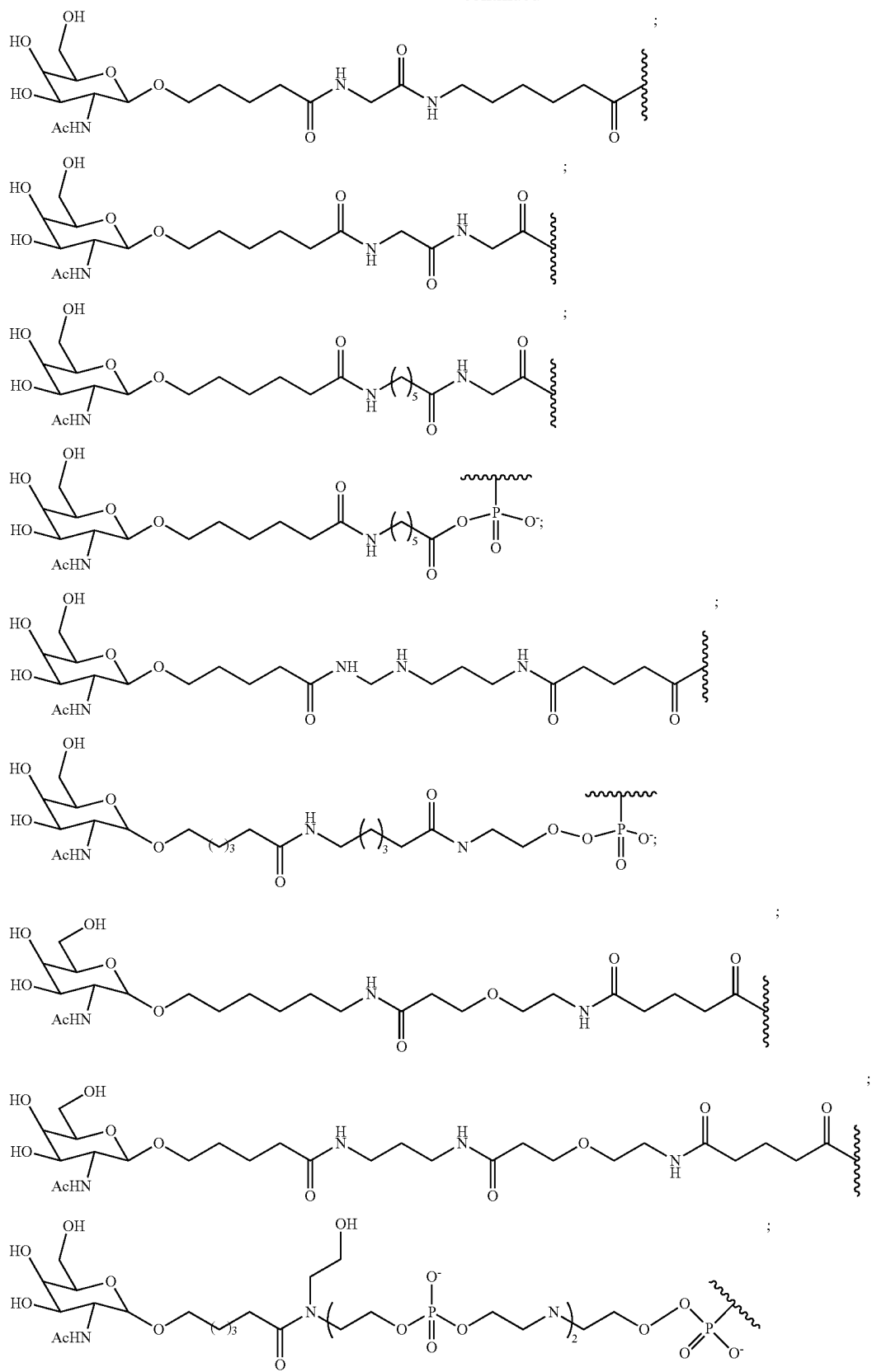
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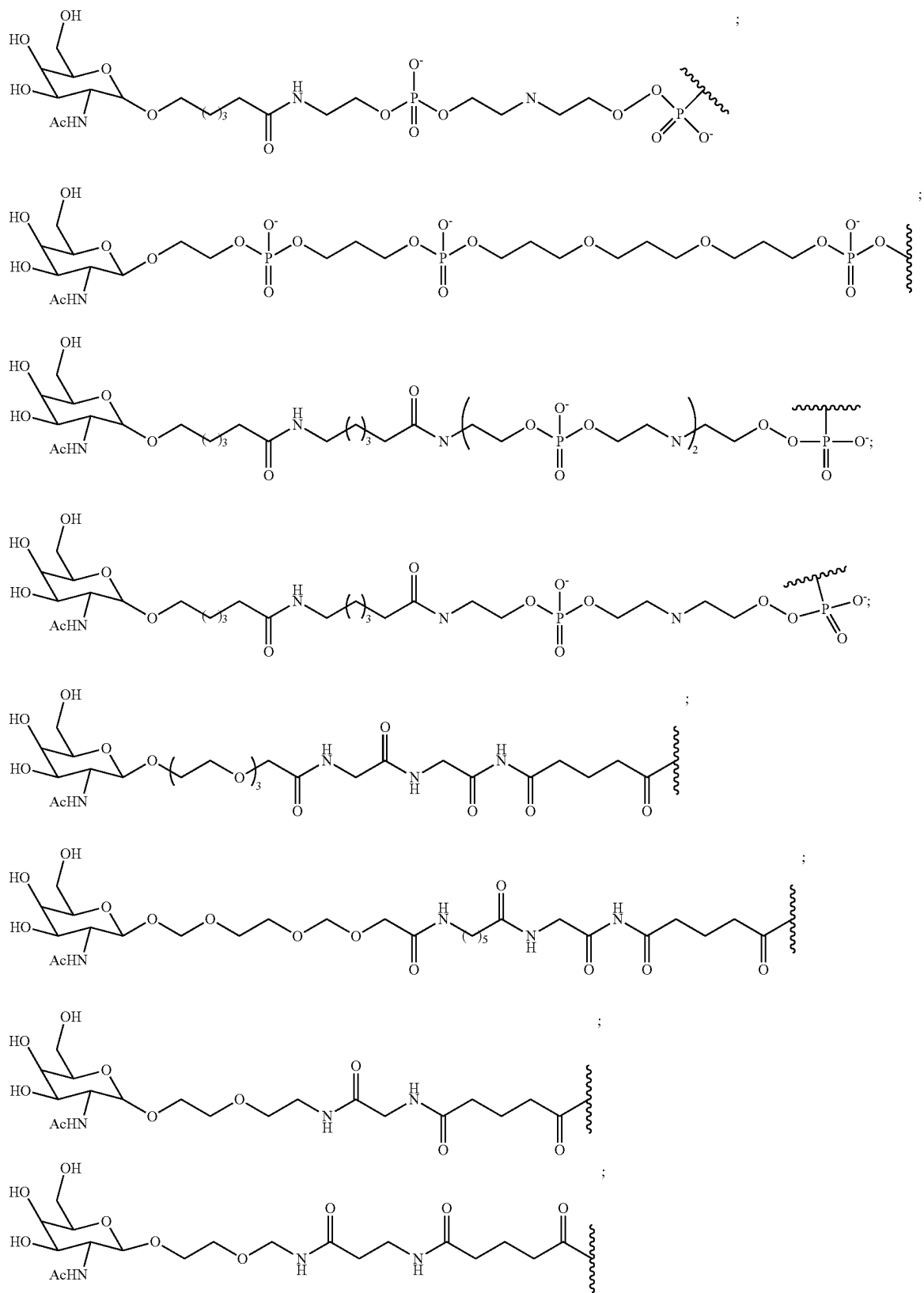
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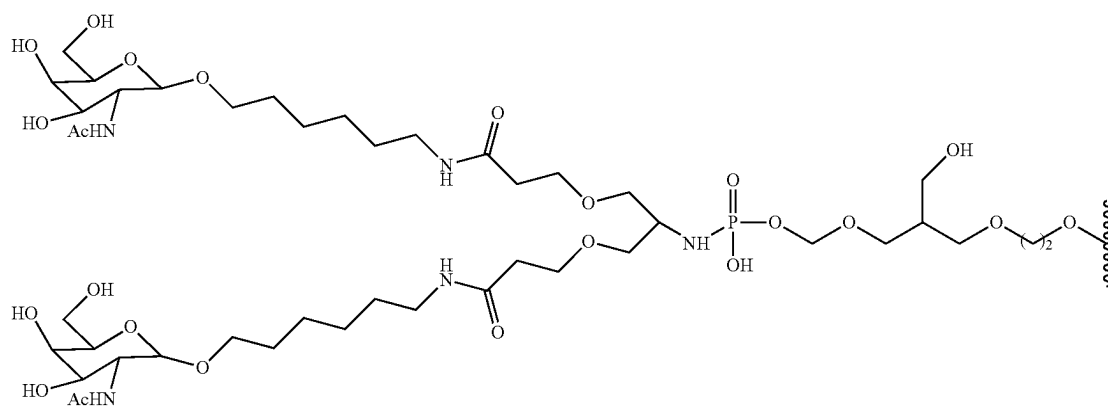
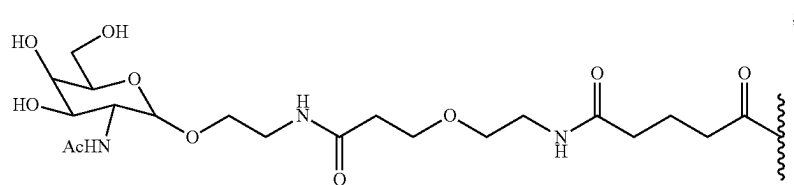
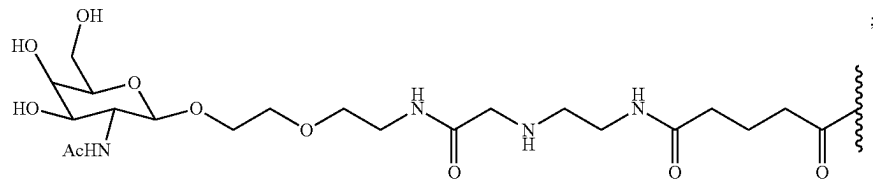
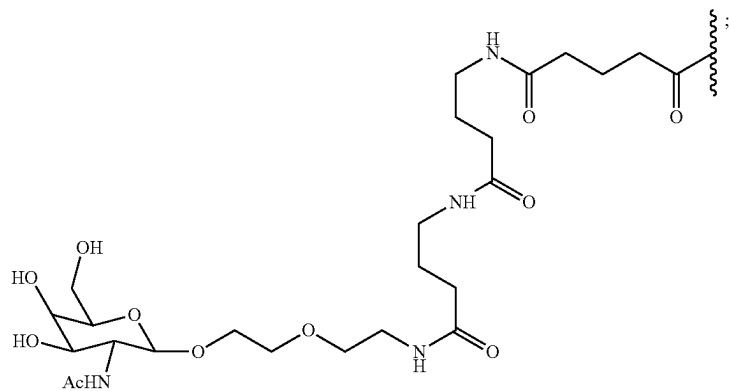
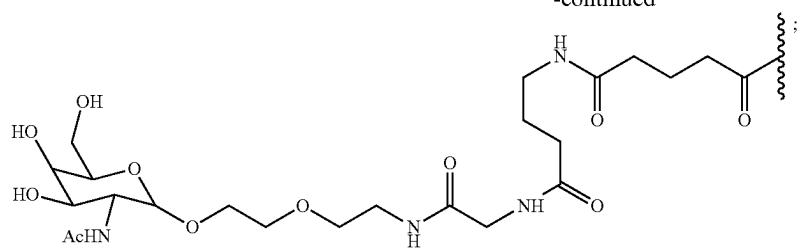
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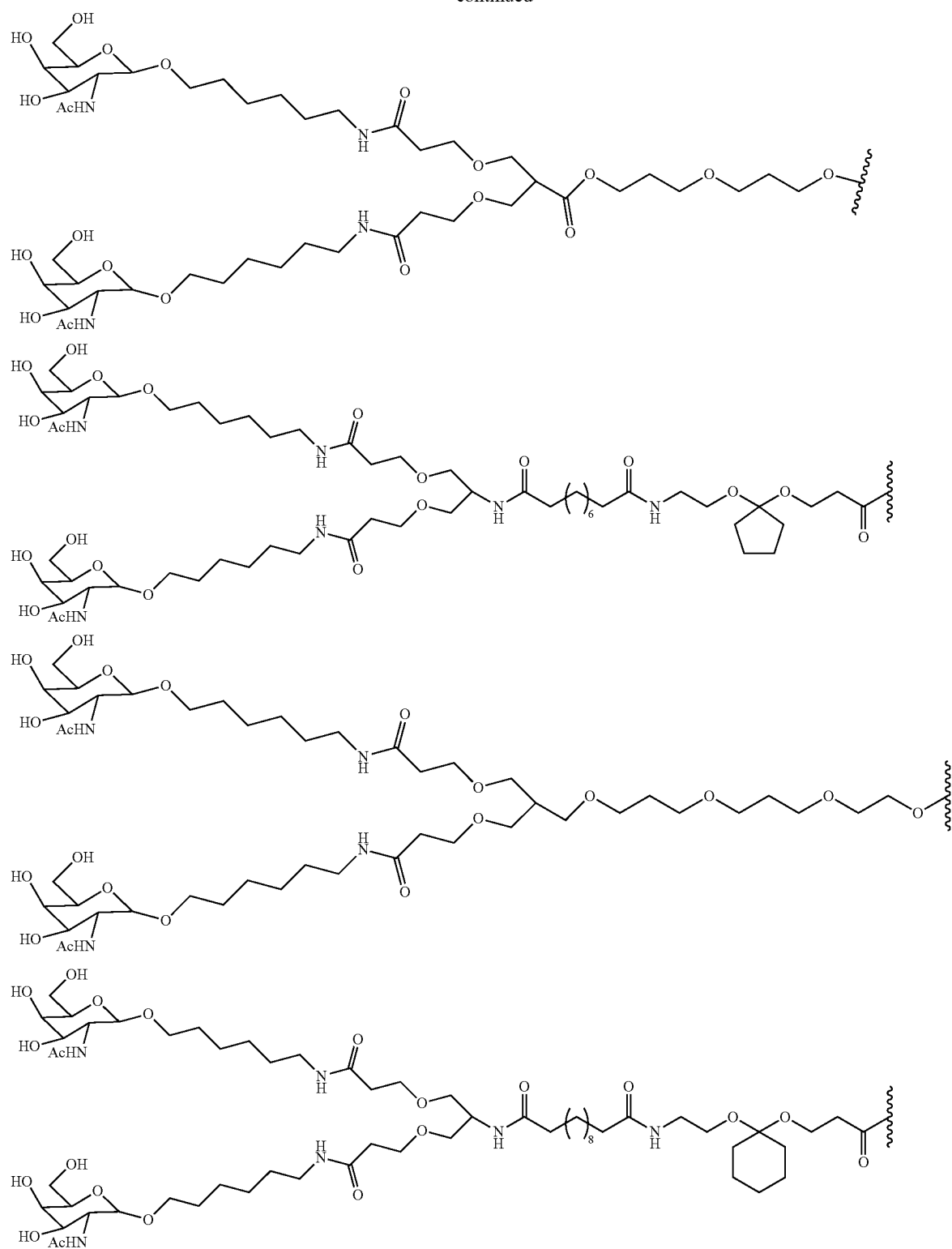
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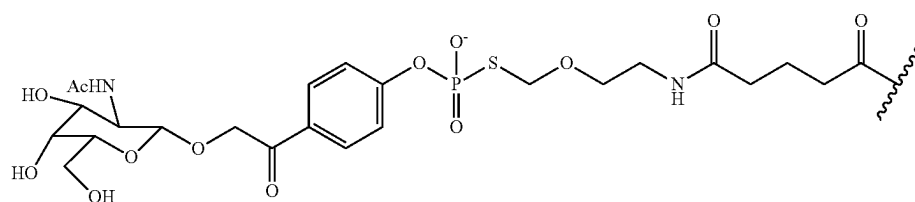
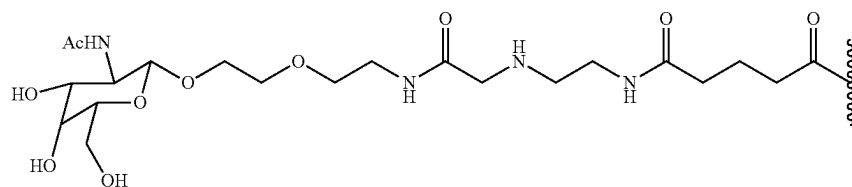
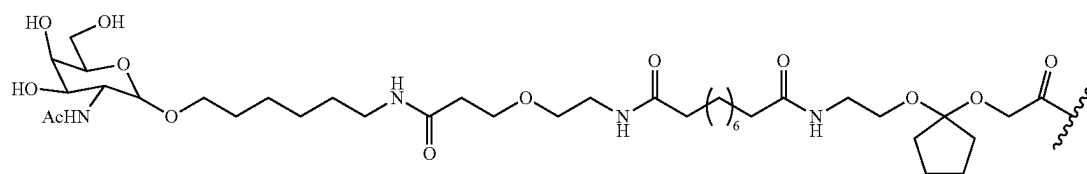
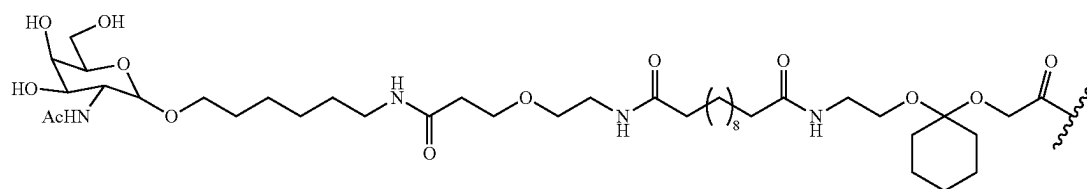
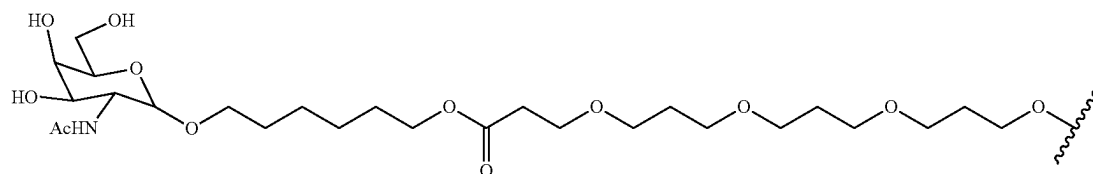
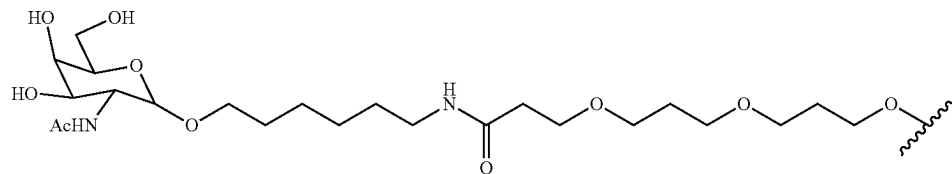
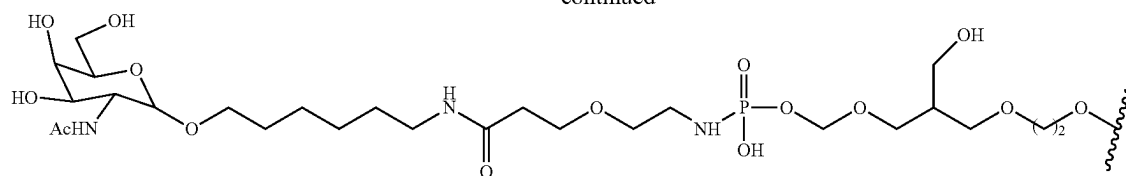
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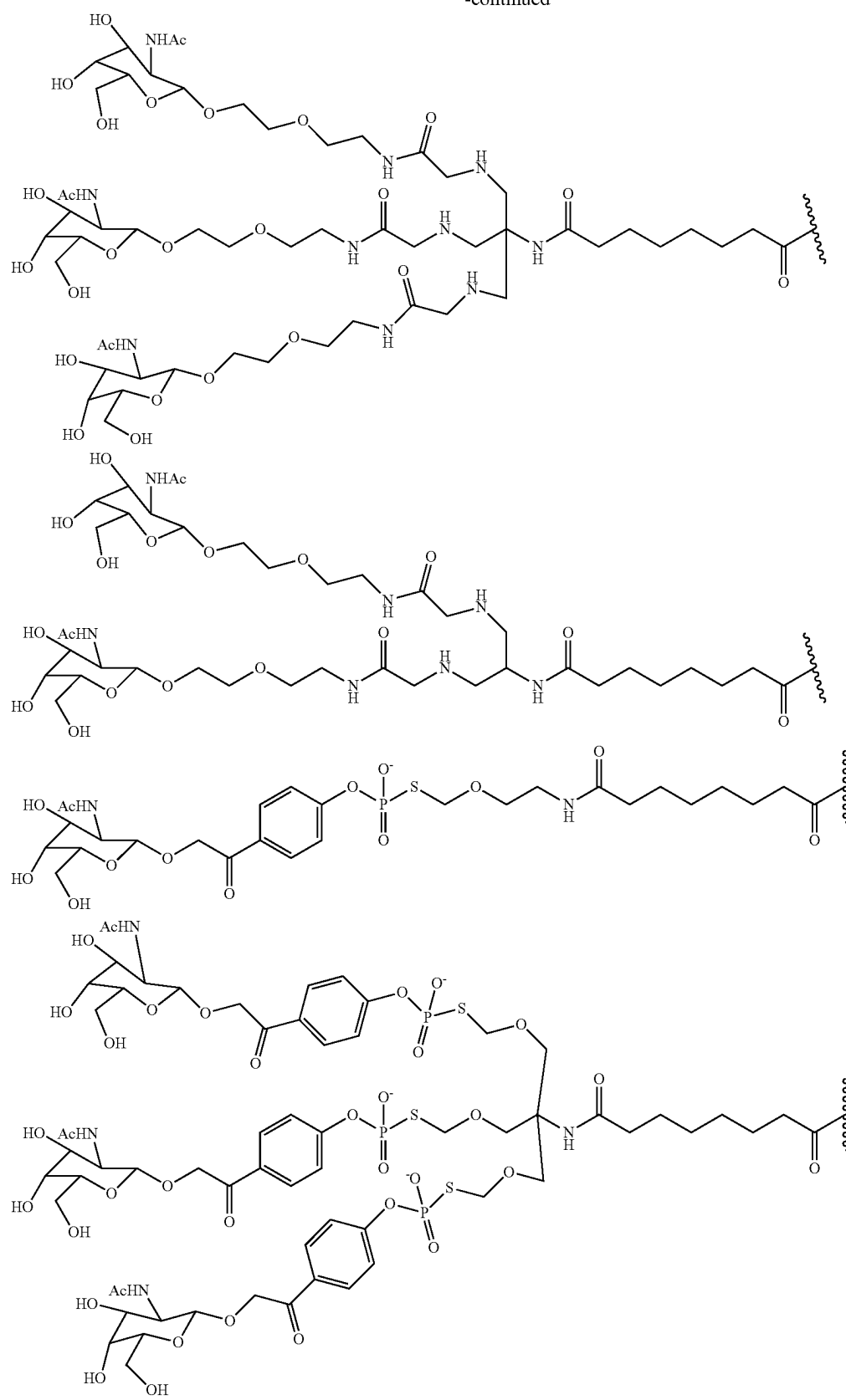
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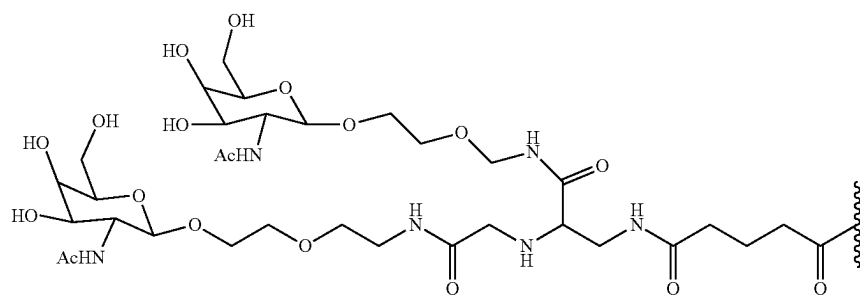
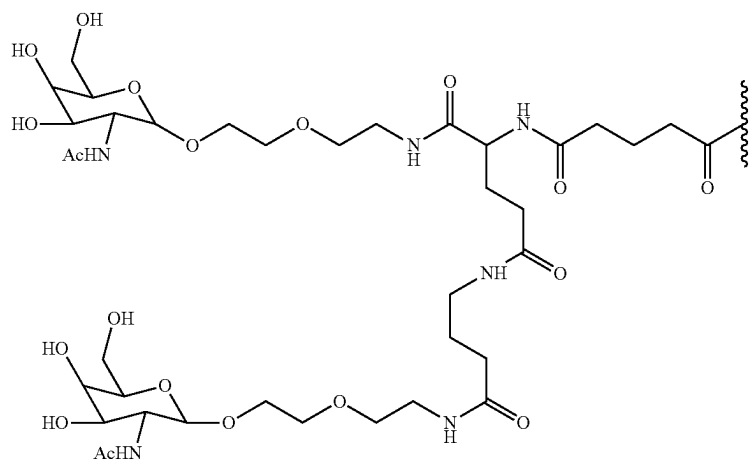
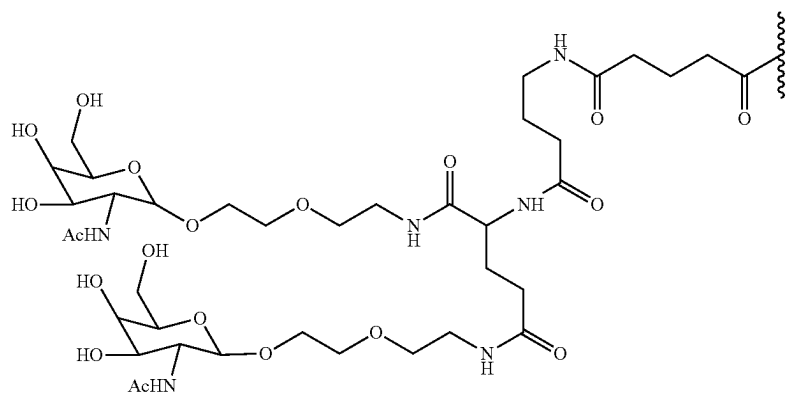
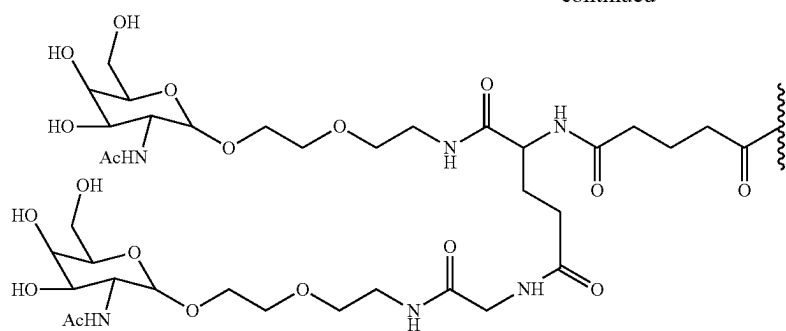
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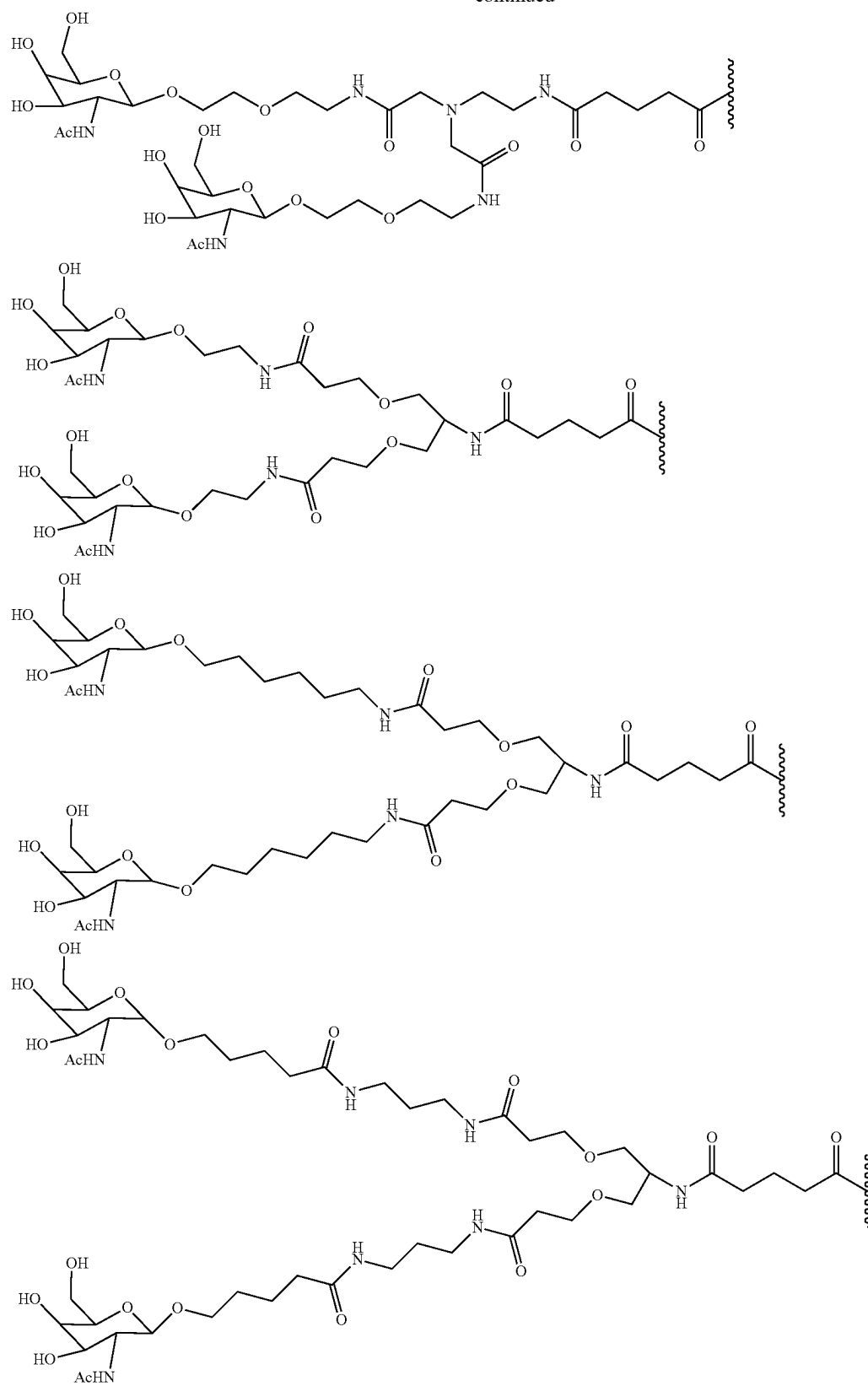
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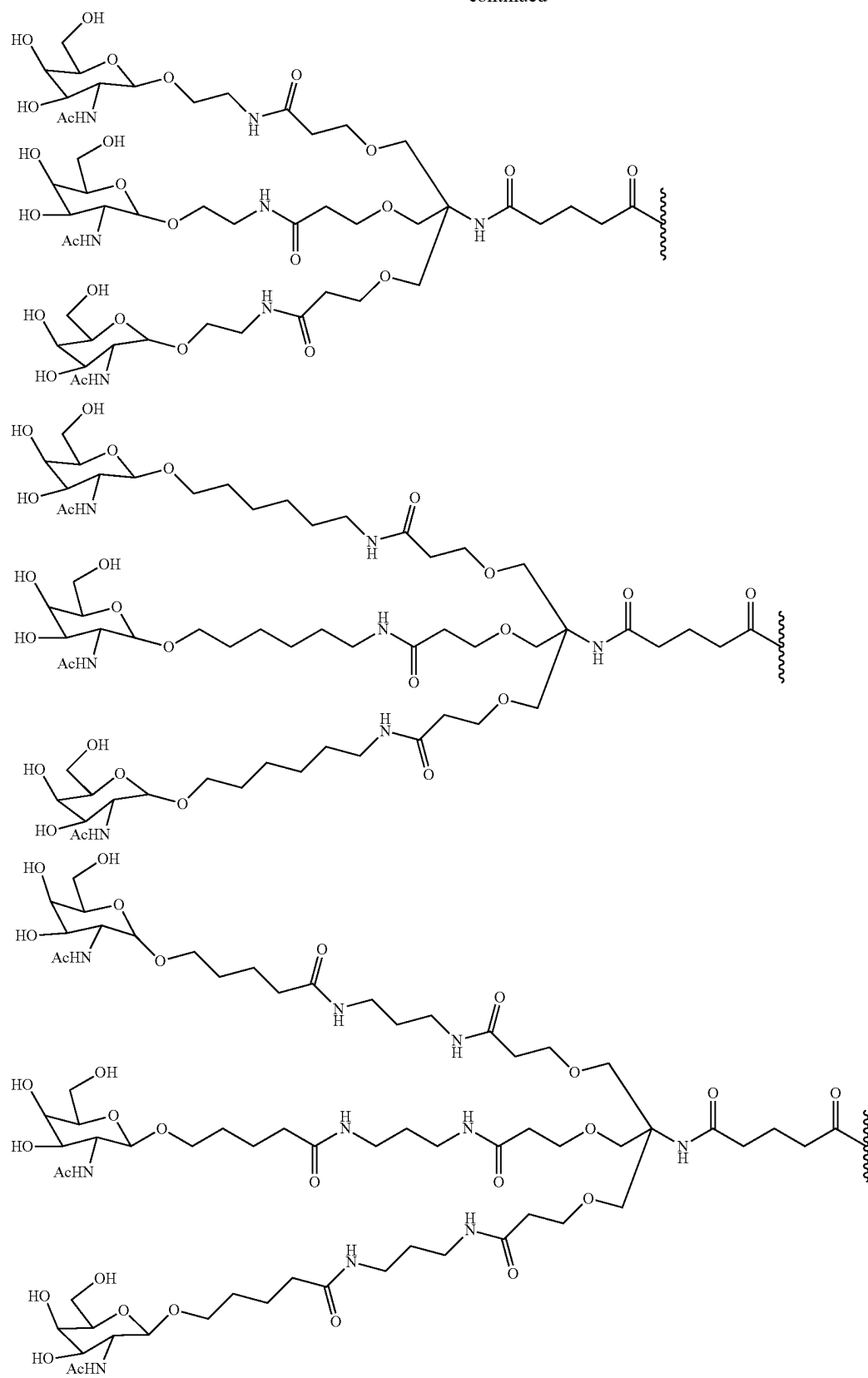
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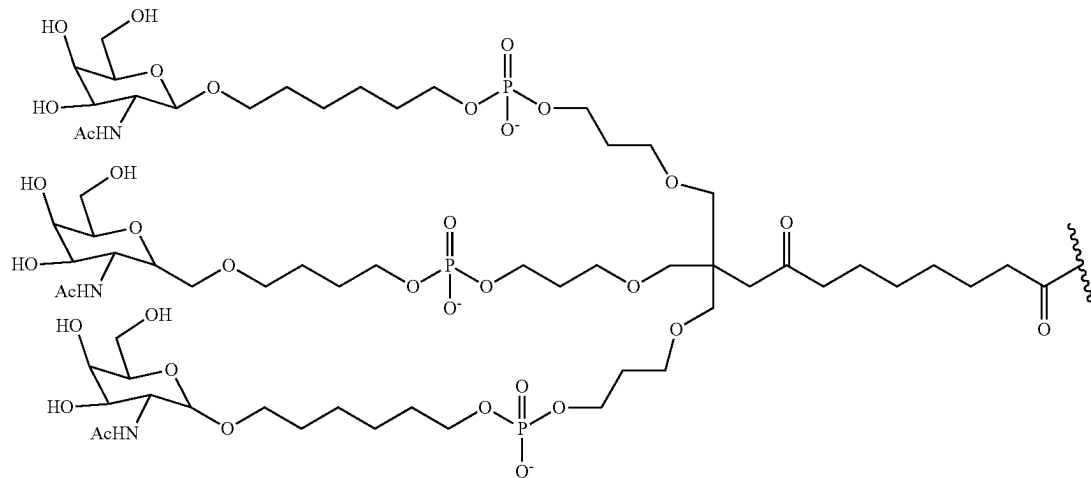
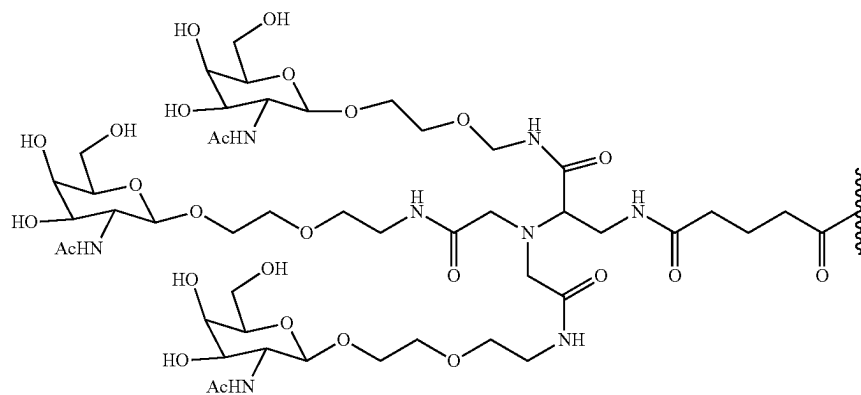
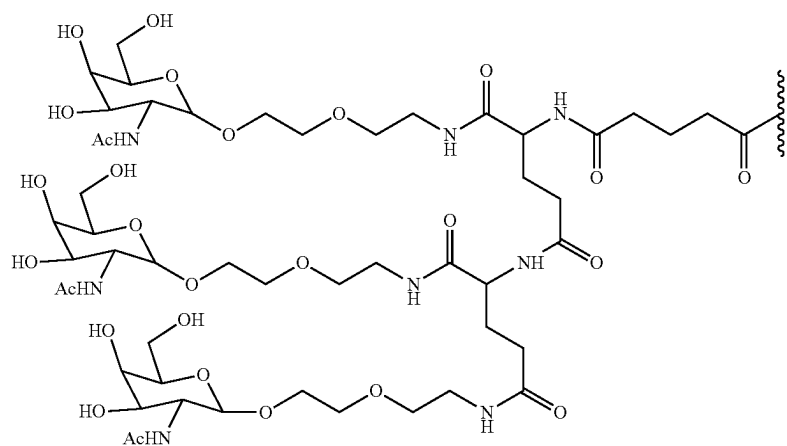
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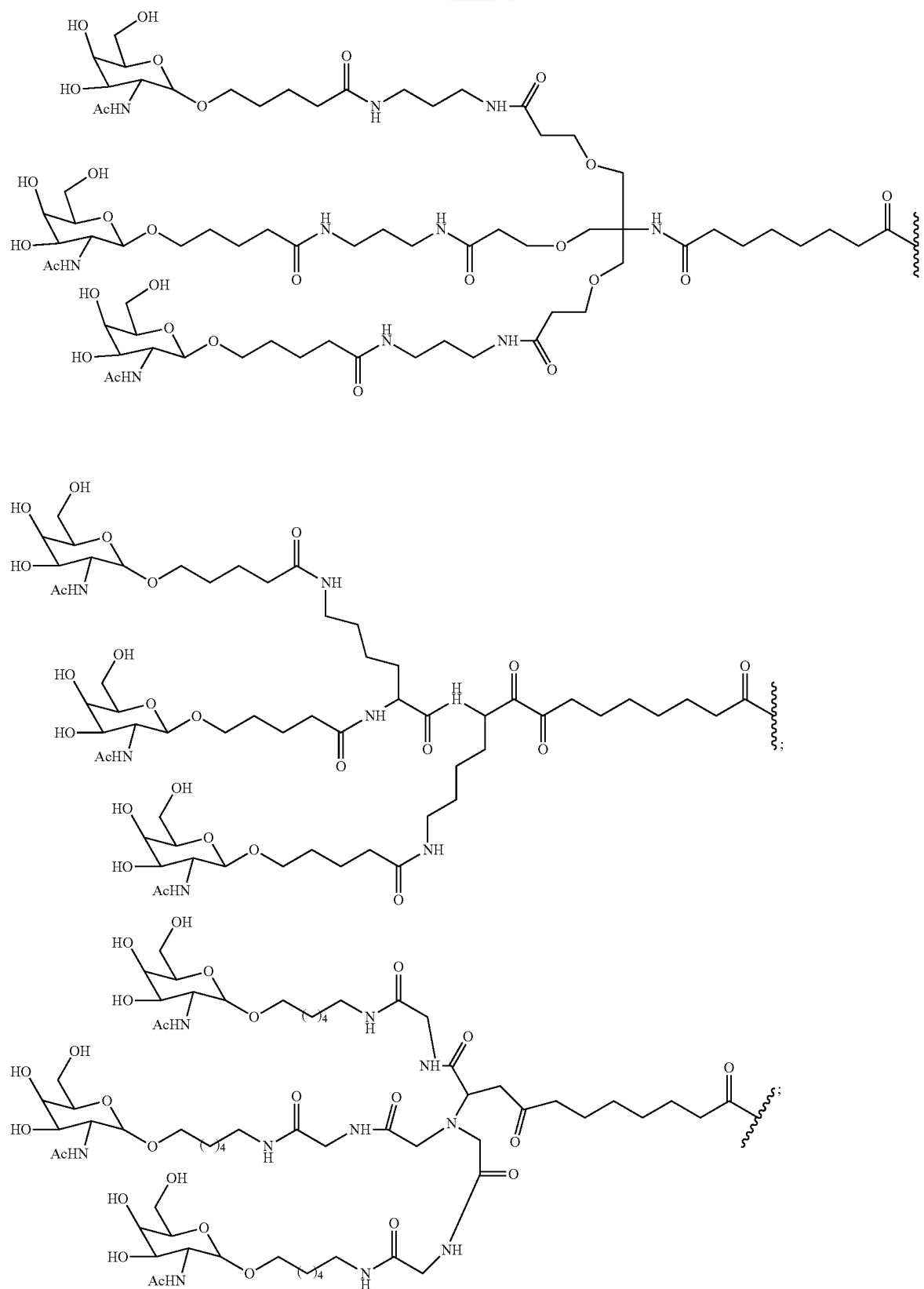
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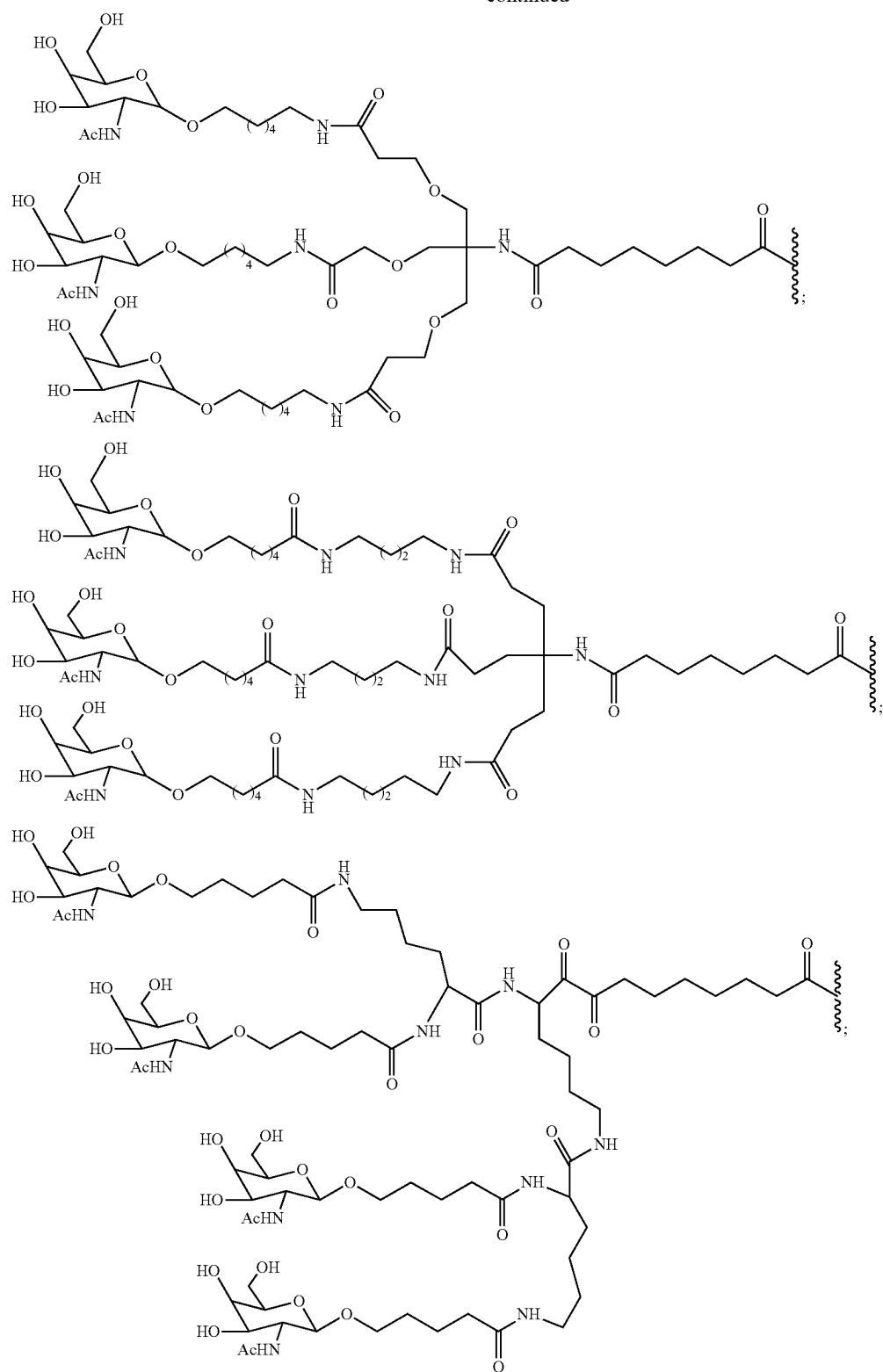
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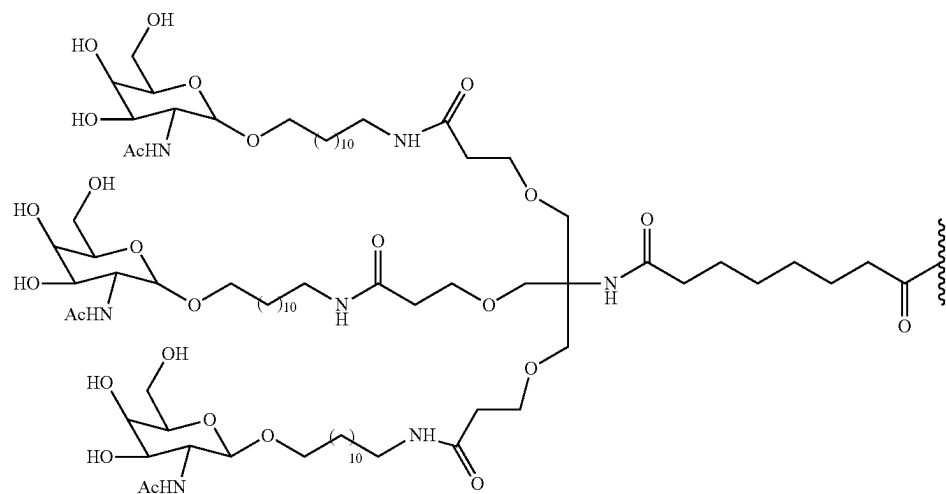
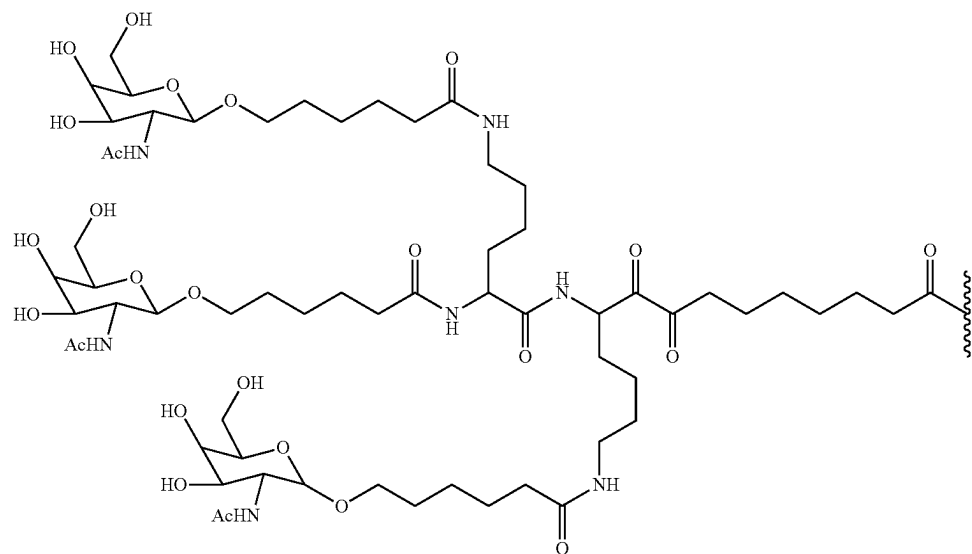
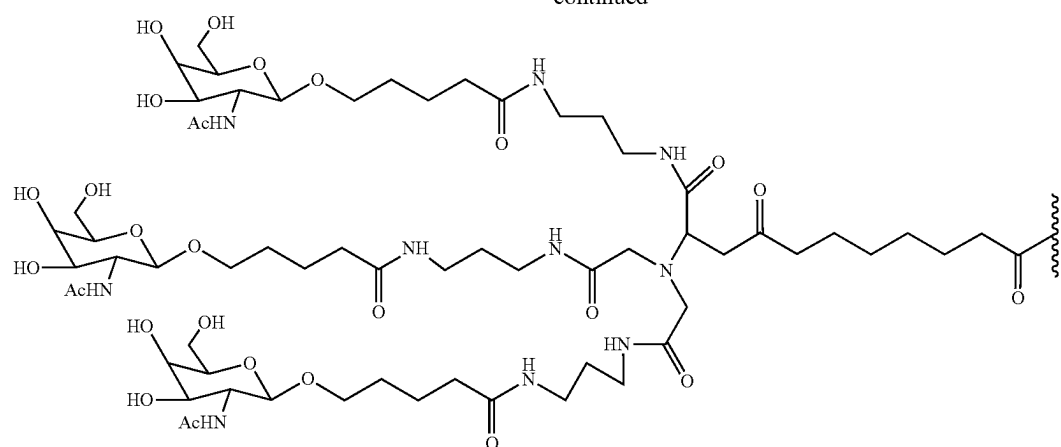
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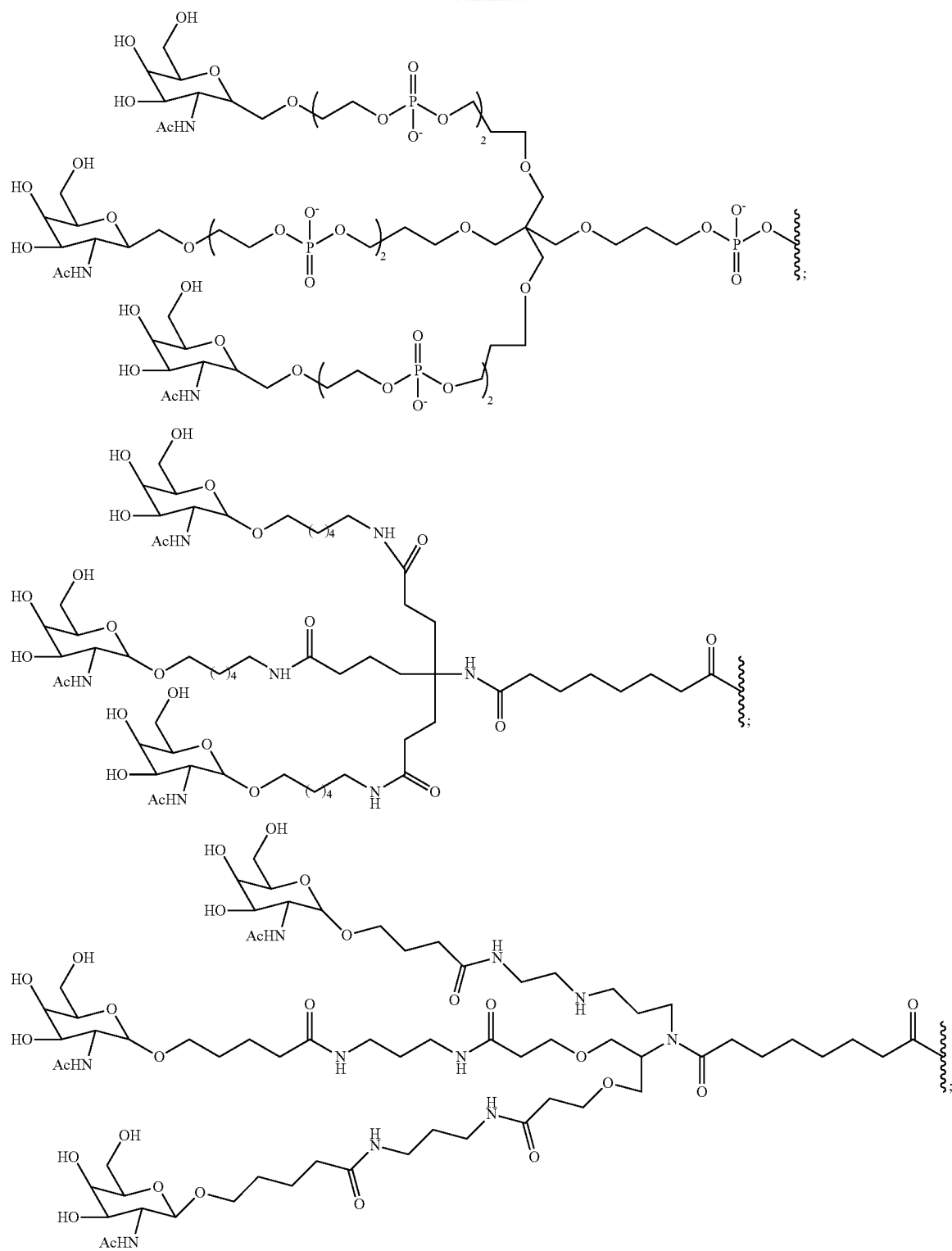
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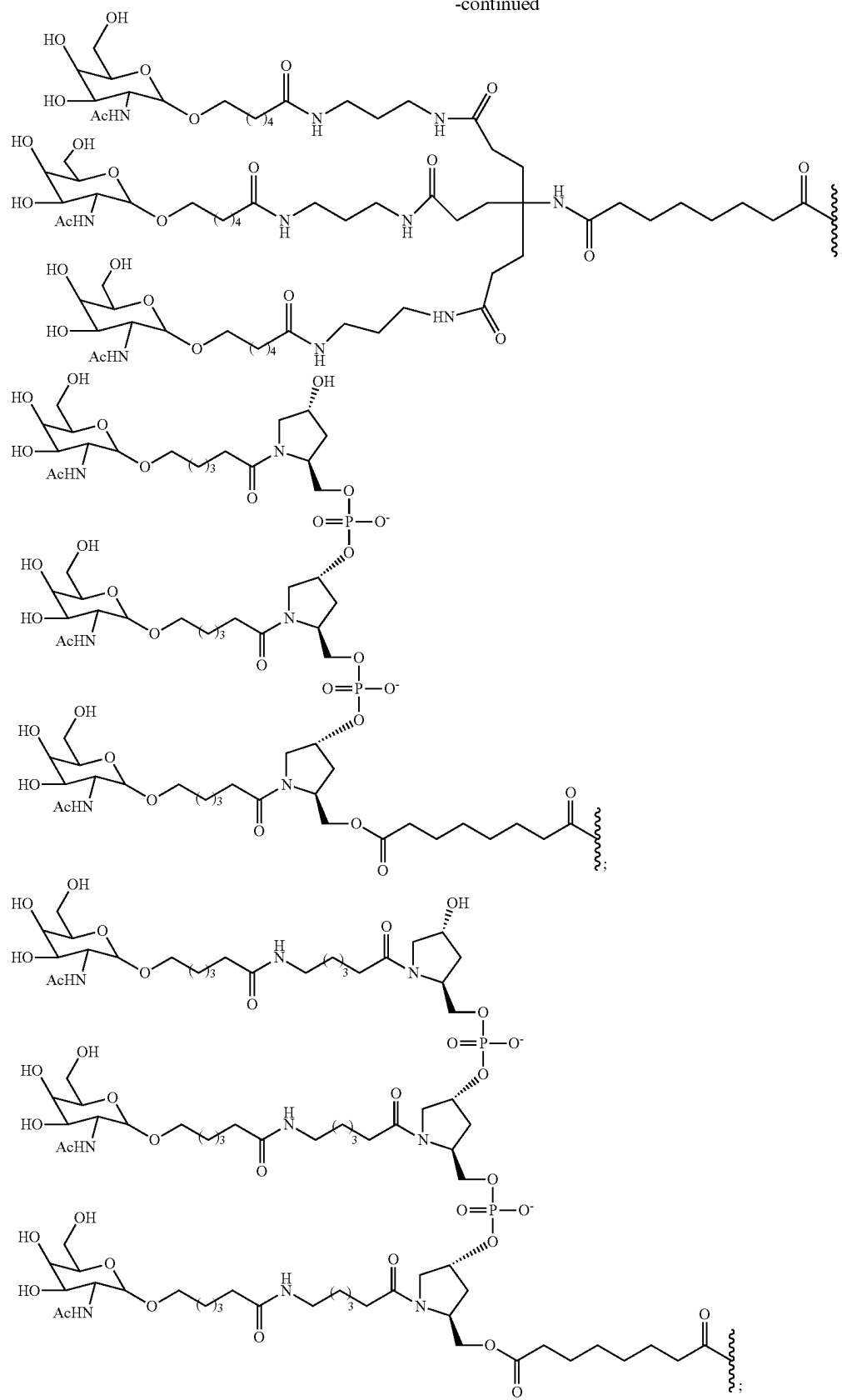
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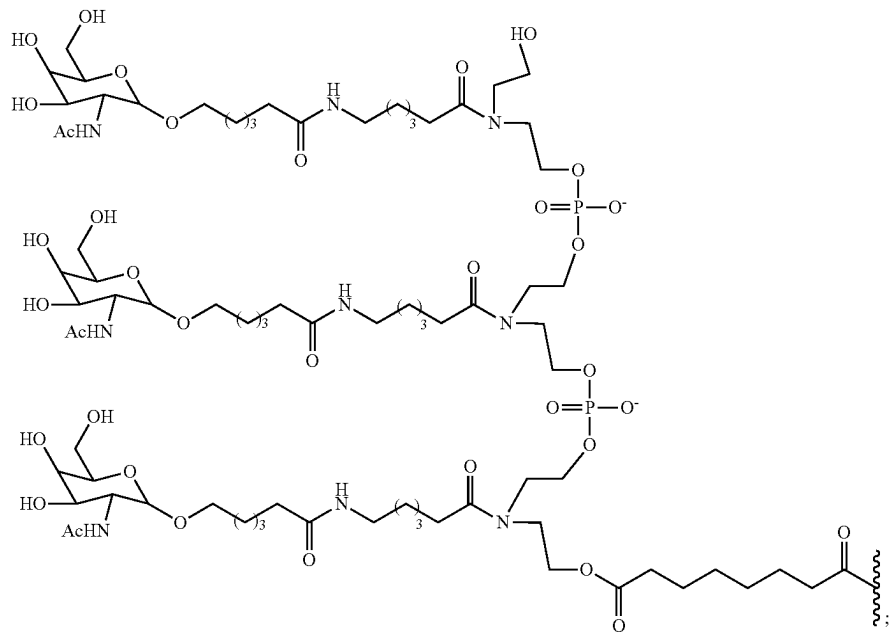
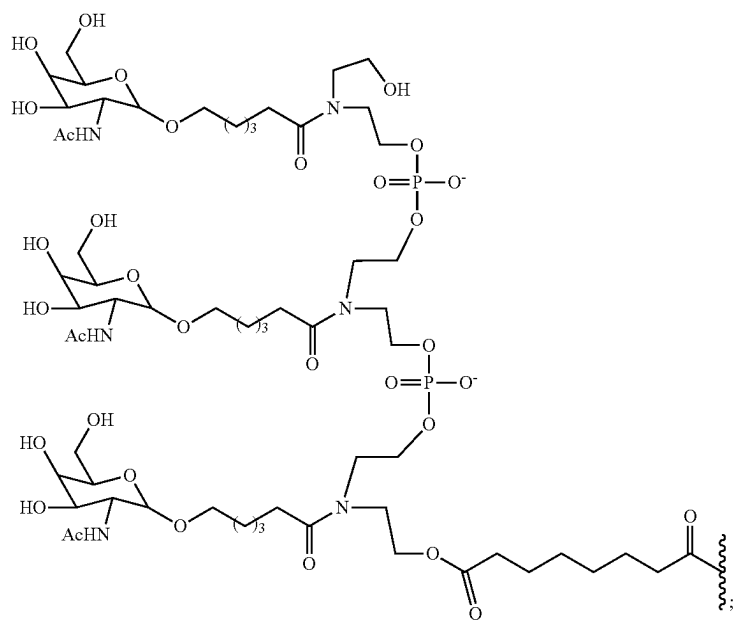
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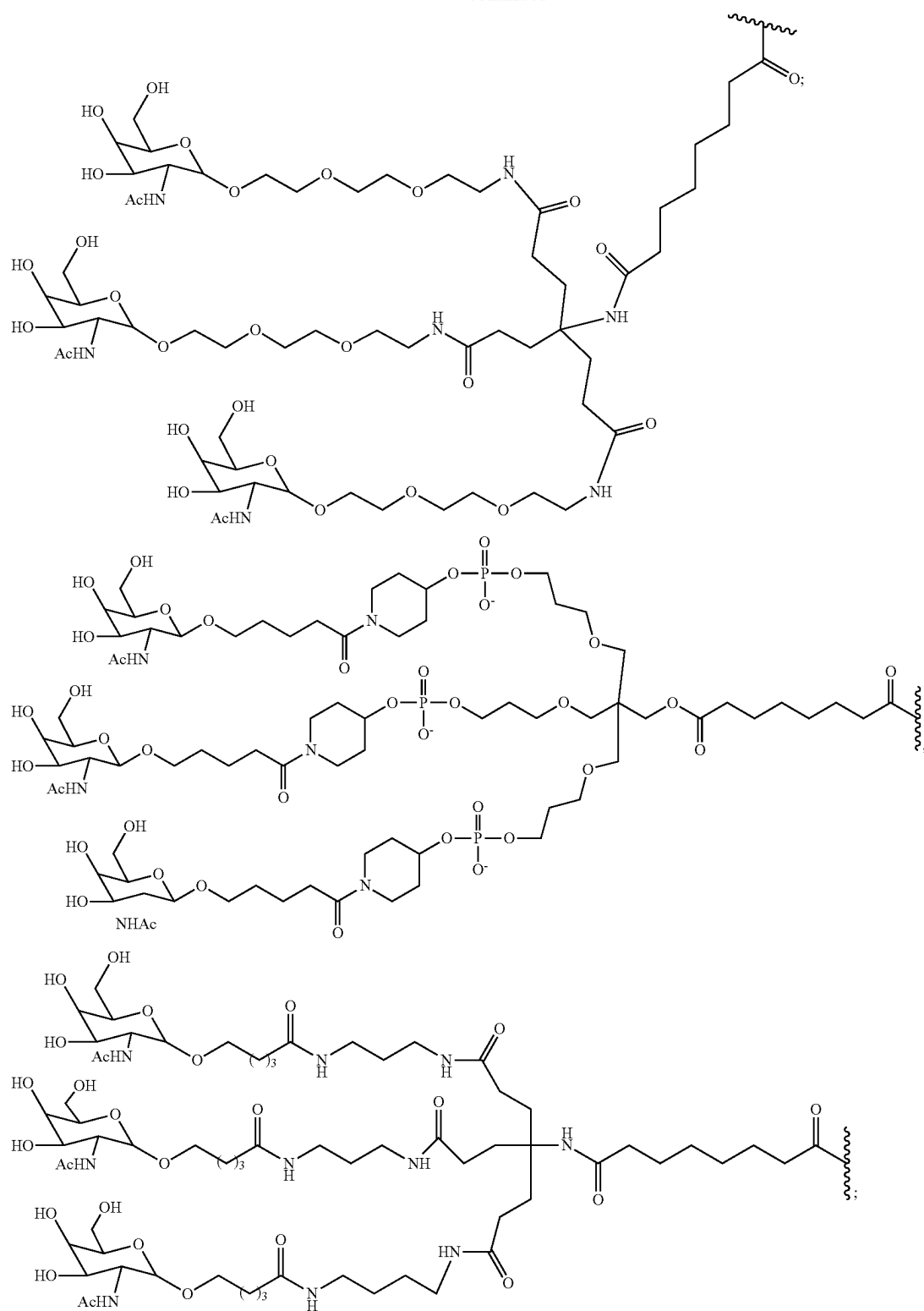
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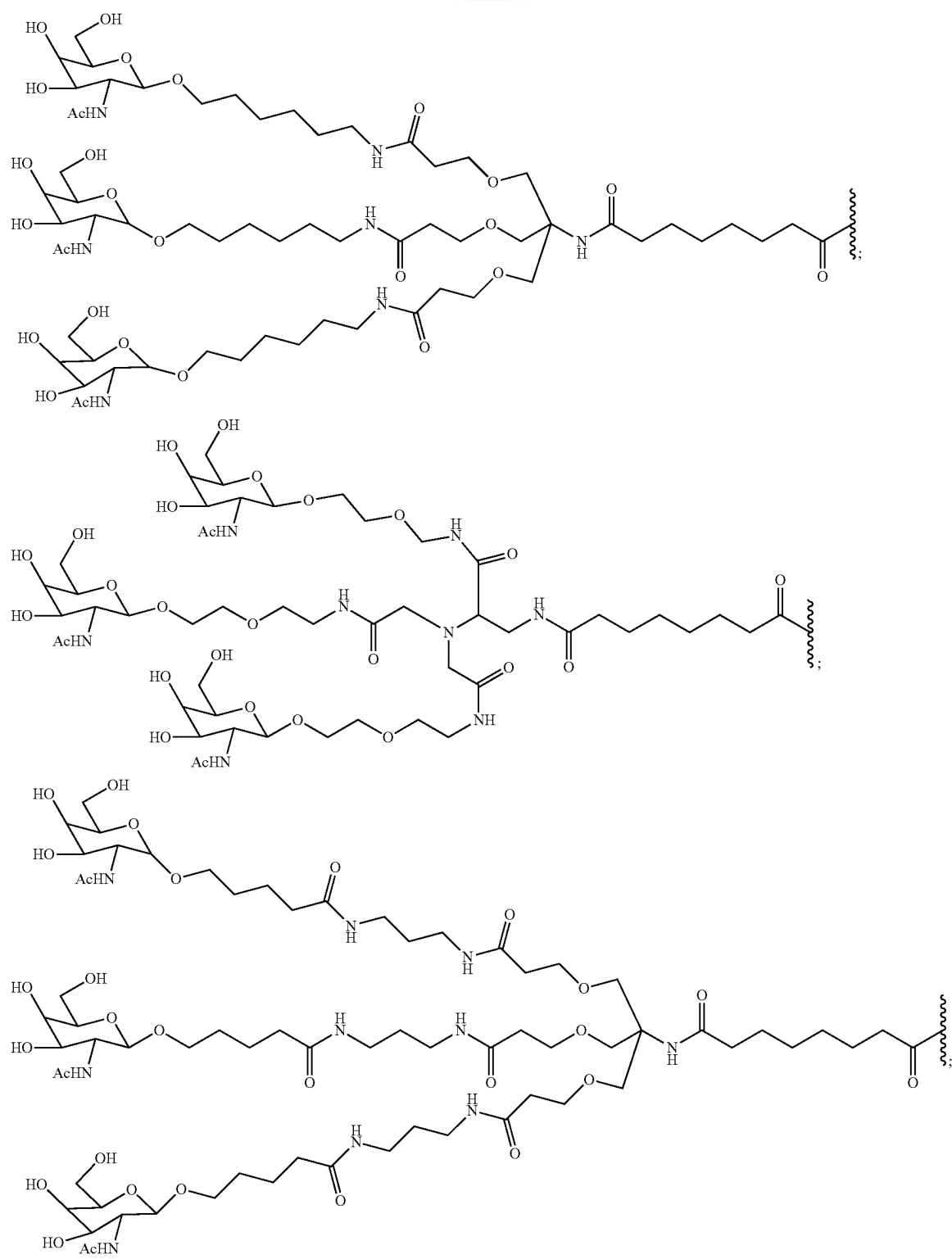
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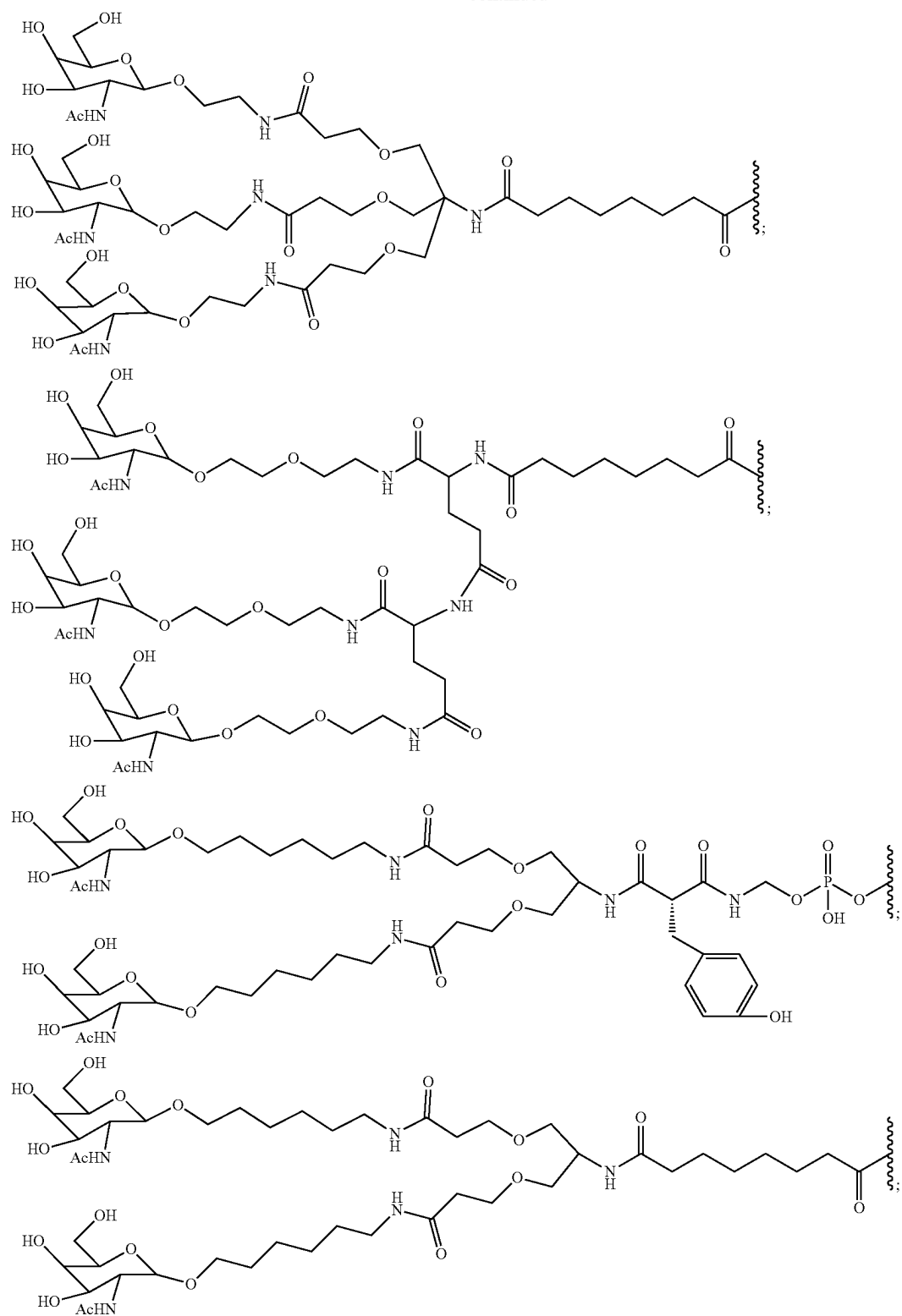
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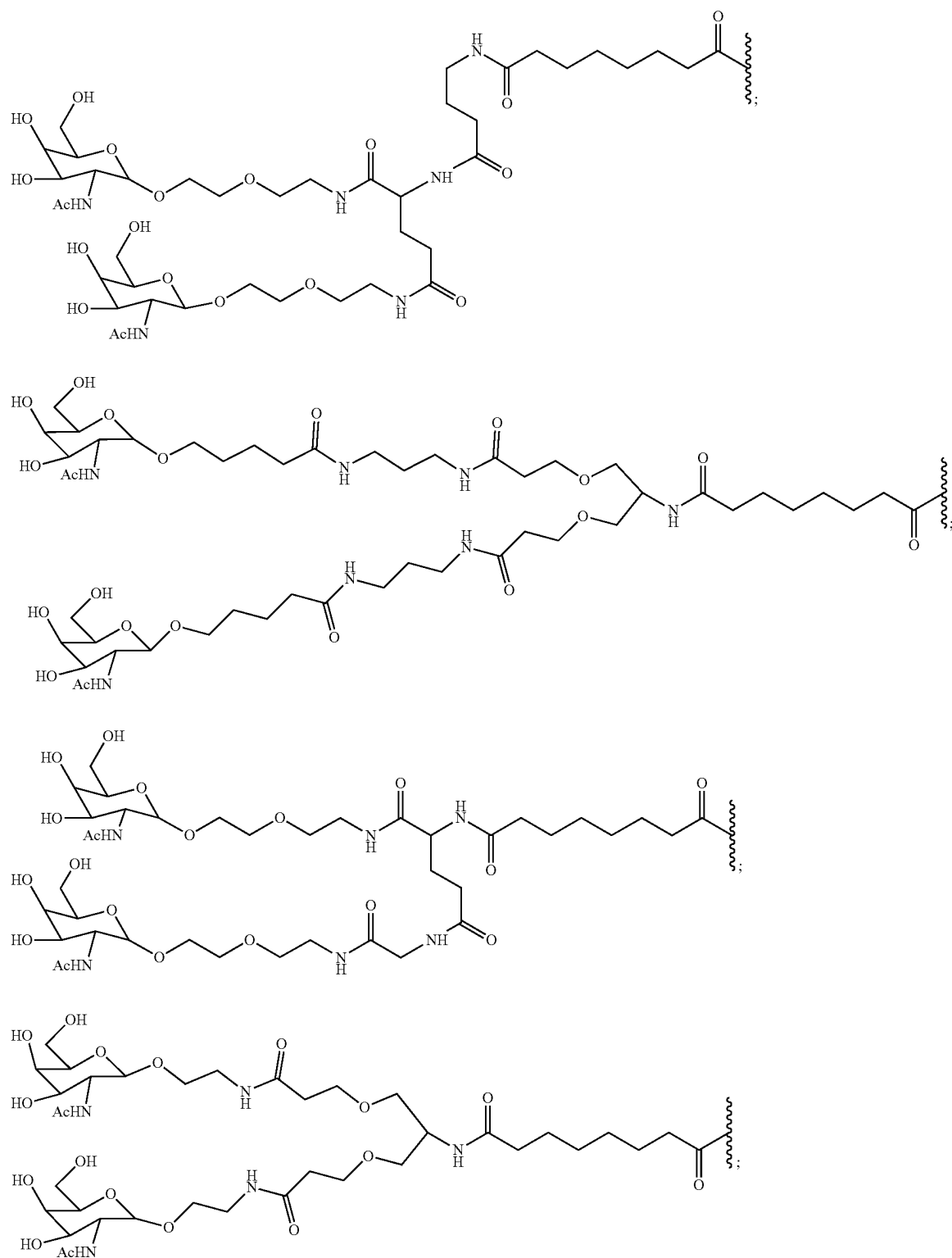
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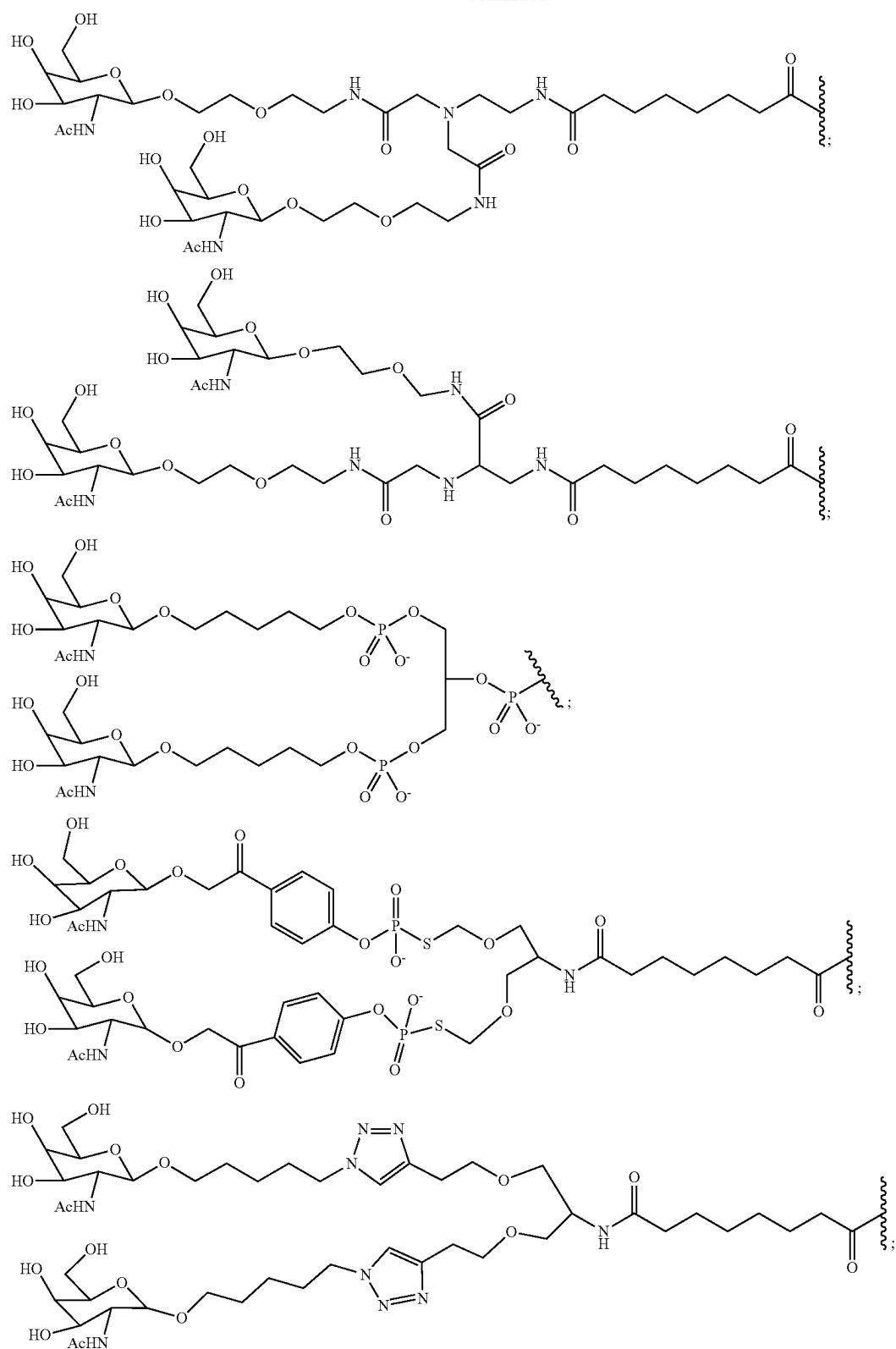
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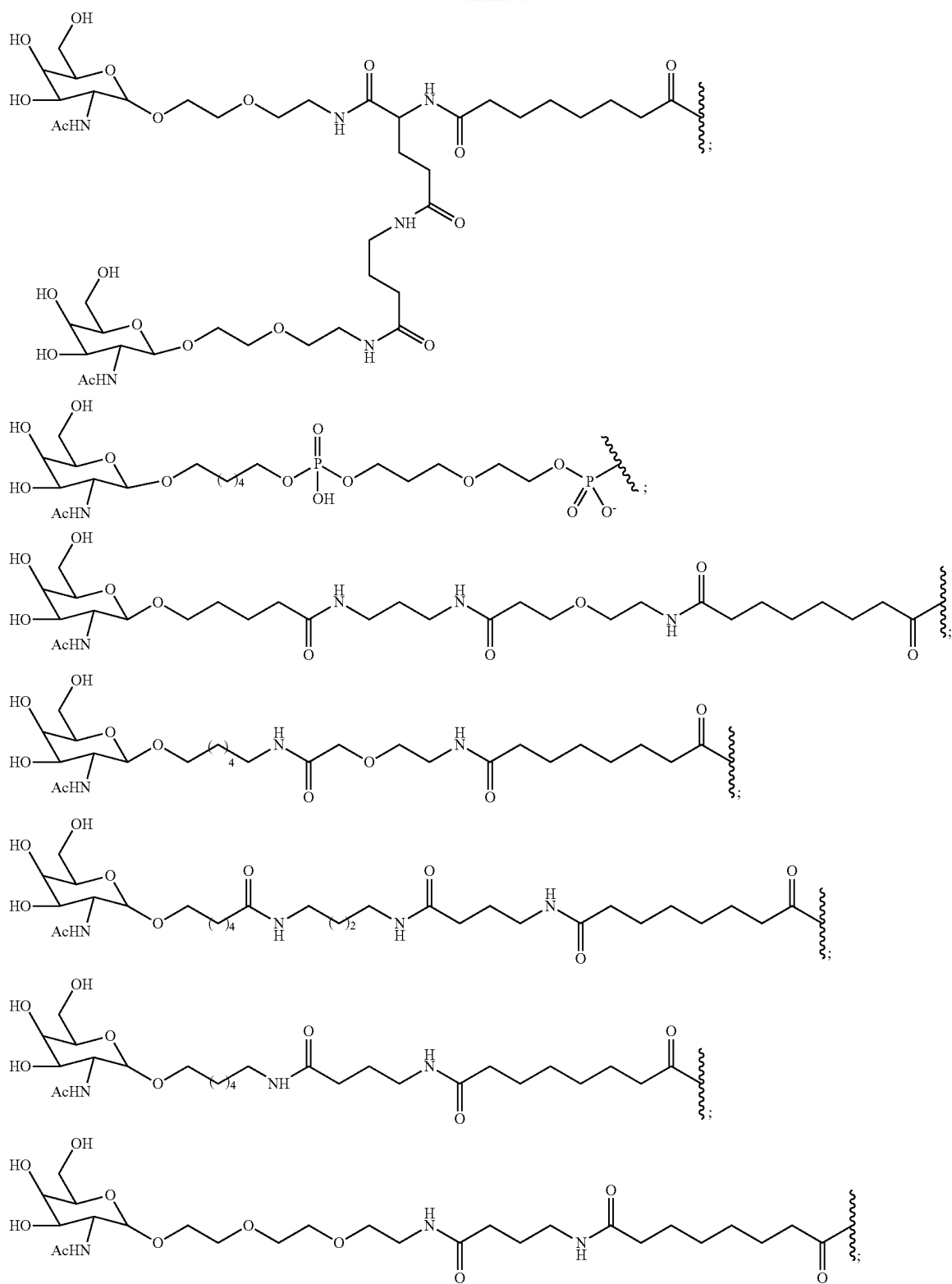
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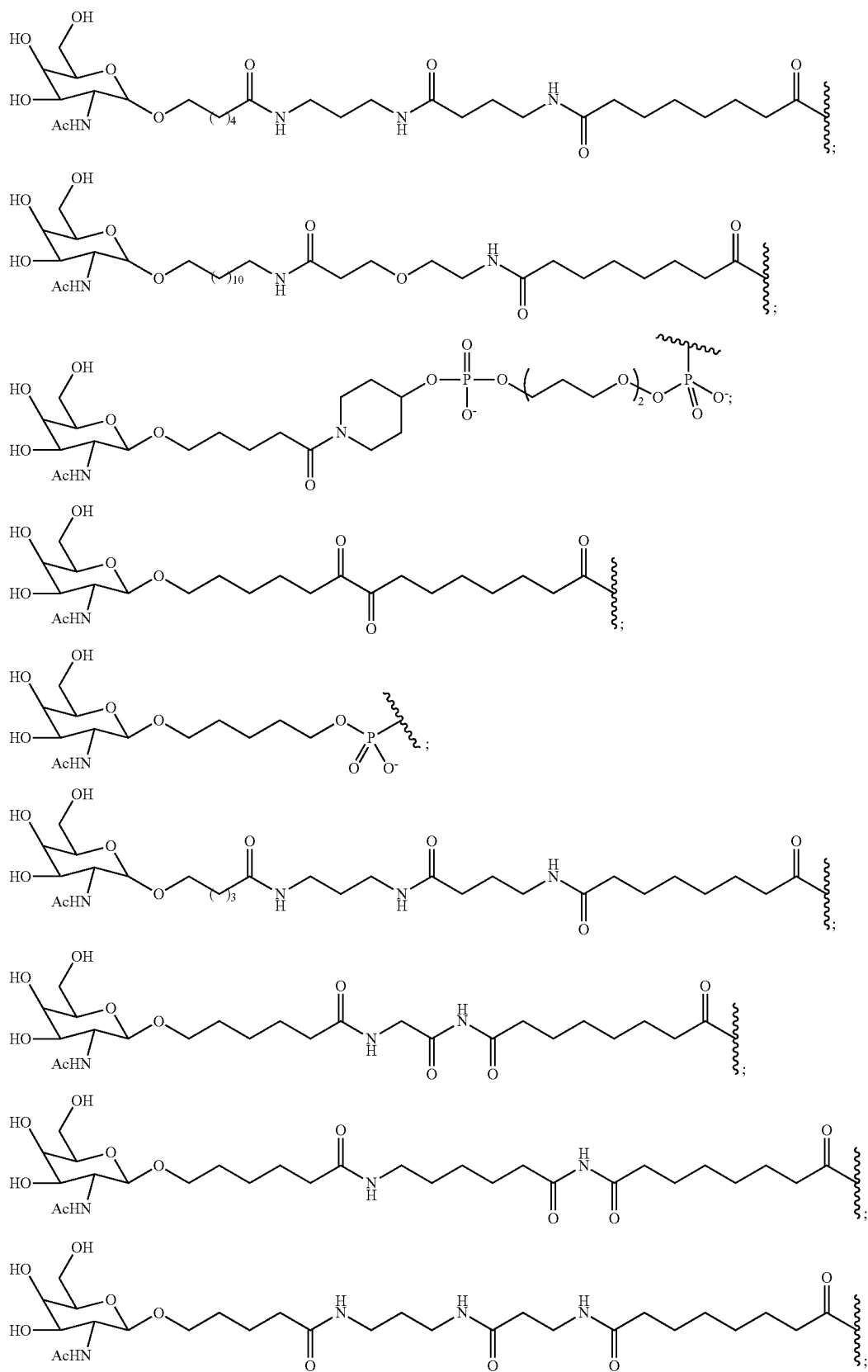
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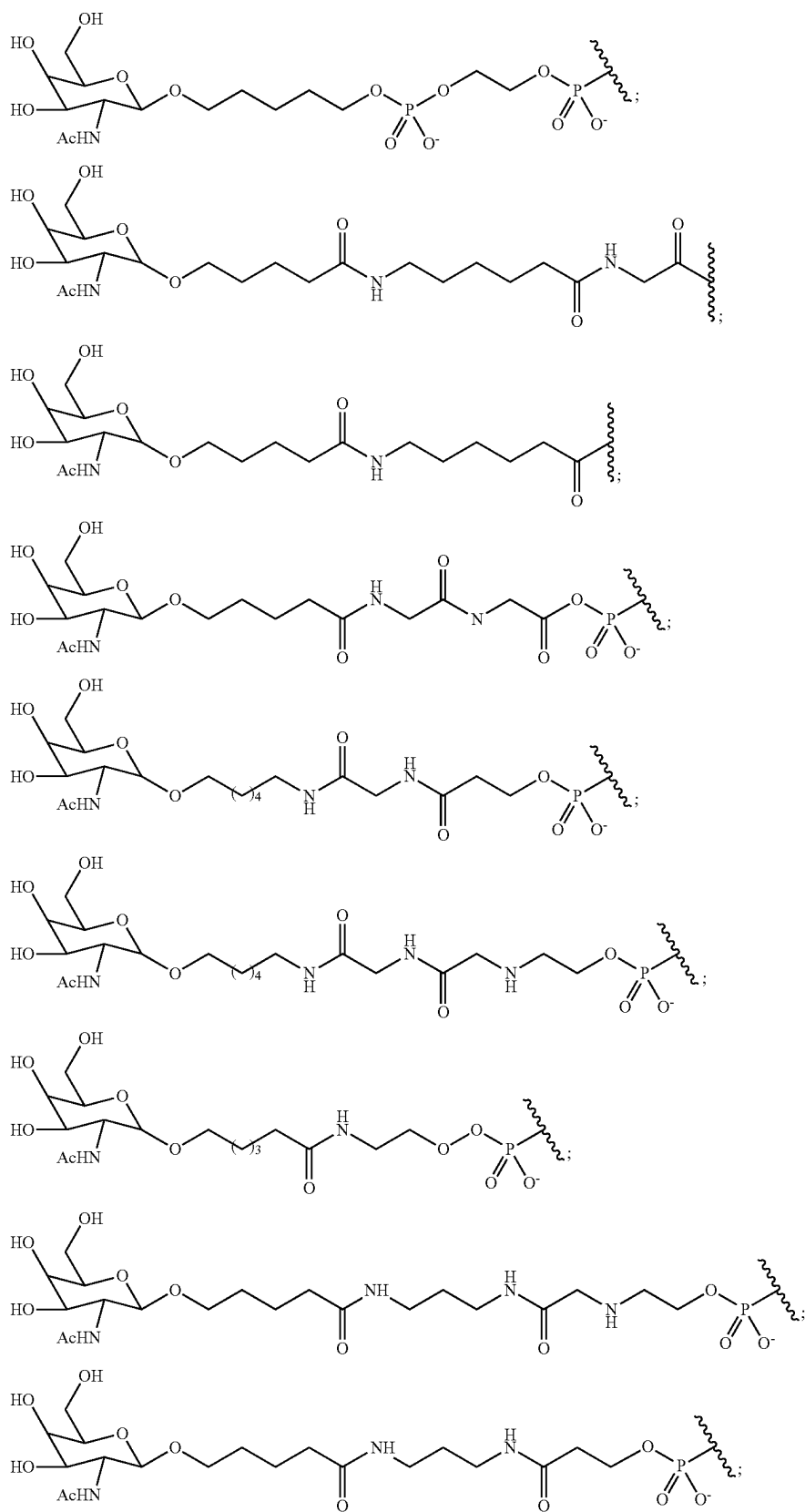
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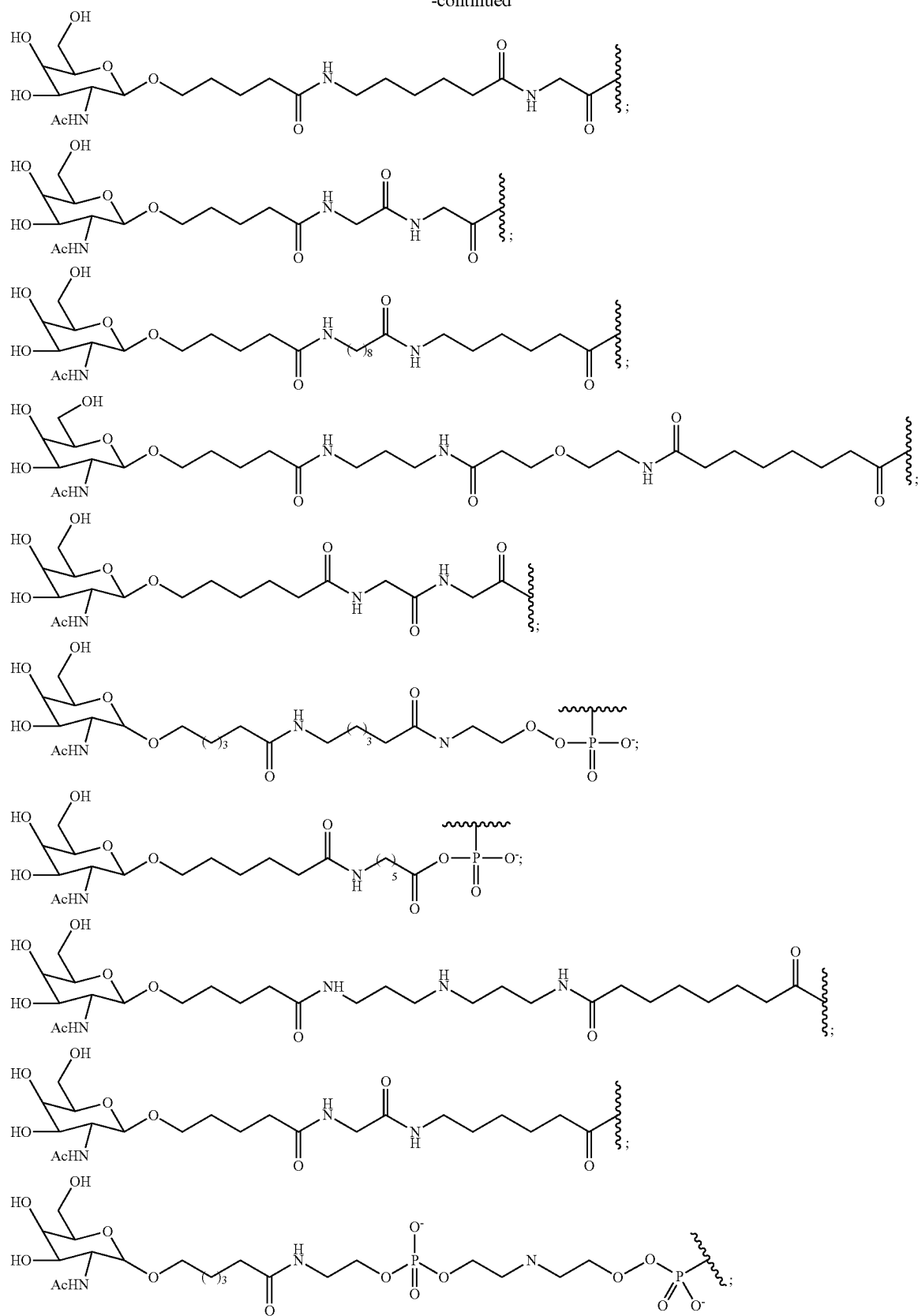
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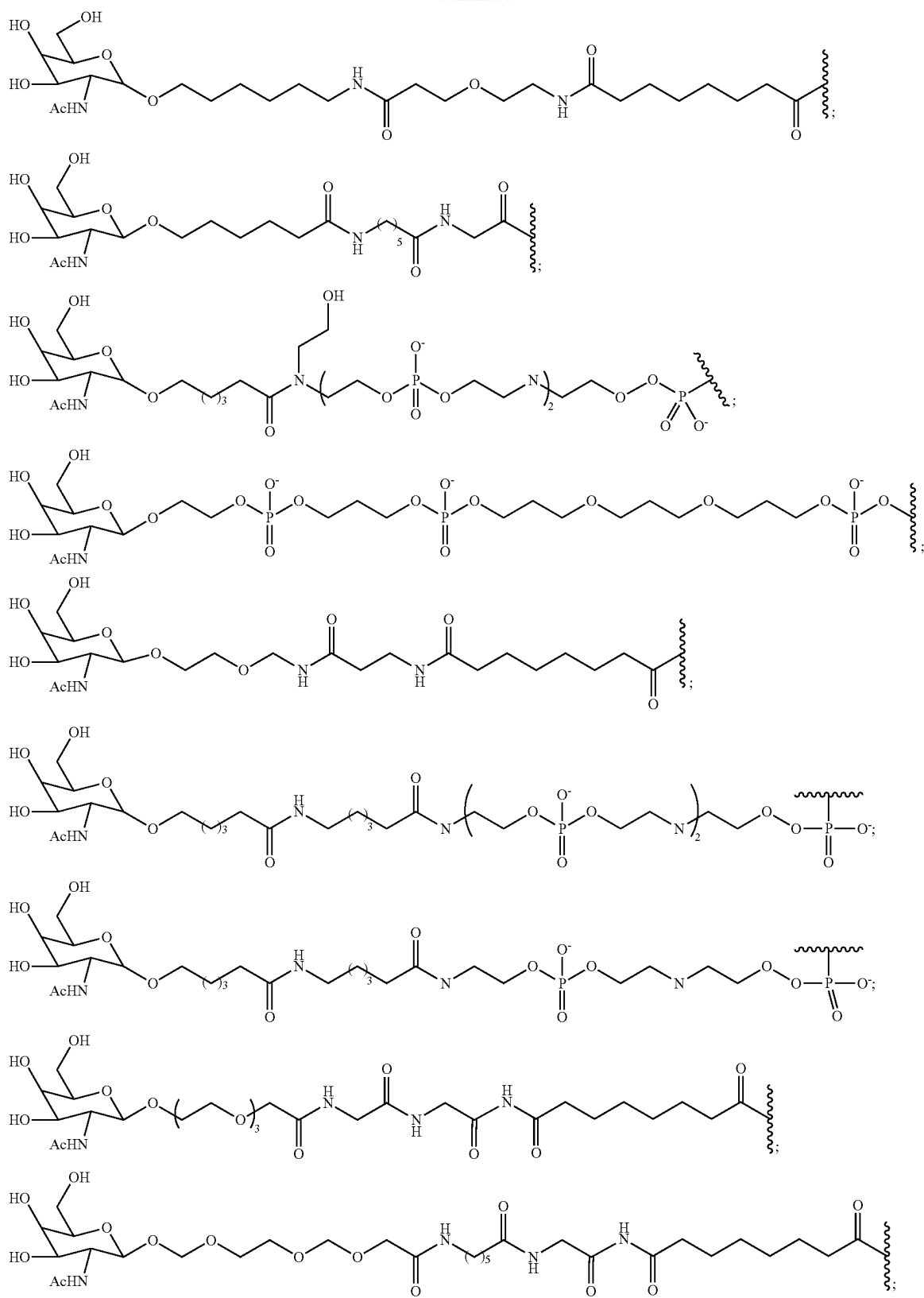
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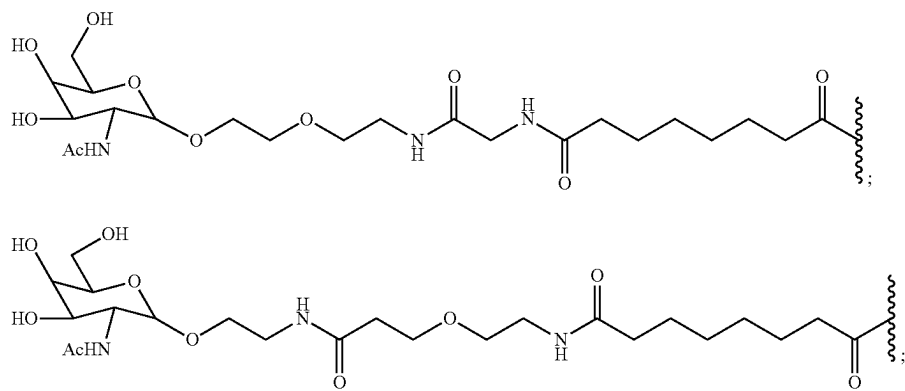
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[0181] In some embodiments, the X, L, D, and B are the same or different within the respective 5' MVIP and 3' MVIP or between the 5' MVIP and 3' MVIP.

[0182] In some embodiments, in the 5' MVIP structure is selected from the structures (X-L)-B-D-shown in Table 8:

TABLE 8

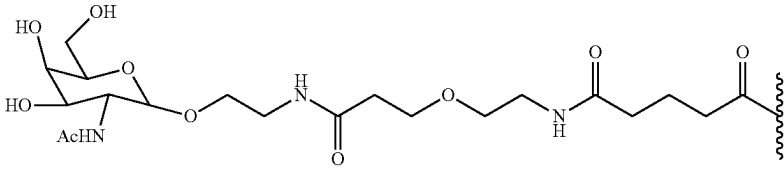
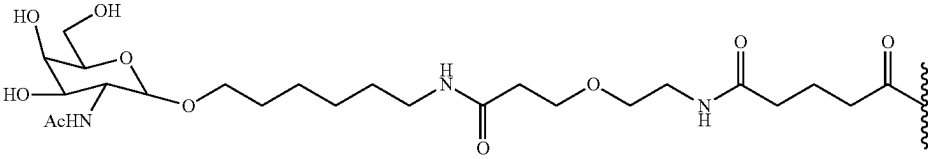
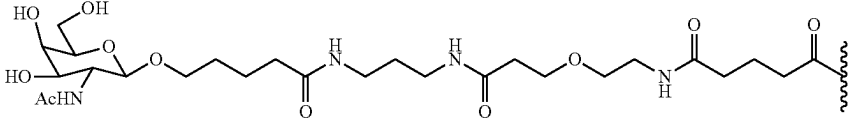
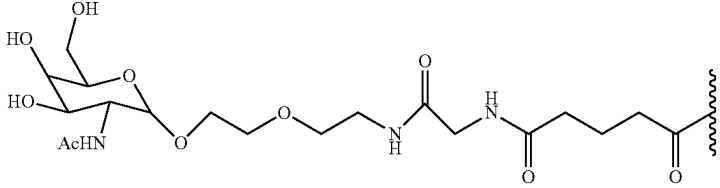
Structures of (X-L) _n -B-D- in 5' MVIP		
Serial num-ber	Code	Structural formula
1	5'YICdd-01	
2	5'YICd-01	
3	5'YICc-01	
4	5'YICa-01	

TABLE 8-continued

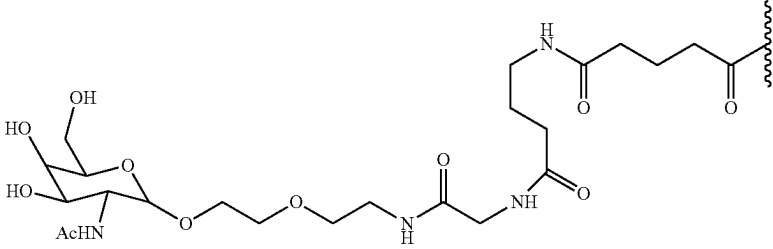
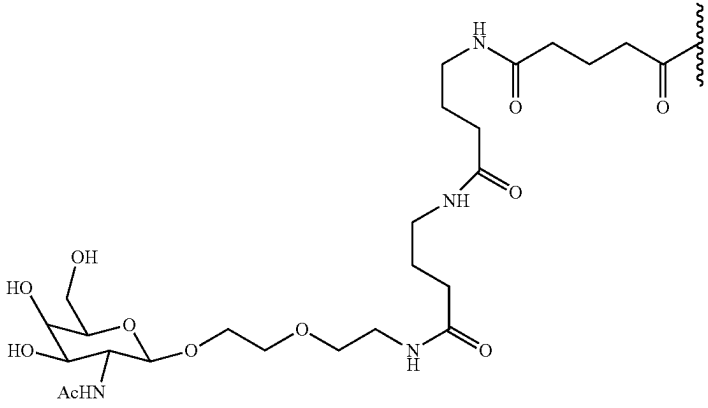
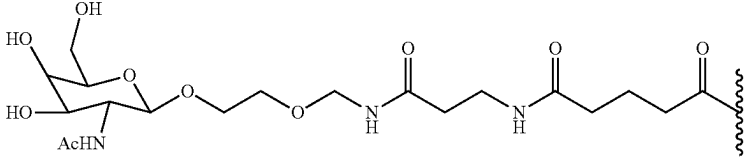
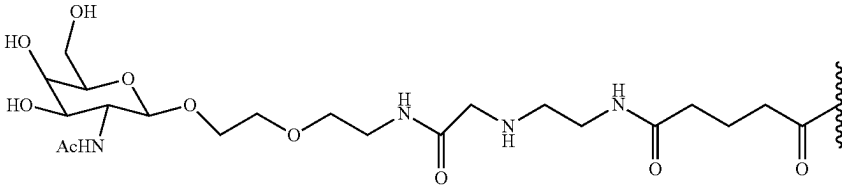
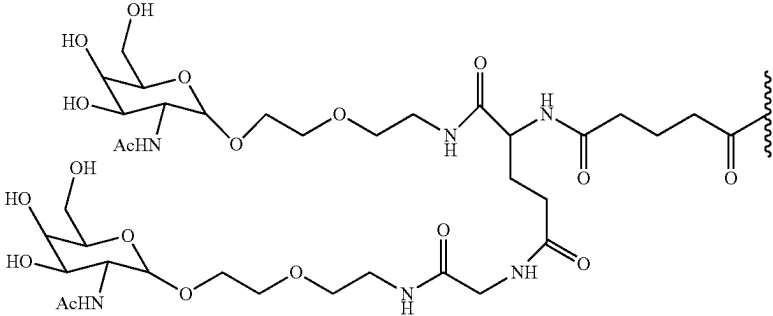
Structures of (X-L) _n -B-D- in 5'MVIP		
Serial number	Code	Structural formula
5	5'YICa-02	
6	5'YICa-03	
7	5'YICa-04	
8	5'YICa-05	
9	5'ERCa-01	

TABLE 8-continued

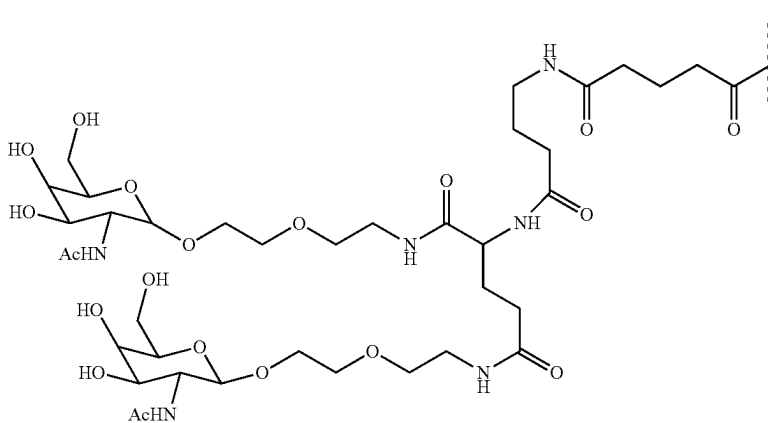
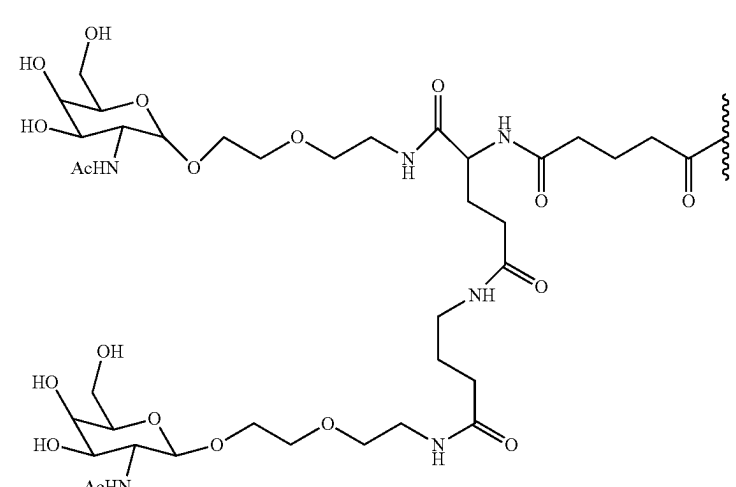
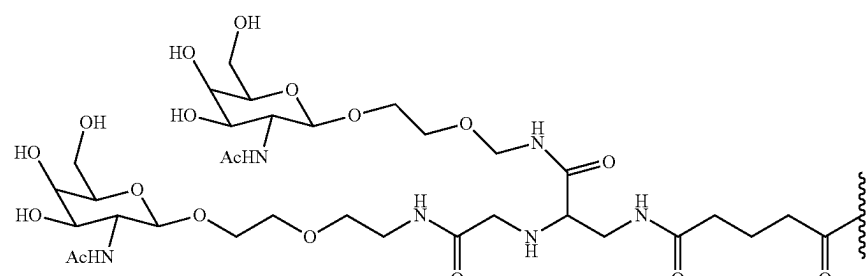
Structures of (X-L) _n -B-D- in 5'MVIP		
Serial number	Code	Structural formula
10	5'ERCa-02	
11	5'ERCa-03	
12	5'ERCa-04	

TABLE 8-continued

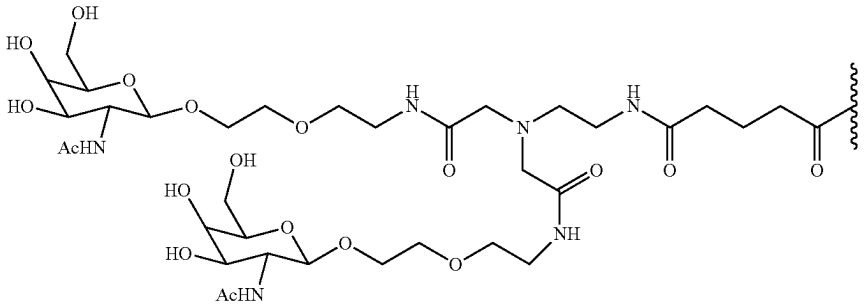
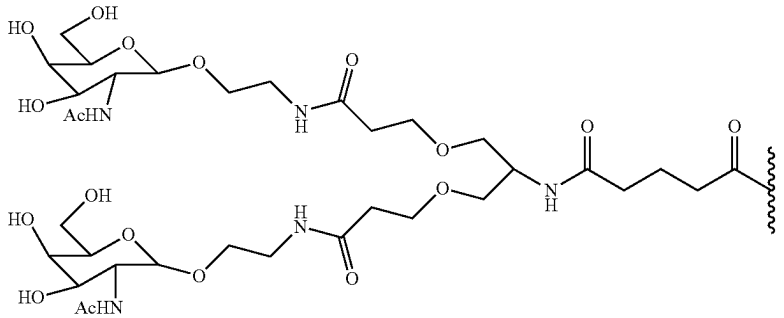
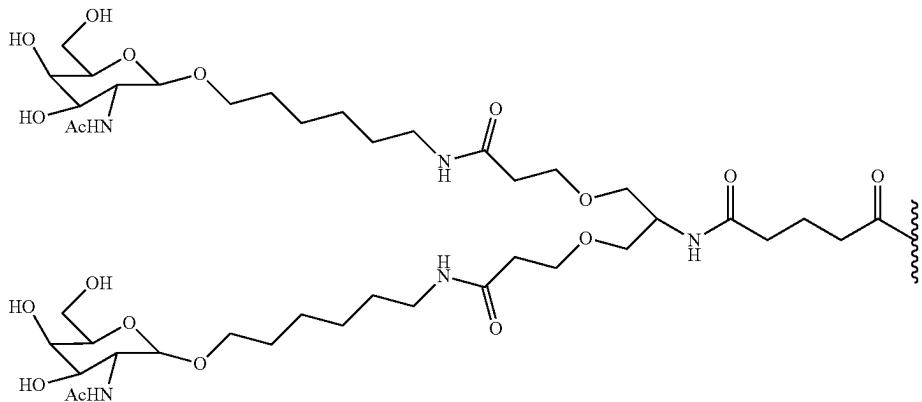
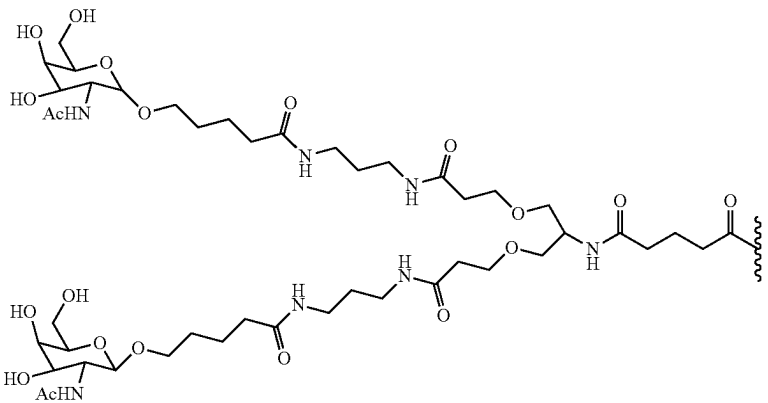
Structures of (X-L) _n -B-D- in 5'MVIP		
Serial number	Code	Structural formula
13	5'ERCa-05	
14	5'ERCdd-01	
15	5'ERCd-01	
16	5'ERCc-01	

TABLE 8-continued

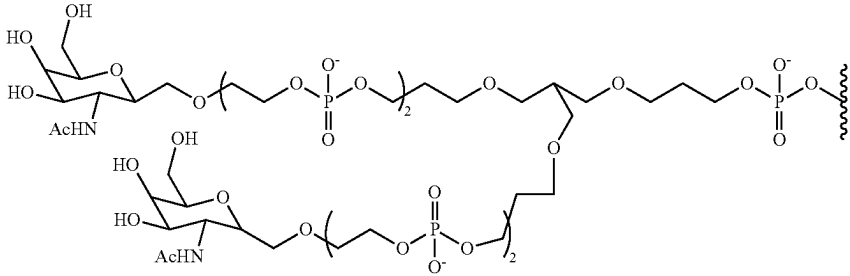
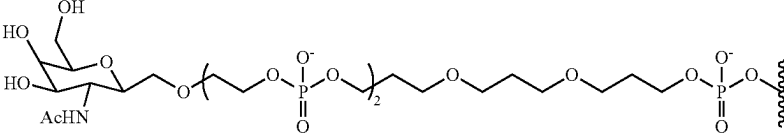
Structures of (X-L) _n -B-D- in 5'MVIP		
Serial number	Code	Structural formula
17	5'SANCdd-01	
18	5'SANCd-01	
19	5'SANCC-01	

TABLE 8-continued

Structures of (X-L)_n-B-D- in 5'MVII

Serial number	Code	Structural formula
20	5'SANCa-01	
21	5'SANCa-02	
22	5'SANCa-03	

TABLE 8-continued

Structures of (X-L) _m -B-D- in 5'MVIP		
Serial num-ber	Code	Structural formula
23	5'ERCr-06	
24	5'YICr-06	

[0183] In some embodiments, 5'MVIP may not exist, in which case m may be an integer of 2-4.

[0184] In some embodiments, (X-L)_m-B-D- in the 3'MVIP structure is selected from the structures shown in Table 9:

TABLE 9

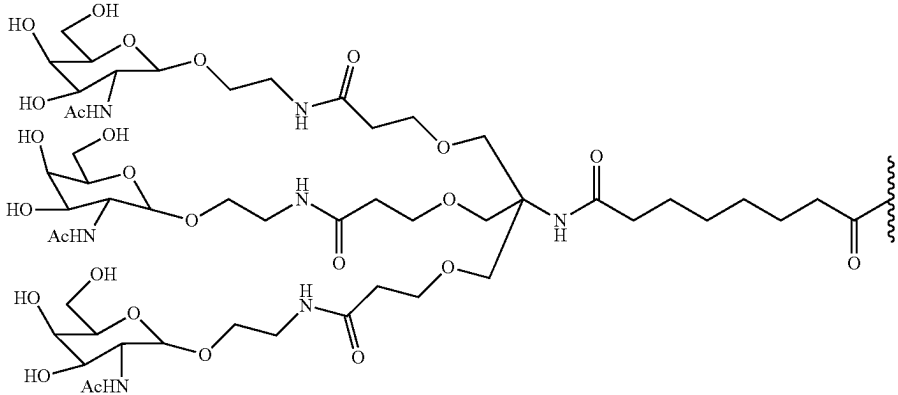
Structures of (X-L) _m -B-D- in 3'MVIP		
Serial num-ber	code	structure
1	3'SANCDd-01	

TABLE 9-continued

Structures of (X-L) _m -B-D- in 3'MVIP		
Serial num-ber	code	structure
2	3'SANCD-01	
3	3'SANCC-01	
4	3'SANCA-01	

Structures of (X-L)_m-B-D- in 3'MVIP

Structures of (X-L) _m -B-D- in 3'MVIP		
Serial number	code	structure
5	3'SANCa-02	
6	3'SANCa-03	
7	3'ERCdd-01	
8	3'ERCd-01	

TABLE 9-continued

Structures of (X-L) _m -B-D- in 3'MVIP		
Serial number	code	structure
9	3'ERCc-01	
10	3'ERCa-01	
11	3'ERCa-02	

TABLE 9-continued

Structures of (X-L) _m -B-D- in 3'MVIP		
Serial number	code	structure
12	3'ERCa-03	
13	3'ERCa-04	
14	3'ERCa-05	
15	3'YICa-01	

TABLE 9-continued

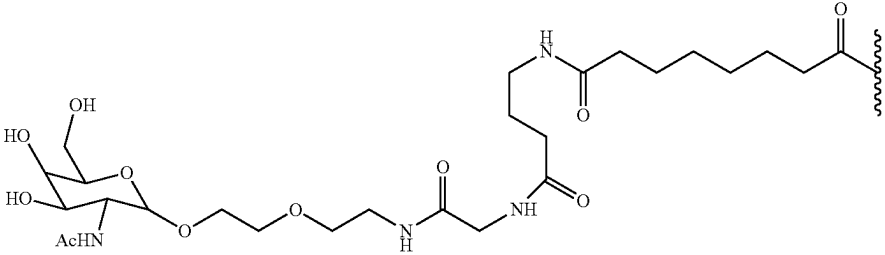
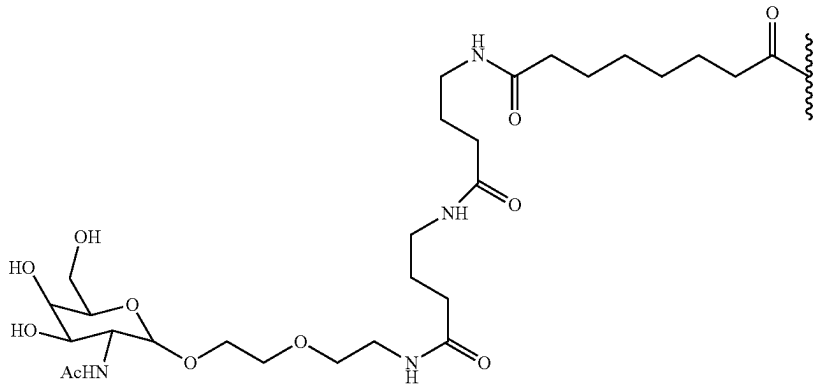
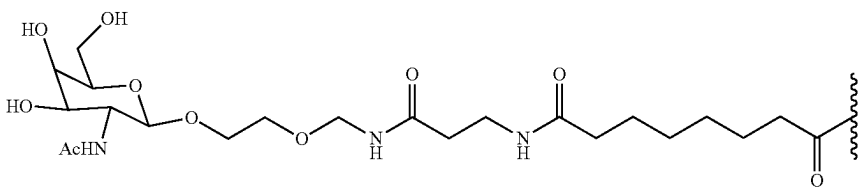
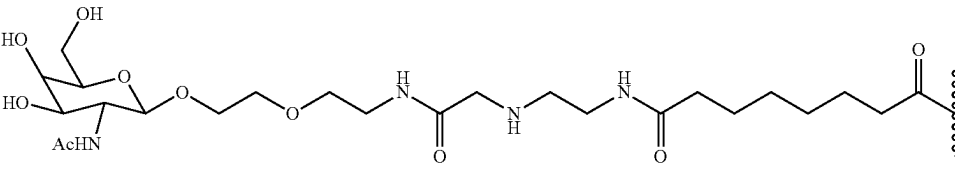
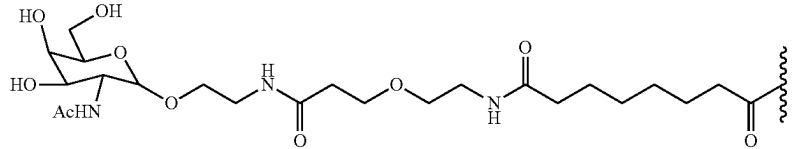
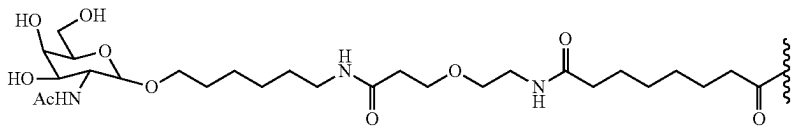
Structures of (X-L) _m -B-D- in 3'MVIP		
Serial number	code	structure
16	3'YICa-02	
17	3'YICa-03	
18	3'YICa-04	
19	3'YICa-05	
20	3'YICdd-01	
21	3'YICd-01	

TABLE 9-continued

Structures of (X-L) _n -B-D- in 3'MVIP		
Serial number	code	structure
22	3'YICc-01	

[0185] In some embodiments, the combinations of R₁ and (X-L)_n-B-D- in the 5'MVIP ligand structures are shown in Table 10.

TABLE 10

Combinations of R ₁ and (X-L) _n -B-D- in 5'MVIP			
Serial number	(X-L) _n -B-D- code	R ₁	5'MVIP code
1	5'YICd-01	—NH(CH ₂) ₆ O—	5'MVIP01
2	5'YICc-01	—NH(CH ₂) ₆ O—	5'MVIP02
3	5'YICa-01	—O(CH ₂) ₆ O—	5'MVIP03
4	5'YICa-02	—O(CH ₂) ₆ O—	5'MVIP04
5	5'YICa-03	—S(CH ₂) ₆ O—	5'MVIP05
6	5'YICa-04	—NH(CH ₂) ₆ S—	5'MVIP06
7	5'YICa-05	—NH(CH ₂) ₈ O—	5'MVIP07
8	5'YICr-06	—NH(CH ₂) ₈ O—	5'MVIP08
9	5'ERCd-01	—NH(CH ₂) ₆ O—	5'MVIP09
10	5'ERCc-01	—NH(CH ₂) ₆ O—	5'MVIP10
11	5'ERCa-01	—NH(CH ₂) ₅ CH(CH ₂ CH ₃)O—	5'MVIP11
12	5'ERCa-02	—O(CH ₂) ₆ O—	5'MVIP12
13	5'ERCa-03	—S(CH ₂) ₆ O—	5'MVIP13
14	5'ERCa-04	—O(CH ₂) ₆ O—	5'MVIP14
15	5'ERCa-05	—O(CH ₂) ₆ O—	5'MVIP15
16	5'ERCr-06	—S(CH ₂) ₄ CH(CH ₃)O—	5'MVIP16
17	5'SANCD-01	—NH(CH ₂) ₆ O—	5'MVIP17
18	5'SANCC-01	—NH(CH ₂) ₆ O—	5'MVIP18
19	5'ERCd-01		5'MVIP19
20	5'ERCd-01		5'MVIP20
21	5'YICd-01		5'MVIP21
22	5'SANCD-01		5'MVIP22

[0186] In some embodiments, 3'MVIP may not exist, in which case n may be 2-4.

[0187] In some embodiments, the combinations of R₂ and (X-L)_m-B-D- in the 3'MVIP ligand structures are shown in Table 11.

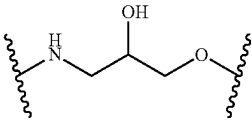
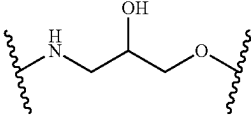
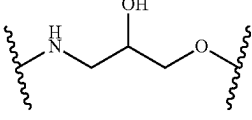
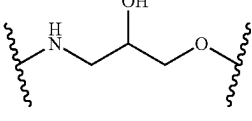
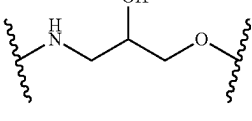
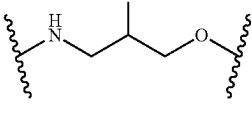
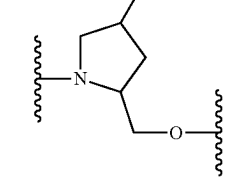
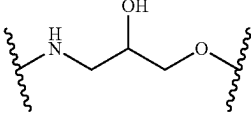
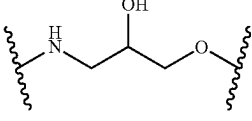
TABLE 11			
Combinations of R ₂ and (X-L) _m -B-D- in 3'MVIP			
Serial number	(X-L) _m -B-D- code	R ₂	3'MVIP code
1	3'YICd-01		3'MVIP01
2	3'YICc-01		3'MVIP02
3	3'YICa-01		3'MVIP03
4	3'YICa-02		3'MVIP04
5	3'YICa-03		3'MVIP05
6	3'YICa-04		3'MVIP06
7	3'YICa-05		3'MVIP07
8	3'YICr-06		3'MVIP08
9	3'ERCd-01		3'MVIP09

TABLE 11-continued

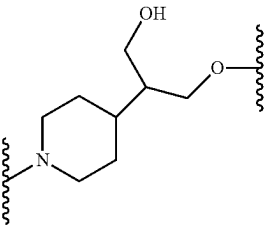
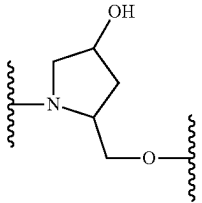
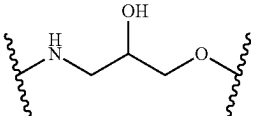
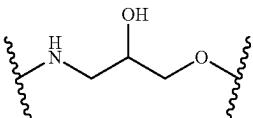
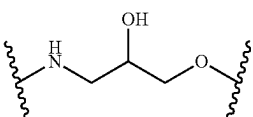
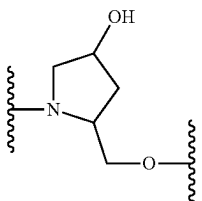
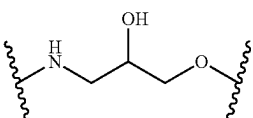
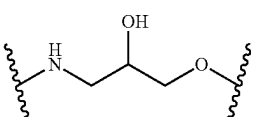
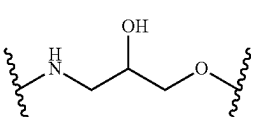
Combinations of R ₂ and (X-L) _m -B-D- in 3'MVIP			
Serial number	(X-L) _m -B-D- code	R ₂	3'MVIP code
10	3'ERCc-01		3'MVIP10
11	3'ERCa-01		3'MVIP11
12	3'ERCa-02		3'MVIP12
13	3'ERCa-03		3'MVIP13
14	3'ERCa-04		3'MVIP14
15	3'ERCa-05		3'MVIP15
16	3'ERCr-06		3'MVIP16
17	3'SANCD-01		3'MVIP17
18	3'SANCC-01		3'MVIP18

TABLE 11-continued

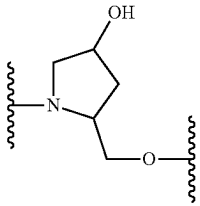
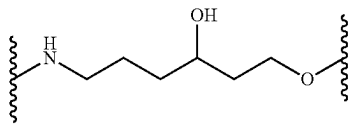
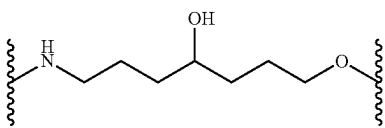
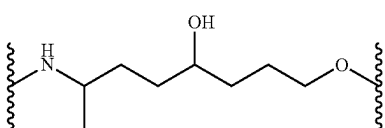
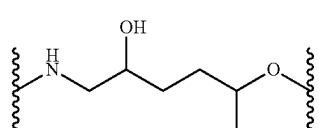
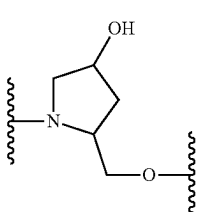
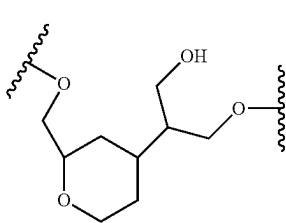
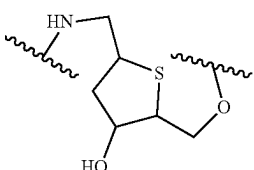
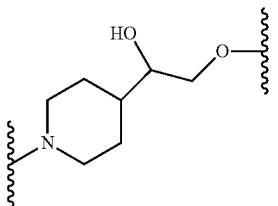
Combinations of R ₂ and (X-L) _m -B-D- in 3'MVIP			
Serial number	(X-L) _m -B-D- code	R ₂	3'MVIP code
19	3'SANCa-01		3'MVIP19
20	3'ERCd-01		3'MVIP20
21	3'ERCd-01		3'MVIP21
22	3'ERCd-01		3'MVIP22
23	3'ERCd-01		3'MVIP23
24	3'ERCd-01		3'MVIP24
25	3'ERCd-01		3'MVIP25
26	3'ERCd-01		3'MVIP26

TABLE 11-continued

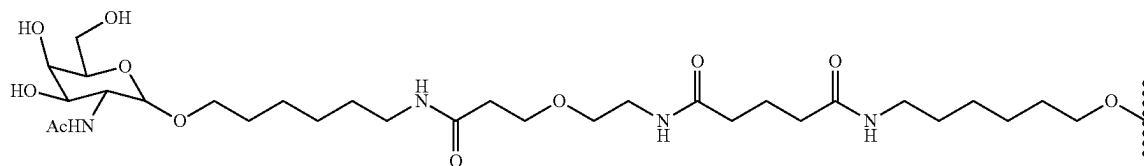
Combinations of R ₂ and (X-L) _m -B-D- in 3'MVIP			
Serial number	(X-L) _m -B-D- code	R ₂	3'MVIP code
27	3'ERCd-01		3'MVIP27

[0188] In some embodiments, the 5'MVIP is selected from any one of 5'MVIP01 to 5'MVIP22 in Table 10.

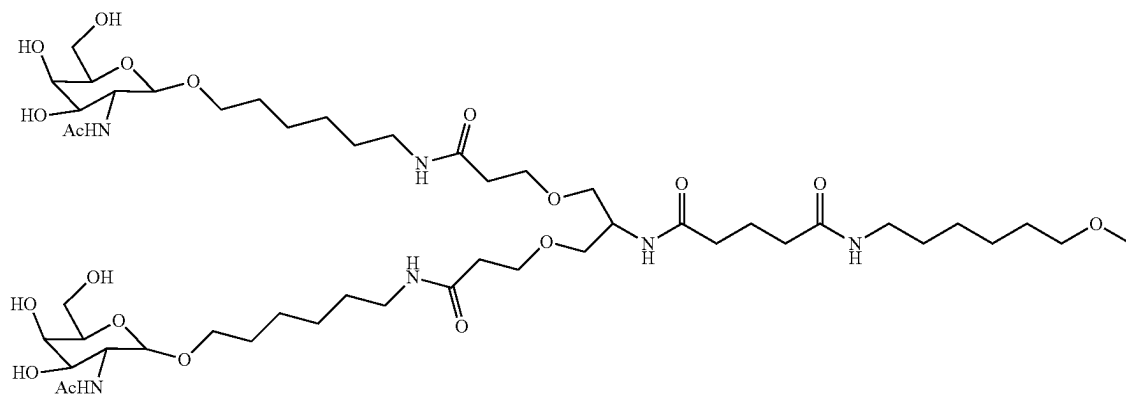
[0189] In some embodiments, the 5'MVIP is selected from:

the 5' end of the sense strand has 5' MVIP, its 3' end can also have 3' MVIP; When the 5' end of the antisense strand has 5' MVIP, its 3' end can also have 3' MVIP. Alternatively, the 5' MVIP can be placed at the 5' ends of both the sense strand

5'MVIP01



5'MVIP09



[0190] In some embodiments, the 3'MVIP is selected from any one of 3'MVIP01 to 3'MVIP27 in Table 11.

[0191] In the RNAi agent of the present application, when there is 3'MVIP on one of its single strands, the other single strand complementary to this single strand correspondingly has a 5' MVIP or 3'MVIP or none; When there is a 5' MVIP on one single strand of the RNA inhibitor, the other single strand complementary to this single strand correspondingly has a 3' MVIP or 5' MVIP or none. The 5' MVIP and 3' MVIP can also be connected to the corresponding ends of the sense strand or the antisense strand simultaneously, that is, when

and the antisense strand. Alternatively, 3' MVIP can be placed at the 3' ends of both the sense strand and the antisense strand.

[0192] In some embodiments, different combinations of 5'MVIP and 3'MVIP in Table 12 are selected to be incorporated into different positions of the sense and/or antisense strands of the RNAi agent, including the terminal, or (X-L)_m-B-D- is introduced into the 2' position of A, U, G, and/or C glycosyl or the appropriate position of base through chemical reaction in advance to form corresponding phosphoramidite monomers, which are introduced into the middle of the sense and/or antisense strands, wherein the middle refers to any position other than the terminal.

TABLE 12

Combinations of 5'MVIP and 3'MVIP		
Serial number	5'MVIP code	5'MVIP structure
1	5'MVIP01	
2	5'MVIP09	
3	5'MVIP17	
4	5'MVIP01	
5	5'MVIP01	
6	5'MVIP09	

TABLE 12-continued

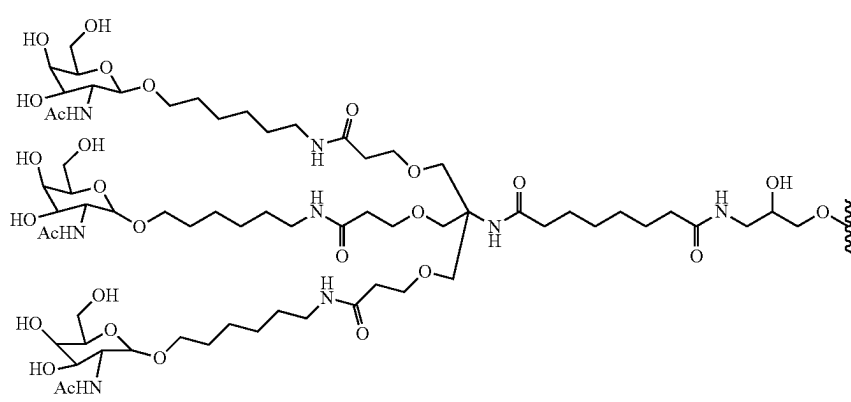
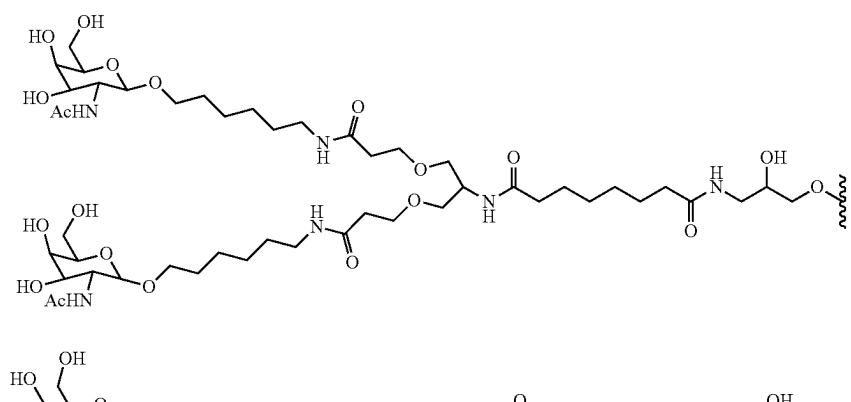
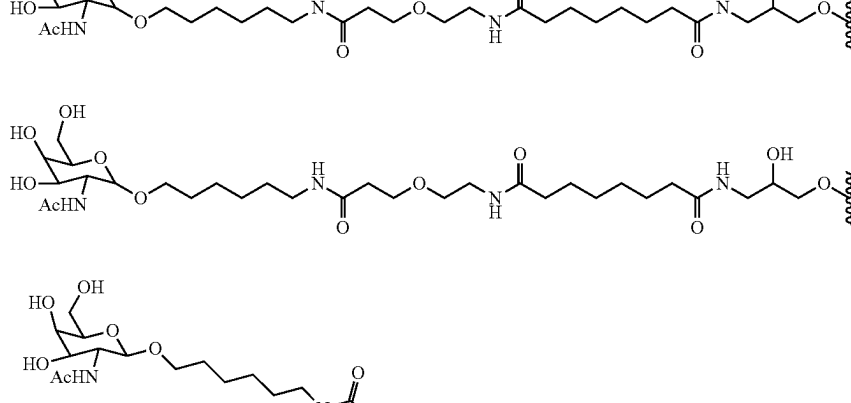
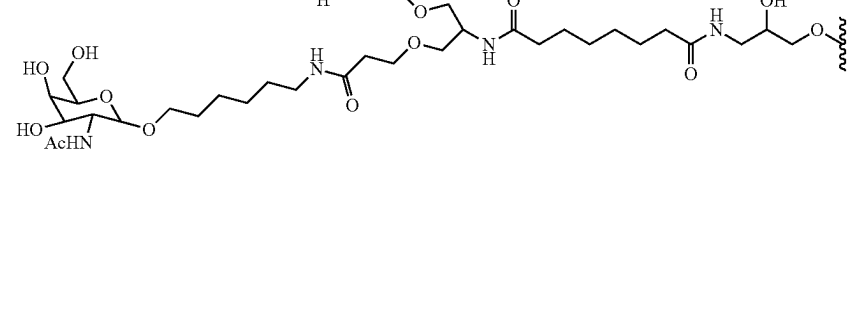
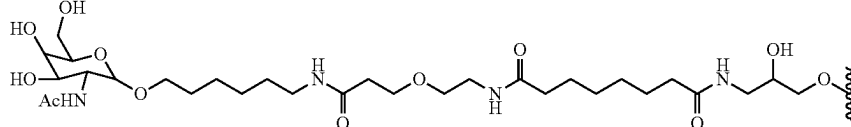
Combinations of 5'MVIP and 3'MVIP		
Serial number	3'MVIP code	3'MVIP structure
1	3'MVIP17	
2	3'MVIP09	
3	3'MVIP01	
4	3'MVIP01	
5	3'MVIP09	

TABLE 12-continued

Combinations of 5'MVIP and 3'MVIP	
6	3'MVIP01
	

[0193] In some embodiments, the RNA inhibitor or its pharmaceutically acceptable salts of the present application may be prepared or synthesized in the form of carboxylate, sodium, triethylamine, or other pharmaceutically acceptable salts.

[0194] In some embodiments, the RNA inhibitor is selected from Table 13.

TABLE 13

RNA inhibitors containing the 5' MVIP09/3' MVIP09 combinations				
Single strand code	Sense strand sequence 5'→3'	Single strand code	Antisense strand sequence 5'→3'	RNA inhibitor
S261	5' MVIP09- UsGsGUfAAfUfGfGACAGAG GUUAUsUsU	AS279	AsUsAACUfCUGUCfCAfU UACCAsUsU-3' MVIP09	Ky11-DS67
S262	5' MVIP09- AsCsAGfAGfUfUfAUCGAG GCUCASTsT	AS280	UsGsAGCCfUCGAUfAAfC UCUGUsUsU-3' MVIP09	Ky11-DS68
S263	5' MVIP09- UsGsGUAAfUfGfGACAGAf GUUAUsUsU	AS281	AsfUsAACfUCUGUCCAfU UfACCAsUsU-3' MVIP09	Ky11-DS69
S264	5' MVIP09- GsUsAAUGfGfAfCAGAGfU UAUCGsAsU	AS282	CsGsAUAAfCfUfCUGUCf CAUUAcsCsA-3' MVIP09	Ky11-DS70
S265	5' MVIP09- GsUsAAfUGfGfAfCAGAGU UAUCGsAsU	AS283	CsGsAUAAfCUCUGfUCfC AUUAcsCsA-3' MVIP09	Ky11-DS71
S266	5' MVIP09- GsCsCCfCUfUfGfGUGUUA UACsAsA	AS284	UsUsGUUfAACACfCAfA GGGGCsUsG-3' MVIP09	Ky11-DS72
S267	5' MVIP09- GsCsCCfCUfUGfGUGUUAf UACsAsA	AS285	UsUsGUfAUfAAfCACCAf AGGGCsUsG-3' MVIP09	Ky11-DS73
S268	5' MVIP09- GsCsCCfCUfUfGfGUGfUfU AfUACsAsA	AS286	UsfUsGfUAfUAfACACCAf AGfGGCsUsG-3' MVIP09	Ky11-DS74
S269	5' MVIP09- AsCsfAGfAGfUfUfAUfCGfA GfGCUCAsTsT	AS287	UsfGsAfGcfCUfCGAUAAf CUfCUGUsUsU-3' MVIP09	Ky11-DS75
S270	5' MVIP09- GsCsUfCfCUfUfAfUfUfGUfU AfUACsGsA	AS288	UsfCsGfUAfUAfACAAUf AGfGAGCsUsG-3' MVIP09	Ky11-DS76
S271	5' MVIP09- fGsUsfAAfUGfGfAfCAGfAG fUUfAUCGsAsU	AS289	CsfGsAfUAACfUCUGUCf CAfUUfACsCsA-3' MVIP09	Ky11-DS77

[0195] In some embodiments, the 5' and/or 3' ends of the sense strand GsUsAAfUGfGfAfCAGAGUUAUCGsAsU of the RNA inhibitor (SEQ ID NO. 460) are linked to the 5' MVIP and/or 3' MVIP of different structures, the sense strand is selected from Table 14 below:

TABLE 14

The 5' end and/or 3' end of the sense strand is linked to the 5' MVIP and/or 3' MVIP of the different structures	
Single strand code	Sense strand sequence 5'→3'
S265	5' MVIP09-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S272	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP17
S273	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP09
S274	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP01
S275	5' MVIP17-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S276	5' MVIP01-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S277	5' MVIP01-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP01
S278	5' MVIP09-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP09
S279	5' MVIP17-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP17
S280	5' MVIP01-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP17
S281	5' MVIP17-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP01
S282	5' MVIP01-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP09
S283	5' MVIP09-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP01
S284	5' MVIP09-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP17
S285	5' MVIP17-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP09
S286	5' MVIP12-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S287	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP19
S288	5' MVIP16-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP16
S289	5' MVIP04-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S290	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP18
S291	5' MVIP03-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S292	5' MVIP08-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S293	5' MVIP16-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S294	5' MVIP13-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP06
S295	5' MVIP04-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP06
S296	5' MVIP11-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S297	5' MVIP11-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP14
S298	5' MVIP15-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S299	5' MVIP02-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S300	5' MVIP05-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S301	5' MVIP06-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S302	5' MVIP07-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU

TABLE 14-continued

The 5' end and/or 3' end of the sense strand is linked to the 5' MVIP and/or 3' MVIP of the different structures	
Single strand code	Sense strand sequence 5'→3'
S303	5' MVIP10-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S304	5' MVIP14-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S305	5' MVIP18-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S306	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP02
S307	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP03
S308	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP04
S309	5' MVIP04-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP04
S310	5' MVIP03-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP19
S311	5' MVIP18-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP18
S312	5' MVIP08-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP18
S313	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP05
S314	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP07
S315	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP10
S316	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP11
S317	5' MVIP11-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP11
S318	5' MVIP15-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP15
S319	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP06
S320	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP08
S321	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP12
S322	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP13
S323	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP14
S324	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP15
S325	GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP16
S326	5' MVIP19-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S327	5' MVIP20-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S328	5' MVIP21-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
S329	5' MVIP22-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU

[0196] In some embodiments, the sense strand of the RNA inhibitor of the present application is a sequence having at least 15 consecutive nucleotides identical to each sequence in Table 14, or a sequence having no more than three different nucleotides from each of the sequences in Table 14.

[0197] In some embodiments, the 5' and/or 3' ends of the antisense strand CsGsAUAAfCUCUGfUCfCAUUAACsCsA (SEQ ID NO. 496) of the RNA inhibitor are linked to the 5' MVIP and/or 3' MVIP of different structures, the antisense strand being selected from Table 15 below.

TABLE 15

The 5' end and/or 3' end of the antisense strand are linked to the 5' MVIP and/or 3' MVIP of the different structures	
Single strand code	Antisense strand sequence 5'→3' (21 mer)
AS283	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP09
AS290	5' MVIP17-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS291	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP01
AS292	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP17
AS293	5' MVIP01-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP01
AS294	5' MVIP09-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP09
AS295	5' MVIP17-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP17
AS296	5' MVIP01-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP17
AS297	5' MVIP17-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP01
AS298	5' MVIP01-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP09
AS299	5' MVIP09-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP01
AS300	5' MVIP09-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP17
AS301	5' MVIP17-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP09
AS302	5' MVIP12-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS303	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP19
AS304	5' MVIP16-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP16
AS305	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP17
AS306	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP18
AS307	5' MVIP03-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS308	5' MVIP08-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS309	5' MVIP16-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS310	5' MVIP13-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP06
AS311	5' MVIP04-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP06
AS312	5' MVIP11-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS313	5' MVIP11-CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP14
AS314	5' MVIP15-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS315	5' MVIP02-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS316	5' MVIP05-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS317	5' MVIP06-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS318	5' MVIP07-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS319	5' MVIP10-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS320	5' MVIP14-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS321	5' MVIP18-CsGsAUAAfCUCUGfUCfCAUUACsCsA
AS322	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP02
AS323	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP03
AS324	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP04

TABLE 15-continued

The 5' end and/or 3' end of the antisense strand are linked to the 5' MVIP and/or 3' MVIP of the different structures	
Single strand code	Antisense strand sequence 5'→3' (21 mer)
AS325	5'MVIP04-CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP04
AS326	5'MVIP03-CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP19
AS327	5'MVIP18-CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP18
AS328	5'MVIP08-CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP18
AS329	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP05
AS330	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP07
AS331	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP10
AS332	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP11
AS333	5'MVIP11-CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP11
AS334	5'MVIP15-CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP15
AS335	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP06
AS336	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP08
AS337	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP12
AS338	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP13
AS339	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP14
AS340	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP15
AS341	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP16
AS342	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP24
AS343	CsGsAUAafCUCUGfUCfCAUUACsCsA-3'MVIP27
AS344	5'MVIP19-CsGsAUAafCUCUGfUCfCAUUACsCsA
AS345	5'MVIP20-CsGsAUAafCUCUGfUCfCAUUACsCsA
AS346	5'MVIP21-CsGsAUAafCUCUGfUCfCAUUACsCsA
AS347	5'MVIP22-CsGsAUAafCUCUGfUCfCAUUACsCsA
AS348	5'MVIP01-CsGsAUAafCUCUGfUCfCAUUACsCsA
AS349	5'MVIP09-CsGsAUAafCUCUGfUCfCAUUACsCsA

[0198] In some embodiments, the antisense strand of the RNA inhibitor of the present application is a sequence having at least 15 consecutive nucleotides identical to each sequence in Table 15, or a sequence having no more than three different nucleotides from each of the sequences in Table 15.

[0199] In some embodiments, the RNA inhibitors described in the present application are formed by random pairing of the sense strands of Table 14 or a sequence that differs by one, two or three nucleotides from these sense

strands and the antisense strands in Table 15 or a sequence that differs by one, two or three nucleotides from these antisense strands.

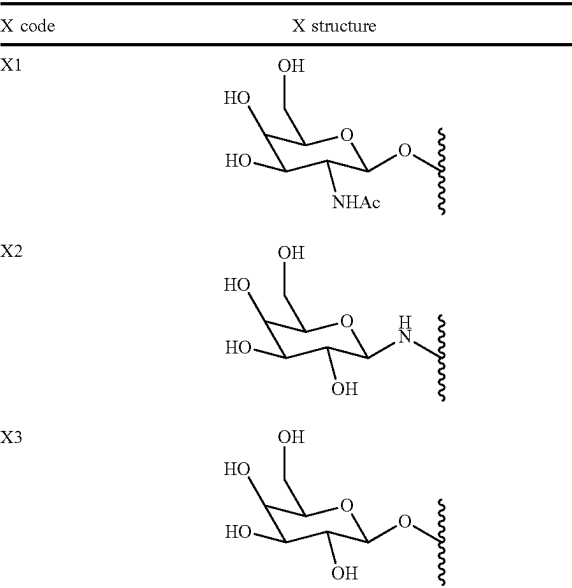
[0200] In some embodiments, the 5' and/or 3' ends of the sense strand GsUsAAfUGfGfAfCAGAGUUAUCGsAsU (SEQ ID NO. 460) of the RNA inhibitor are linked to the 5' MVIP and/or 3' MVIP of different structures, while the mer number of the sequence is adjusted. In some embodiments, the RNA inhibitors of the present application are selected from Table 16.

TABLE 16

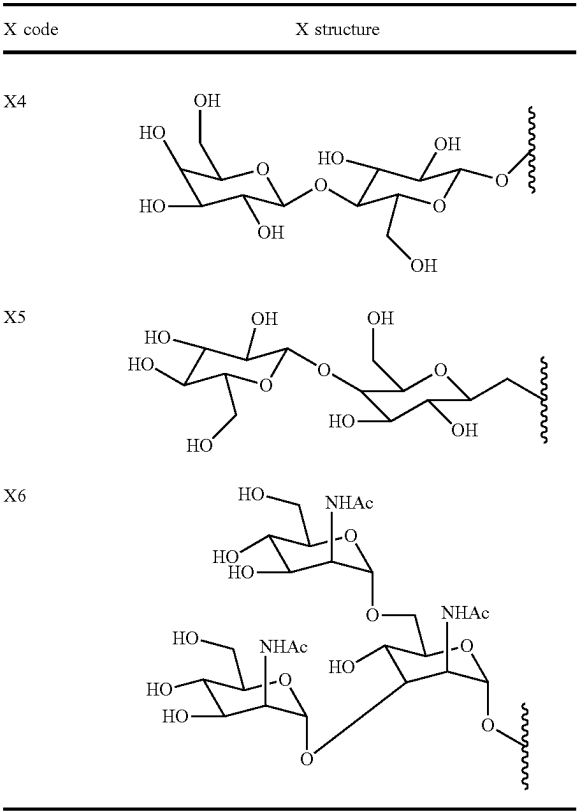
RNA inhibitor codes, single strands contained and their sequences		
RNA inhibitor code	Single strand code	Sequence 5'→3'
Ky11-DS71	S265	5' MVIP09-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
	AS283	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP09
Ky11-DS7101	S330	5' MVIP09-GsUsAAfUGfGfAfCAGAGUUAUsCsG
	AS283	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP09
Ky11-DS7102	S283	5' MVIP09-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP01
	AS236	C ₃ G ₃ AUAAfCUCUGfUCfCAUUAC ₃ C ₃ A
Ky11-DS7103	S333	5' MVIP09-GsUsAAfUGfGfAfCAGAGUUAUsCsG-3' MVIP01
	AS236	C ₃ G ₃ AUAAfCUCUGfUCfCAUUAC ₃ C ₃ A
Ky11-DS7104	S282	5' MVIP01-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU-3' MVIP09
	AS236	C ₃ G ₃ AUAAfCUCUGfUCfCAUUAC ₃ C ₃ A
Ky11-DS7105	S332	5' MVIP01-GsUsAAfUGfGfAfCAGAGUUAUsCsG-3' MVIP09
	AS236	C ₃ G ₃ AUAAfCUCUGfUCfCAUUAC ₃ C ₃ A
Ky11-DS7106	S276	5' MVIP01-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
	AS291	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP01
Ky11-DS7107	S331	5' MVIP01-GsUsAAfUGfGfAfCAGAGUUAUsCsG
	AS291	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP01
Ky11-DS7108	S276	5' MVIP01-GsUsAAfUGfGfAfCAGAGUUAUCGsAsU
	AS292	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP17
Ky11-DS7109	S331	5' MVIP01-GsUsAAfUGfGfAfCAGAGUUAUsCsG
	AS292	CsGsAUAAfCUCUGfUCfCAUUACsCsA-3' MVIP17

[0201] The patent CN113171371B provides a detailed examination of the effects of different X, L, B, D, R₁, and R₂ in the structures of 5'MVIP and/or 3'MVIP on the activity of RNA inhibitors. The full text of it is introduced into this specification. When one of X, L, B, D, R₁, and R₂ is different, the other parts of the corresponding 5'MVIP and/or 3'MVIP are the same as those of 5'MVIP 09/3'MVIP 09, and the interfering nucleic acid conjugated is the same.

[0202] When X is galactose, galactosamine, N-acetylgalactosamine and their derivatives respectively, the resulting RNA inhibitor preferably uses N-acetylgalactosamine and its derivatives as the ligand:

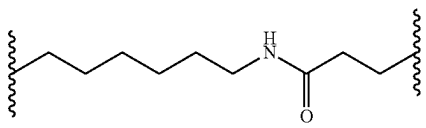
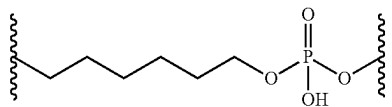
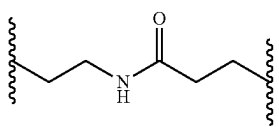
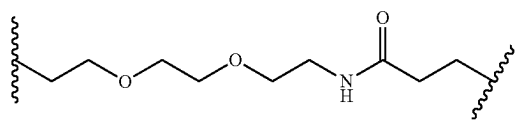
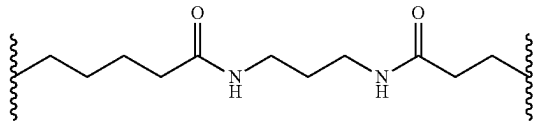
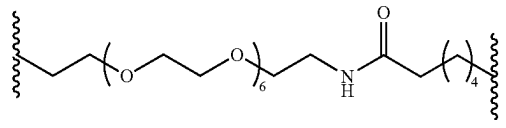
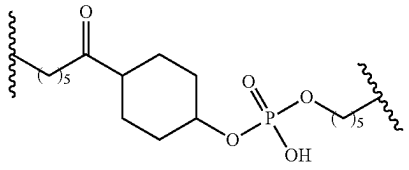
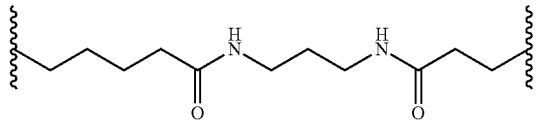
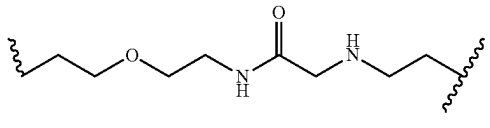
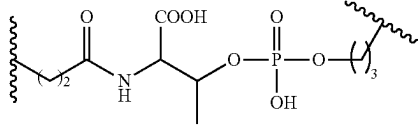


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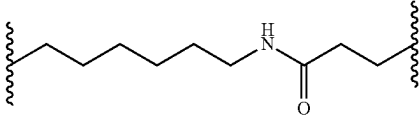
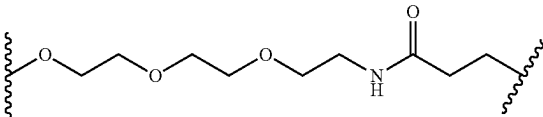
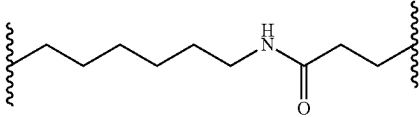
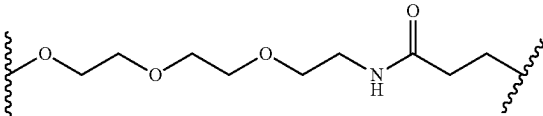
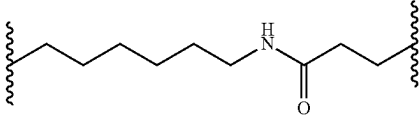
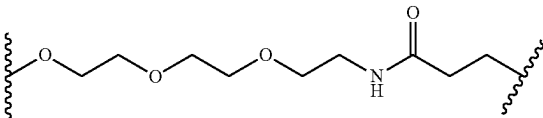
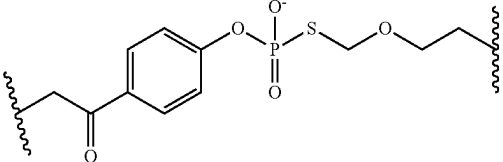


[0203] The length of L has a great impact on the effect of RNA inhibitors. When L contains-NH—, C=O, O, S, amido, phosphoryl, thiophosphoryl, aliphatic carbocyclyl such as cyclohexane, or a combination of these groups, or in

the same 5'MVIP, 3'MVIP structure or 5'MVIP and 3'MVIP with different L structures from each other, the activities of the resulting RNA inhibitors do not differ significantly in the range of chain length of C7-C18.

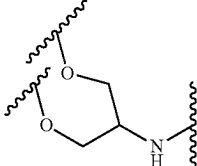
L code	L structure
L1	
L2	
L3	
L4	
L5	
L6	
L7	
L8	
L9	
L10	

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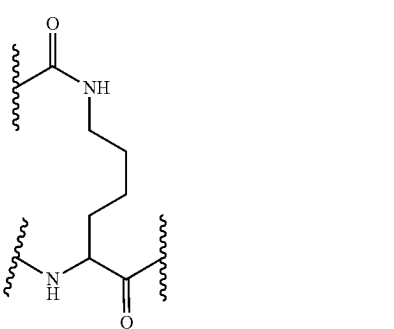
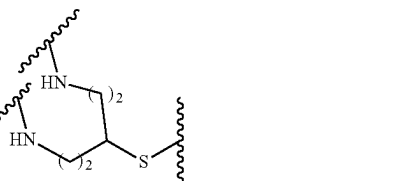
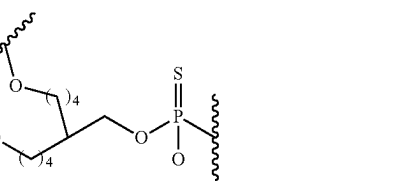
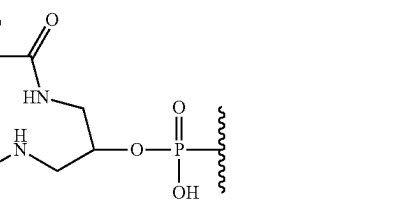
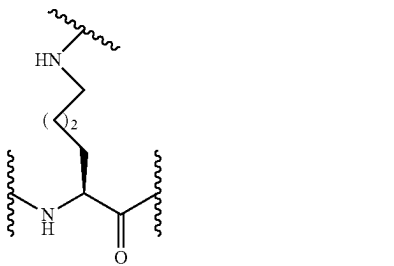
L code	L structure
L11	L in the S of 5' MVIP:  L in the AS of 3' MVIP: 
L12	L in the S of 5' MVIP:  L in the AS of 3' MVIP: 
L13	L in the S of 5' MVIP:  L in the AS of 3' MVIP: 
L14	

[0204] Except for the change in the structure of linker B, when X, L, D and R₁/R₂ are consistent with those in the combination 5'MVIP09/3'MVIP09, A1 and A2 in the general formula of linker B are each independently C, O, S,

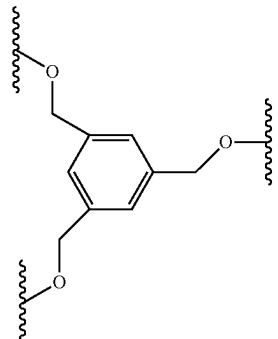
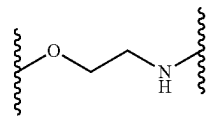
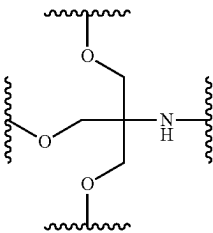
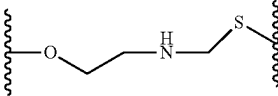
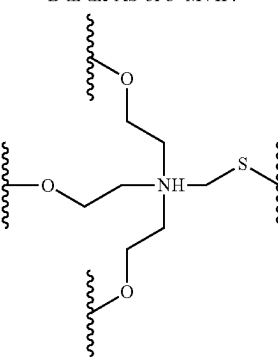
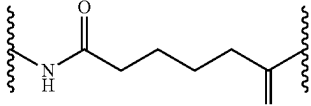
—NH—, carbonyl, amido, phosphoryl or thiophosphoryl, and r is an integer from 0 to 4, and when linker B is the same or different between 5'MVIP and 3'MVIP, the inhibitory activity of resulting RNA does not differ significantly.

B code	B structure
B1	

-continued

B code	B structure
B2	
B3	
B4	
B5	
B6	

-continued

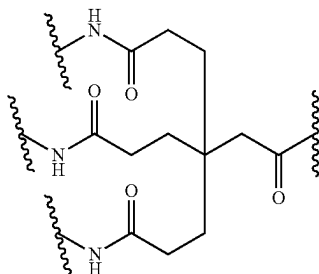
B code	B structure
B7	
B8	B in the S of 5' MVIP:  B in the AS of 3' MVIP: 
B9	B in the S of 5' MVIP:  B in the AS of 3' MVIP: 
B10	B in the S of 5' MVIP: 

-continued

B code

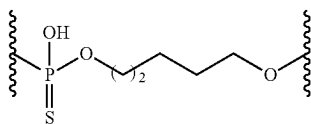
B structure

B in the AS of 3' MVIP:

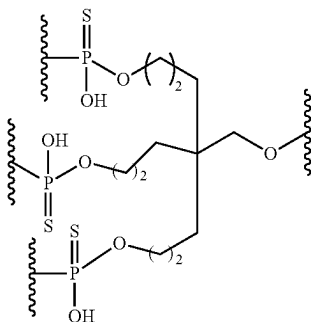


B11

B in the S of 5' MVIP:

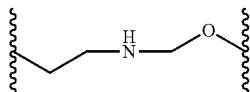


B in the AS of 3' MVIP:

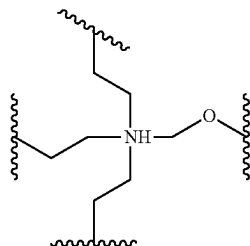


B12

B in the S of 5' MVIP:

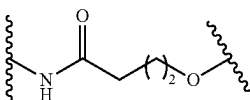


B in the AS of 3' MVIP:



B13

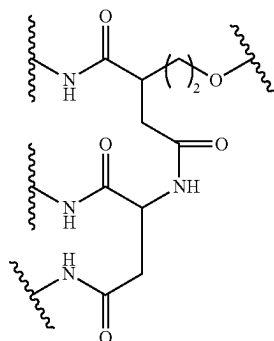
B in the S of 5' MVIP:



-continued

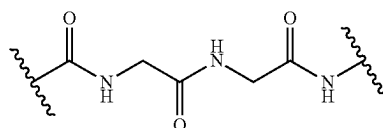
B code	B structure
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15
16	16
17	17
18	18
19	19
20	20
21	21
22	22
23	23
24	24
25	25
26	26
27	27
28	28
29	29
30	30
31	31
32	32
33	33
34	34
35	35
36	36
37	37
38	38
39	39
40	40
41	41
42	42
43	43
44	44
45	45
46	46
47	47
48	48
49	49
50	50
51	51
52	52
53	53
54	54
55	55
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60	60
61	61
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63	63
64	64
65	65
66	66
67	67
68	68
69	69
70	70
71	71
72	72
73	73
74	74
75	75
76	76
77	77
78	78
79	79
80	80
81	81
82	82
83	83
84	84
85	85
86	86
87	87
88	88
89	89
90	90
91	91
92	92
93	93
94	94
95	95
96	96
97	97
98	98
99	99
100	100

B in the AS of 3' MVIP:

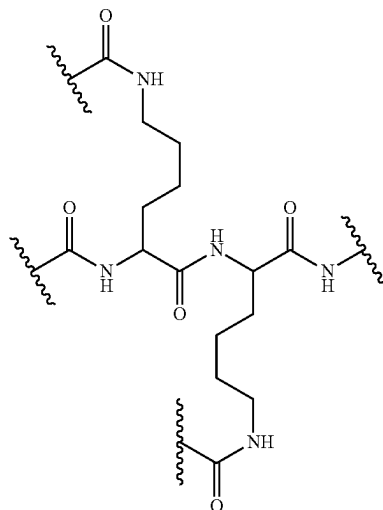


B14

B in the S of 5' MVIP:

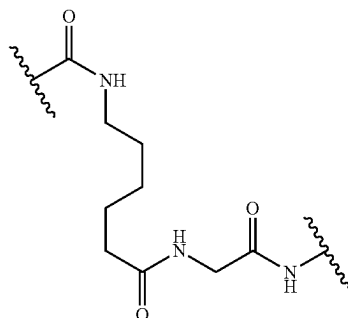


B in the AS of 3' MVIP:



B15

B in the S of 5' MVIP:

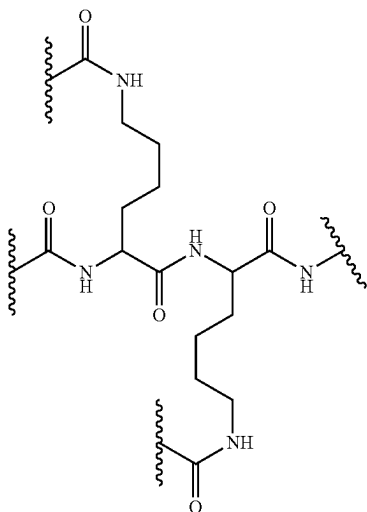


-continued

B code

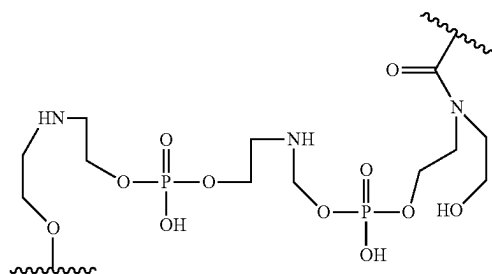
B structure

B in the AS of 3' MVIP:

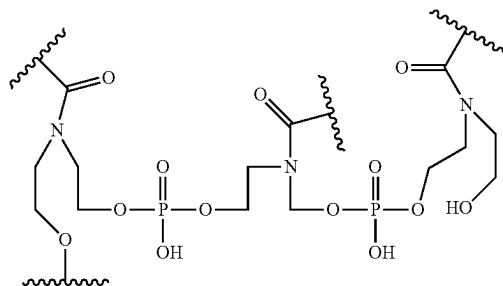


B16

B in the S of 5' MVIP:

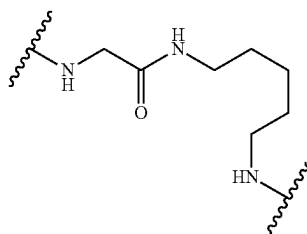


B in the AS of 3' MVIP:



B17

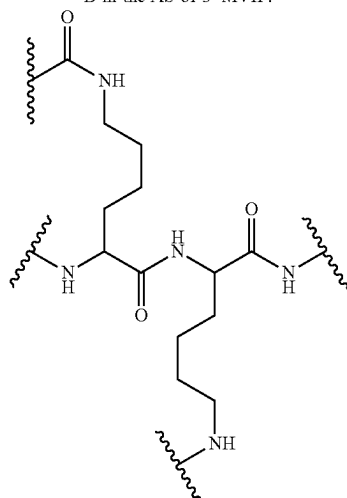
B in the S of 5' MVIP:



-continued

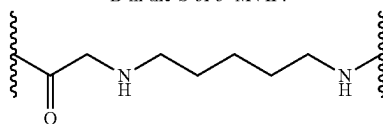
B code	B structure
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B in the AS of 3' MVIP:

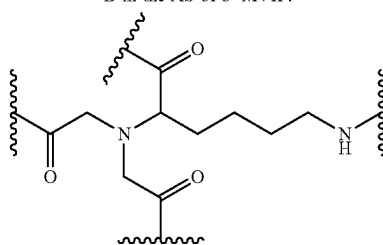


B18

B in the S of 5' MVIP:

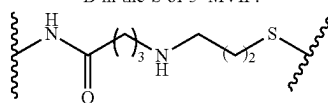


B in the AS of 3' MVIP:

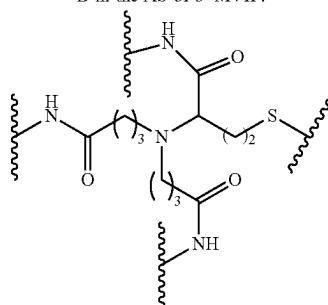


B19

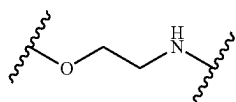
B in the S of 5' MVIP:



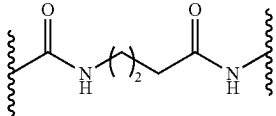
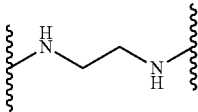
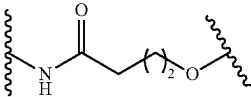
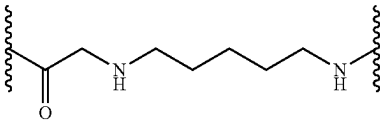
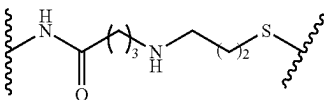
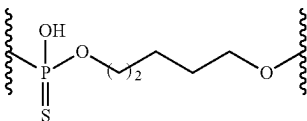
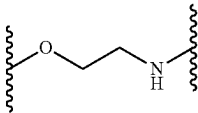
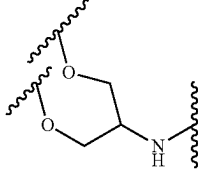
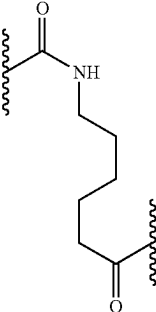
B in the AS of 3' MVIP:



B20



-continued

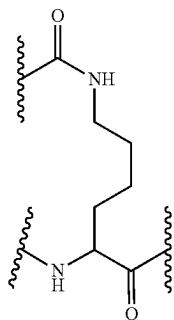
B code	B structure
B21	
B22	
B23	
B24	
B25	
B26	
B27	B in the S of 5' MVIP:  B in the AS of 3' MVIP: 
B28	B in the S of 5' MVIP: 

-continued

B code

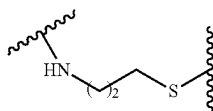
B structure

B in the AS of 3' MVIP:

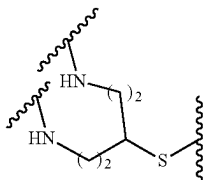


B29

B in the S of 5' MVIP:

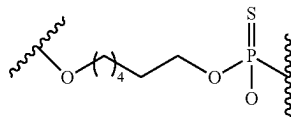


B in the AS of 3' MVIP:

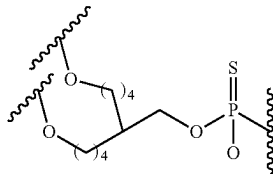


B30

S 的 5'MVIP 中 B:

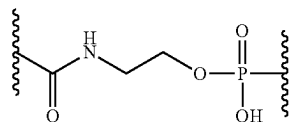


AS 的 3'MVIP 中 B:

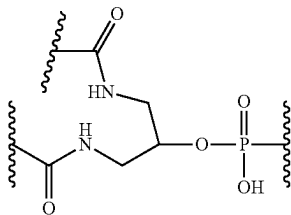


B31

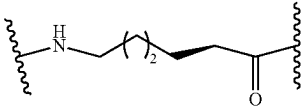
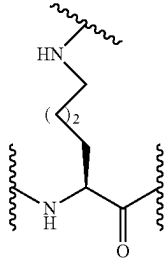
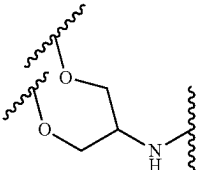
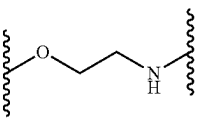
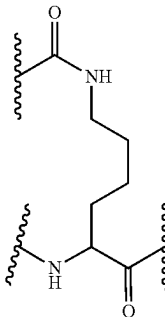
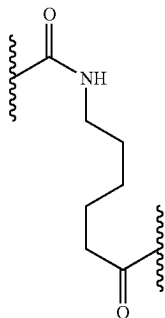
S 的 5'MVIP 中 B:



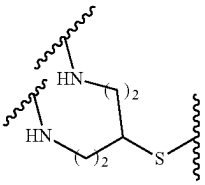
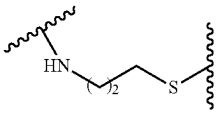
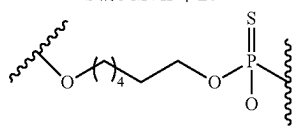
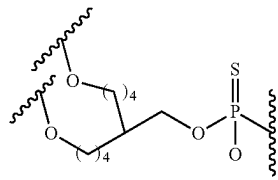
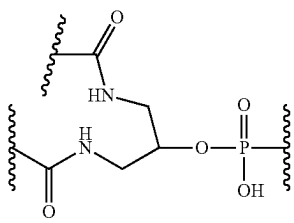
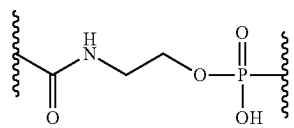
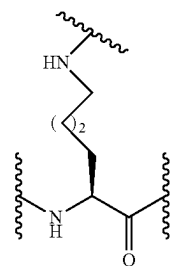
AS 的 3'MVIP 中 B:



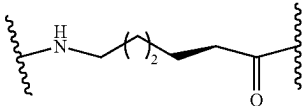
-continued

B code	B structure
B32	B in the S of 5' MVIP:  B in the AS of 3' MVIP: 
B33	B in the S of 5' MVIP:  B in the AS of 3' MVIP: 
B34	B in the S of 5' MVIP:  B in the AS of 3' MVIP: 

-continued

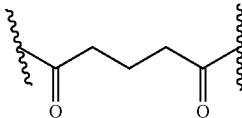
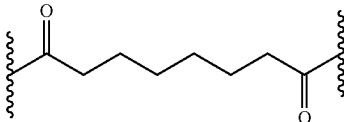
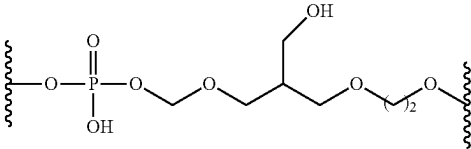
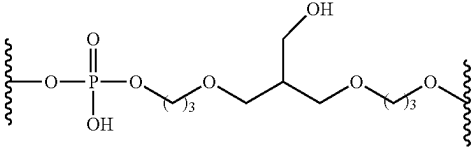
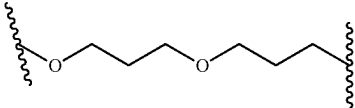
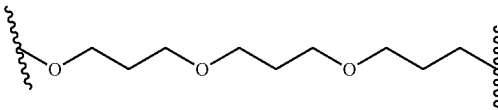
B code	B structure
B35	B in the S of 5' MVIP:  B in the AS of 3' MVIP: 
B36	S 的 5'MVIP 中 B:  AS 的 3'MVIP 中 B: 
B37	S 的 5'MVIP 中 B:  AS 的 3'MVIP 中 B: 
B38	B in the S of 5' MVIP: 

-continued

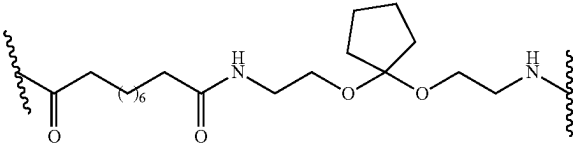
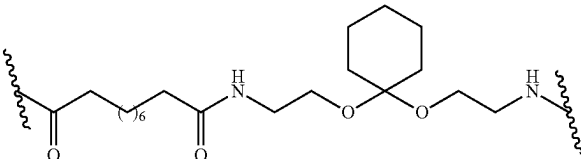
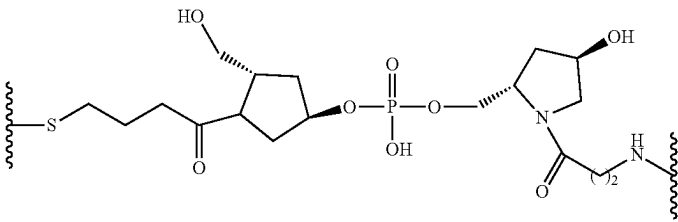
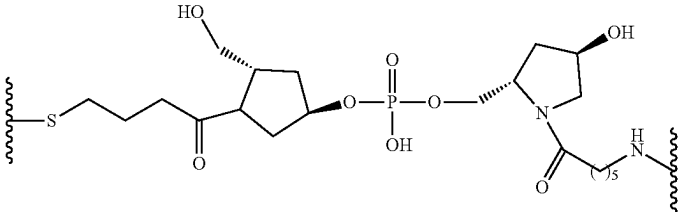
B code	B structure
	B in the AS of 3' MVIP: 

[0205] Except for the change in the structure of linking chain D, when X, L, D and R_1/R_2 are consistent with those in the combination 5'MVIP09/3'MVIP09, different linking

chain D will affect the activity of RNA inhibitor, among which D1, D2, and D4 have similar effects and are better than D3.

D code	D structure
D1	D in the S of 5' MVIP:  D in the AS of 3' MVIP: 
D2	D in the S of 5' MVIP:  D in the AS of 3' MVIP: 
D3	D in the S of 5' MVIP:  D in the AS of 3' MVIP: 

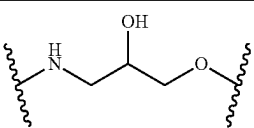
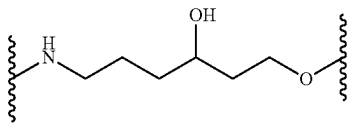
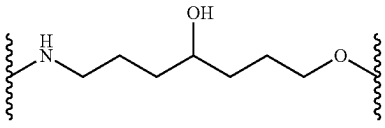
-continued

D code	D structure
D4	D in the S of 5' MVIP: 
	D in the AS of 3' MVIP: 
D5	D in the S of 5' MVIP: 
	D in the AS of 3' MVIP: 

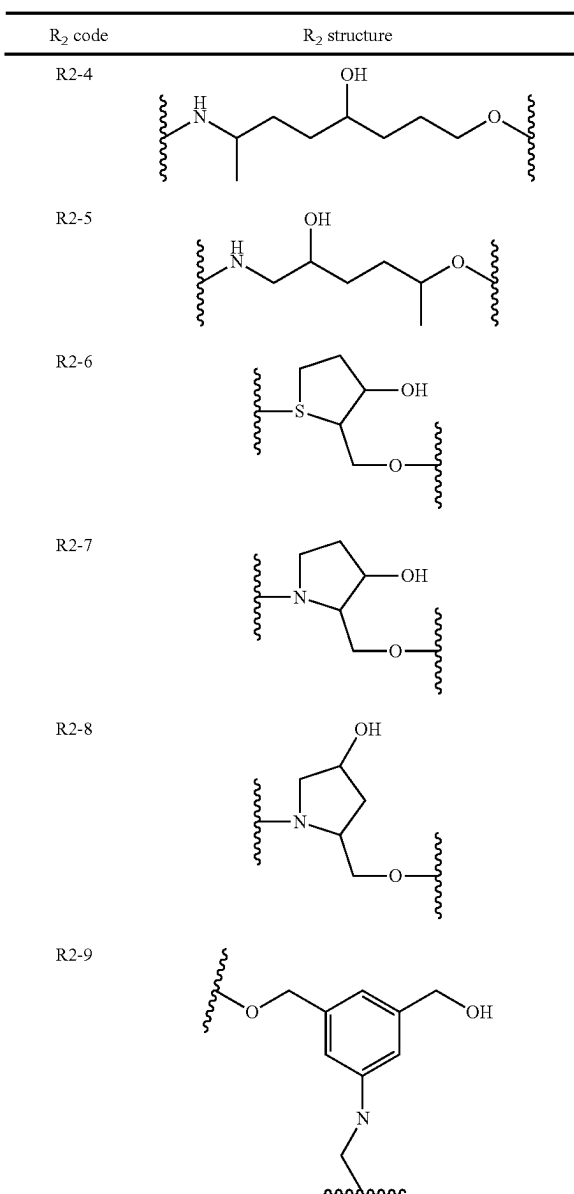
[0206] Different transition point R_1 will affect the activity of RNA inhibitor, among which the obtained RNA inhibitory activity using R1-1 as the transition point is the best.

R_1 code	R_1 structure
R1-1	$-\text{NH}(\text{CH}_2)_6\text{O}-$
R1-2	$-\text{O}(\text{CH}_2)_6\text{O}-$
R1-3	$-\text{S}(\text{CH}_2)_6\text{O}-$
R1-4	$-\text{NH}(\text{CH}_2)_8\text{O}-$
R1-5	$-\text{NH}(\text{CH}_2)_5\text{CH}(\text{CH}_2\text{CH}_3)\text{O}-$
R1-6	$-\text{S}(\text{CH}_2)_4\text{CH}(\text{CH}_3)\text{O}-$

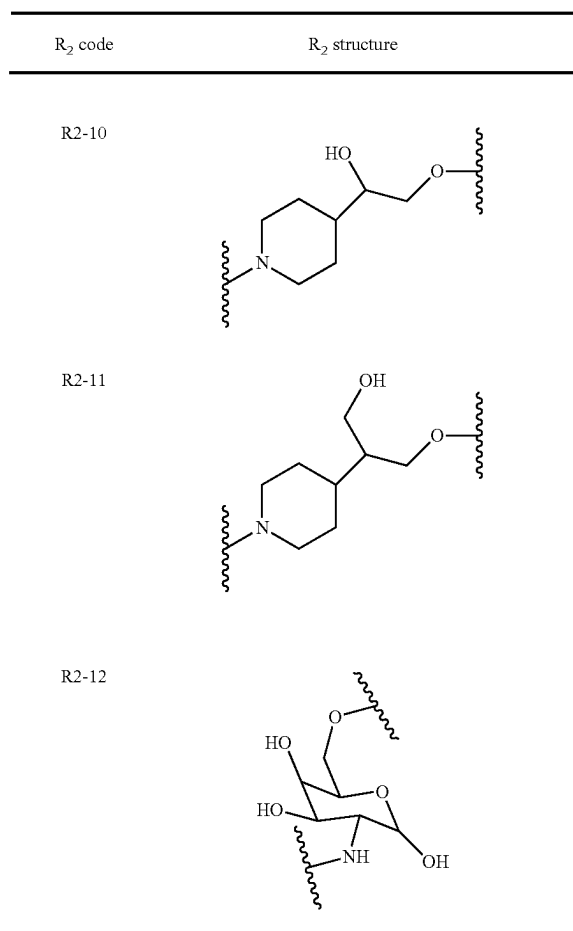
[0207] Different transition point R_2 will affect the activity of RNA inhibitor, and the obtained RNA inhibitor is the best when using R2-1 as the transition point.

R_2 code	R_2 structure
R2-1	
R2-2	
R2-3	

-continued



-continued



[0208] In some embodiments, the n+m of the RNA inhibitors are 2, 3, 4, 5 and 6, respectively. The positions for 5'MVIP and/or 3'MVIP conjugating include the 5' end and/or 3' end of antisense strand, the 5' end and/or 3' end of sense strand, the 5' end of antisense strand and 3' end of sense strand, and the 5' end of and 3' end of sense strand. The antisense strand in Table 15 and the sense strand in Table 14 are randomly paired and annealed, where n+m=2, 3, 4, 5 and 6, the RNA inhibitors are shown in Table 17.

TABLE 17

RNA inhibitors containing combinations of 5' MVIP and 3' MVIP						
Single strand code	Ligand structure	Single strand code	Ligand structure	Double strand code	n + m	The conjugation position of ligand structure and siRNA(S:AS)
S276	5'MVIP01	AS291	3' MVIP01	Ky11-DS7106	2	n:m
S328	5'MVIP21	AS291	3' MVIP01	Ky11-DS7110	2	n:m
S276	5'MVIP01	AS330	3' MVIP07	Ky11-DS7111	2	n:m
S277	5'MVIP01/3'MVIP01	AS236	/	Ky11-DS7112	2	nm:/
S276	5'MVIP01	AS283	3'MVIP09	Ky11-DS7113	3	n:m
S292	5' MVIP08	AS341	3'MVIP15	Ky11-DS7114	3	n:m
S265	5'MVIP09	AS291	3'MVIP01	Ky11-DS7115	3	n:m
S326	5' MVIP19	AS331	3' MVIP07	Ky11-DS7116	3	n:m

TABLE 17-continued

RNA inhibitors containing combinations of 5' MVIP and 3' MVIP						
Single strand code	Ligand structure	Single strand code	Ligand structure	Double strand code	n + m	The conjugation position of ligand structure and siRNA(S':AS)
S282	5'MVIP01/3'MVIP09	AS236	/	Ky11-DS7104	3	nm:/
S283	5'MVIP09/3'MVIP01	AS236	/	Ky11-DS7102	3	nm:/
S316	3'MVIP11	AS335	5'MVIP06	Ky11-DS7117	3	m:n
S276	5'MVIP01	AS292	3' MVIP17	Ky11-DS7108	4	n:m
S328	5' MVIP21	AS292	3' MVIP17	Ky11-DS7118	4	n:m
S275	5'MVIP17	AS291	3' MVIP01	Ky11-DS7119	4	n:m
S329	5' MVIP22	AS330	3' MVIP07	Ky11-DS7120	4	n:m
S265	5'MVIP09	AS283	3'MVIP09	Ky11-DS71	4	n:m
S327	5' MVIP20	AS331	3' MVIP10	Ky11-DS7121	4	n:m
S326	5' MVIP19	AS340	3'MVIP15	Ky11-DS7122	4	n:m
S303	5' MVIP10	AS337	3'MVIP12	Ky11-DS7123	4	n:m
S278	5'MVIP09/3'MVIP09	AS236	/	Ky11-DS7124	4	nm:/
S274	3'MVIP01	AS290	5' MVIP17	Ky11-DS7125	4	m:n
S306	3'MVIP02	AS321	5' MVIP18	Ky11-DS7126	4	m:n
S265	5'MVIP09	AS292	3' MVIP17	Ky11-DS7127	5	n:m
S327	5'MVIP20	AS303	3' MVIP19	Ky11-DS7128	5	n:m
S275	5'MVIP17	AS303	3'MVIP09	Ky11-DS7129	5	n:m
S329	5'MVIP22	AS332	3'MVIP11	Ky11-DS7130	5	n:m
S284	5'MVIP09/3'MVIP17	AS236	/	Ky11-DS7131	5	nm:/
S285	5'MVIP17/3'MVIP09	AS236	/	Ky11-DS7132	5	nm:/
S287	3'MVIP19	AS309	5' MVIP16	Ky11-DS7133	5	m:n
S273	3'MVIP09	AS290	5' MVIP17	Ky11-DS7134	5	m:n
S324	3' MVIP15	AS290	5' MVIP17	Ky11-DS7135	5	m:n
S275	5'MVIP17	AS292	3' MVIP17	Ky11-DS7136	6	n:m
S329	5' MVIP22	AS303	3' MVIP19	Ky11-DS7137	6	n:m
S279	5'MVIP17/3'MVIP17	AS236	/	Ky11-DS7138	6	nm:/
S272	3'MVIP17	AS290	5' MVIP17	Ky11-DS7139	6	m:n

Pharmaceutical Composition

[0209] The present invention also includes a pharmaceutical composition comprising the RNAi agent or a pharmaceutically acceptable salt thereof of the present invention.

[0210] In one embodiment, a pharmaceutical composition comprising the RNAi agent of the present invention and a pharmaceutically acceptable pharmaceutical adjuvants is provided herein. The pharmaceutical composition containing the RNAi agent can be used to prevent and/or treat LPA related disorders, such as hypertension. Such pharmaceutical compositions are formulated according to the mode of delivery. An embodiment is to formulate as a composition for systemic administration by parenteral delivery, such as, subcutaneous (SC), intramuscular (IM), or intravenous (IV) delivery. The pharmaceutical composition of the present invention can be administered at a dose sufficient to inhibit LPA gene expression.

[0211] The pharmaceutically acceptable “adjuvants” or “excipients” are pharmaceutically acceptable solvents, suspensions or any other pharmaceutically inert vehicles used to deliver one or more nucleic acids to animals. The excipients may be liquid or solid and are selected taking account of the intended mode of administration to provide the required volume, consistency, etc. when combined with nucleic acids and other components in a given pharmaceutical composition. The RNAi agents can be delivered in a manner that targets specific tissues (e.g., hepatocytes).

[0212] In some embodiments, the pharmaceutical composition further comprises a delivery vehicle (such as nanoparticles, dendrimers, polymers, liposomes, or cation delivery

systems). In some embodiments, the delivery vehicle includes the amphoteric lipid compounds M10C1.

Purpose

[0213] In another aspect, the present application provides the use of the aforementioned RNA inhibitor or a pharmaceutically acceptable salt thereof, as well as the aforementioned pharmaceutical composition for inhibiting LPA gene expression in the preparation of drugs for preventing or treating diseases or conditions or reducing the risk of diseases or conditions.

[0214] In some embodiments, the disease or condition includes a disease or condition associated with elevated Lp (a) and/or apo (a) levels.

[0215] In some embodiments, the disease or condition comprises a hepatogenic disease.

[0216] In some embodiments, the disease or condition comprises inflammatory, cardio-cerebrovascular, or metabolic diseases.

[0217] In some embodiments, the cardio-cerebrovascular diseases include hyperlipidemia, stroke, atherosclerosis, thrombosis, coronary heart disease, or aortic valve stenosis.

[0218] In yet another aspect, the present application provides a method for preventing or treating diseases, conditions or syndromes, which includes administering to a subject in need an effective amount of the aforementioned RNA inhibitor, a pharmaceutically acceptable salt thereof, or the aforementioned pharmaceutical composition for inhibiting LPA gene expression.

[0219] The RNA inhibitors, the pharmaceutically acceptable salts thereof, or the aforementioned pharmaceutical

compositions for inhibiting LPA gene expression is administered to subject via subcutaneous, intravenous, oral, aerosol inhalation, rectal, or intraperitoneal route.

[0220] Cells suitable for treatment using the method of the present application can be any cells expressing LPA gene, such as liver cells, brain cells, gallbladder cells, heart cells or kidney cells, but preferably liver cells. Cells suitable for use in the method of the present application can be mammalian cells. When in contact with cells expressing LPA gene, RNAi agent inhibits the expression of LPA gene (for example, human, primate, non-primate or rat LPA gene) by at least about 50%, for example, as determined by PCR or methods based on branched DNA (bDNA), or methods based on protein, such as immunofluorescence analysis, Western blotting or flow cytometry.

[0221] The inhibition of LPA gene expression can be represented by a decrease in the amount of mRNA expressed by the first cell or cell population that inhibit LPA gene expression (such cells may be, for example, present in a sample derived from a subject) in which the LPA gene is transcribed and treated (for example, by contacting one or more cells with the RNAi agent of the present invention, or by administering the RNAi agent of the present invention to a subject in which the cell is present) compared to a second cell or cell population (control cells not treated with RNAi agents or RNAi agents targeting the gene of interest) that is essentially the same as that first cell or cell population but not so treated. In a preferred embodiment, inhibition is evaluated in a cell line that highly expresses LPA by the method provided in Example 2 using an appropriate concentration of siRNA, and the mRNA level in the interfered cells is expressed as a percentage of the mRNA levels in the uninterfered control cells.

[0222] In other embodiments, the inhibition of LPA gene expression can be evaluated by the reduction of parameters functionally related to LPA gene expression, such as LPA protein levels in the blood or serum of subjects.

[0223] The inhibition of LPA protein expression can be manifested by the reduction of LPA protein levels expressed by cells or cell populations or subject samples (e.g., protein levels in blood samples derived from subjects). As mentioned above, for the evaluation of mRNA inhibition, the inhibition of protein expression levels of treated cells or cell populations can be similarly shown as a percentage of protein levels of control cells or cell populations, or changes in protein levels in subject samples (e.g., blood or serum derived from it). LPA gene silencing can be assayed in any LPA expressing cell (endogenous or exogenous from the expression constructs) and by any assay known in the art.

[0224] LPA mRNA levels expressed by cells or cell populations can be determined using any method known in the art for evaluating mRNA expression. For example, qRT-PCR is used to evaluate the reduction of gene expression. The reduction in protein production can be evaluated by any method known in the art, for example, ELISA. In some embodiments, the liver biopsy sample is used as tissue material to monitor the reduction of LPA gene or protein expression. In other embodiments, blood samples are used as subject samples for monitoring the reduction of LPA proteins expression.

[0225] Specifically, the present application also discloses the following specific embodiments:

[0226] 1. An RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression, com-

prising a complementary region of an antisense strand, the complementary region comprises at least 15 consecutive nucleotides, wherein the antisense strand comprises the following nucleotide sequence: any sequence of SEQ ID NO. 414-418, or a sequence having at least 15 consecutive nucleotides identical to that in SEQ ID NO. 414-418, or a sequence having no more than three different nucleotides from the above sequence.

[0227] 2. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 1, wherein the complementary region contains 12-25 consecutive nucleotide base pairs.

[0228] 3. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 1, comprising an antisense strand, wherein the antisense strand comprises the following nucleotide sequence: any sequence of SEQ ID NO. 414-418, or a sequence having at least 15 consecutive nucleotides identical to that in SEQ ID NO. 414-418, or a sequence having no more than three different nucleotides from the above sequence.

[0229] 4. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 1, further comprising a sense strand, wherein at least about 80% of the bases are complementary between the sense strand and the antisense strand.

[0230] 5. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 4, wherein the sense and antisense strands are each independently 15-30 nucleotides.

[0231] 6. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 5, wherein the sense and antisense strands are each independently 17-25 nucleotides.

[0232] 7. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 6, wherein the sense and antisense strands are each independently 19-23 nucleotides.

[0233] 8. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 1, wherein the sense strand comprises the following nucleotide sequence: any one of SEQ ID NO. 396-400, or a sequence having at least 15 consecutive nucleotides identical to that in SEQ ID NO. 396-400, or a sequence having no more than three different nucleotides from the above sequence.

[0234] 9. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 1, wherein the sense and antisense strands are selected from one or more of the following combinations:

- 1) The sense strand: (SEQ ID NO. 396)
 ugguaauggacagauuuu,
 the antisense strand: (SEQ ID NO. 414)
 auaacucuguccauuaccuu;
- 2) The sense strand: (SEQ ID NO. 397)
 acagaguuauaggagcucatt,

-continued
the antisense strand: (SEQ ID NO. 415)
ugagccucgauaacucuguuu;
3) The sense strand: (SEQ ID NO. 398)
guaauggacagaguuaucgau,
the antisense strand: (SEQ ID NO. 416)
cgauaacucuguccauuacca;
4) The sense strand: (SEQ ID NO. 399)
gccccuugguguuauacaa,
the antisense strand: (SEQ ID NO. 417)
uuguauaacaccaagggcgug;
And
5) The sense strand: (SEQ ID NO. 400)
gcuccuauuguuauacga,
the antisense strand: (SEQ ID NO. 418)
ucguauaacaauaggagcgug.

[0235] 10. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to any one of embodiments 1-9, wherein one or more nucleotides on the sense strand and/or antisense strand are modified to form a modified nucleotide sequence.

[0236] 11. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 10, wherein the modified nucleotides are selected from the group consisting of deoxyribo-nucleotides, nucleotide mimics, abasic nucleotides, 2'-modified nucleotides, 3'-3'-linkage (inverted) nucleotides, nucleotides containing unnatural bases, bridged nucleotides, peptide nucleic acids (PNA), unlocked nucleobase analogues, locked nucleotides, 3'-O-methoxy (2' inter-nucleo-side linkage) nucleotides, 2'-F-arabinose nucleotides, 5'-Me/2'-Fluoro-nucleotides, morpholino nucleotides, vinyl phosphonate deoxyribonucleotides, nucleotides containing vinyl phosphonate and nucleotides containing cyclopropyl phosphonate.

[0237] 12. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 11, wherein the 2'-modified nucleotides include: 2'-O-methyl nucleotides, 2'-deoxy-2'-fluoronucleotides, 2'-deoxynucleotides, 2'-methoxyethyl nucleotides, 2'-amino nucleotides and/or 2'-alkyl nucleotides.

[0238] 13. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 10, wherein part or all of the —OH at the 2' positions of the nucleotide glycosyls of the sense and antisense strands are substituted, and the substituent are fluorine or methoxy.

[0239] 14. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 13, wherein at least one of the —OH at the 2' positions of the nucleotide glycosyls at positions 2, 4, 6, 7, 8, 12, 14 and 16 starting from the 5' end of the antisense strand are fluorinated.

[0240] 15. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression

according to embodiment 14, wherein the —OH at the 2' positions of the nucleotide glycosyls at positions 7, 12, and 14 starting from the 5' end of the antisense strand are fluorinated, or the —OH at the 2' positions of the nucleotide glycosyls at positions 2, 4, 6, 8, 14, and 16 starting from the 5' end of the antisense chain are fluorinated.

[0241] 16. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 13, wherein except for nucleotides at positions 7, 12, 14 starting from the 5' end of the antisense strand, at least one of the —OH at the 2' position of the remaining nucleotide glycosyls are substituted with methoxy; or except for nucleotides at positions 2, 4, 6, 8, 14, 16 starting from the 5' end of the antisense strand, at least one of the —OH at the 2' position of the remaining nucleotide glycosyls are substituted with methoxy.

[0242] 17. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 13, wherein the —OH at the 2' positions of the nucleotide glycosyls at positions 7, 12, 14 starting from the 5' end of the antisense strand are fluorinated and the —OH at the 2' positions of the remaining nucleotide glycosyls are substituted with methoxy; or the —OH at positions 2, 4, 6, 8, 14, 16 starting from the 5' end of the antisense strand are fluorinated, and the —OH at the 2' positions of the remaining nucleotide glycosyls are substituted with methoxy.

[0243] 18. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 13, wherein at least one of the —OH at the 2' positions of the nucleotide glycosyls at positions 3, 5, 7, 8, 9, 11, 13, 15 starting from the 5' end of the sense strand are fluorinated.

[0244] 19. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 18, wherein the —OH at the 2' positions of the nucleotide glycosyls at positions 5, 7, 8, and 9 starting from the 5' end of the sense strand are fluorinated, or the —OH at the 2' positions of the nucleotide glycosyls at positions 3, 5, 7, 8, 9, 11, 13, 15 starting from the 5' end of the sense strand are fluorinated.

[0245] 20. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 13, wherein except for nucleotides at positions 5, 7, 8, 9 starting from the 5' end of the sense strand, at least one of the —OH at the 2' positions of the remaining nucleotide glycosyls are substituted with methoxy; or except for nucleotides at positions 3, 5, 7, 8, 9, 11, 13, 15 starting from the 5' end of the sense strand, at least one of the —OH at the 2' positions of the remaining nucleotide glycosyls are substituted with methoxy.

[0246] 21. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 20, wherein the —OH at the 2' positions of the nucleotide glycosyls at positions 5, 7, 8, 9 starting from the 5' end of the sense strand is fluorinated, and the —OH at the 2' positions of the remaining nucleotide glycosyls are substituted with methoxy; or the —OH at the 2' positions of the nucleotide glycosyls at positions 3, 5, 7, 8, 9, 11, 13, 15 starting from the 5' end of the sense strand are fluorinated, and the —OH at the 2' position of the remaining nucleotide glycosyls are substituted with methoxy.

[0256] 31. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 30, wherein the ligand is conjugated to the 5' end and 3' ends of the antisense strand, the ligand is conjugated to the 5' end of the antisense strand, or the ligand is conjugated to the 3' end of the antisense strand.

[0257] 32. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 31, wherein the ligand is conjugated to the 5' end and 3' end of the sense strand, the ligand is conjugated to the 5' end of the sense strand, or the ligand is conjugated to the 3' end of the sense strand.

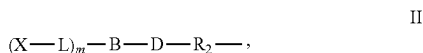
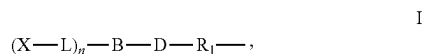
[0258] 33. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 32, wherein the ligand is conjugated to the 5' end of the antisense strand, and the ligand is conjugated to the 3' end of the sense strand.

[0259] 34. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 32, wherein the ligand is conjugated to the 3' end of the antisense strand, and the ligand is conjugated to the 5' end of the sense strand.

[0260] 35. The RNA inhibitor or a pharmaceutically acceptable salt thereof that inhibits LPA gene expression according to embodiment 32, wherein the ligand is conjugated to the 5' and 3' ends of the sense strand.

[0261] 36. An RNA inhibitor or a pharmaceutically acceptable salt thereof, characterized by comprising: a drug,

a ligand conjugated to the drug, wherein the ligand comprises 5' MVIP (5' Multi Valent Import Platform) and 3' MVIP (3' Multi Valent Import Platform), the 5'MVIP is conjugated at the 5' end of the sense strand and/or antisense strand, the 3'MVIP is conjugated at the 3' end of the antisense strand and/or sense strand, the structure of the 5'MVIP is shown in formula I, and the structure of the 3'MVIP is shown in formula II,



[0262] wherein,

[0263] n and m are each independently any integer from 0 to 4, preferably any integer from 1 to 3,

[0264] X is a targeting specific ligand;

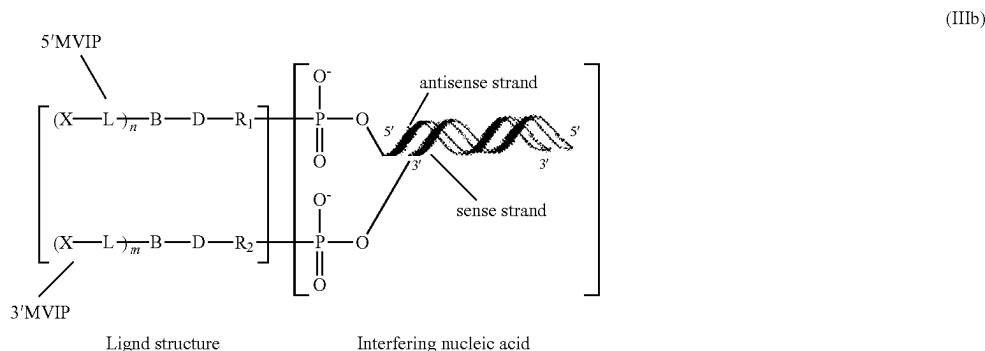
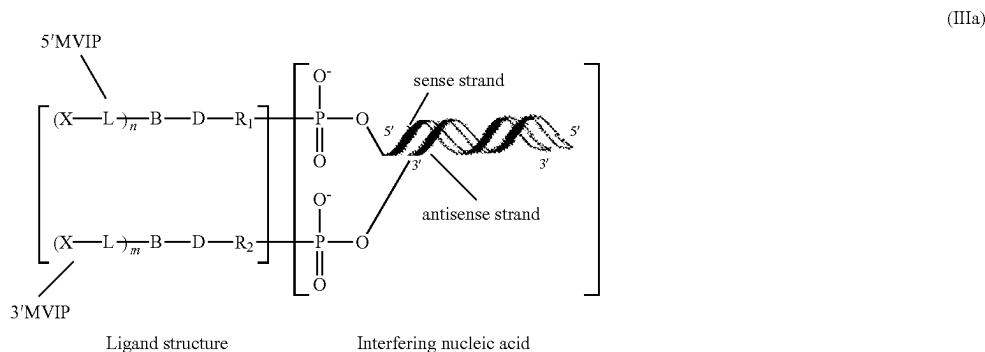
[0265] L is a branched chain;

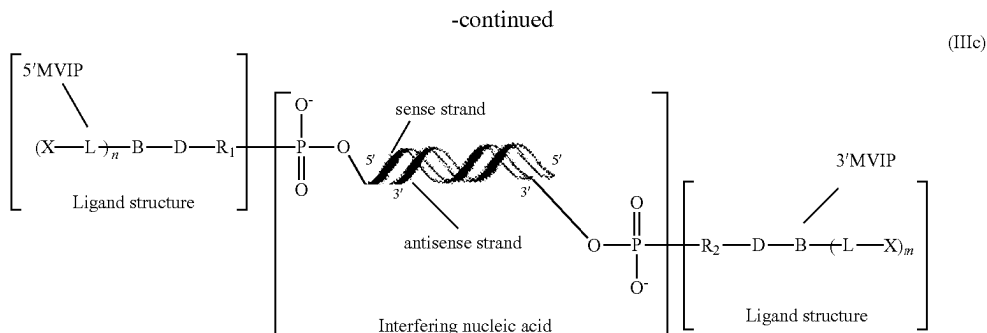
[0266] B is a linker;

[0267] D is a linking chain;

[0268] R₁ and R₂ are transition points; and are linked to the sense or antisense strand through a phosphate or modified phosphate, preferably through a phosphate or phosphorothioate.

[0269] 37. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36, having a structure as shown in formula IIIa, IIIb or IIIc:





[0270] n and m are each independently any integer from 0 to 4.

[0271] 38. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein $n+m$ is an integer of 2-6, preferably $n+m=2, 3$ or 4, more preferably 4.

[0272] 39. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the transition points R_1 and R_2 contain $-\text{NH}-$, $-\text{S}-$, and/or $-\text{O}-$ in their structures, R_1 and R_2 are respectively connected with the linking chain D and the 5' and 3' ends of the sense strand and/or antisense strand through $-\text{NH}-$, $-\text{S}-$, or $-\text{O}-$, and R_1 and R_2 are the same or different.

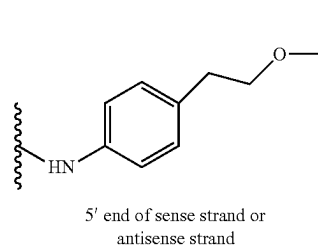
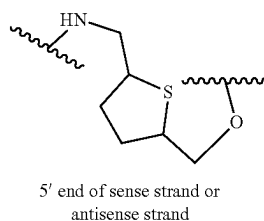
[0273] 40. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the transition points R_1 and R_2 are optionally straight chains, or straight chains or cyclic structures with amido, carboxyl, or alkyl branched chains, and the cyclic structure includes saturated or unsaturated aliphatic carbocyclyl, or five or six membered heterocyclyl or aromatic hydrocarbonyl containing sulfur, oxygen, or nitrogen atoms.

[0274] 41. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the transition points R_1 and/or R_2 are $-\text{E}_1(\text{CH}_2)_x\text{CH}_2\text{E}_2-$, where x is any integer of 3-12, and the groups E_1 and E_2 can be $-\text{NH}-$, $-\text{S}-$, or $-\text{O}-$, respectively.

[0275] 42. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the transition points R_1 and/or R_2 are $-\text{E}_1(\text{CH}_2)_{x1}\text{CH}(\text{OH})(\text{CH}_2)_{x2}\text{E}_2-$, where X_1 or X_2 are each independently any integer of 3-10, and E_1 and E_2 can be $-\text{NH}-$, $-\text{S}-$, or $-\text{O}-$, respectively.

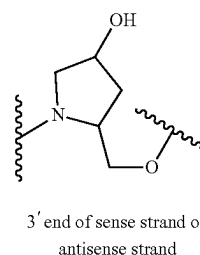
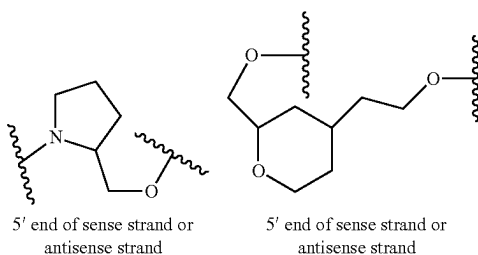
[0276] 43. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the transition point R_1 is a heterocyclic or carbocyclic structure containing N, S or O:

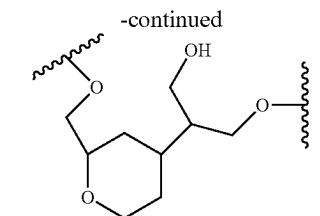
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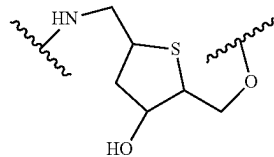
[0277] 44. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the transition point R_1 is $-\text{NH}(\text{CH}_2)_x\text{CH}_2\text{O}-$, where x is any integer of 3-12, preferably any integer of 4-6.

[0278] 45. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the transition point R_2 is a heterocyclic or carbocyclic structure containing N, S or O:

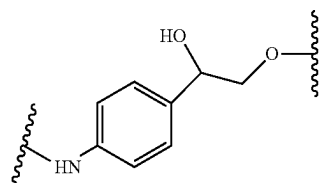




3' end of sense strand or
antisense strand



3' end of sense strand or
antisense strand



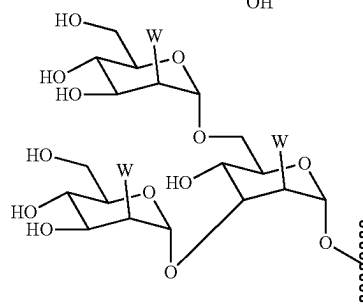
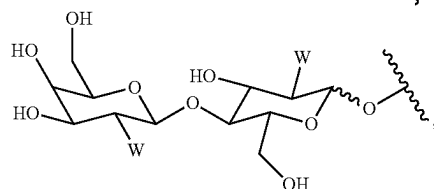
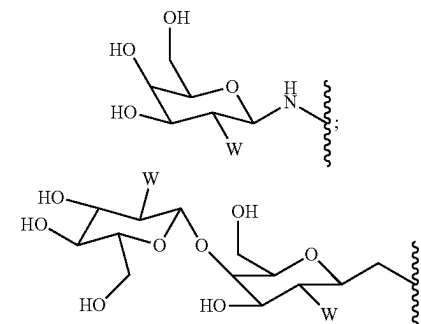
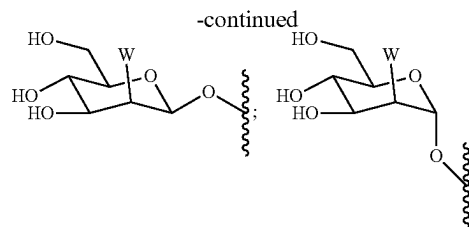
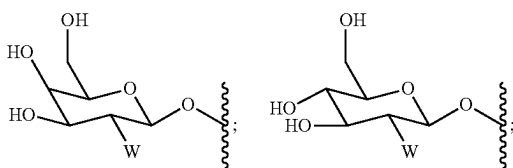
3' end of sense strand or
antisense strand

[0279] 46. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the transition point R_2 is $-\text{NH}(\text{CH}_2)_{x1}\text{CH}(\text{OH})(\text{CH}_2)_{x2}\text{CH}_2\text{O}-$, where x_1 is any integer of 1-4 and x_2 is any integer of 0-4.

[0280] 47. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the liver targeting specific ligand X is selected from structures used to enhance the uptake of RNA inhibitor by hepatocytes.

[0281] 48. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 47, wherein the targeting specific ligand X includes: mannose, galactose, D-arabinose, glucose, fructose, xylose, glucosamine, ribose, mannose derivatives, galactose derivatives, glucose derivatives, ribose derivatives, galactose, galactosamine, N-acetylgalactosamine, galactosamine derivatives, N-acetylgalactosamine derivatives, folic acid, folate derivatives, peptide chain.

[0282] 49. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the targeting specific ligand X is selected from the following structures:



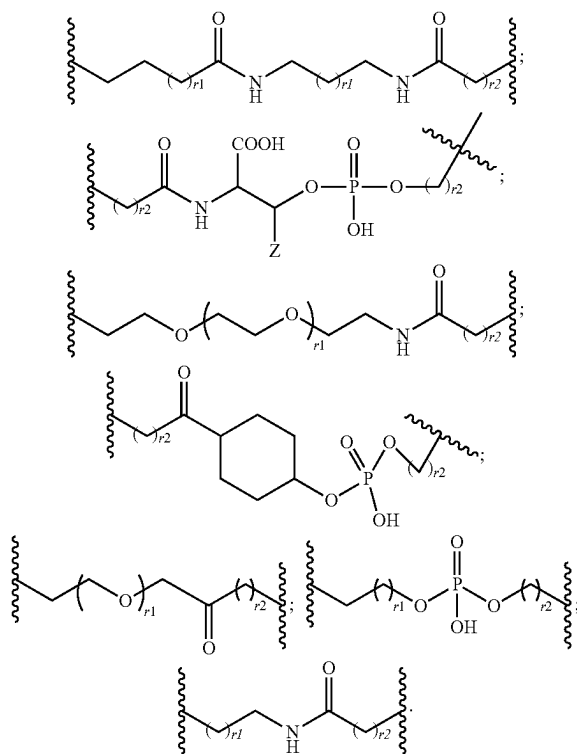
[0283] wherein W is selected from one or two of $-\text{OH}$, $-\text{NHCOOH}$ or $-\text{NHCO}(\text{CH}_2)_q\text{CH}_3$, where q is an integer of 0-4.

[0284] 50. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the branched chain L is a $\text{C}_4\text{-C}_{18}$ straight chain containing $-\text{NH}-$, $-\text{C}(=\text{O})-$, $-\text{O}-$, $-\text{S}-$, amido, phosphoryl, thiophosphoryl, $\text{C}_4\text{-C}_{10}$ aliphatic carbocyclyl, phenyl, or a combination of these groups.

[0285] 51. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 50, wherein the branched chain L also has a side chain of hydroxyethyl or carboxylic acids group.

[0286] 52. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the branched chain L is a $\text{C}_7\text{-C}_{18}$ straight chain containing amido group or six-membered aliphatic carbocyclyl group.

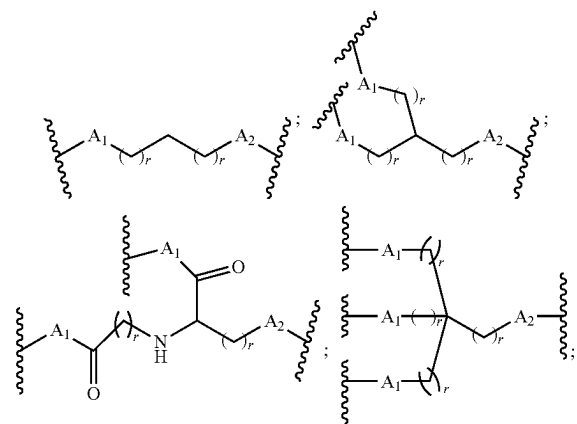
[0287] 53. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the branched chain L is selected from one or more of the following structures:



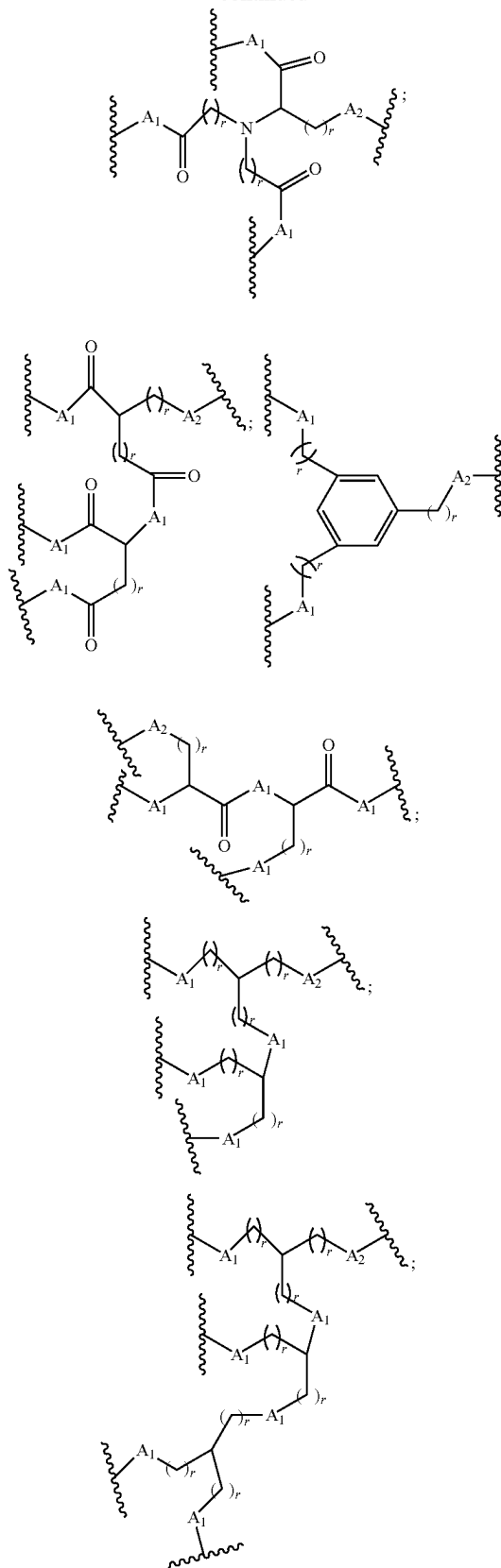
[0288] where r_1 is any integer from 1 to 12, r_2 is any integer from 0 to 20, and Z is H, alkyl or amido group.

[0289] 54. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the structure of the linker B is related to the number of the targeting specific ligand X that can be introduced, and the linker B contains —NH—, C, O, S, amido, phosphoryl, thiophosphoryl, when n or m is 1, it is a straight chain, and when n or m is 2, 3, or 4, the number of branches is 2, 3, or 4, respectively.

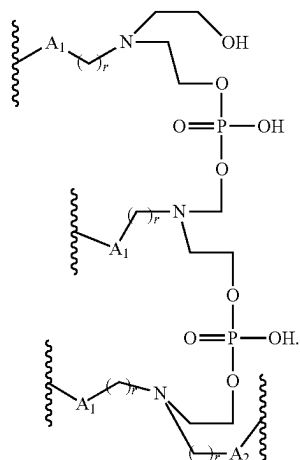
[0290] 55. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the linker B is selected from the following structures:



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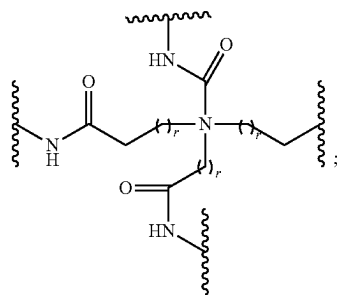
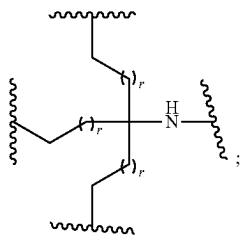
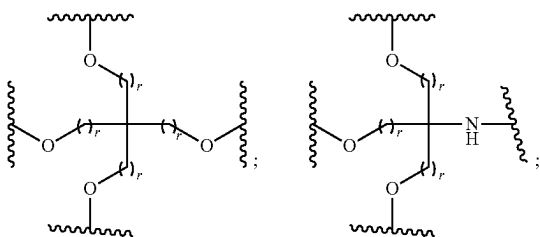


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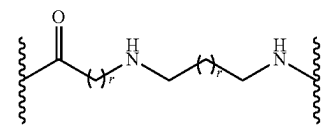
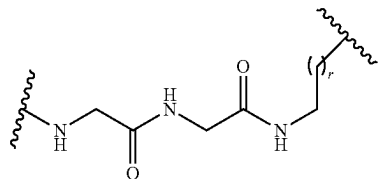
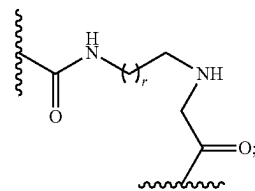
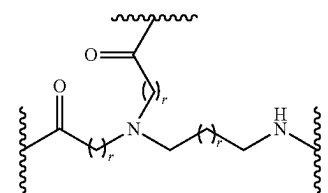
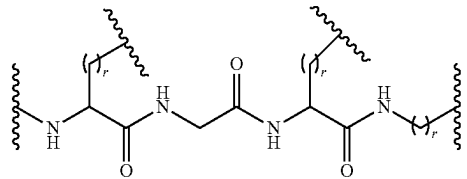
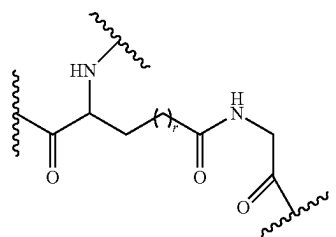
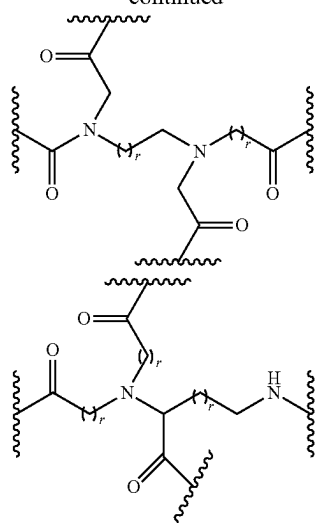


[0291] wherein A_1 and A_2 are each independently C, O, S, —NH—, carbonyl, amido, phosphoryl or thiophosphoryl, and r is an integer of 0-4.

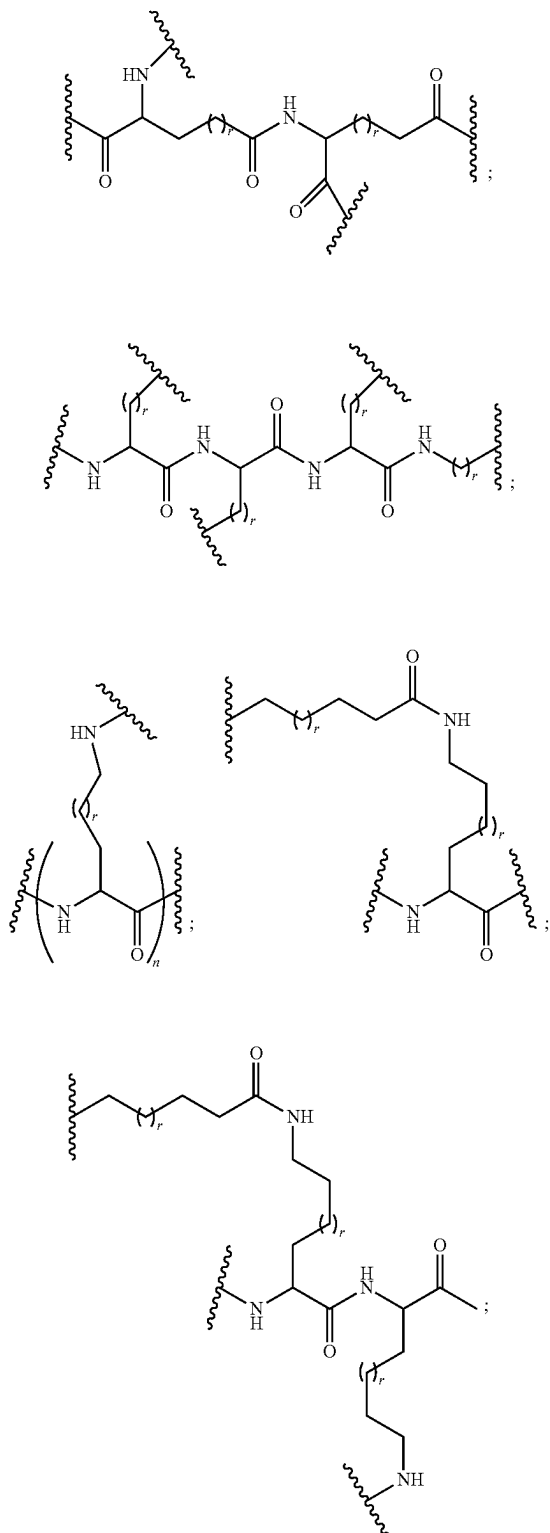
[0292] 56. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the linker B is selected from the following structures:



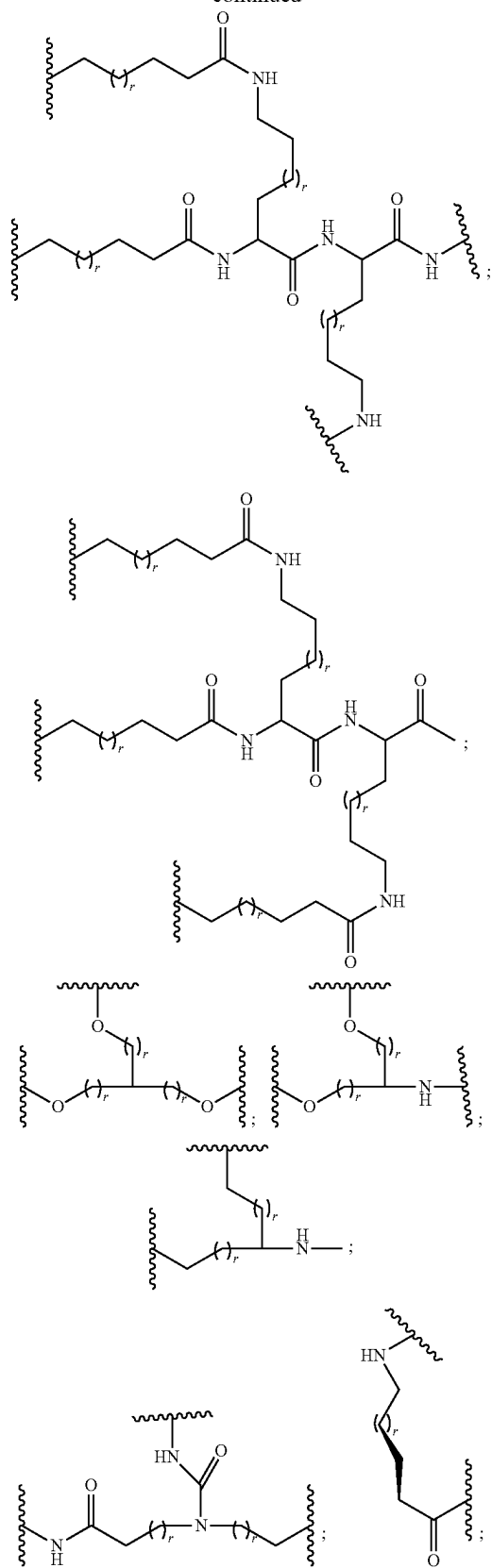
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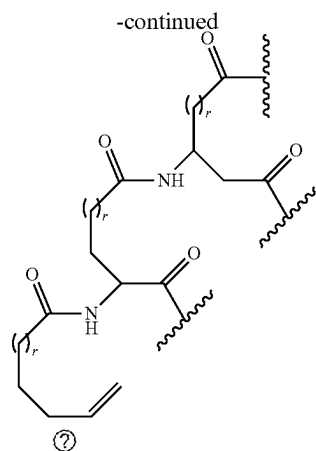


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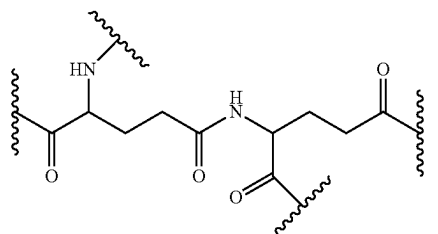
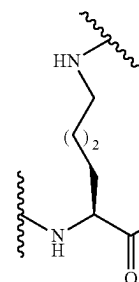
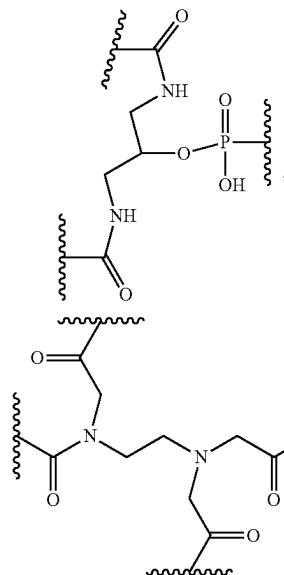
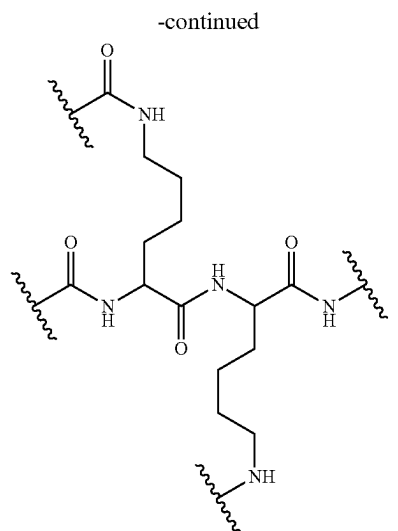
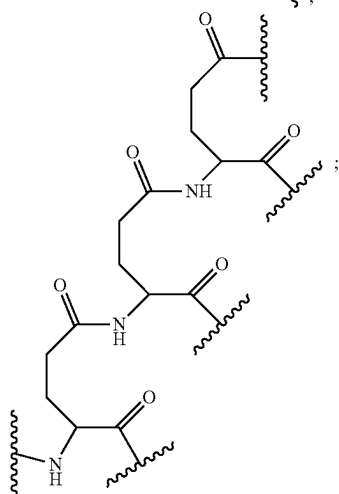
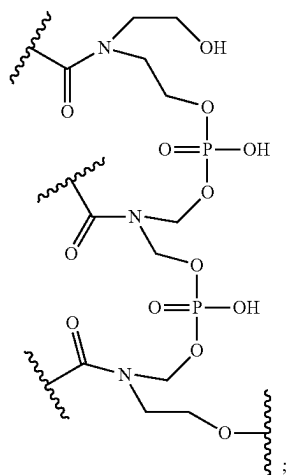




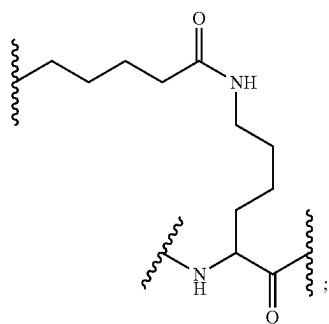
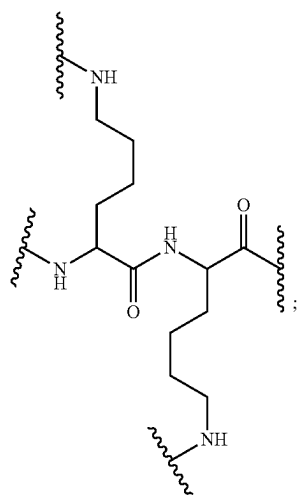
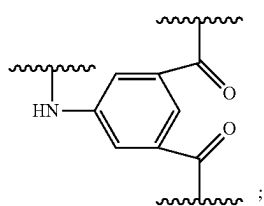
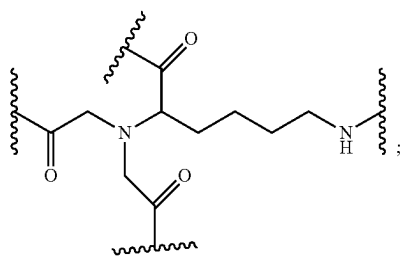
② indicates text missing or illegible when filed

[0293] wherein r is any integer from 0 to 4.

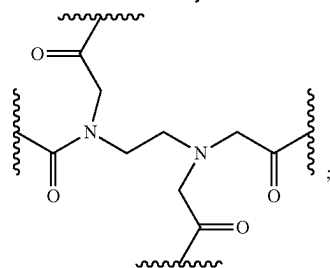
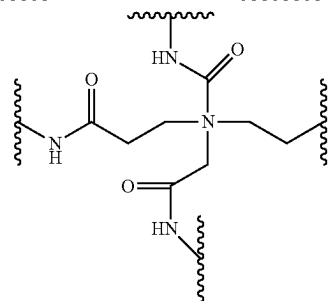
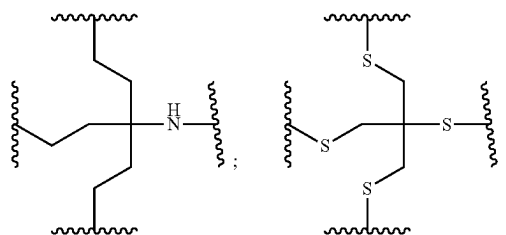
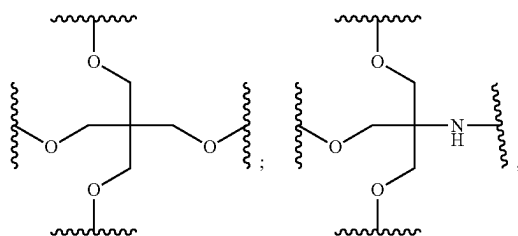
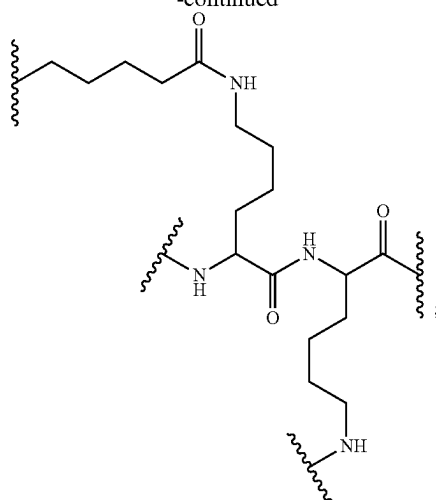
[0294] 57. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the linker B is selected from the following structures:

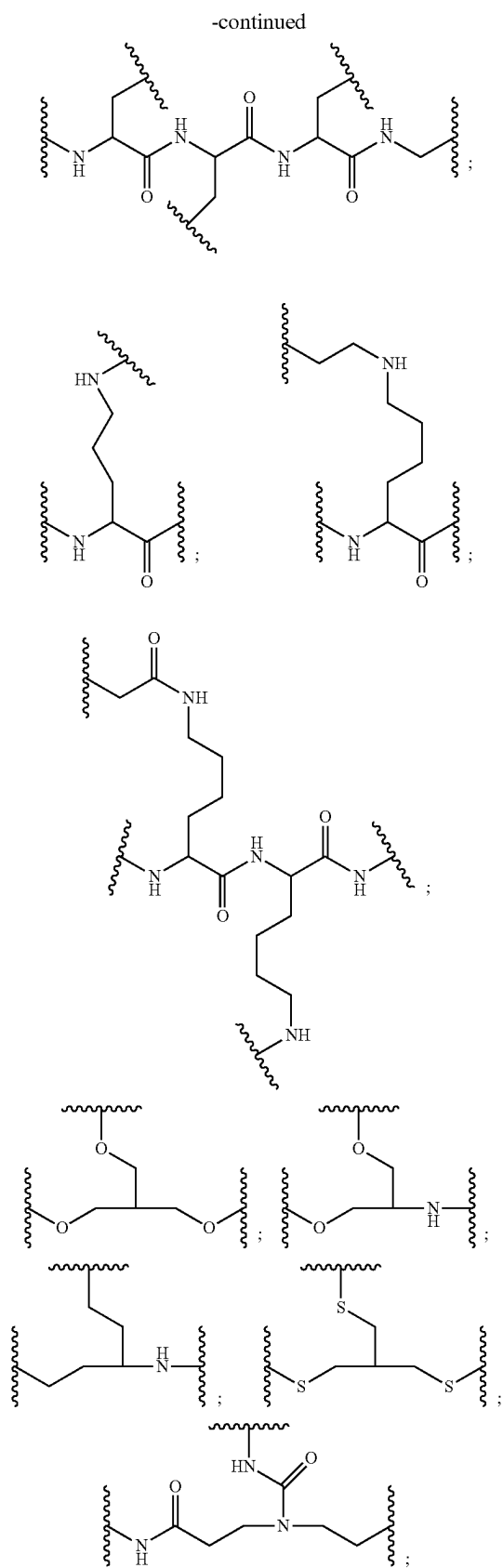
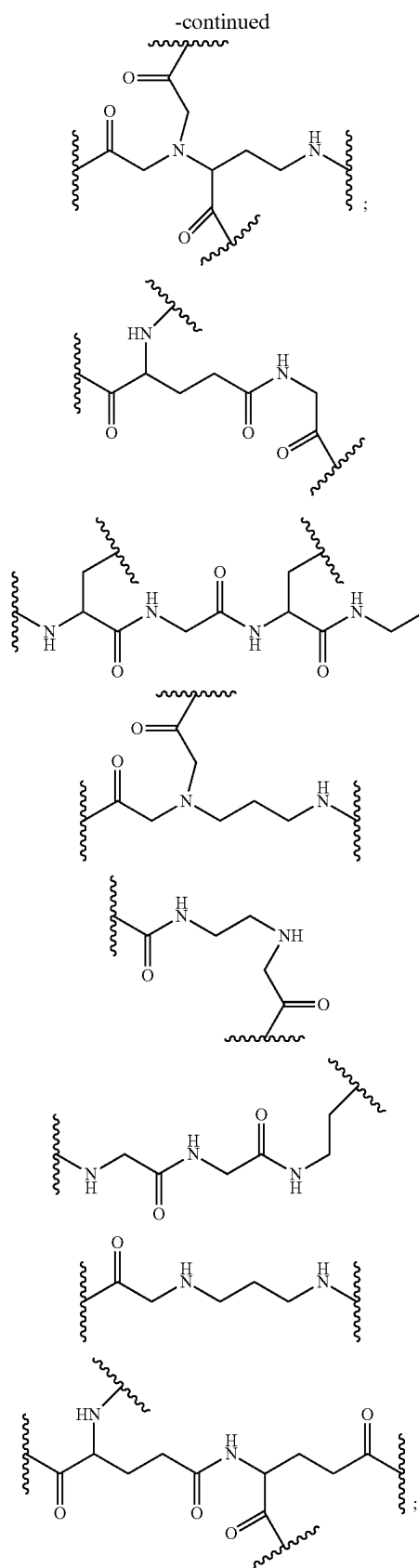


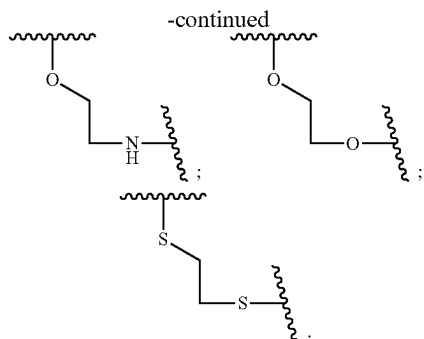
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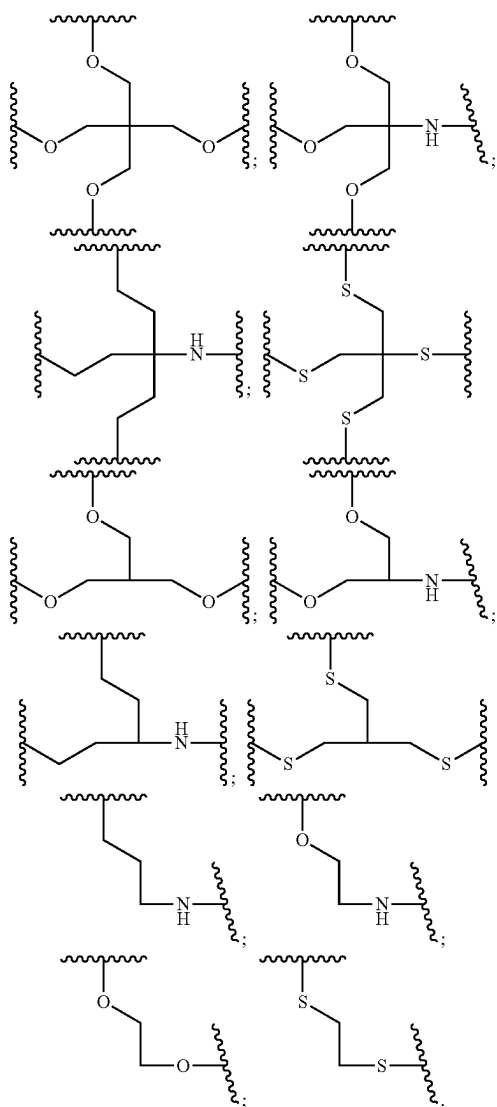
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[0295] 58. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the linker B is selected from the following structures:



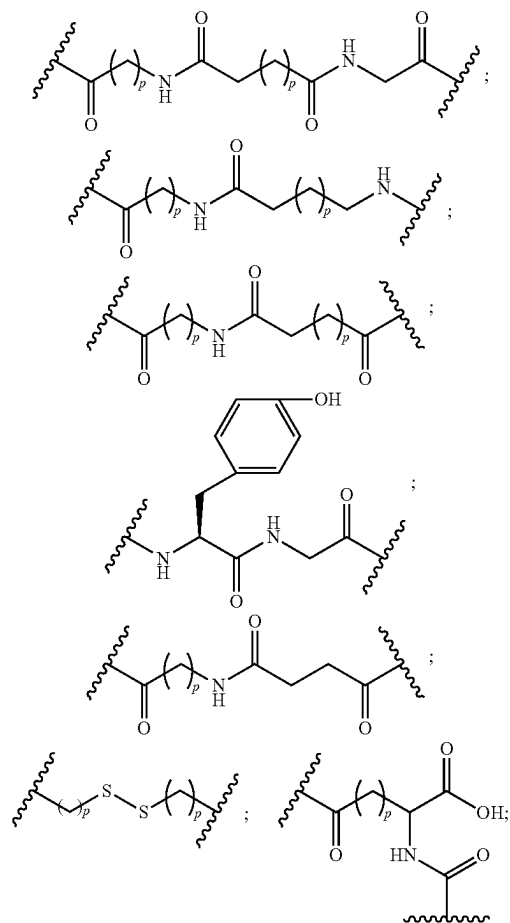
[0296] 59. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the linking chain D is a C3-C18 straight chain containing —NH—, C=O, O, S, amido, phosphoryl, thiophosphoryl, aromatic hydrocarbyl, C4-C10 aliphatic carbocyclyl, five- or six-membered heterocyclyl containing 1-3 nitrogen, or a combination of these groups.

[0297] 60. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the linking chain D also has side chains of hydroxymethyl, methyl tert-butyl, methylphenoyl, and C₅-C₆ aliphatic cyclic groups.

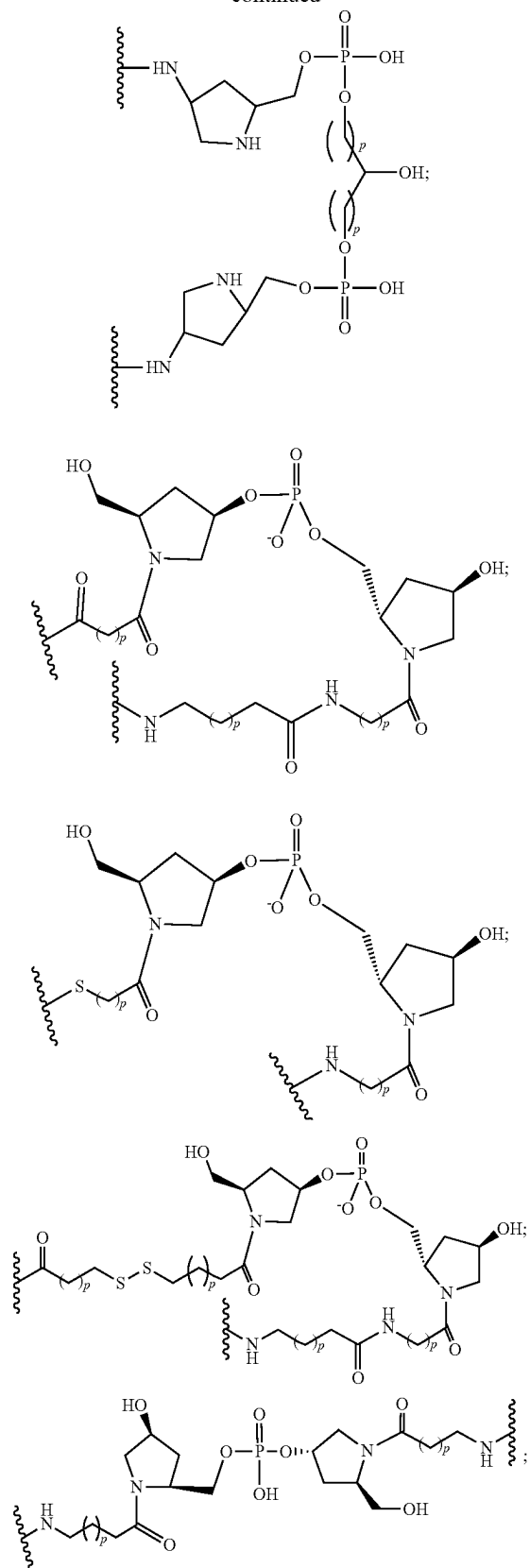
[0298] 61. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the linking chain D is a C₃-C₁₀ straight chain containing two C=O, six-membered aliphatic carbocyclyl or phenyl group.

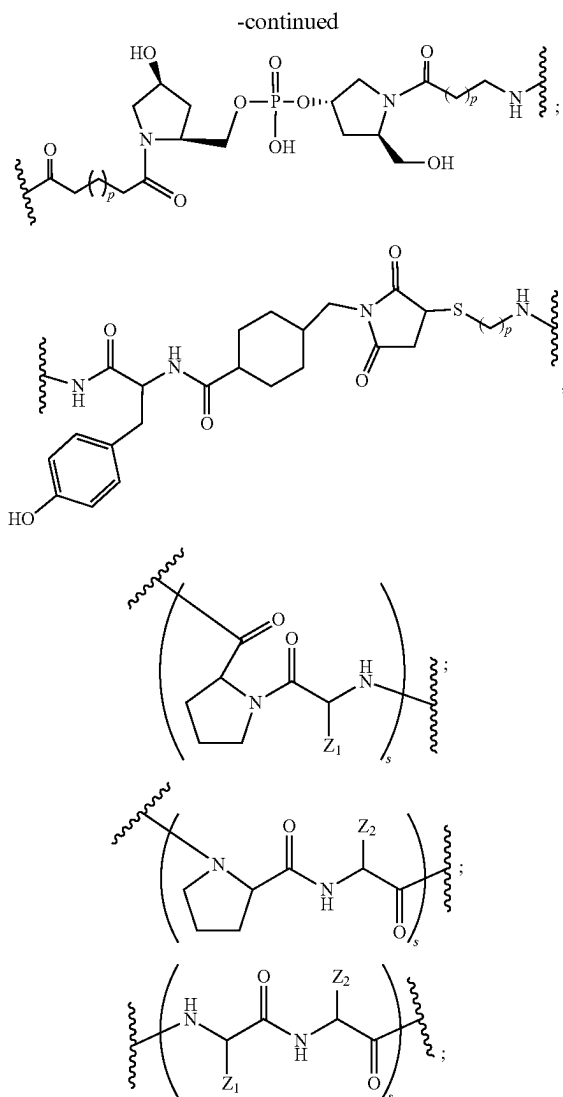
[0299] 62. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the linking chain D is a C₃-C₁₀ straight chain containing two C=O.

[0300] 63. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the linking chain D is selected from the following structures:



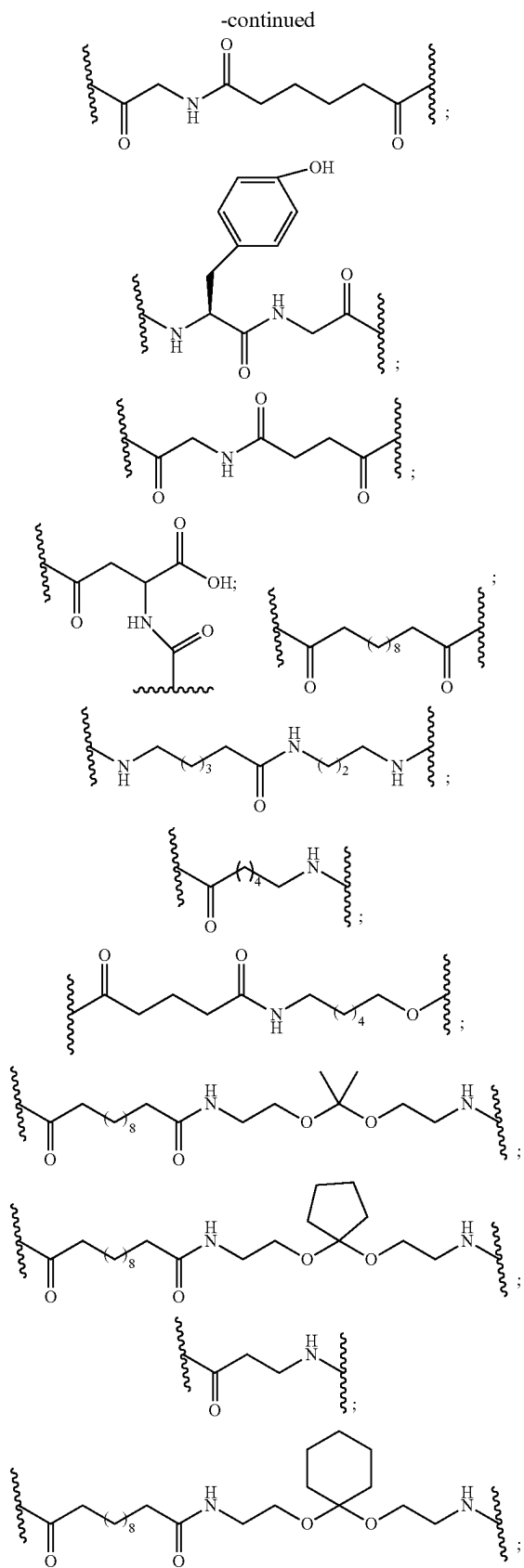
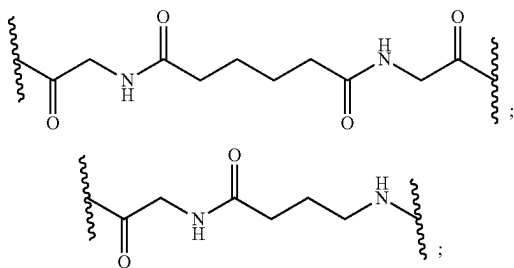
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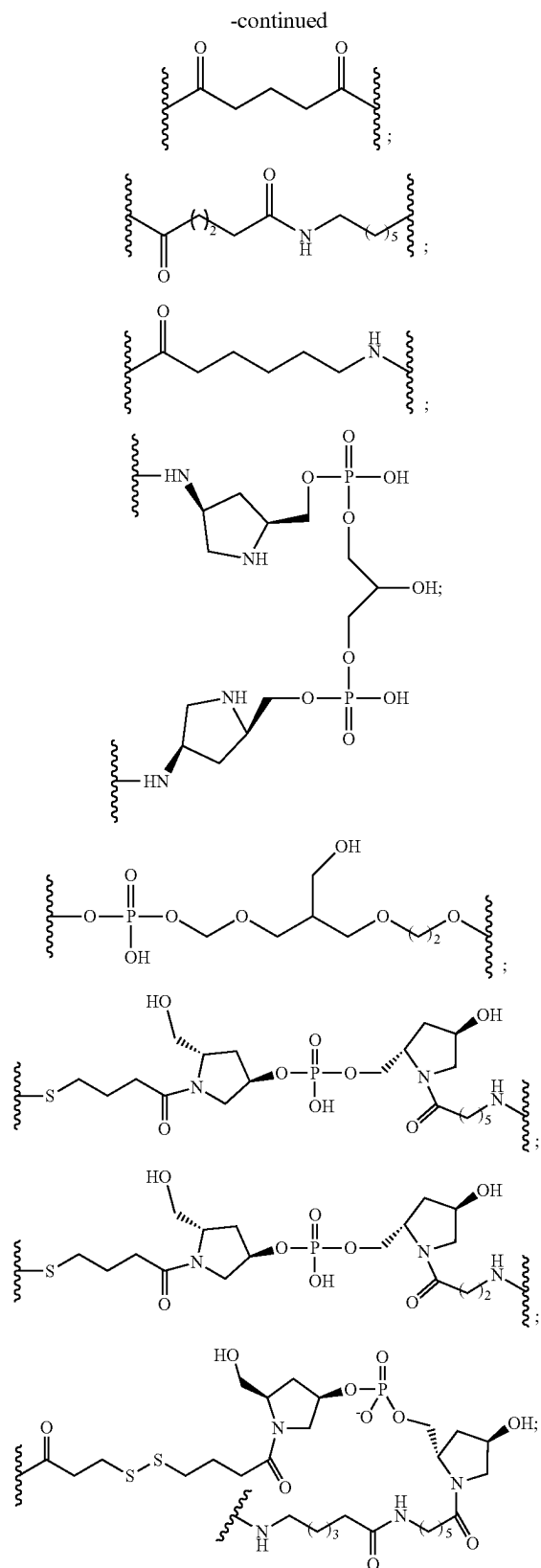


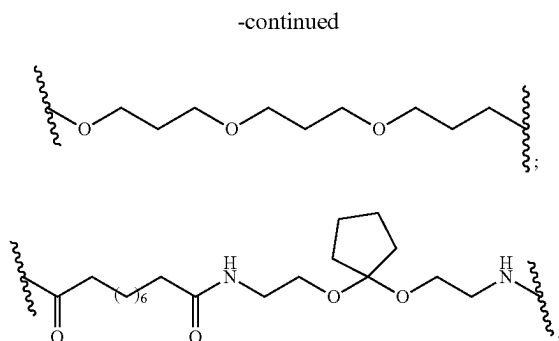
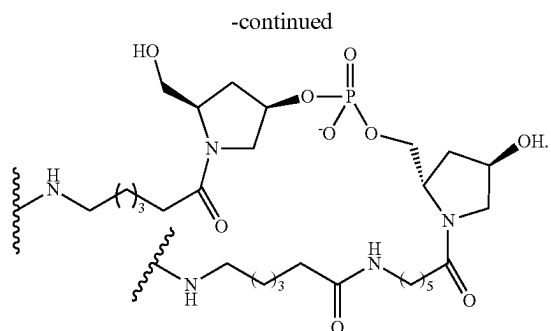


[0301] wherein, each p is independently any integer from 1 to 20; s is an integer of 2-13; Z_1 and Z_2 are the same or different substituents.

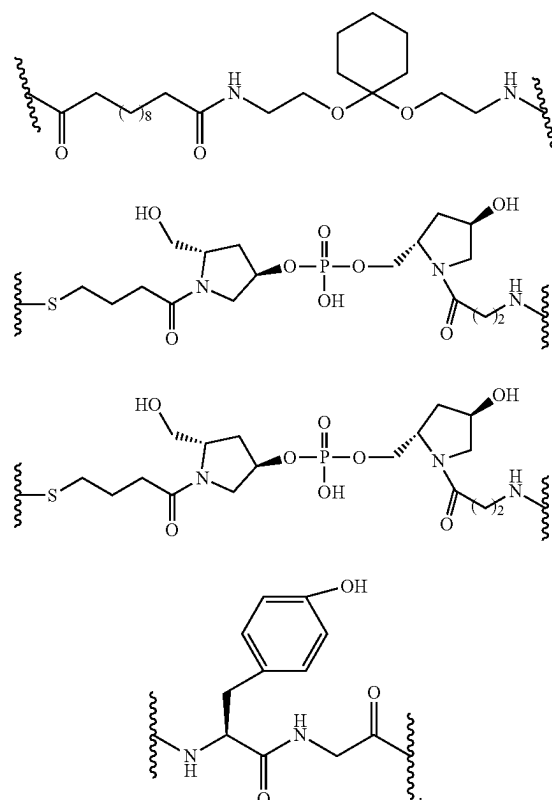
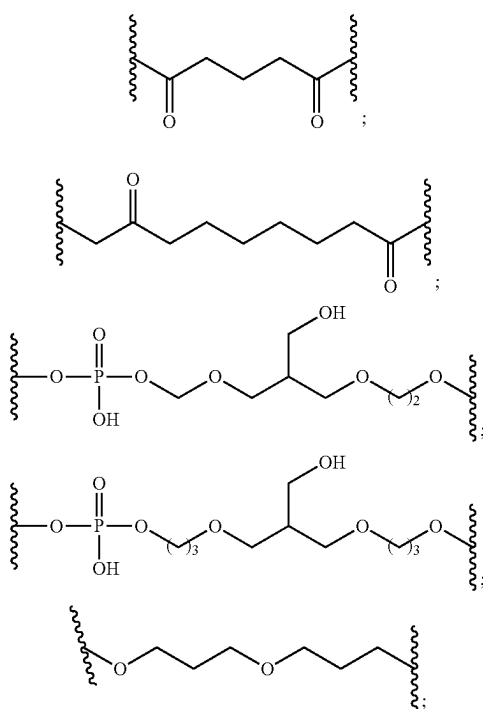
[0302] 64. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the linking chain D is selected from the following structures:



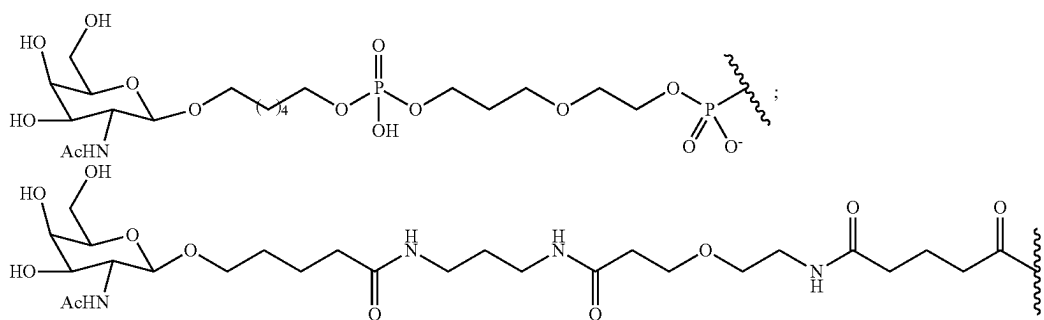




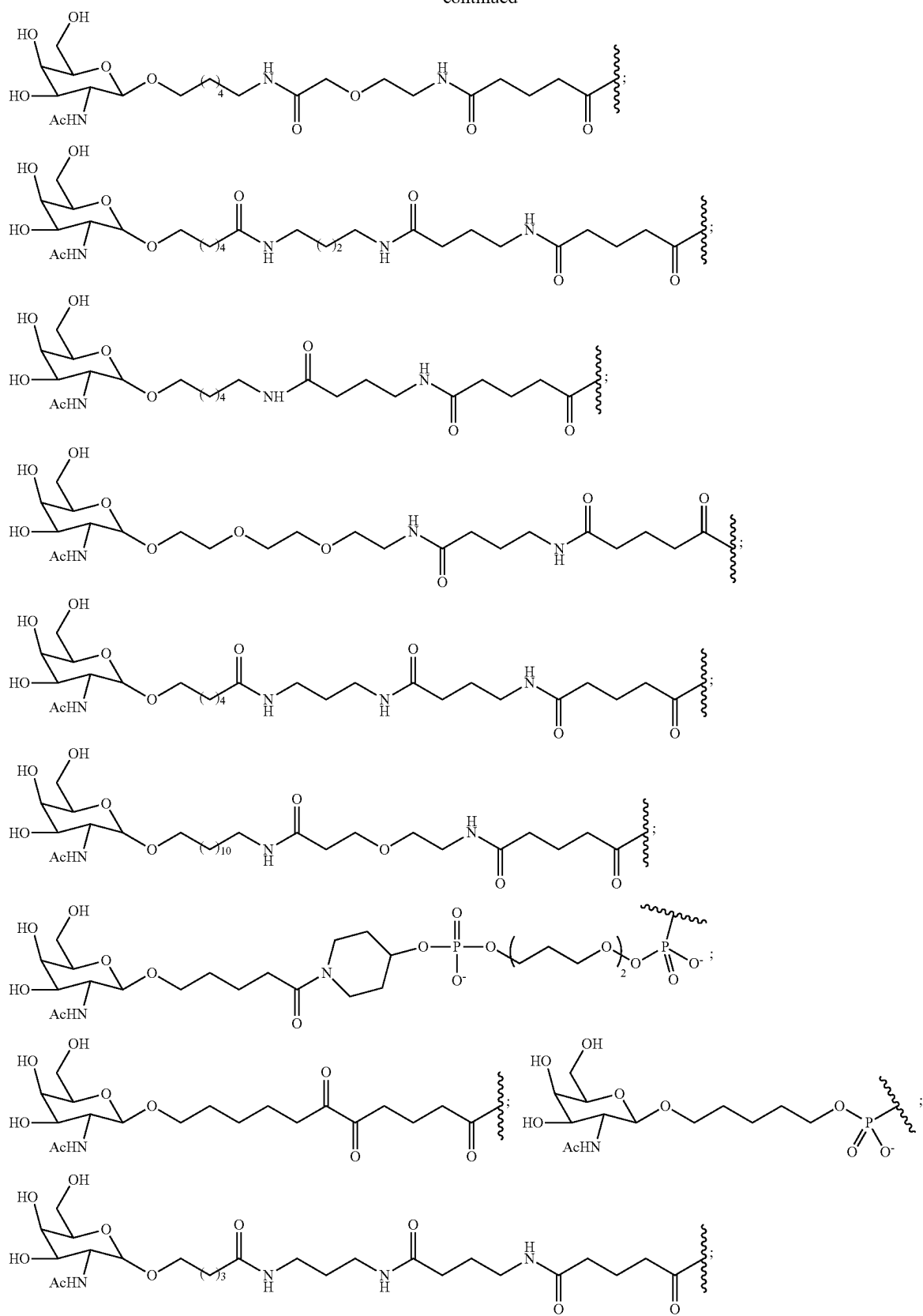
[0303] 65. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the linking chain D is selected from the following structures:

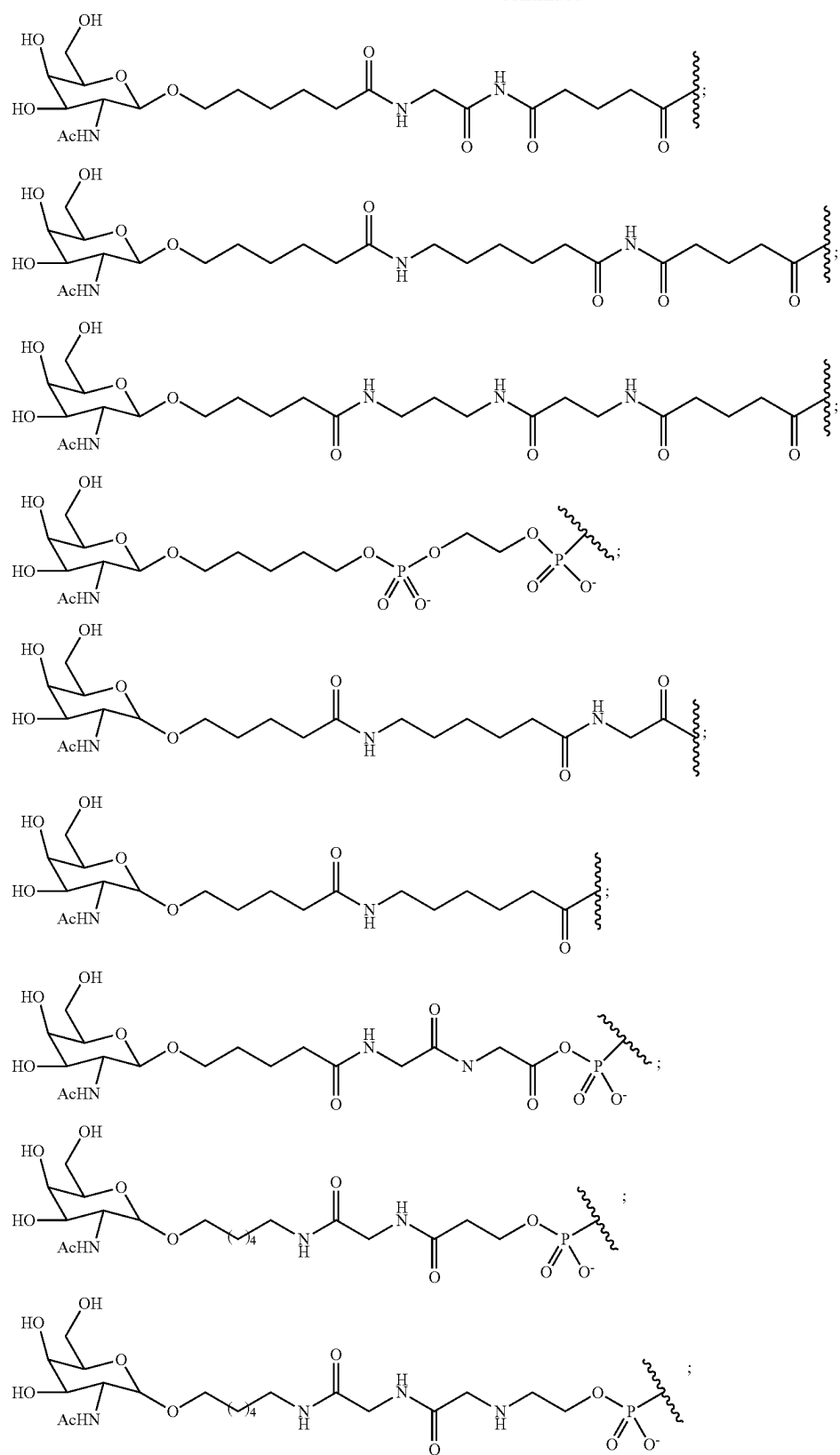


[0304] 66. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein (X-L)_n-B-D- in the 5'MVIP structure and (X-L)_m-B-D- in the 3'MVIP structure are selected from one or more of the following structures:

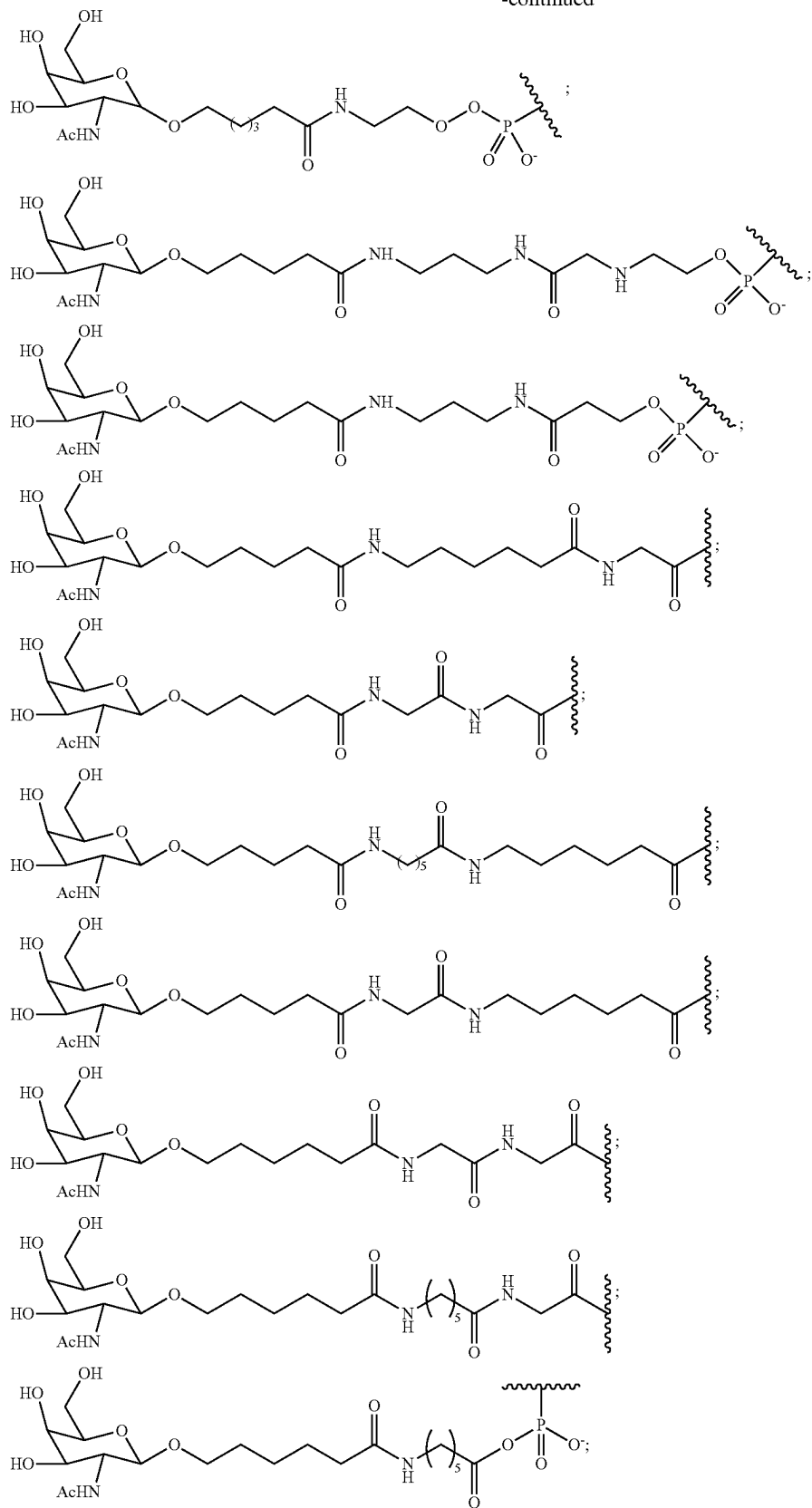


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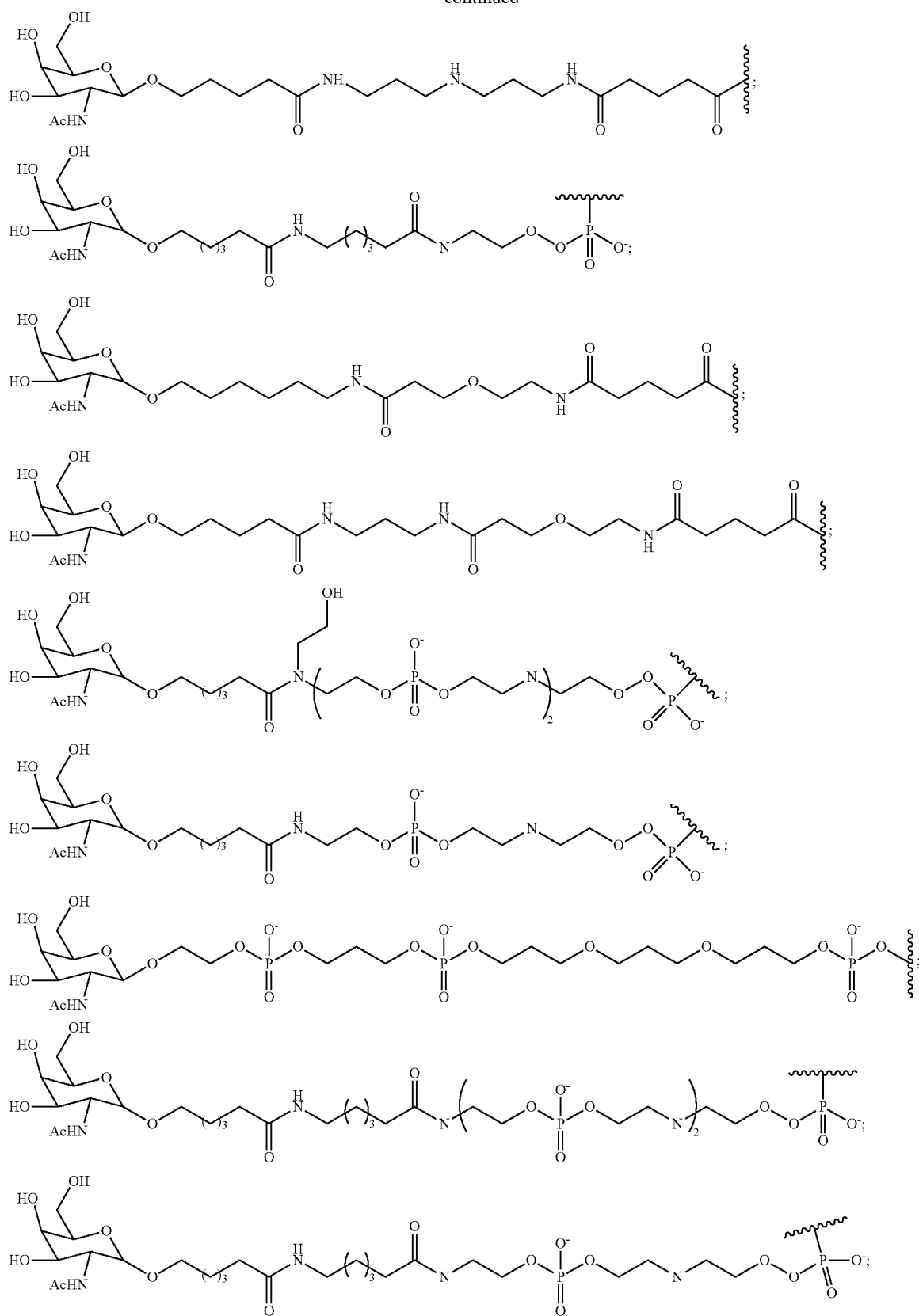




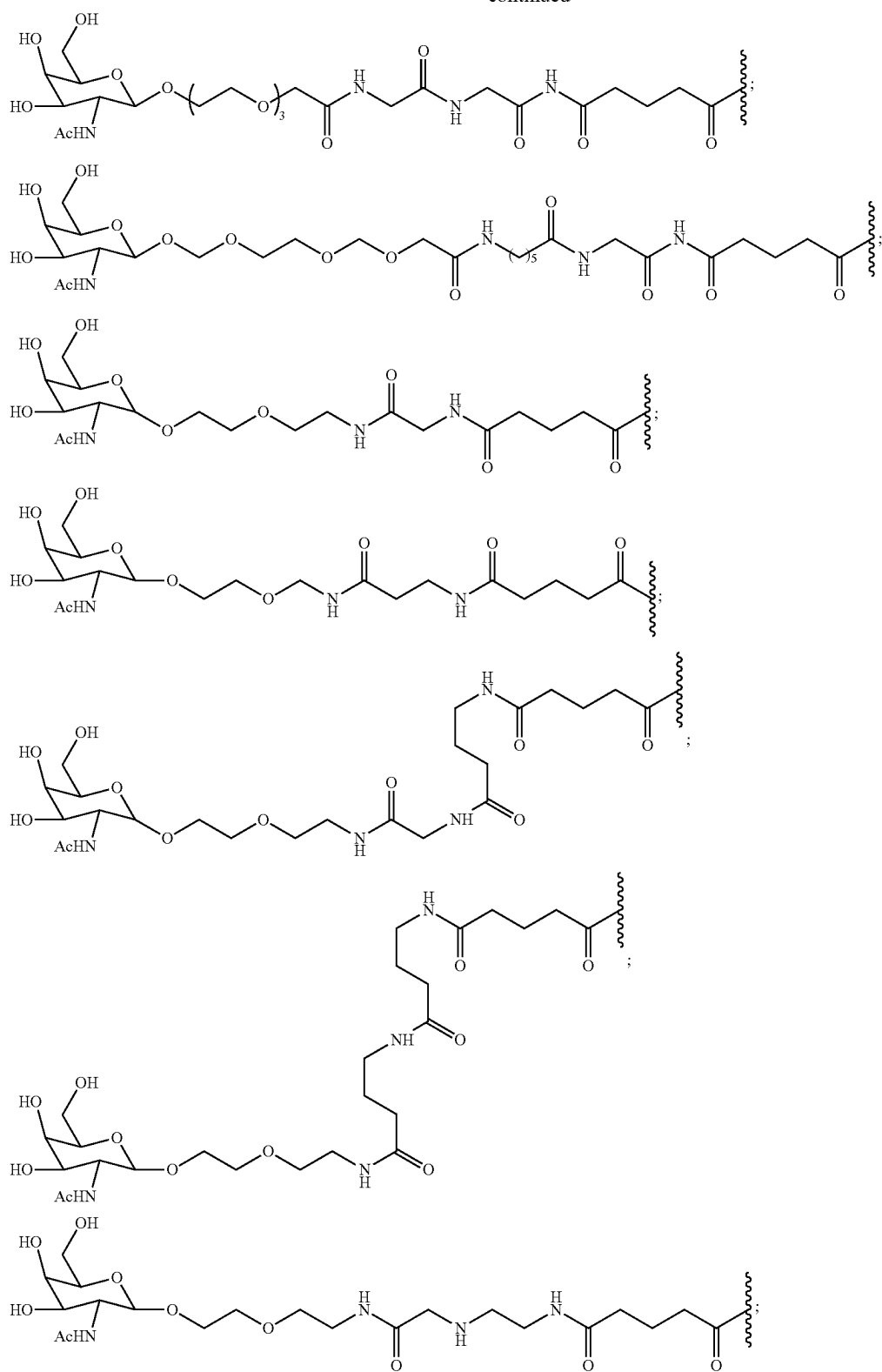
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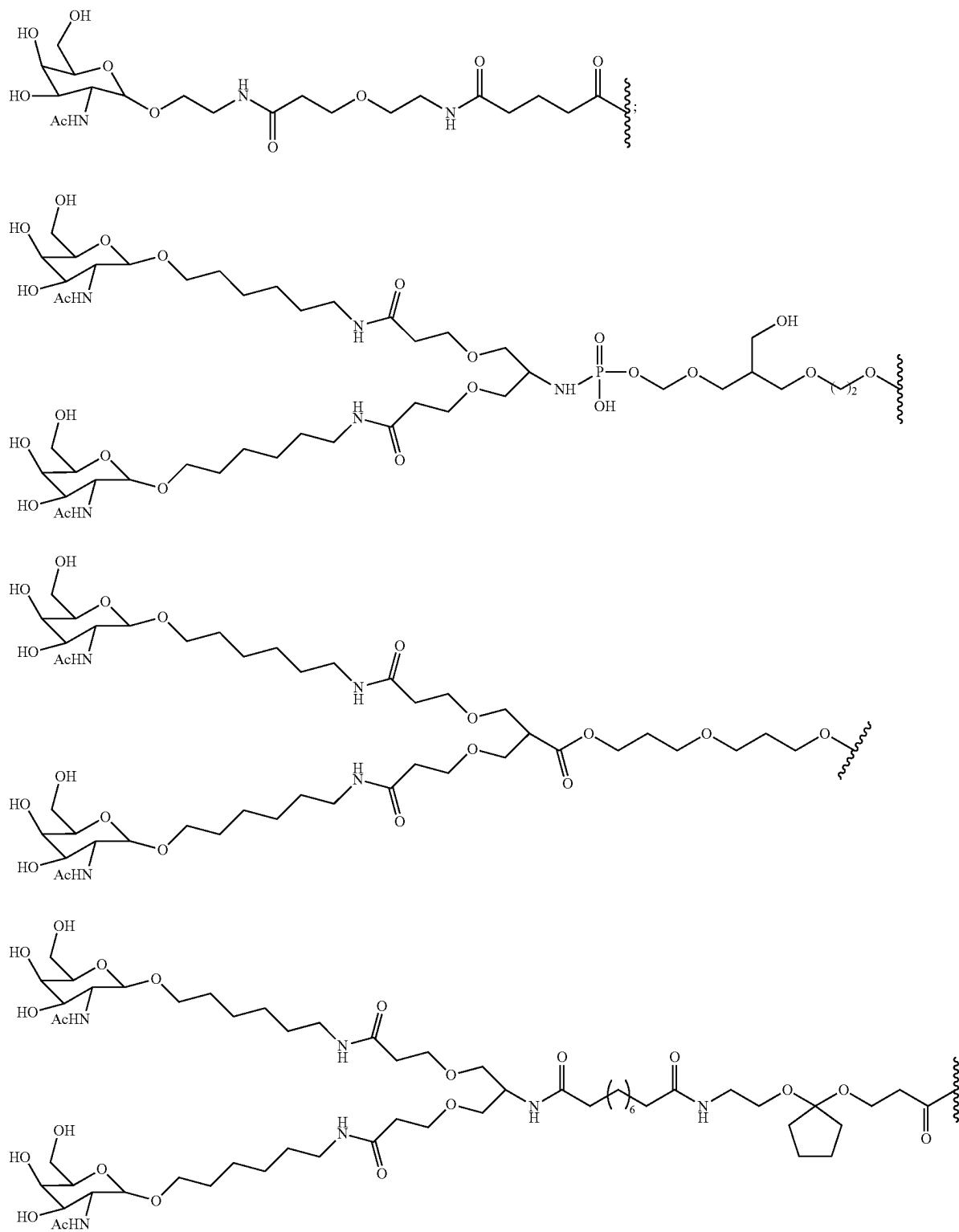
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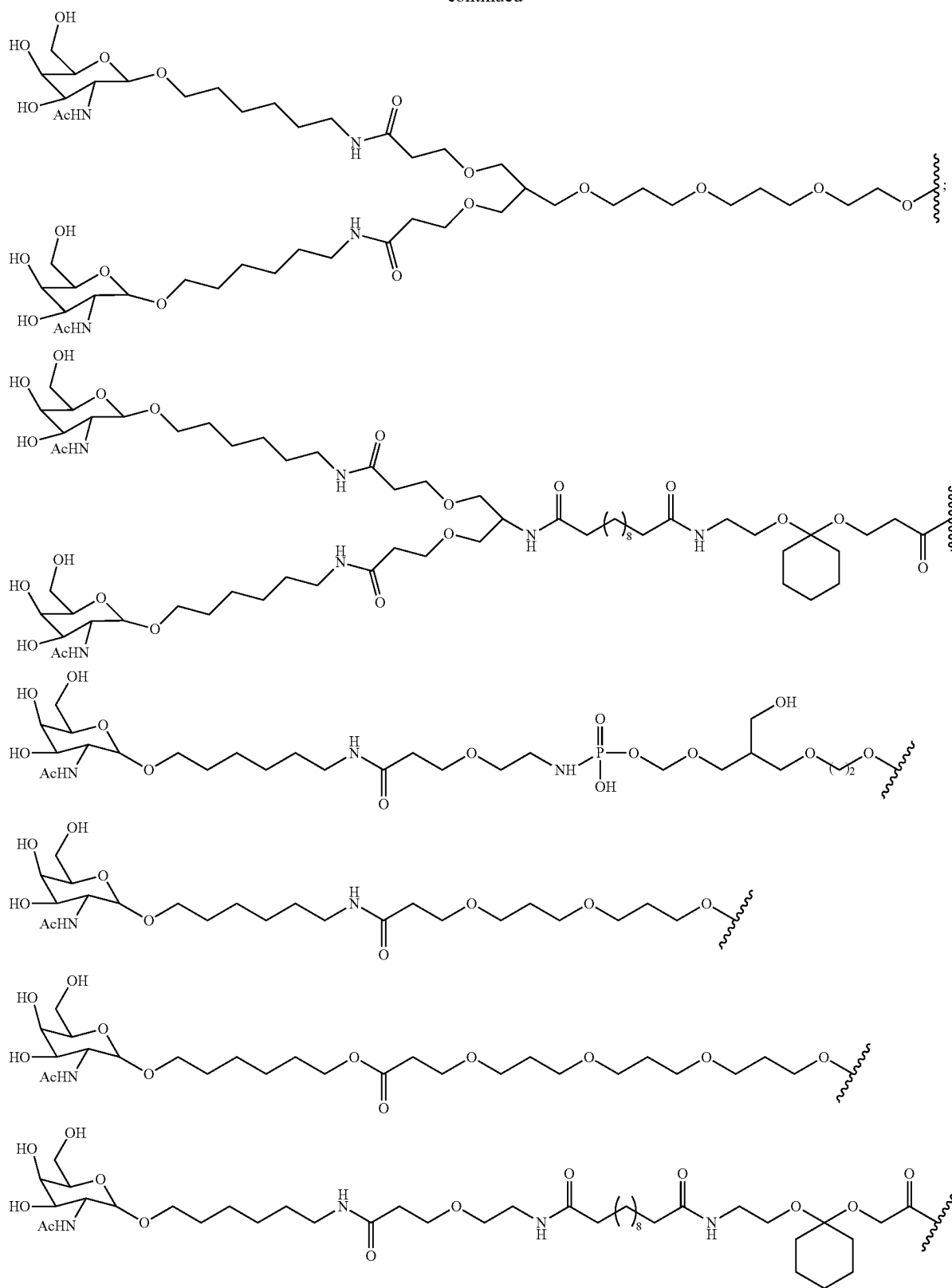
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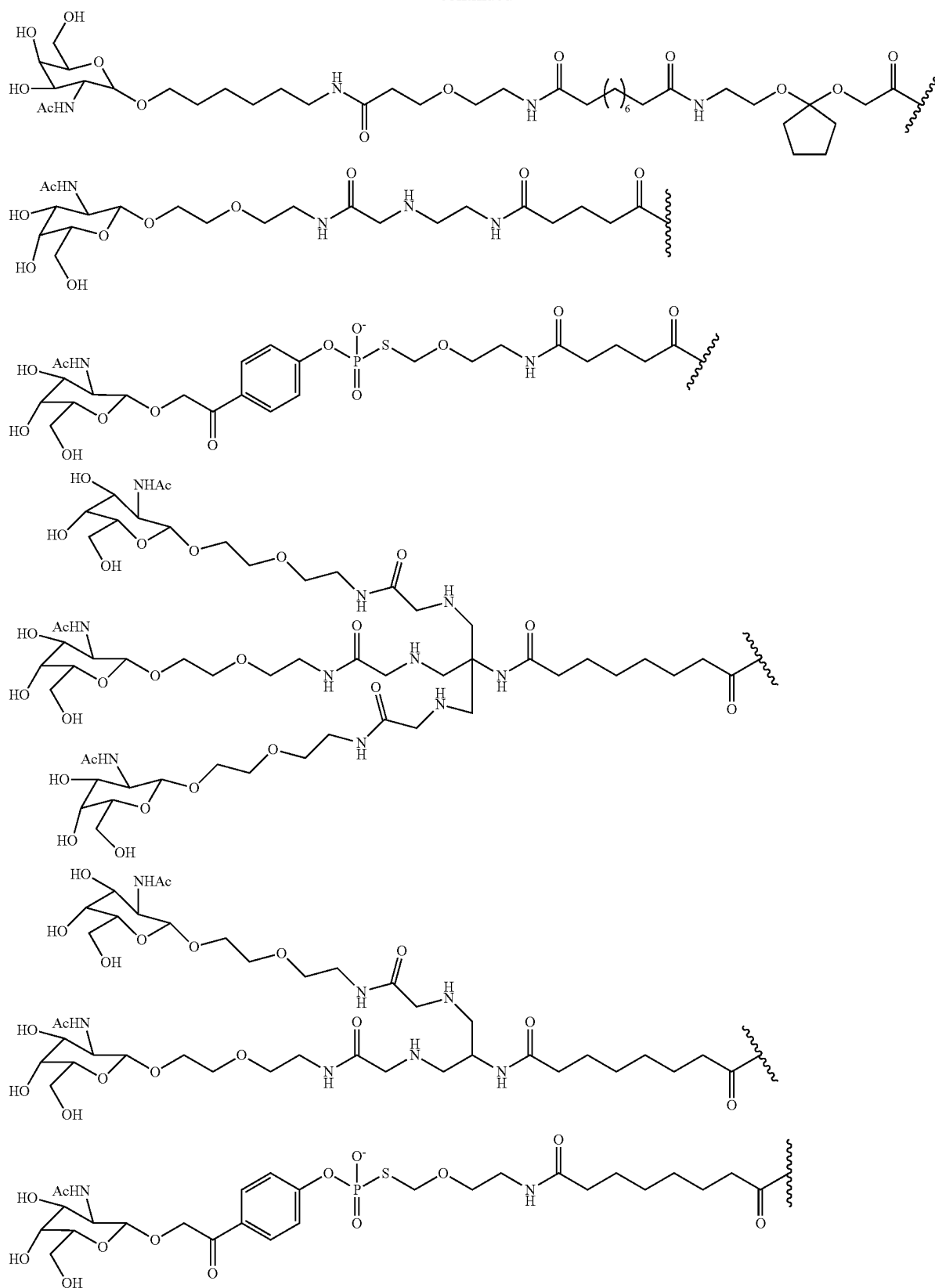
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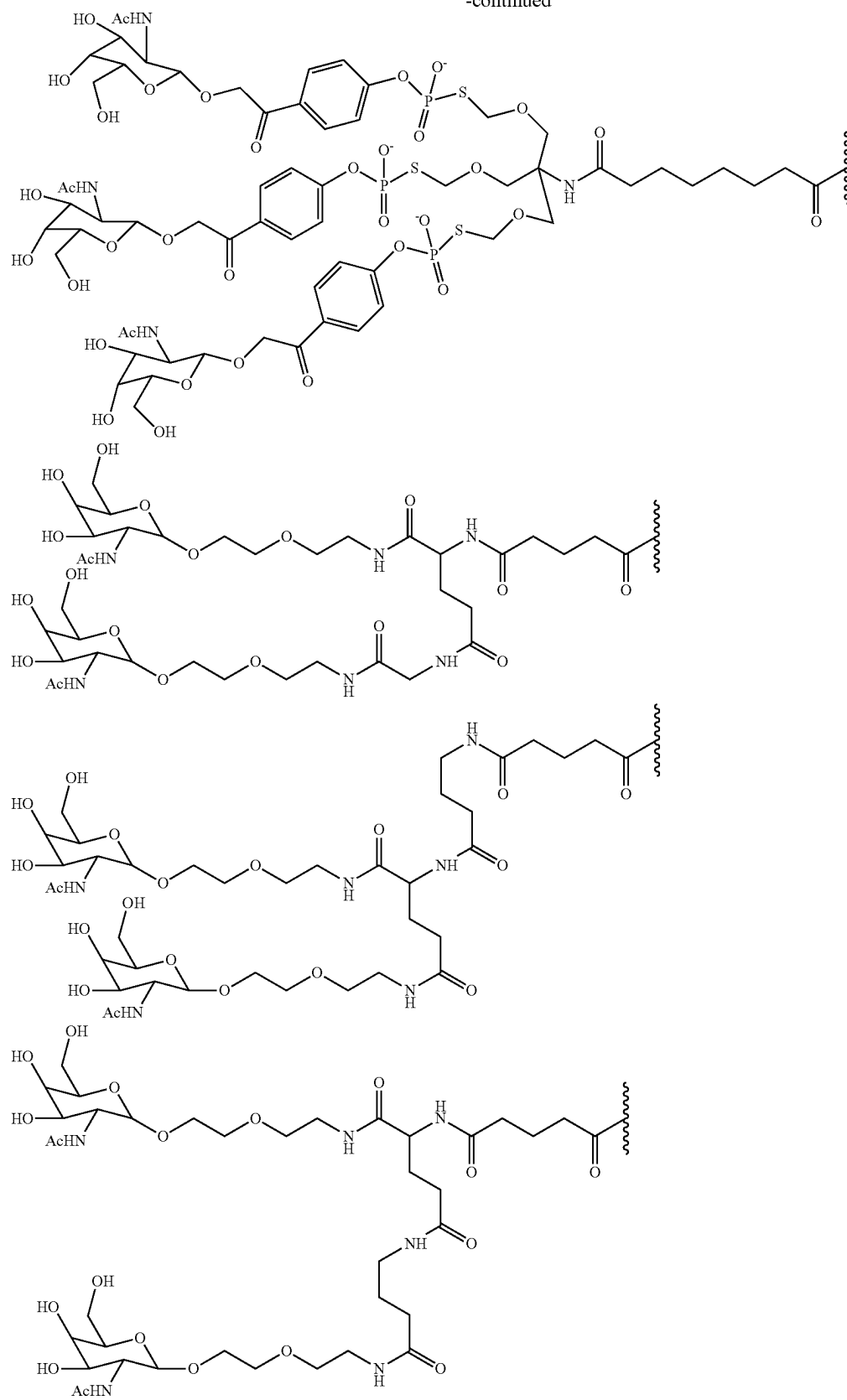
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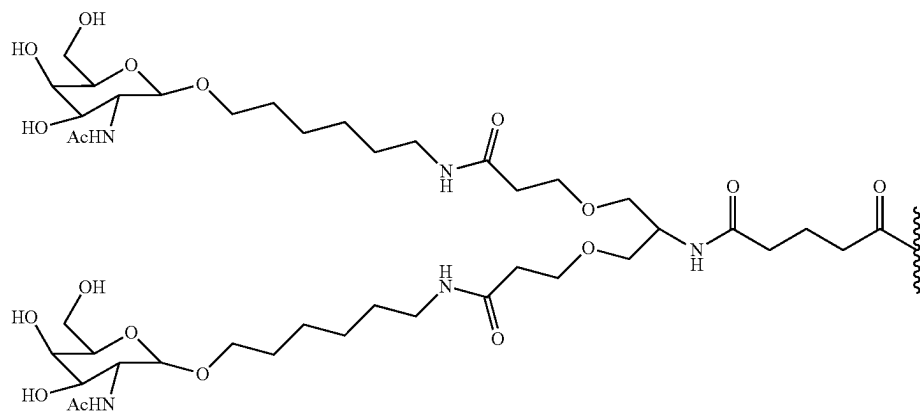
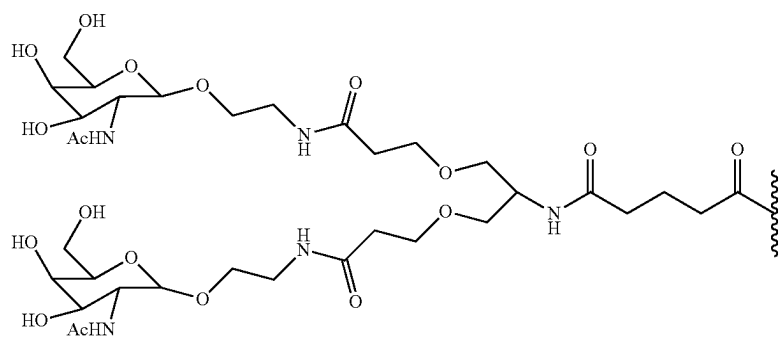
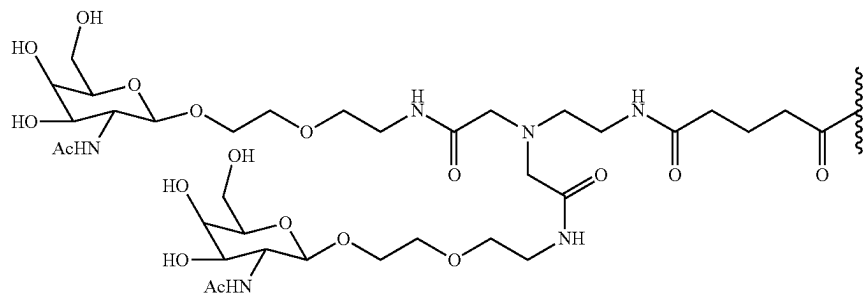
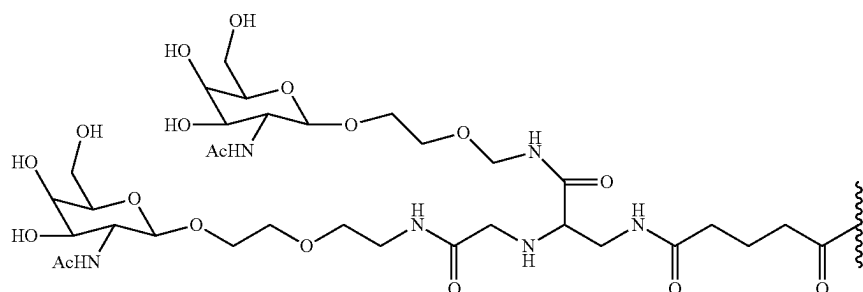
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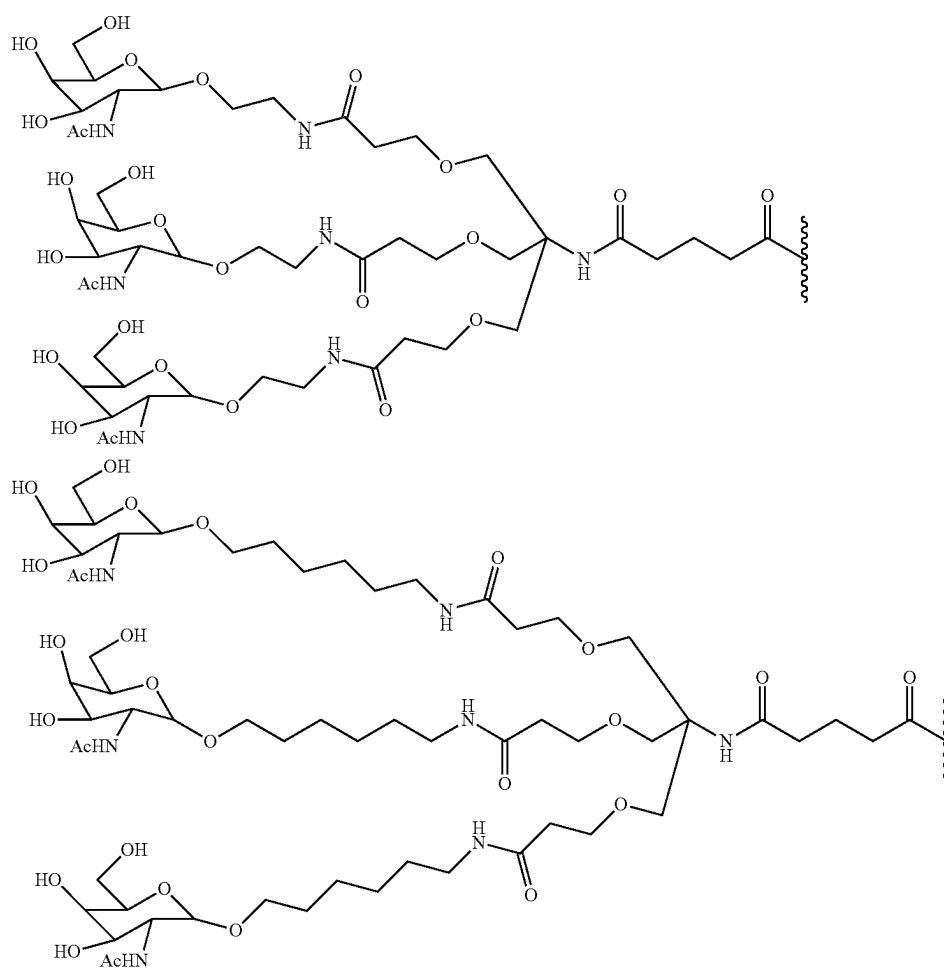
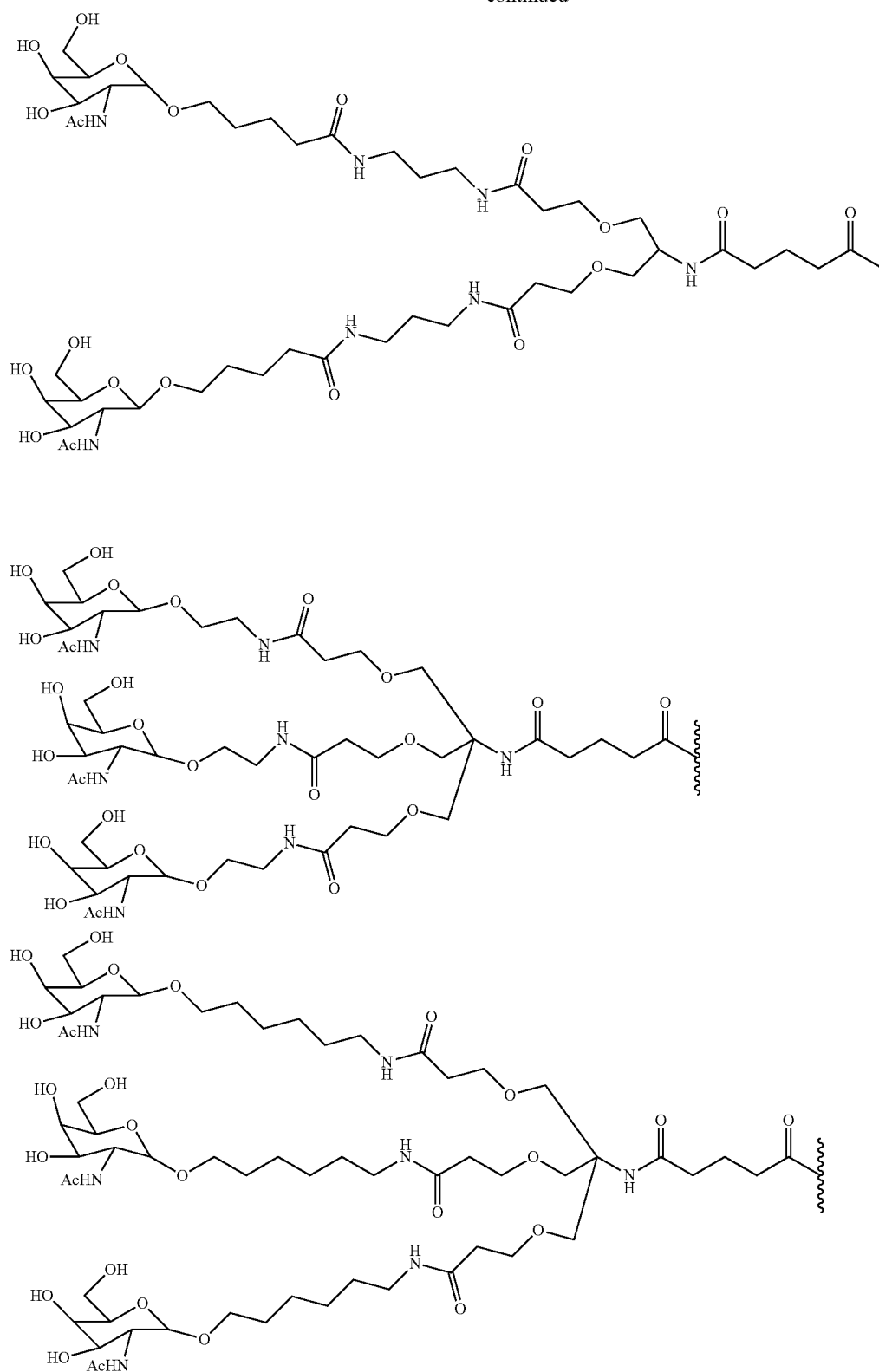
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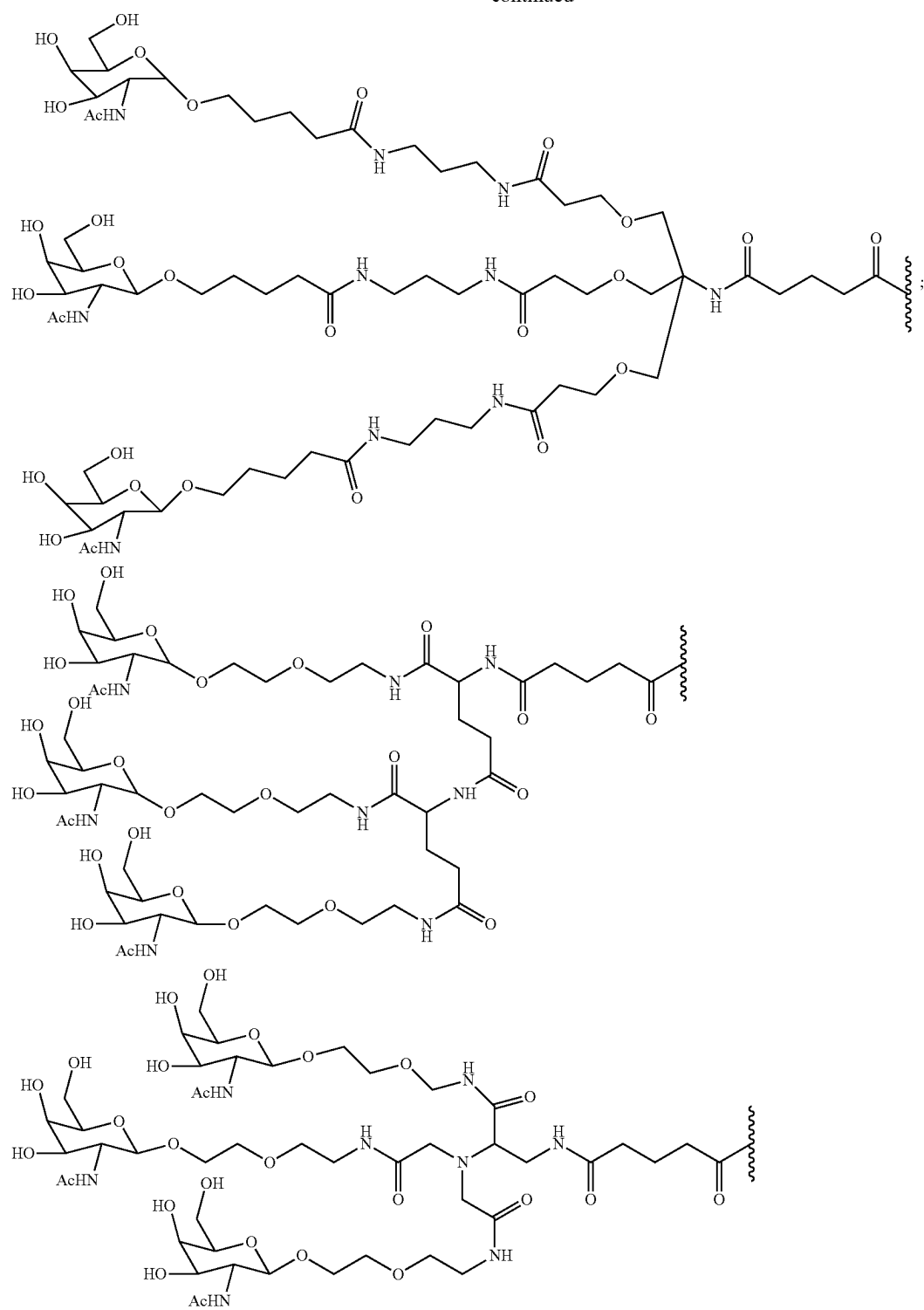
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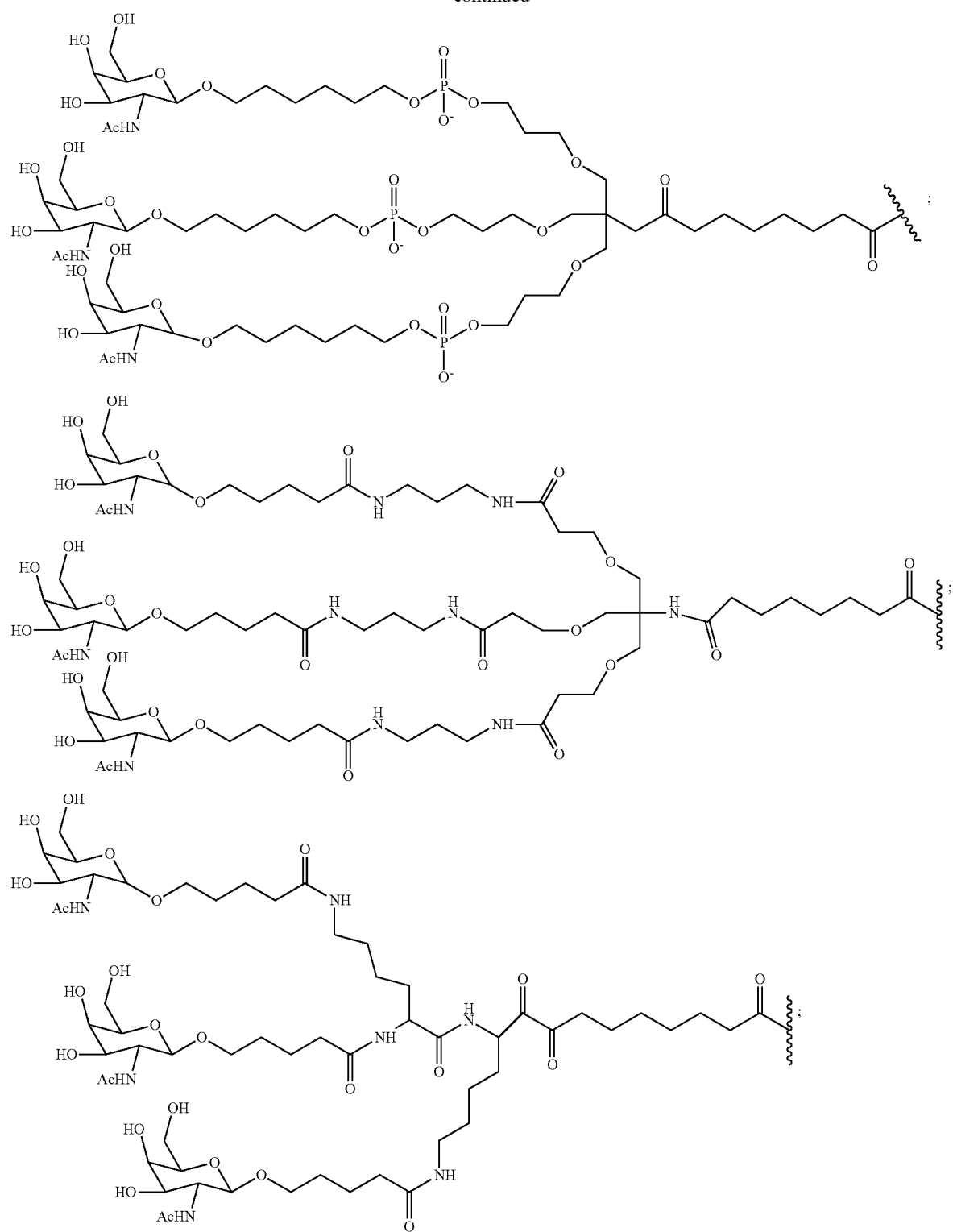
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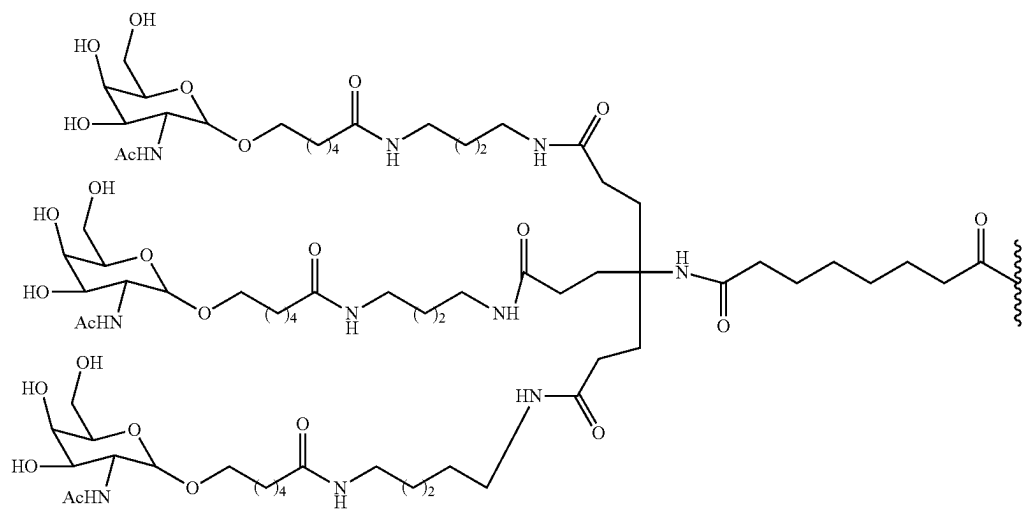
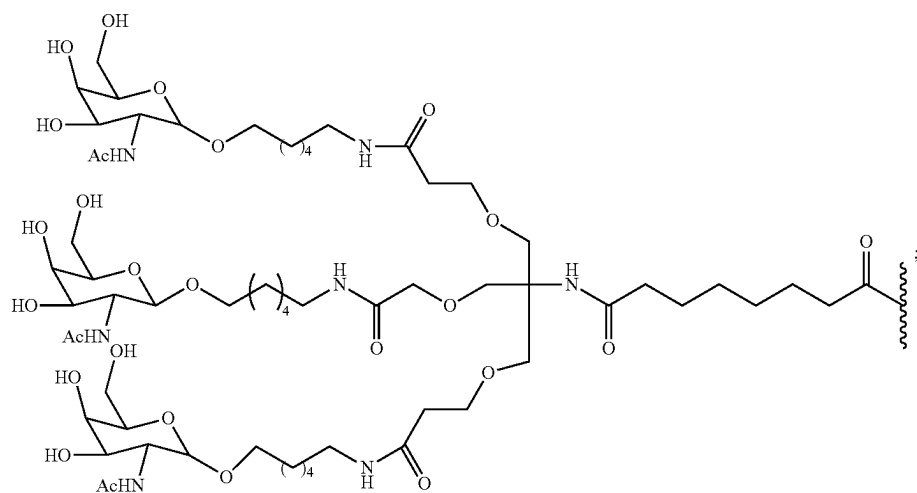
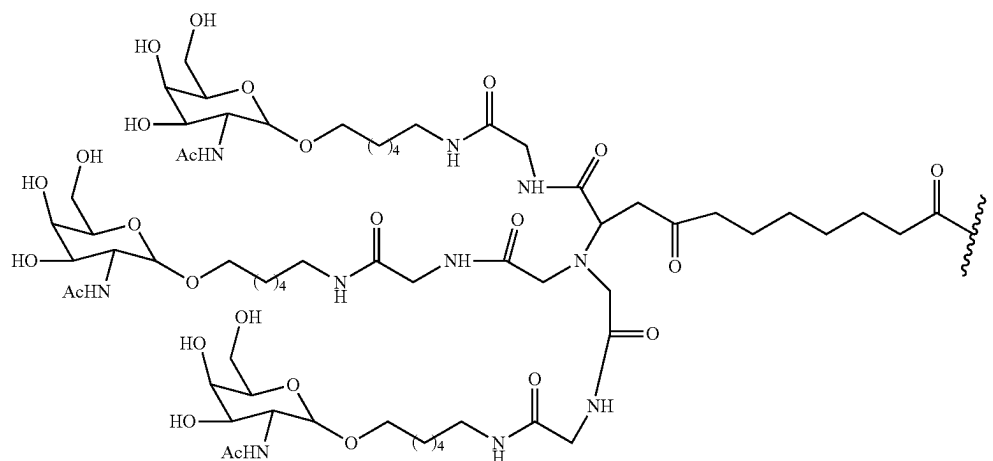
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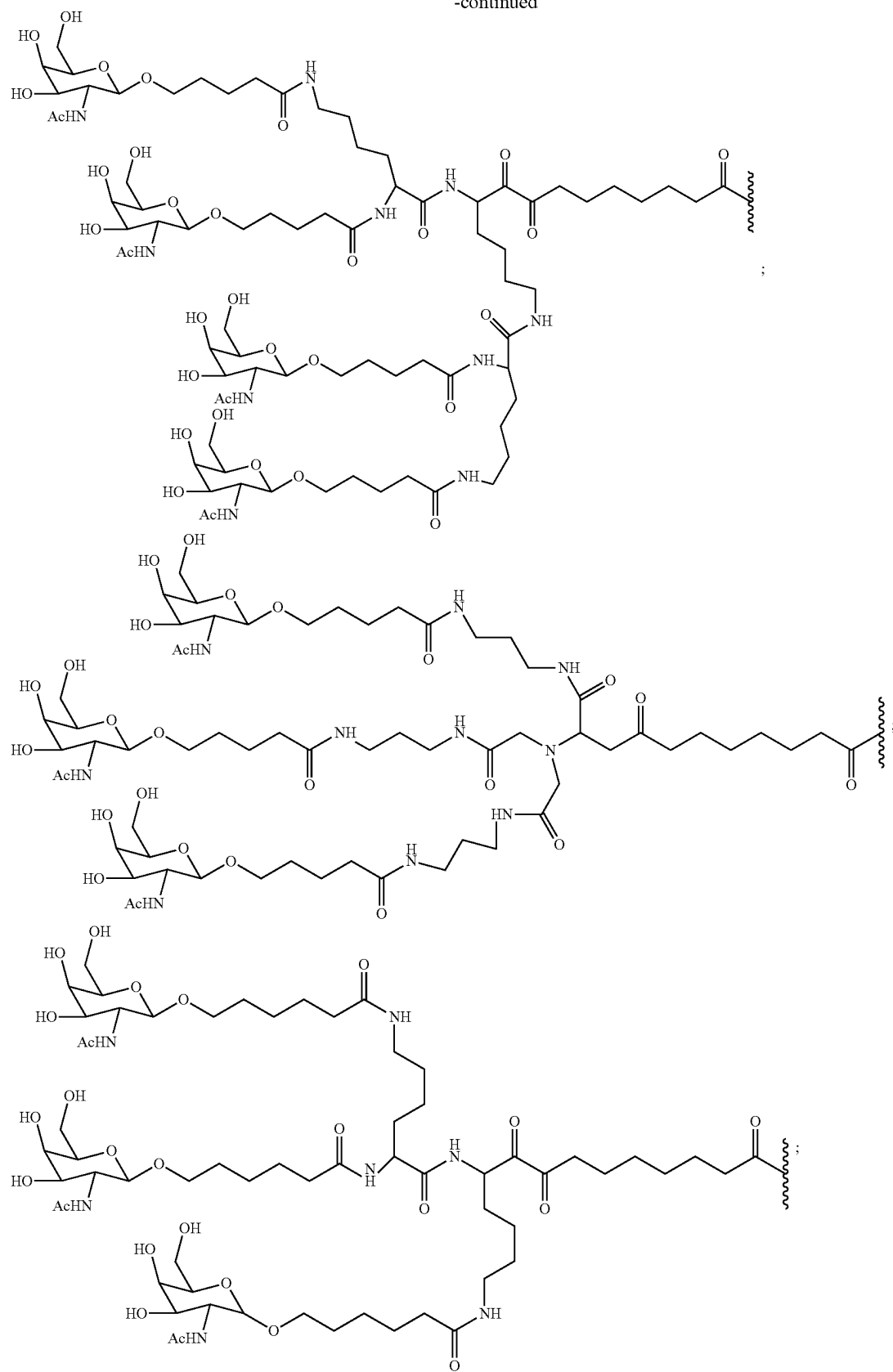
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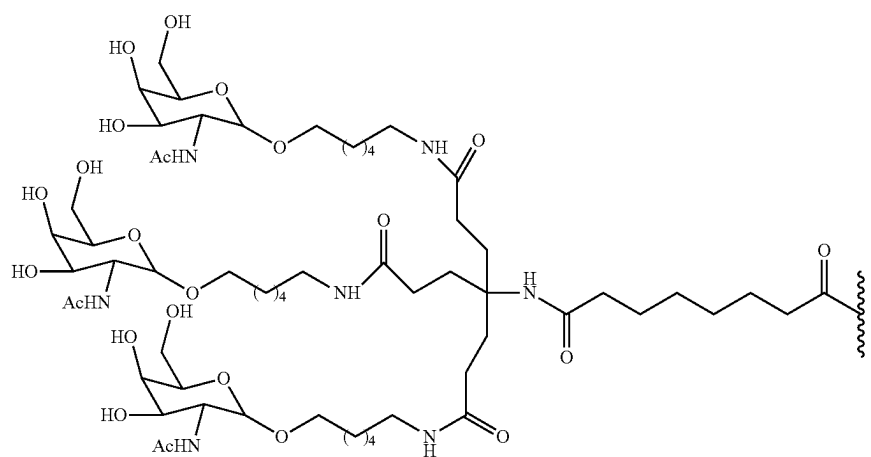
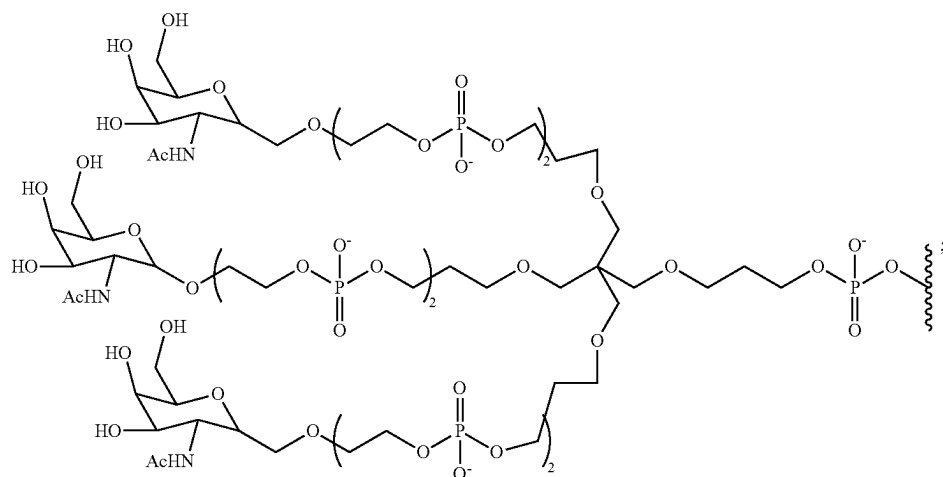
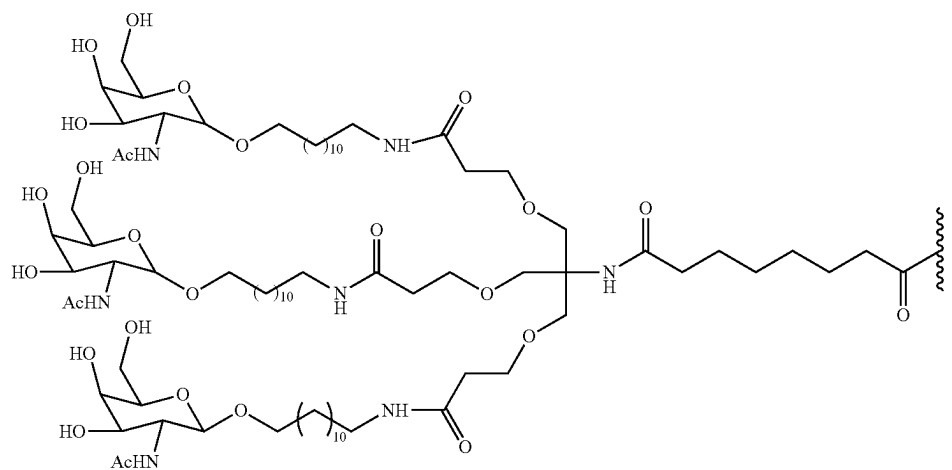
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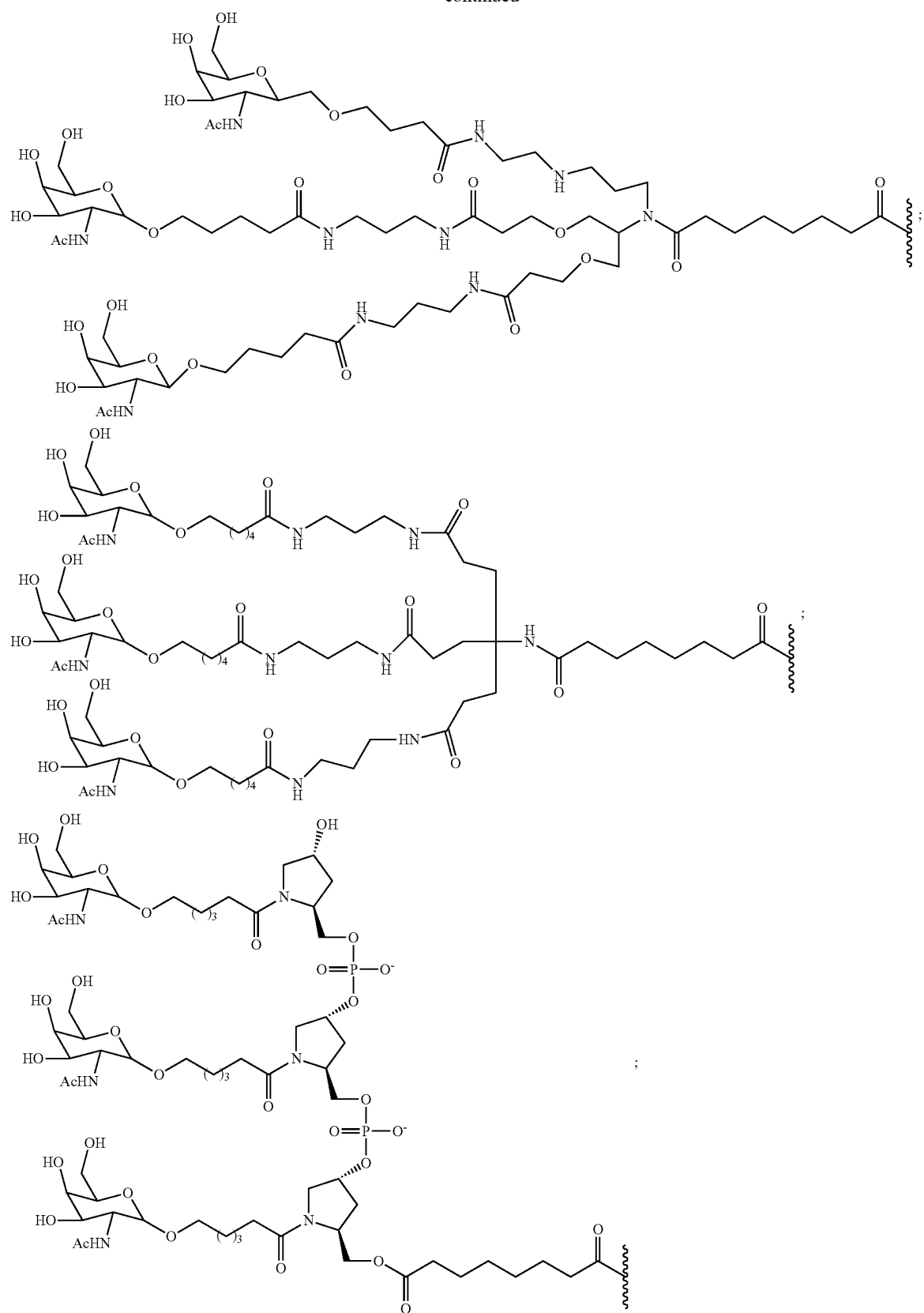
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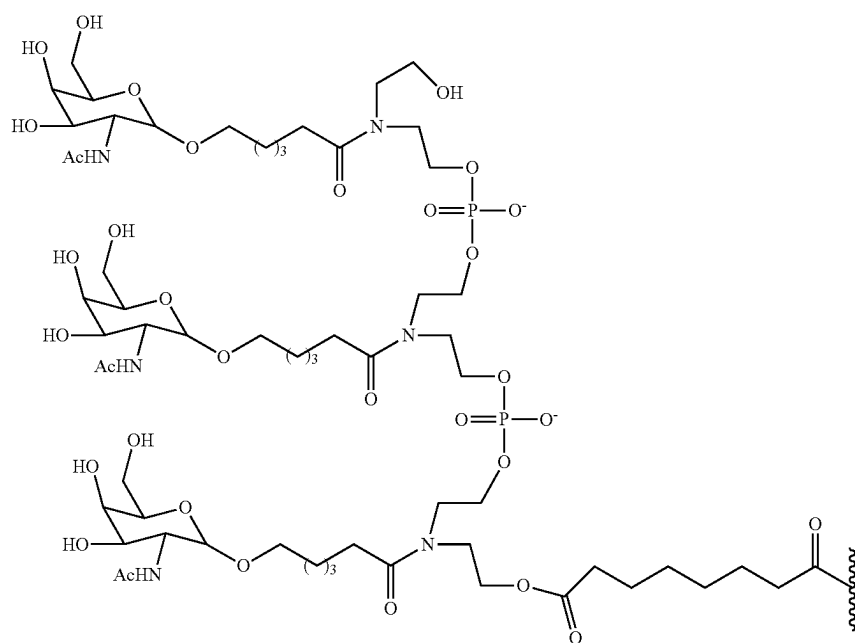
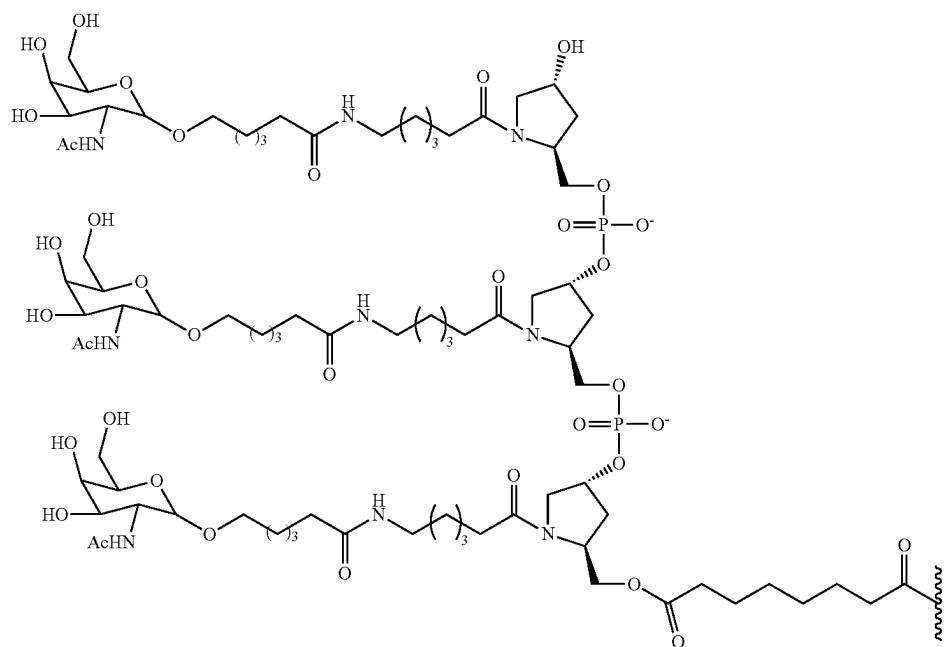
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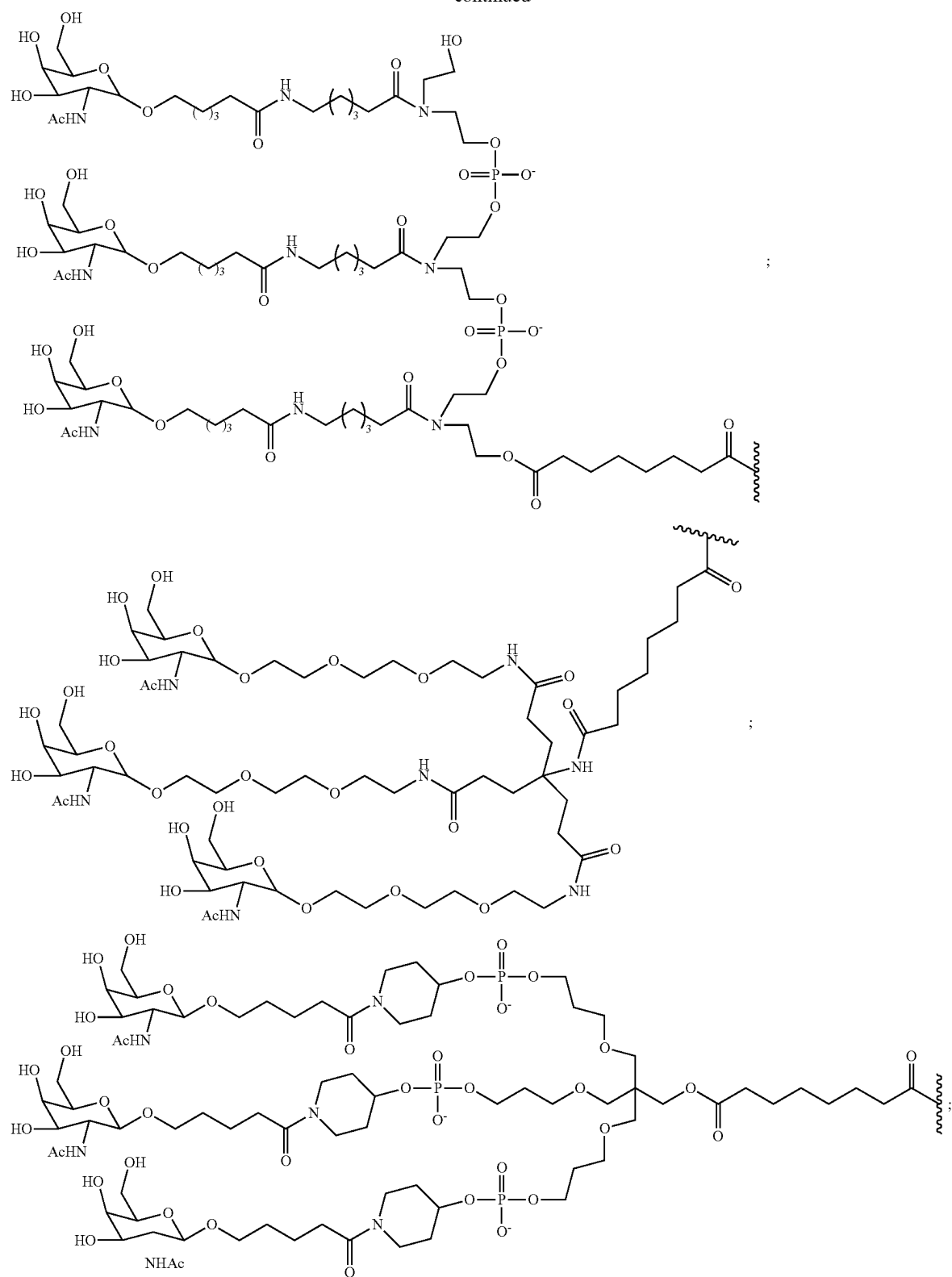
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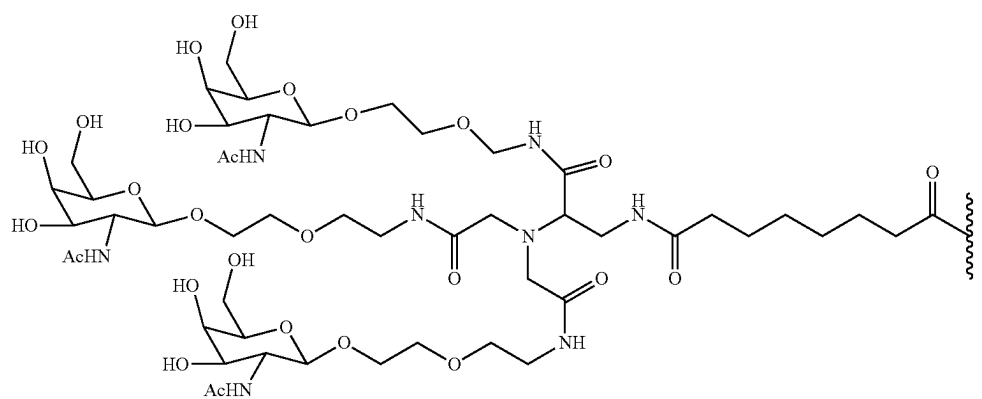
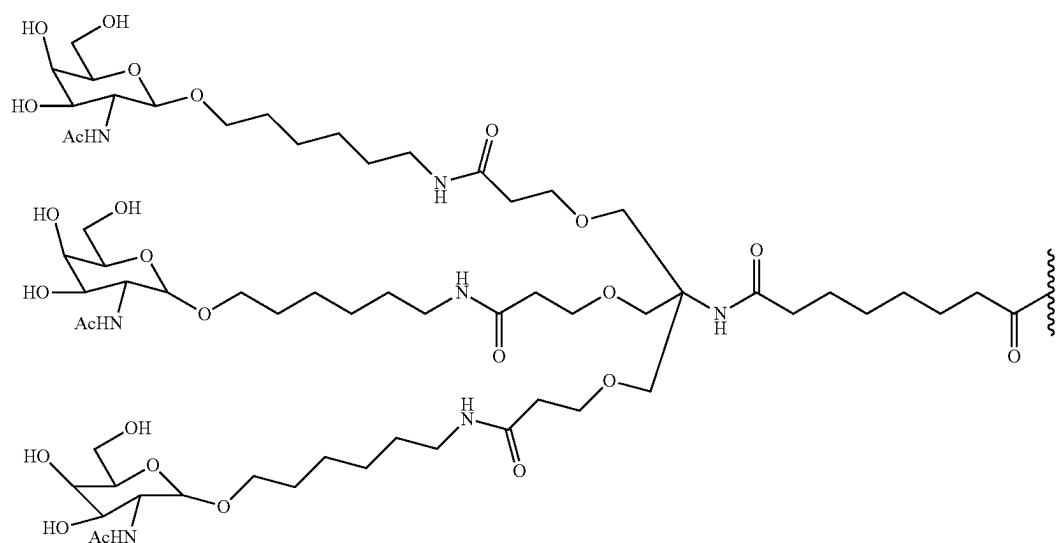
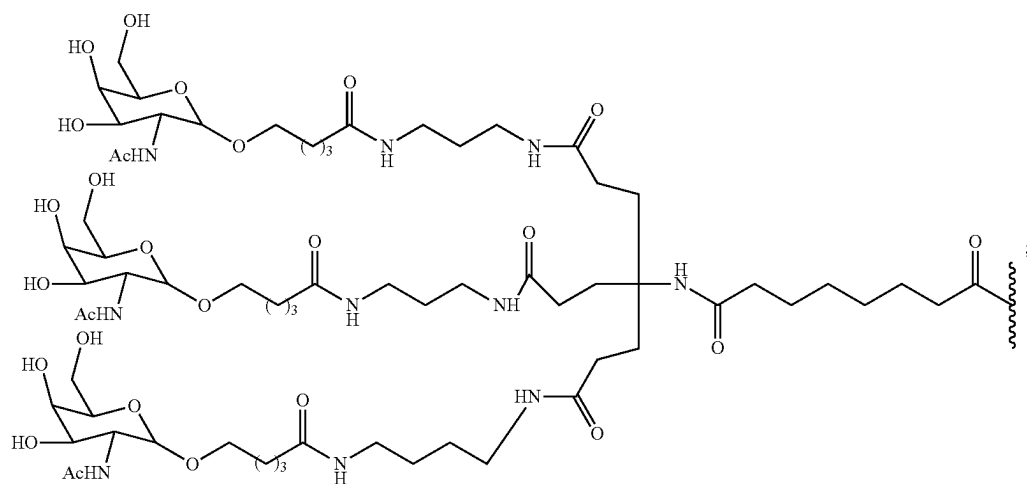
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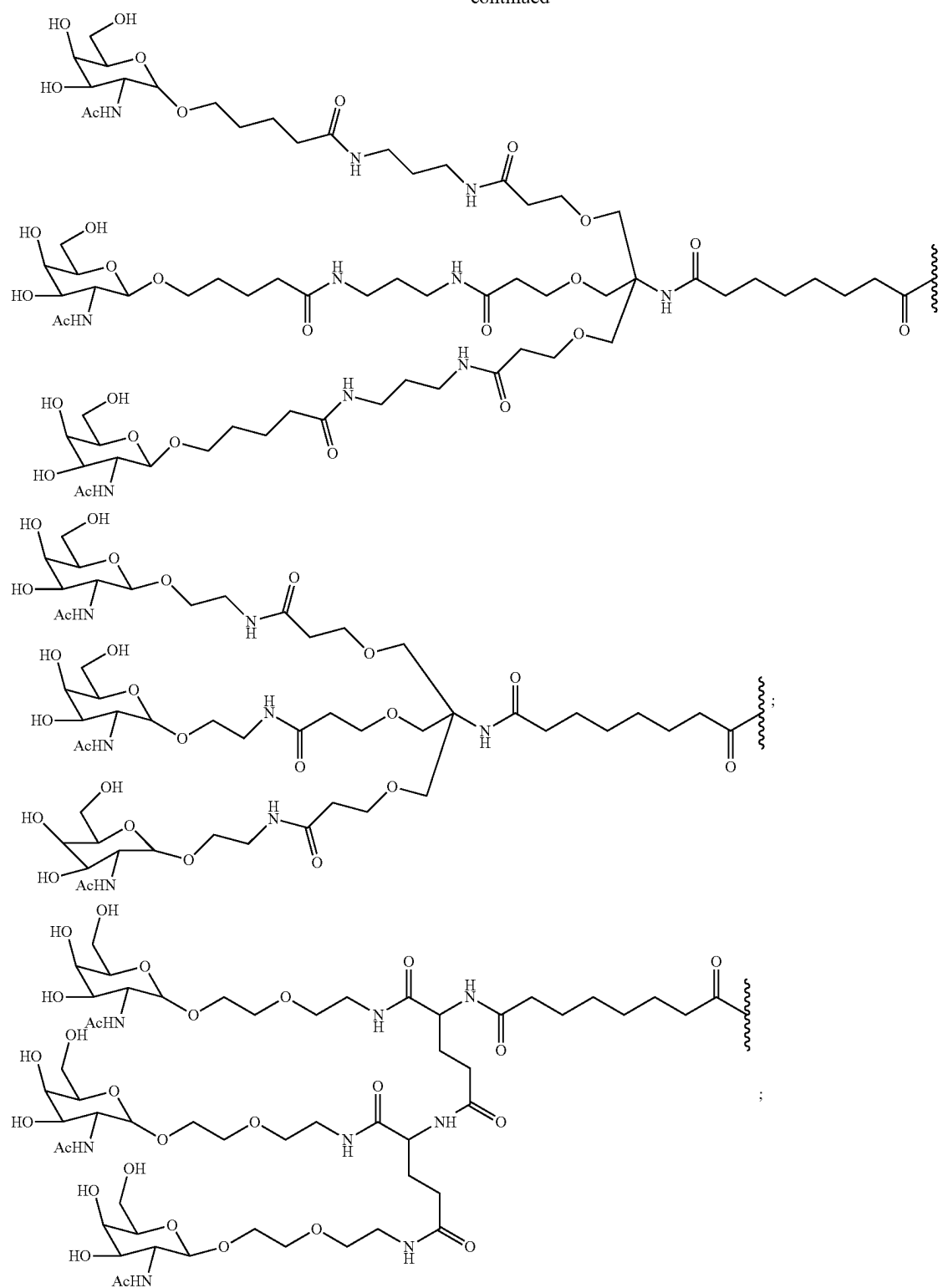
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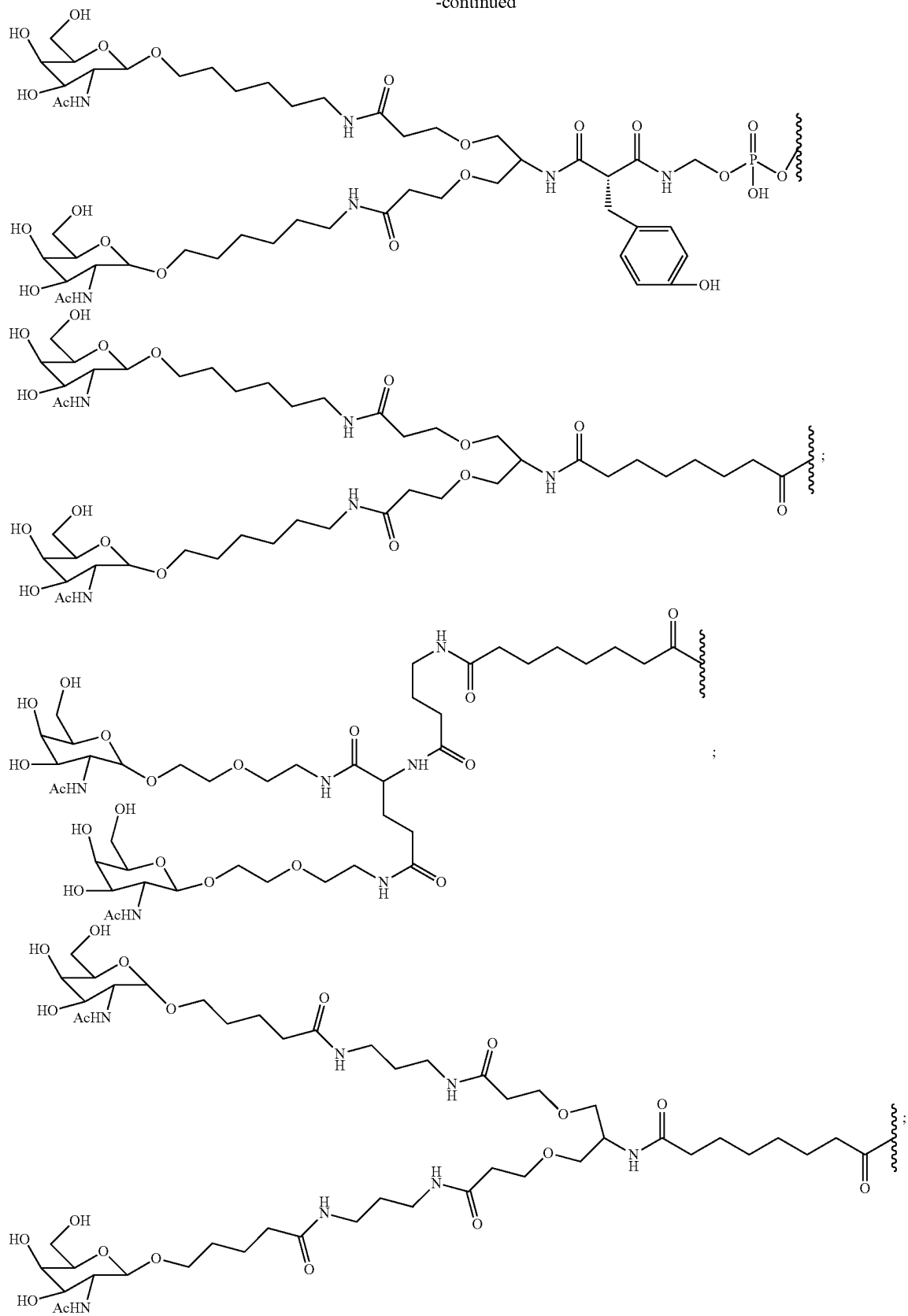
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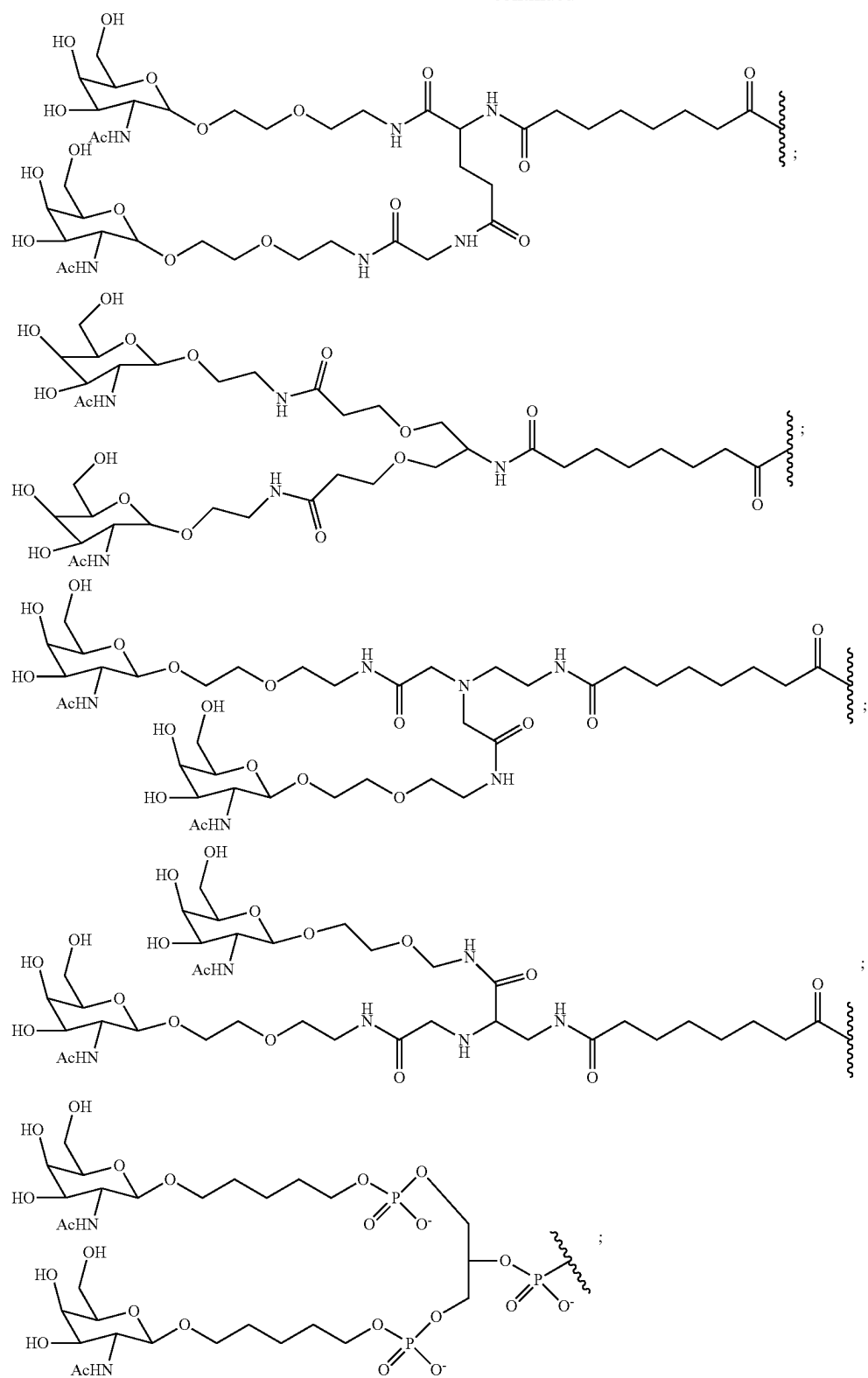
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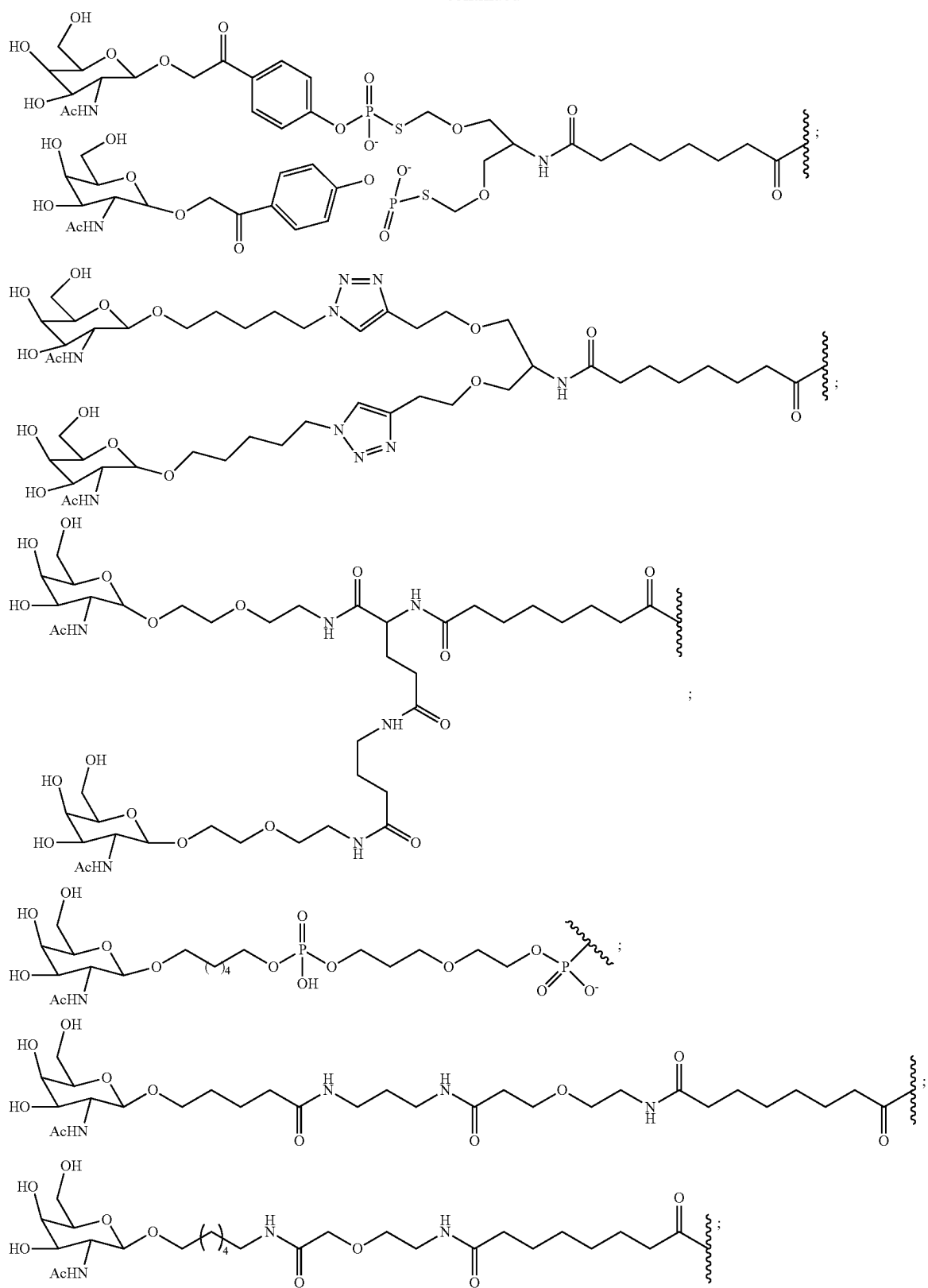
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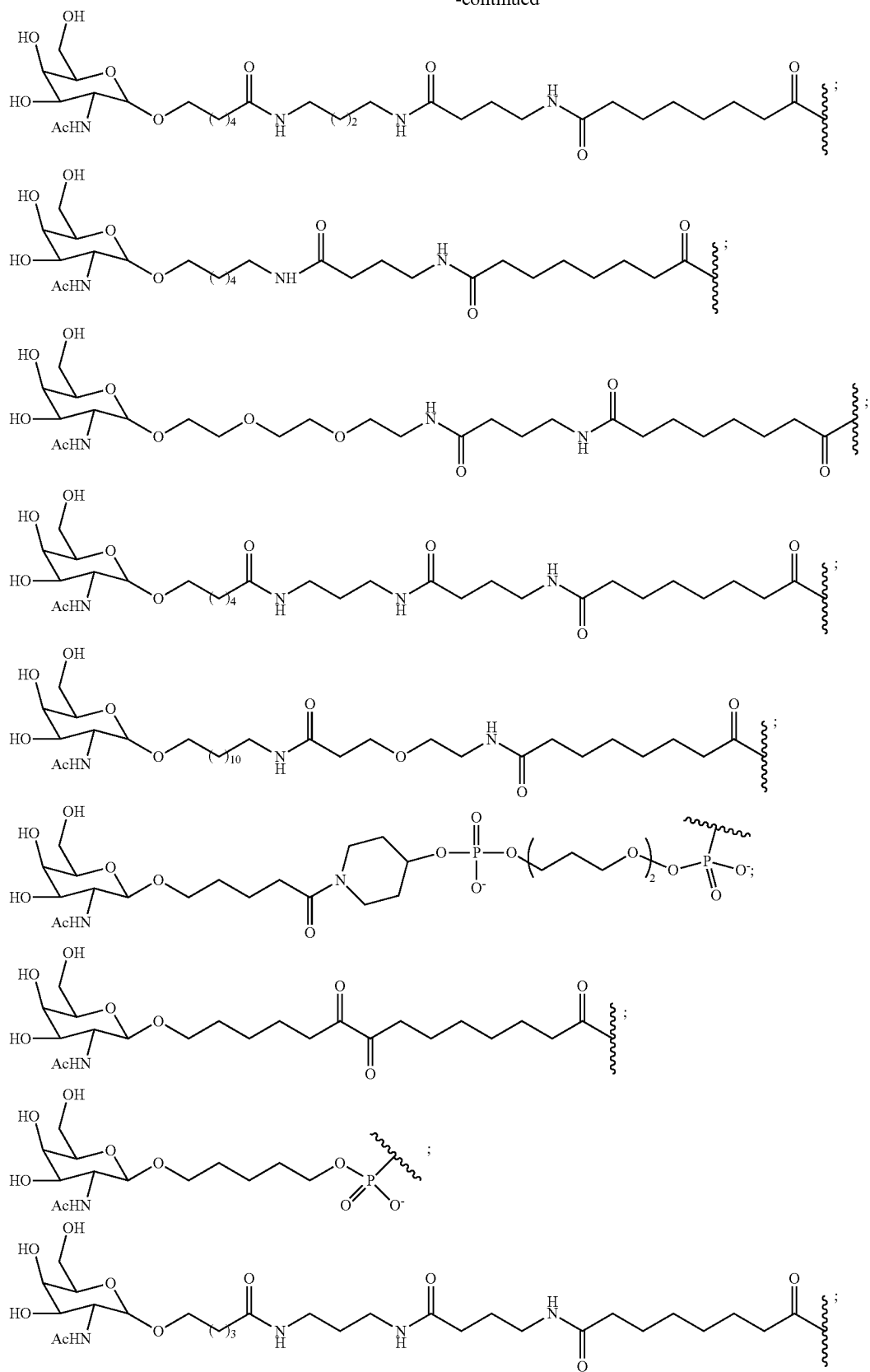
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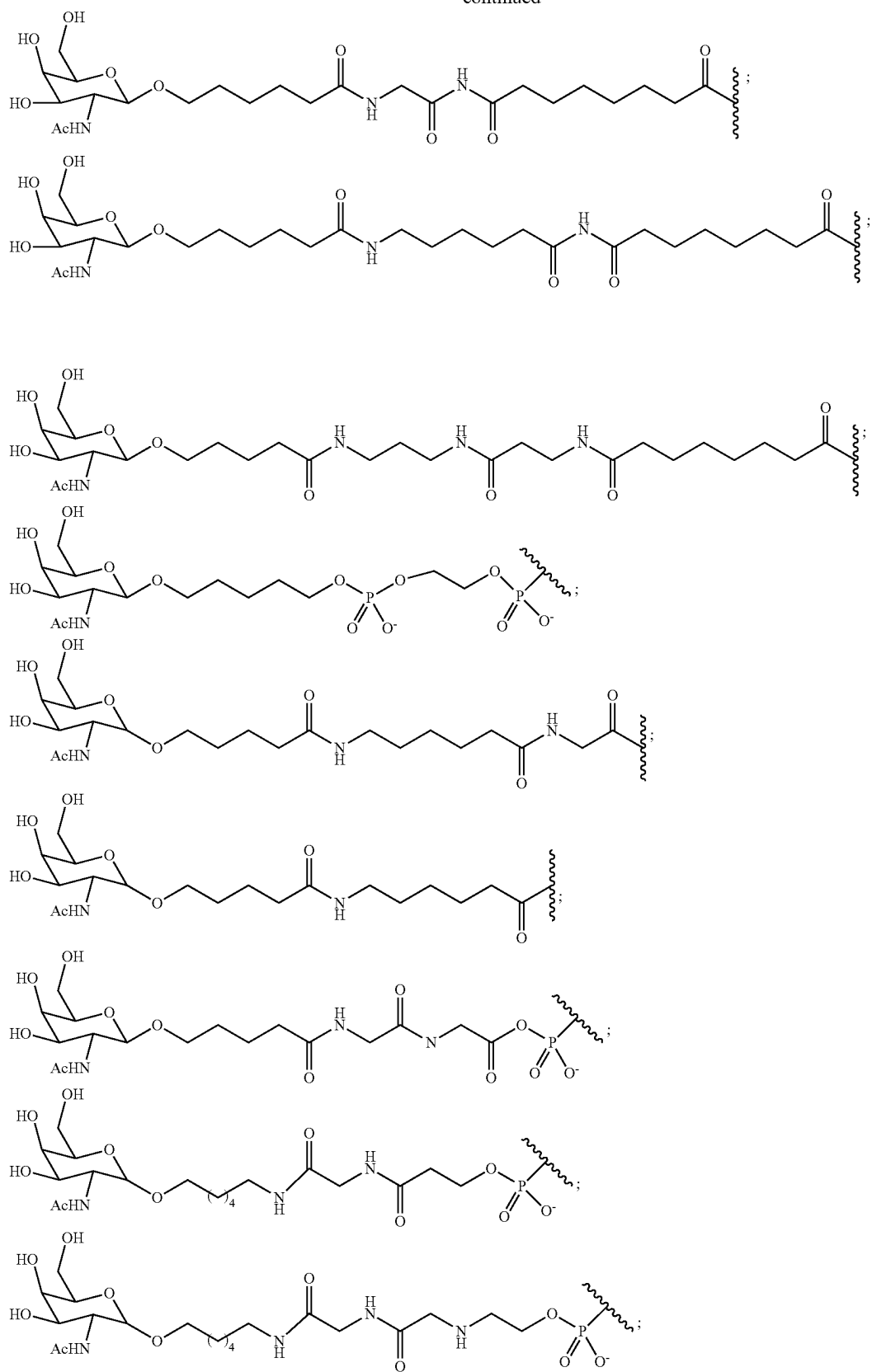
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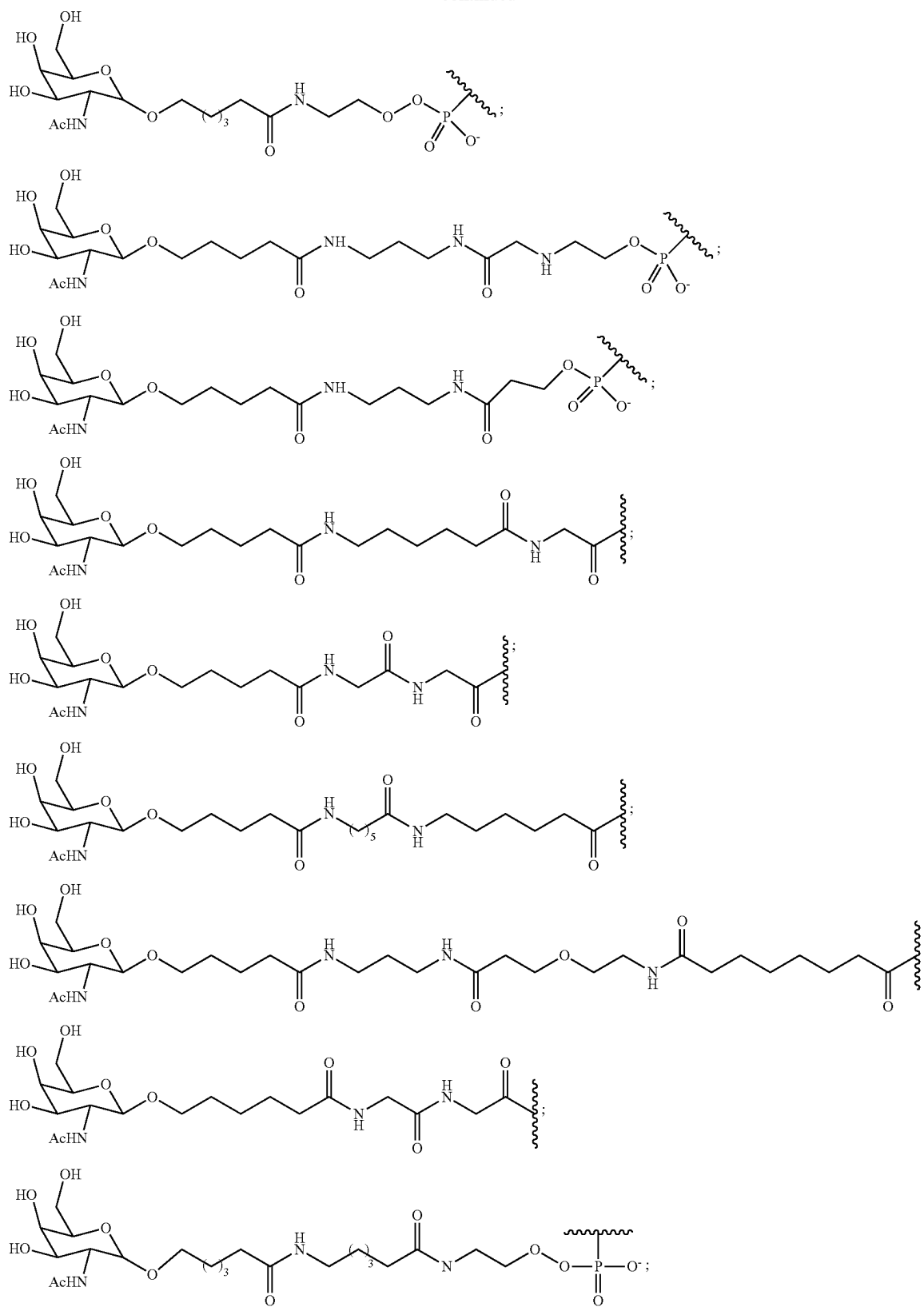
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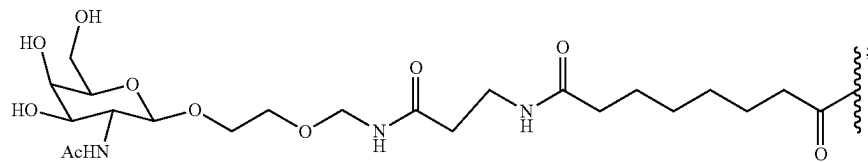
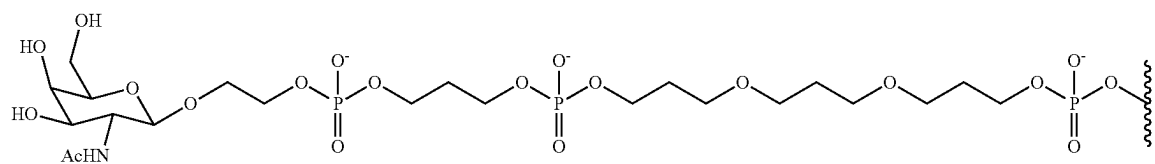
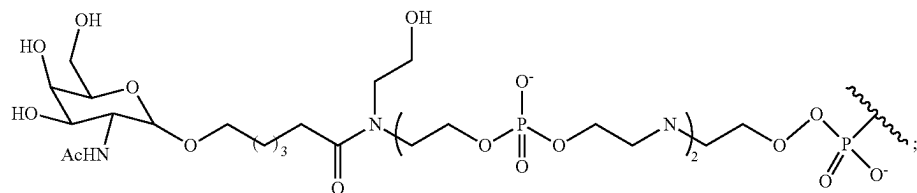
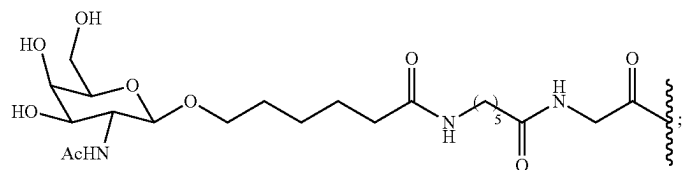
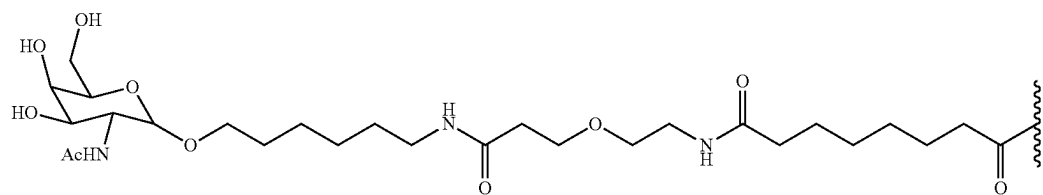
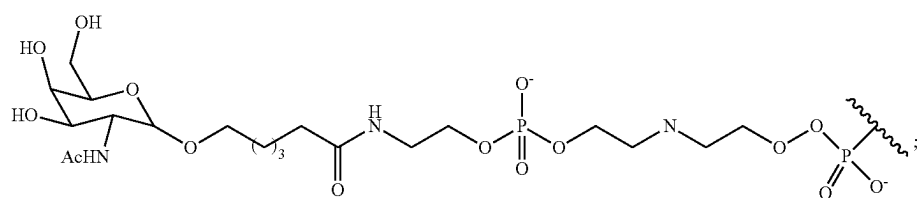
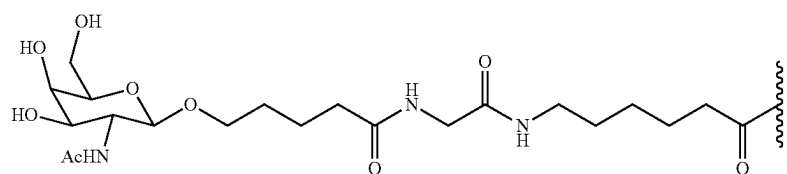
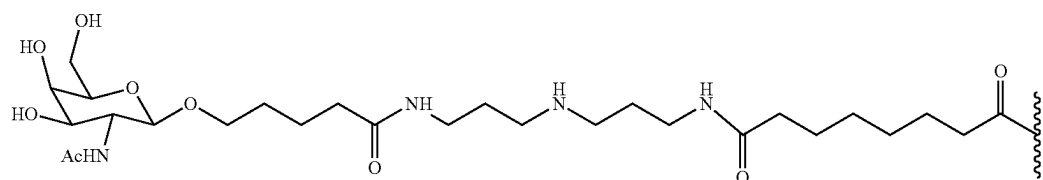
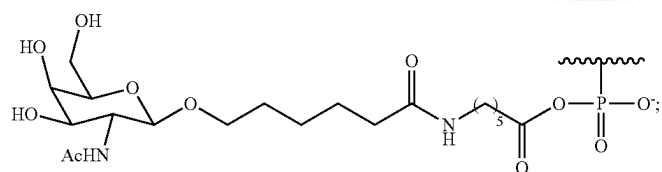
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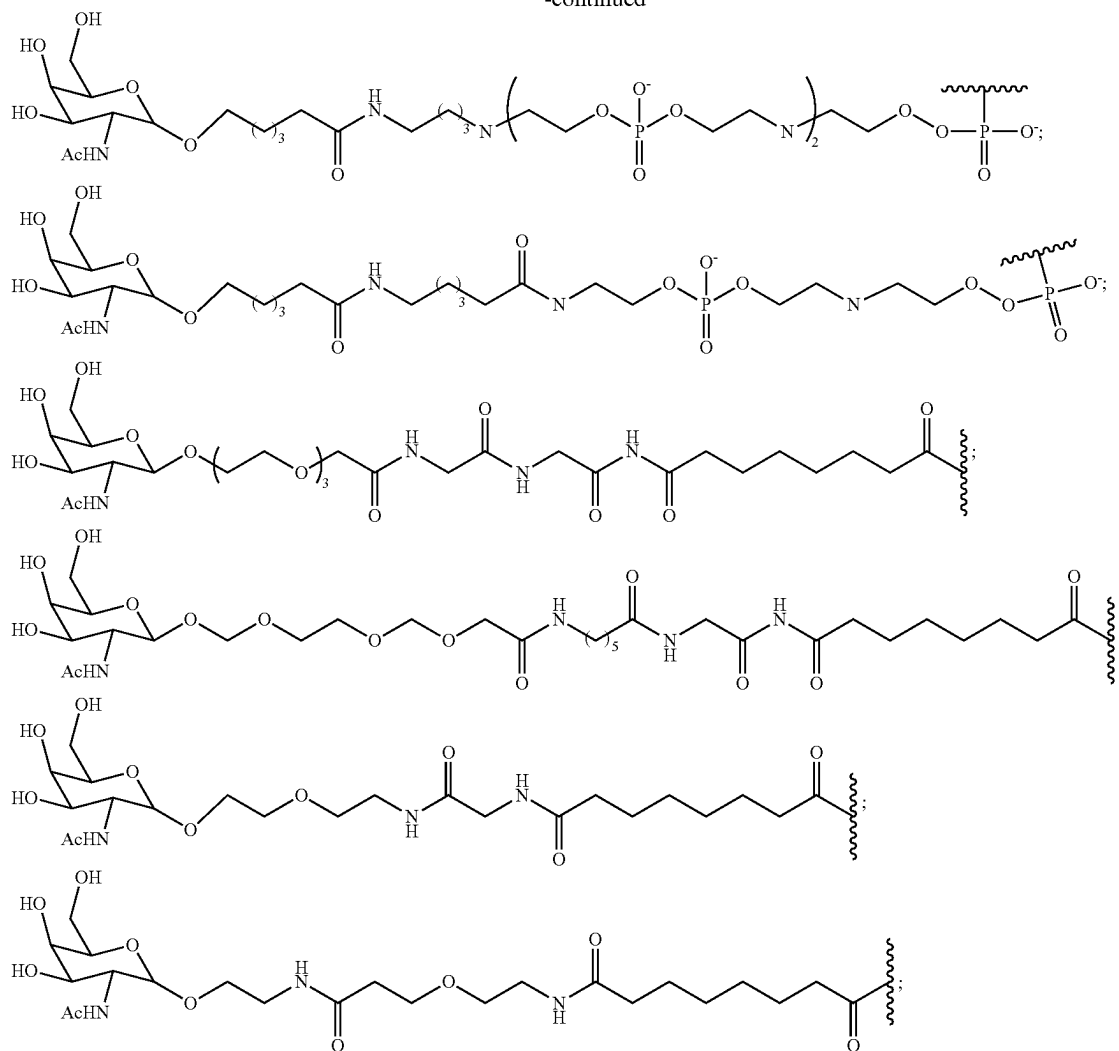
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[0305] 67. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the targeting specific ligand X, the branched chain L, the linking chain D, and the linker B are the same or different within the respective 5' MVIP and 3' MVIP or between the 5' MVIP and 3' MVIP.

[0306] 68. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the 5'MVIP is selected from any one of 5'MVIP01 to 5'MVIP22.

[0307] 69. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the 5'MVIP is selected from any one of 5'MVIP 01, 5'MVIP 09, 5'MVIP 17.

[0308] 70. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the 3'MVIP is selected from any one of 3'MVIP01 to 3'MVIP27.

[0309] 71. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the 3'MVIP is selected from any one of 3'MVIP01, 3'MVIP09, 3'MVIP17.

[0310] 72. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the combination of the sense strand 5'MVIP and the antisense strand 3'MVIP is 5'MVIP01/3'MVIP01, 5'MVIP01/3'MVIP17 or 5'MVIP09/3'MVIP09; or the combination of the antisense strand 5'MVIP and the sense strand 3'MVIP is 5'MVIP01/3'MVIP09 or 5'MVIP09/3'MVIP01.

[0311] 73. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the RNA inhibitor is selected from any one of Ky11-DS67, Ky11-DS68, Ky11-DS71, Ky11-DS73, Ky11-DS74, Ky11-DS7101, Ky11-DS7102, Ky11-DS7104, Ky11-DS7106, and Ky11-DS7108.

[0312] 74. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 36 or 37, wherein the drug comprises one or more of: a nucleic acid drug, a chemical drug, an antibody drug.

[0313] 75. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 74, wherein the nucleic acid drug is an siRNA drug.

[0314] 76. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 75, wherein the antisense strand is selected from any one or more in Table 15.

[0315] 77. The RNA inhibitor or a pharmaceutically acceptable salt thereof according to embodiment 75, wherein the sense strand is selected from any one or more in Table 14.

[0316] 78. A pharmaceutical composition comprising the RNA inhibitor according to any one of embodiments 1-77, further comprising a delivery vehicle, and/or a pharmaceutically acceptable excipient, and/or carrier, and/or diluent.

[0317] 79. The pharmaceutical composition according to embodiment 78, wherein the delivery vehicle comprises one or more of: nanoparticles, dendrimers, polymers, liposomes, or cation delivery systems.

[0318] 80. The use according to embodiment 78, wherein the disease or condition includes a disease or condition associated with elevated Lp (a) or apo (a) level.

[0319] 81. The use according to embodiment 78, wherein the disease or condition comprises a hepatogenic disease.

[0320] 82. The use according to embodiment 78, wherein the disease or condition comprises inflammatory, cardio-cerebrovascular, or metabolic diseases.

[0321] 83. The use according to embodiment 78, wherein the cardio-cerebrovascular diseases include hyperlipidemia, stroke, atherosclerosis, thrombosis, coronary heart disease, or aortic valve stenosis.

[0322] 84. A method of preventing or treating a disease, condition or syndrome, the method comprising administering to a subject in need an effective amount of the RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting LPA gene expression according to any one of embodiments 1-37, or the RNA inhibitor according to any one of embodiments 38-77, or the pharmaceutical composition according to any one of embodiments 78-79.

[0323] 85. The method according to embodiment 84, wherein the RNA inhibitor, the pharmaceutically acceptable salts thereof, or the pharmaceutical composition is administered to subject via subcutaneous, intravenous, oral, aerosol inhalation, rectal, or intraperitoneal route.

[0324] 86. A method for inhibiting the expression of LPA mRNA or Lp (a), comprising administering to a subject in need an effective amount of the RNA inhibitor or the pharmaceutically acceptable salt thereof for inhibiting LPA gene expression according to any one of embodiments 1-37, or the RNA inhibitor or the pharmaceutically acceptable salt thereof according to any one of embodiments 38-77, or the pharmaceutical composition according to any one of embodiments 78-79.

[0325] Without being bound by any theory, the examples in the following are only for the purpose of explaining the RNA inhibitor, preparation method and use of the present application, and are not used to limit the scope of the invention of the application.

EXAMPLES

Description

- [0326] The name of DMSO is dimethyl sulfoxide;
- [0327] The name of DMF is N, N-dimethylformamide;
- [0328] The name of HOBt is 1-hydroxybenzotriazole;
- [0329] The name of HBTU is O-benzotriazole-tetramethylurea hexafluorophosphate;

[0330] The name of DIPEA (DIEA) is N, N-diisopropylethylamine;

[0331] The name of DCM is dichloromethane;

[0332] The name of DMAP is 4-dimethylaminopyridine;

[0333] The name of DMT-CL is 4,4'-dimethoxytriphenylchloromethane;

[0334] The name of MEOH is methanol;

[0335] The name of TBTU is O-benzotriazole-N,N,N', N'-tetramethylurea tetrafluoroboric acid;

[0336] The name of  is solid-phase support, such as macroporous aminomethyl resin (Resin).

Example 1 Synthesis of RNA Inhibitors

[0337] The sense and antisense strands were synthesized by the solid-phase phosphoramidite method well known in the art. Complementary annealing of the sense strand and the corresponding antisense strand yielded the corresponding RNA inhibitor.

[0338] The basic steps of the solid-phase phosphoramidite method include:

[0339] 1) Deprotection: removing the hydroxyl protecting group (DMTr) of starting monomer Solid Support;

[0340] 2) Conjugation: adding the first phosphoramidite monomer to conjugate in the 3' to 5' direction;

[0341] 3) Oxidation: oxidizing the resulting nucleoside phosphite to a more stable nucleoside phosphate (that is, trivalent phosphorus is oxidized to pentavalent phosphorus);

[0342] 4) Blocking: adding cap to unreacted 5'-OH of the nucleotide monomer in the previous step to prevent further reaction; repeating the above steps until the last phosphoramidite monomer is introduced; then cleaving the ester bond between Solid Support and the starting monomer with methylamine aqueous solution and ammonia water, and removing the protective groups cyanoethyl (P), benzoyl (mA, fA), acetyl (mC), etc. on each base and phosphate on the resulting oligonucleotide; after separation and purification by HPLC, filtering to sterilize and lyophilizing to obtain the corresponding sense strand or antisense strand.

[0343] Annealing: Accurately determining the concentration of the solution in which the sense strand and antisense strand are reconstituted, mixing them at an equimolar concentration, then adding 1/20 volume of 1 M PBS solution and mixing well again, heating the mixed system to 95° C. for 5 minutes, then cooling naturally for 3 hours until it reaches 40° C. or room temperature. Conducting HPLC detection. If the residual single-strand is less than 5%, the reaction can be considered complete.

Example 2 Experiment of Liposome-Delivered RNAi Agent to Inhibit LPA Gene Expression In Vitro

[0344] Took the RNA inhibitor (Ky11-DS01~ Ky11-DS18 in Table 3) prepared by the method described in Example 1. The aqueous solution of RNA inhibitor and organic solution of DOTMA were mixed to form water-insoluble precipitate, which was dissolved in chloroform after separation and drying, and further mixed with chloroform solution of other lipid, including M10C1 and PEG600-cholesterol. The mixture was evaporated and dried overnight by vacuum centrifugation to obtain nanolipid-encapsulated RNAi agent, in

which the weight ratios of DOTMA, M10C1 and PEG600-cholesterol to RNA inhibitor were 1~1.6, 1.5~2.5 and 2.5~3.5.

[0345] DMEM containing 10% fetal bovine serum was used to prepare the nanolipid-encapsulated RNA inhibitor sample solution with corresponding concentration. HepG 2 cells were seeded at a density of 10^5 cells. After incubation for 24 h in DMEM medium supplemented with 10% fetal bovine serum, 37° C., 5% CO₂, samples with 10 nM, 1 nM, 0.1 nM were added for interference. After incubation for 72 h, collected the cell samples, added 1 ml Ezol lysate to the collected cell samples and mixed well by a vortex shaker. Added 0.2 ml trichloromethane, shook vigorously for 10 s, and left at room temperature for 1 min. Centrifuged at 12000×g for 15 min at 4° C. Transferred the upper clear-water phase to another new RNase free centrifuge tube, and added an equal volume of 100% ethanol. Pipetted all of the sample and added it to a mini-spin centrifuge column with a 2 ml collection tube. Centrifuged at 8000×g for 15 seconds at room temperature, and discarded the flow-through liquid. Transferred the remaining sample to a centrifuge column and repeated the previous step. Added 700 µl WB to the centrifuge column, gently closed the cover, centrifuged at 8000×g, room temperature for 15 seconds, discarded the flow-through liquid. Repeated the previous step, and used 500 µL WB to wash the centrifuge column twice. LPA mRNA levels were determined by qRT-PCR. Compared with the supernatant of HepG 2 cells without interference, the relative percentages of LPA mRNA in the interference group were calibrated. The obtained test results were shown in FIG. 1.

[0346] Experimental results: when the dosage was 10 nM, Ky11-DS08~Ky11-DS12 had significant effects on the inhibition of the expression level of LPA mRNA in HepG 2 cell lines.

[0347] As shown in Table 3, the sequences of SEQ ID NO. 411 and SEQ ID NO. 422 have the same 18 consecutive nucleotides and 3 different nucleotides. Furthermore, as evident from FIG. 1's experimental results, Ky11-DS16 corresponding to SEQ ID NO. 422 exhibits a similar inhibitory effect on LPA mRNA expression levels as Ky11-DS05 corresponding to SEQ ID NO. 411; SEQ ID NO. 416 and SEQ ID NO. 419 have the same 17 consecutive nucleotides, although the results of their corresponding Ky11-DS10 and Ky11-DS13 on the inhibition of LPA mRNA expression levels are quite different, the effect of ky11-DS13 is also acceptable; Similar to this situation, SEQ ID NO. 415, 420 and 421 have the same 17 consecutive nucleotides, and their inhibitory effects on LPA mRNA expression levels are also acceptable; the others are not listed here one by one. It can be seen that any sequence of SEQ ID NO. 414-418, a sequence having at least 15 consecutive nucleotides identical to those of SEQ ID NO. 414-418, or a sequence having no more than three different nucleotides from the above sequences, are all within the scope of protection of the present invention.

[0348] It should be noted that this is not the result of a limited number of experiments, but a design idea. Any sequence obtained by transforming multiple nucleotides or any sequence designed by truncating 15 or more consecutive identical nucleotides falls under this design idea and is within the scope of protection of the present invention.

Example 3 Examination of RNA Inhibitor Effects In Vitro Using HepG2 Cells

[0349] The present example examines in vitro activity of the sequences screened in Example 2 after modifications, said modifications comprise: the —OH at the 2' positions of the nucleotide glycosyls at positions 7, 12, 14 starting from the 5' end of the antisense strand are fluorinated, the —OH at the 2' positions of the remaining nucleotide glycosyls of the antisense strand are substituted with methoxy, and there are at least two consecutive phosphorothioate bonds between the three consecutive nucleotides at the end of the antisense strand; the —OH at the 2' positions of the nucleotide glycosyls at positions 5, 7, 8, 9 starting from the 5' end of the sense strand are fluorinated, the —OH at the 2' position of the remaining nucleotide glycosyls of the antisense strand are substituted with methoxy, and there are at least two consecutive phosphorothioate bonds between the three consecutive nucleotides at the end of the sense strand; or

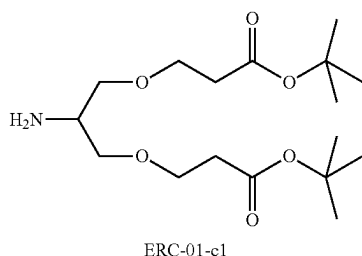
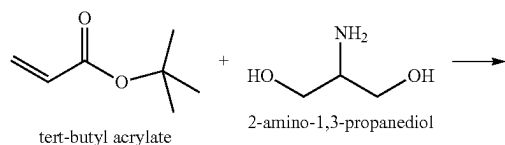
[0350] The —OH at the 2' positions of the nucleotide glycosyls at positions 2, 4, 6, 8, 14, 16 starting from the 5' end of the antisense strand are fluorinated, the —OH at the 2' positions of the remaining nucleotide glycosyls of the antisense strand are substituted with methoxy, and there are at least two consecutive phosphorothioate bonds between the three consecutive nucleotides at the end of the antisense strand; the —OH at the 2' positions of the nucleotide glycosyls at positions 3, 5, 7, 8, 9, 11, 13, 15 starting from the 5' end of the sense strand are fluorinated, the —OH at the 2' positions of the remaining nucleotide glycosyls of the antisense strand are substituted with methoxy, and there are at least two consecutive phosphorothioate bonds between the three consecutive nucleotides at the end of the sense strand.

[0351] Took the RNA inhibitor in Table 7 prepared according to the method described in Example 1. Prepared DMEM medium containing 10% fetal bovine serum. When the cells (HepG2) were cultured in 10 cm culture dishes until 80-90% confluence, seeded them into a 6-well plate. Poured away the culture medium and washed the cells twice with 2 ml PBS. Added 2 ml Trypsin-EDTA solution (Trypsin cell digest solution), mixed well, and placed at 37° C. for 3-5 minutes. Carefully removed the Trypsin solution, added 2 ml of DMEM culture medium containing 10% FBS, and then pipetted to form a single-cell suspension. Counted the cells on a hemocytometer, diluted them to 1.5×10^7 cells/ml. seeded into a 6-well plate at a concentration of 1.5×10^6 cells/well, mixed well and incubated for 24 hours at 37° C., 5% CO₂. Dissolved with 120 µl DEPC-H₂O per 1 OD260 siRNA oligo, the final concentration is approximately 20 µM. Took transfection of 1 well of a 6-well plate as an example. Added 250 µl serum-free DMEM medium into a 1.5 ml EP tube, added 30 µl siRNA oligo, mixed well; Took another 1.5 ml EP tube, added 250 µl serum-free DMEM, added 5 µl GP-Transfect-Mate (GenePharma), mixed well, placed at room temperature for 5 minutes, then combined the content in two tubes and placed at room temperature for 20 minutes. Pipetted the culture medium in the 6-well plate, added the transfection mixture dropwise into the 6-well plate, mixed well and incubated in an incubator for 5 hours. Removed the transfection solution and added 1 ml of DMEM culture medium containing 10% FBS. Kept culturing the cells for 24 or 48 hours at 37° C., 5% CO₂, collected

Description of Synthesis Process:

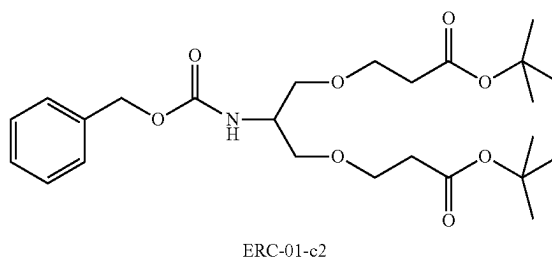
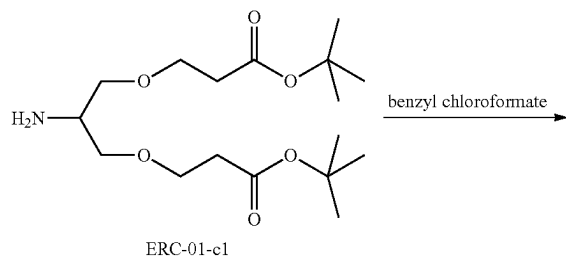
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1.1.1 Synthesis of ERC-01-c1



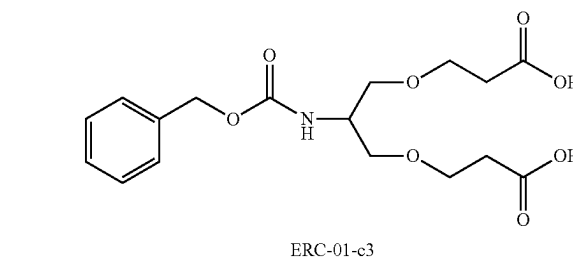
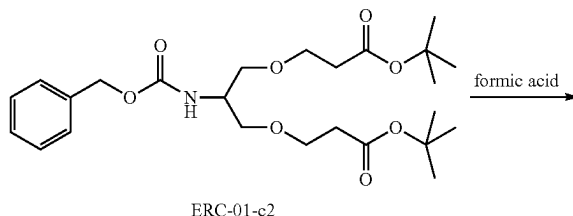
[0362] Weighed 2-amino-1,3-propanediol (5.0 g, 54.9 mmol) and added 50 ml of DMSO, 5 ml of sodium hydroxide solution (1 g/ml), cooled it down to 0° C., added tert-butyl acrylate (20 ml, 137.8 mol) dropwise for 2 hours, let it react at room temperature for 48 h, added petroleum ether (100 ml), washed twice with saturated salt water, and dried the organic layer. Passed it through a chromatography column (eluent: ethyl acetate:petroleum ether=25%-75%), then added 0.05% triethylamine to the column, and obtained 6.2 g of colorless oil.

1.1.2 Synthesis of ERC-01-c2



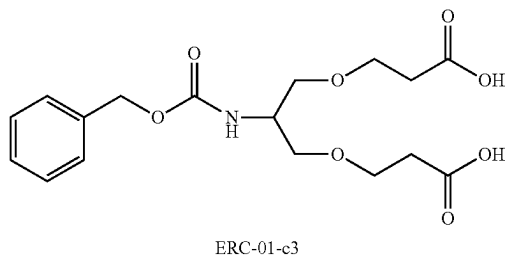
[0363] Weighed ERC-01-c1 (6.2 g, 17.9 mmol), added 50 ml of dichloromethane, 23 ml of sodium carbonate solution (25%), added benzyl chloroformate (8.2 ml, 57.4 mmol) dropwise at room temperature for 2 hours, let it react overnight at room temperature, washed with saturated salt water for three times, dried with anhydrous sodium sulfate, evaporated the solvent, passed it through a chromatography column (ethyl acetate:petroleum ether=5%-30%), and obtained 4.0 g of oil.

1.1.3 Synthesis of ERC-01-c3

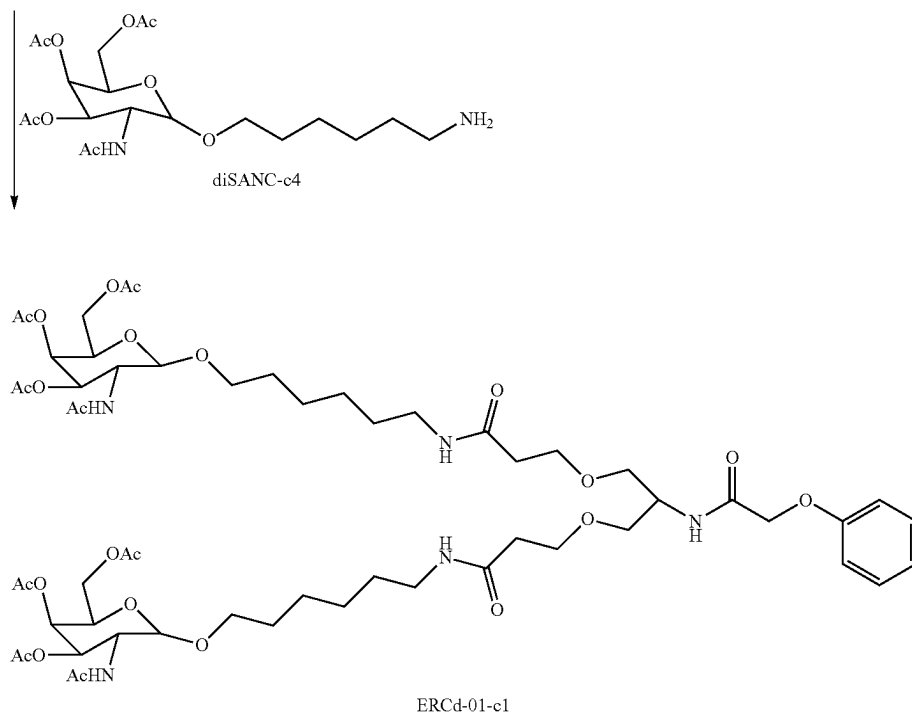


[0364] Weighed ERC-01-c2 (4.0 g, 8.3 mmol), added 12 ml of formic acid, let it react overnight at room temperature, evaporated the solvent under reduced pressure to obtain 2.8 g of product.

1.1.4 Synthesis of ERCd-01-c1



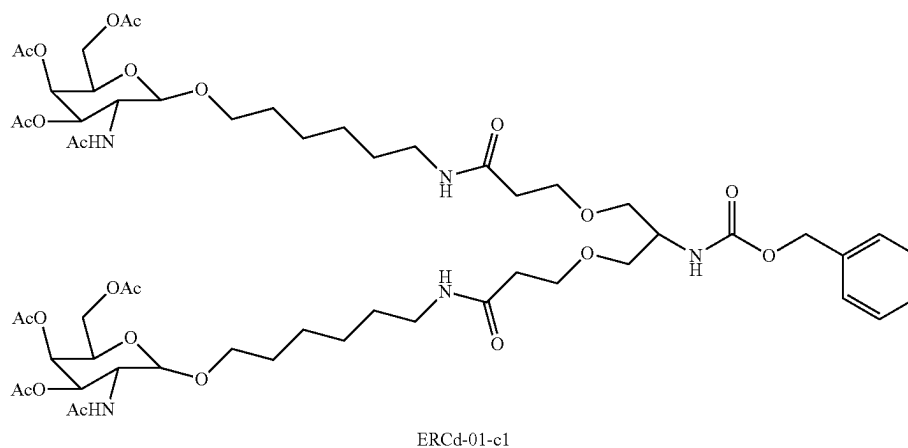
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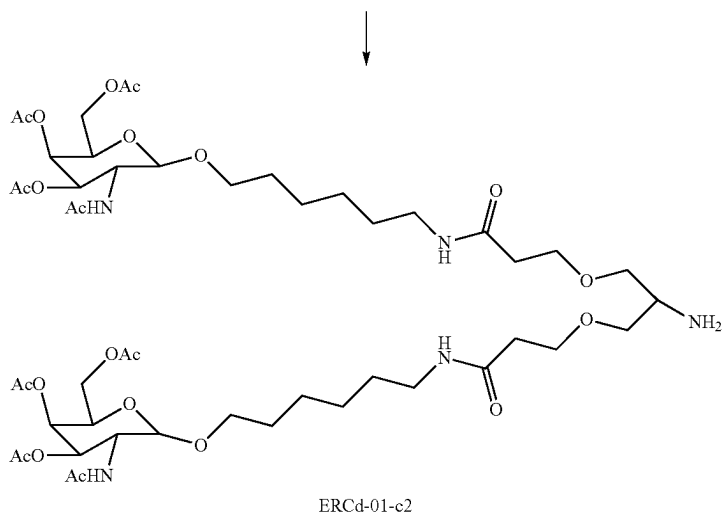
[0365] Added compounds ERC-01-c3 (1.11 g, 3.0 mmol) and diSANC-c4 (3.6 g, 8.04 mmol) to DMF (60 ml), then added HOBt (2.24 g) and HBTU (3.36 g), and then slowly added DIEA (4.16 ml). Stirred the reaction solution for 3 hours at room temperature. Then added water and extracted the aqueous layer with dichloromethane (2×10 ml). Combined the organic layers, and then washed with saturated

sodium bicarbonate (80 ml), water (2×60 ml), and saturated salt water (60 ml). Dried with anhydrous sodium sulfate, evaporated to dryness under reduced pressure, and purified by silica gel column chromatography (eluent: 3-15% MeOH in DCM). Obtained light yellow solid 3.24 g.

1.1.5 Synthesis of ERCd-01-c2



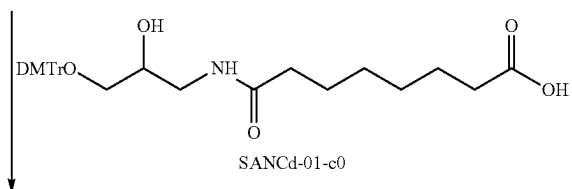
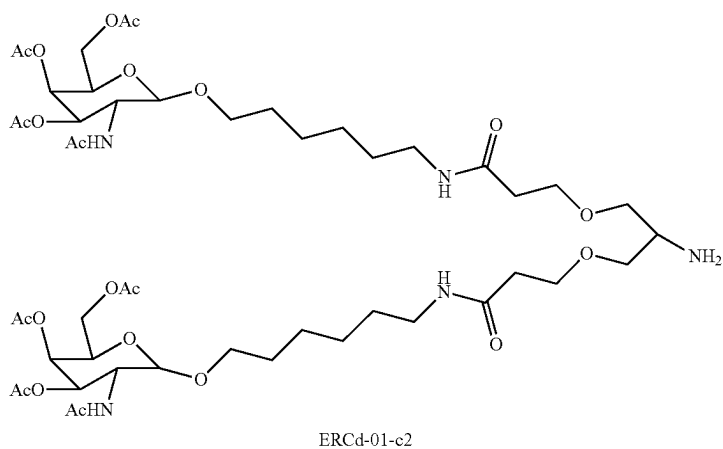
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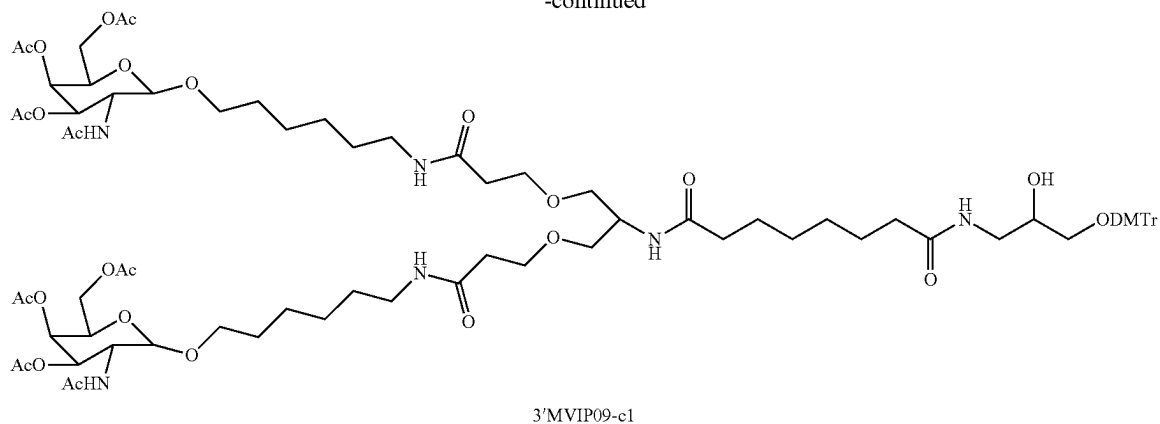
[0366] Dissolved ERCd-01-c1 (3.24 g, 2.6 mmol) in methanol (60 ml), and added 10% palladium carbon (0.3 g) and acetic acid (2.0 ml). Then introduced hydrogen at atmospheric pressure to react overnight. The reaction solution was filtered with diatomite, and the filtrate was evapo-

rated to dryness under reduced pressure to obtain 2.9 g of oil ERCd-01-c2, its high-resolution mass spectrum was shown in FIG. 3.

1.1.6 Synthesis of 3'MVIP09-c1



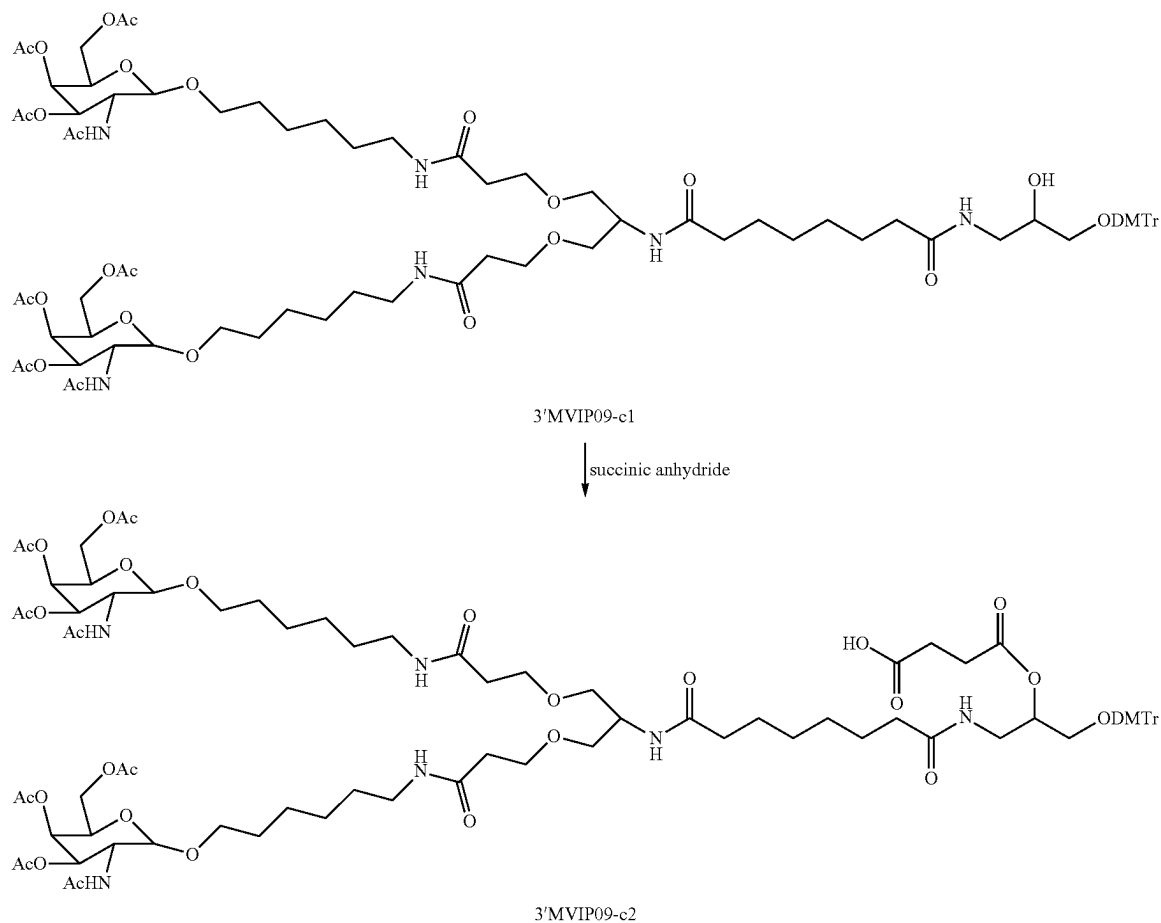
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[0367] Successively added SANCd-01-c0 (0.824 g, 1.5 mmol) and ERCd-01-c2 (1.09 g, 1.0 mmol) into a vial, then added 10 ml of DCM, stirred until dissolved, then successively added TBTU (0.963 g) and DIPEA (0.517 g), let it react overnight, added water, extracted with DCM, washed

the organic phase with saturated salt water, dried, filtered and concentrated, and finally purified by a silica gel column to obtain 1.3 g of product.

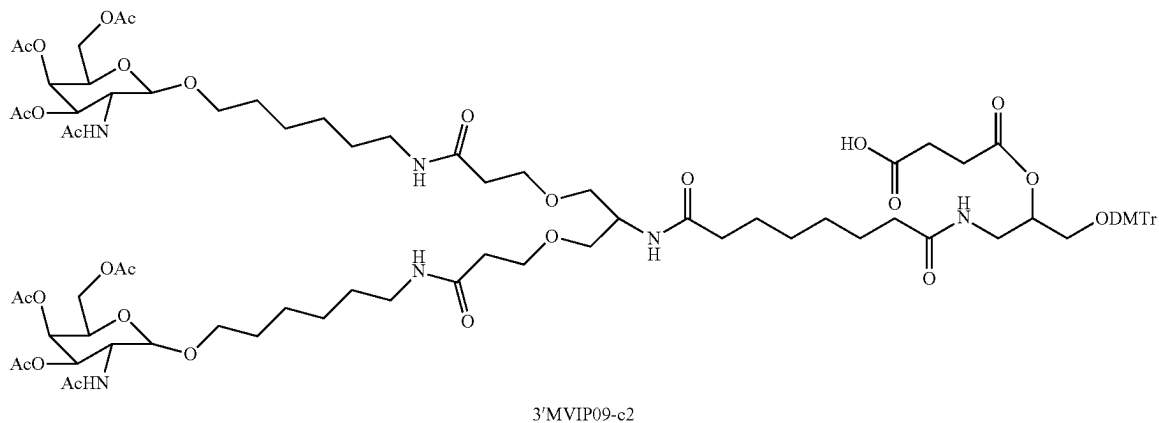
1.1.7 Synthesis of 3'MVIP09-c2



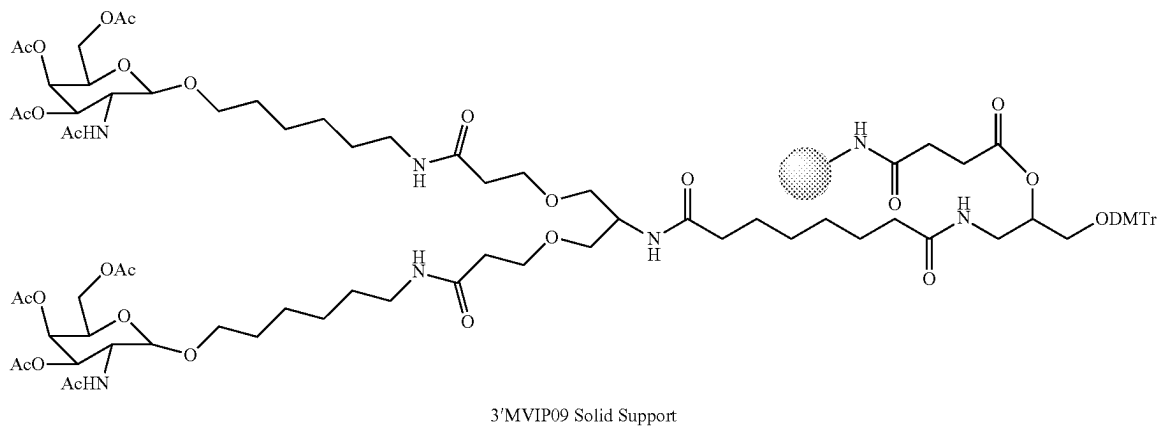
[0368] Added 3'MVIP09-c1 (1.62 g, 1 μ mol) and 10 ml of DCM into a vial in sequence, stirred at room temperature until dissolved, then added DMAP (0.366 g) and succinic anhydride (0.2 g, 3 μ mol) in sequence, stirred at room temperature to react. Concentrated DCM after the reaction is qualified by TLC analysis, added water, extracted with

DCM, washed the organic phase with saturated salt water, dried the organic phase by anhydrous sodium sulfate, filtered and concentrated, and finally purified by a silica gel column to obtain 1.55 g product.

1.1.8 Synthesis of Solid Support of 3'MVIP09

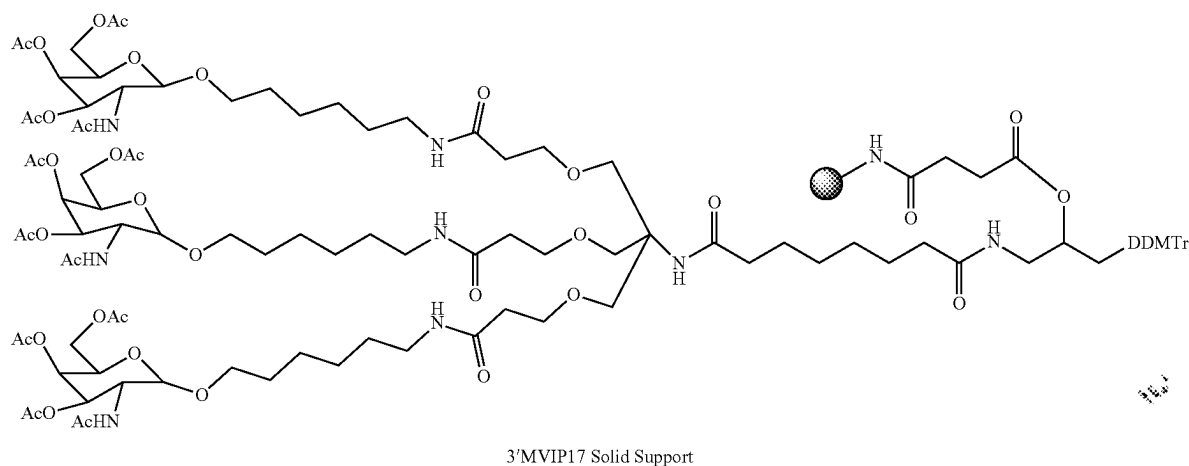


↓ macroporous aminomethyl resin

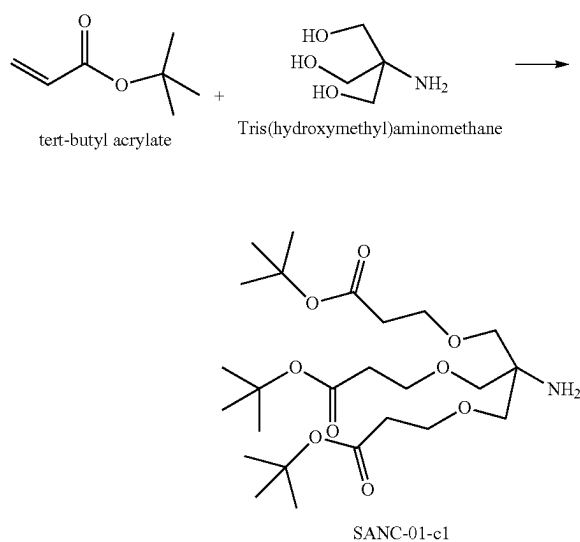


[0369] Added 3'MVIP09-c2 (0.86 g, 0.5 μ mol) and 10 ml DMF into a vial in sequence, after dissolution, added HBTU (0.19 g), DIPEA (0.194 g) and macroporous aminomethyl resin (2.0 g) in sequence, placed on the shaker for 24 h, filtered, washed the resin with 10% methanol/DCM, and then used 25% acetic acid/pyridine for end capping, with a degree of substitution of 150 μ mol/g.

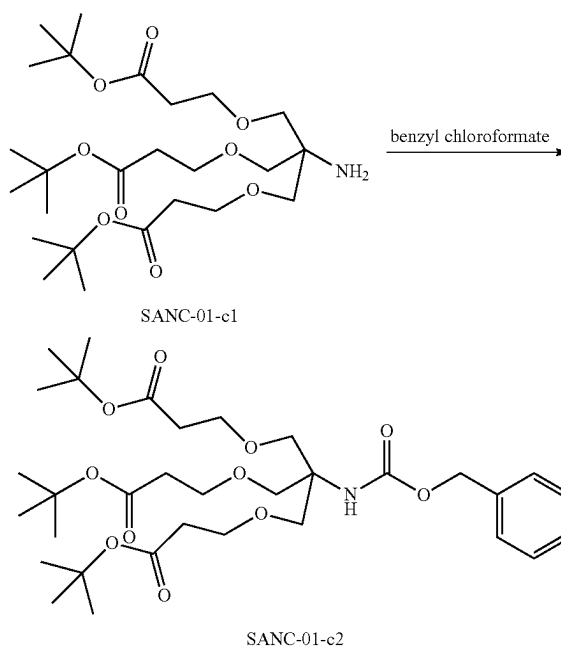
1.2 Synthesis of Solid Support of 3'MVIP17 for the Antisense Strand of RNA Inhibitors Ky11-DS7108, Ky11-DS7109



1.2.1 Synthesis of SANC-01-c1



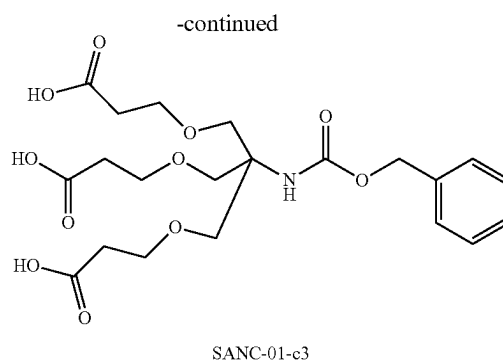
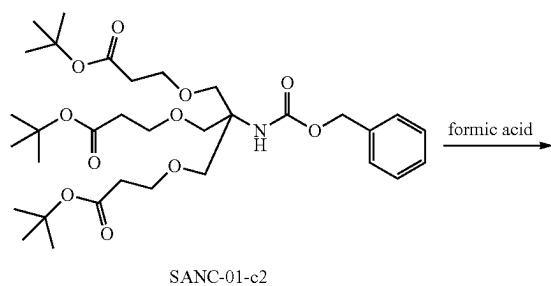
1.2.2 Synthesis of SANC-01-c2



[0370] The synthesis steps are referred to 1.1.1 synthesis of ERC-01-c1.

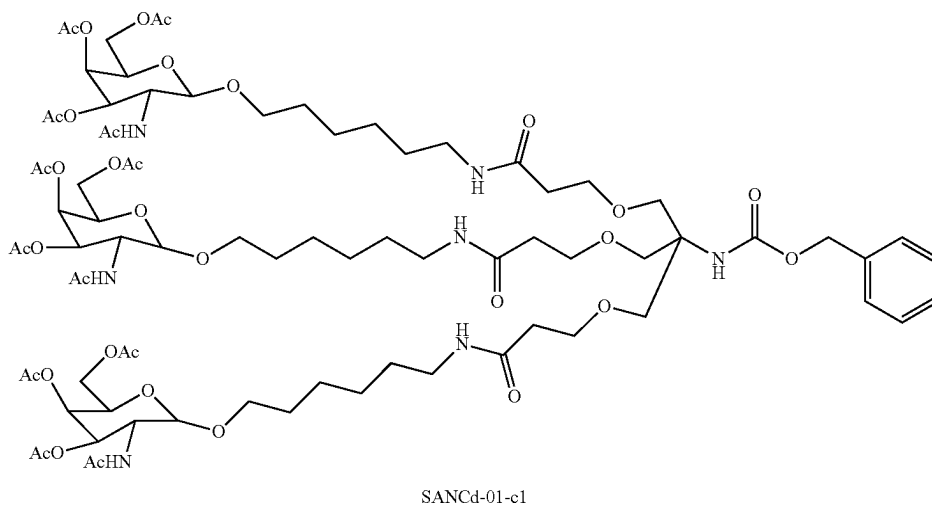
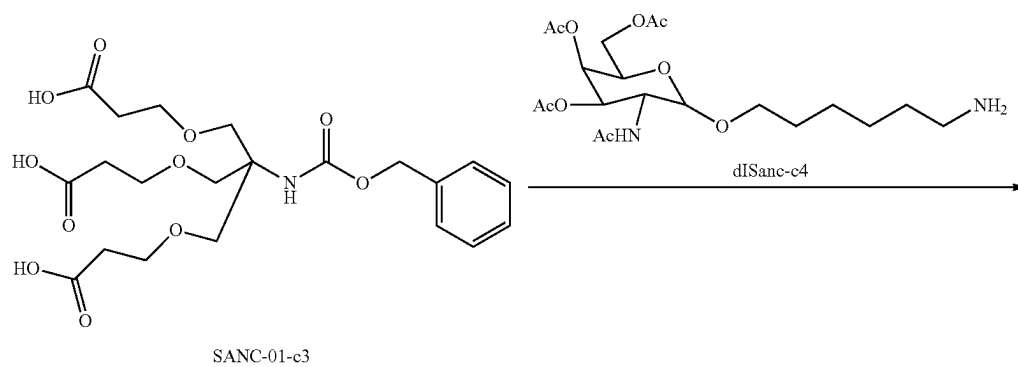
[0371] The synthesis steps are referred to 1.1.2 synthesis of ERC-01-c2.

1.2.3 Synthesis of SANC-01-c3



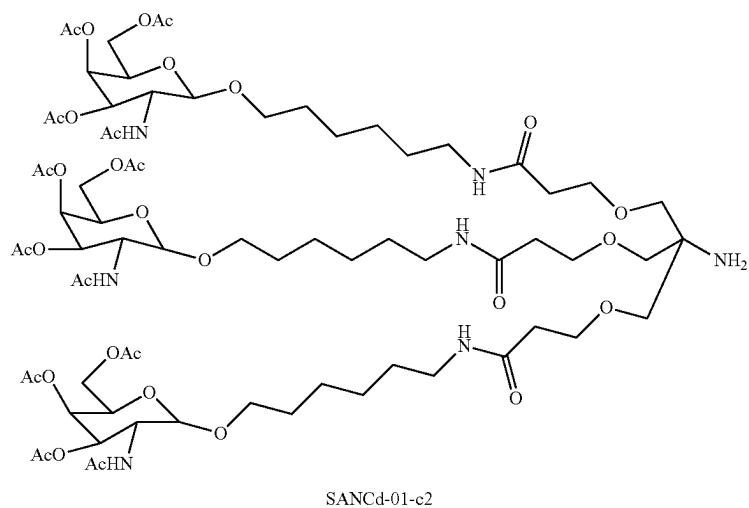
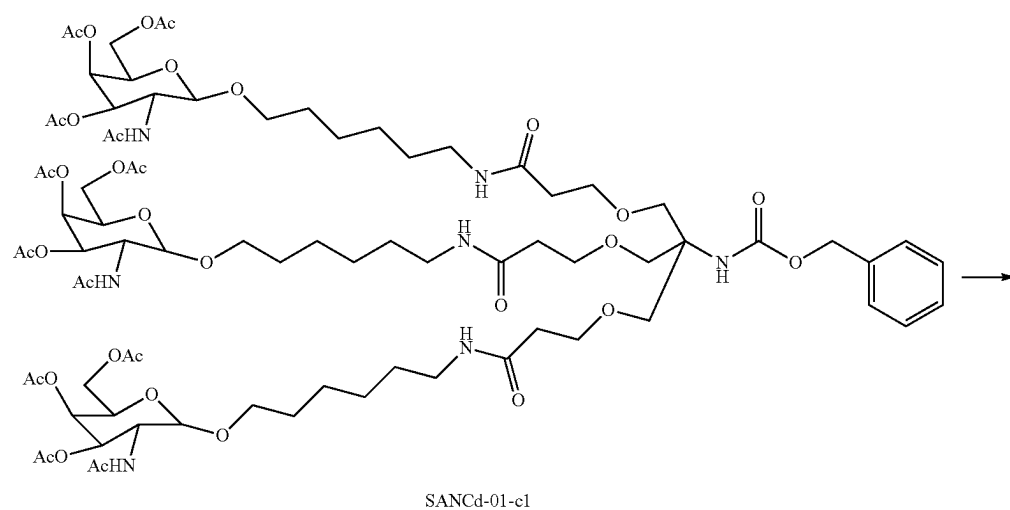
[0372] The synthesis steps are referred to 1.1.3 synthesis of ERC-01-c3.

1.2.4 Synthesis of SANCd-01-c1



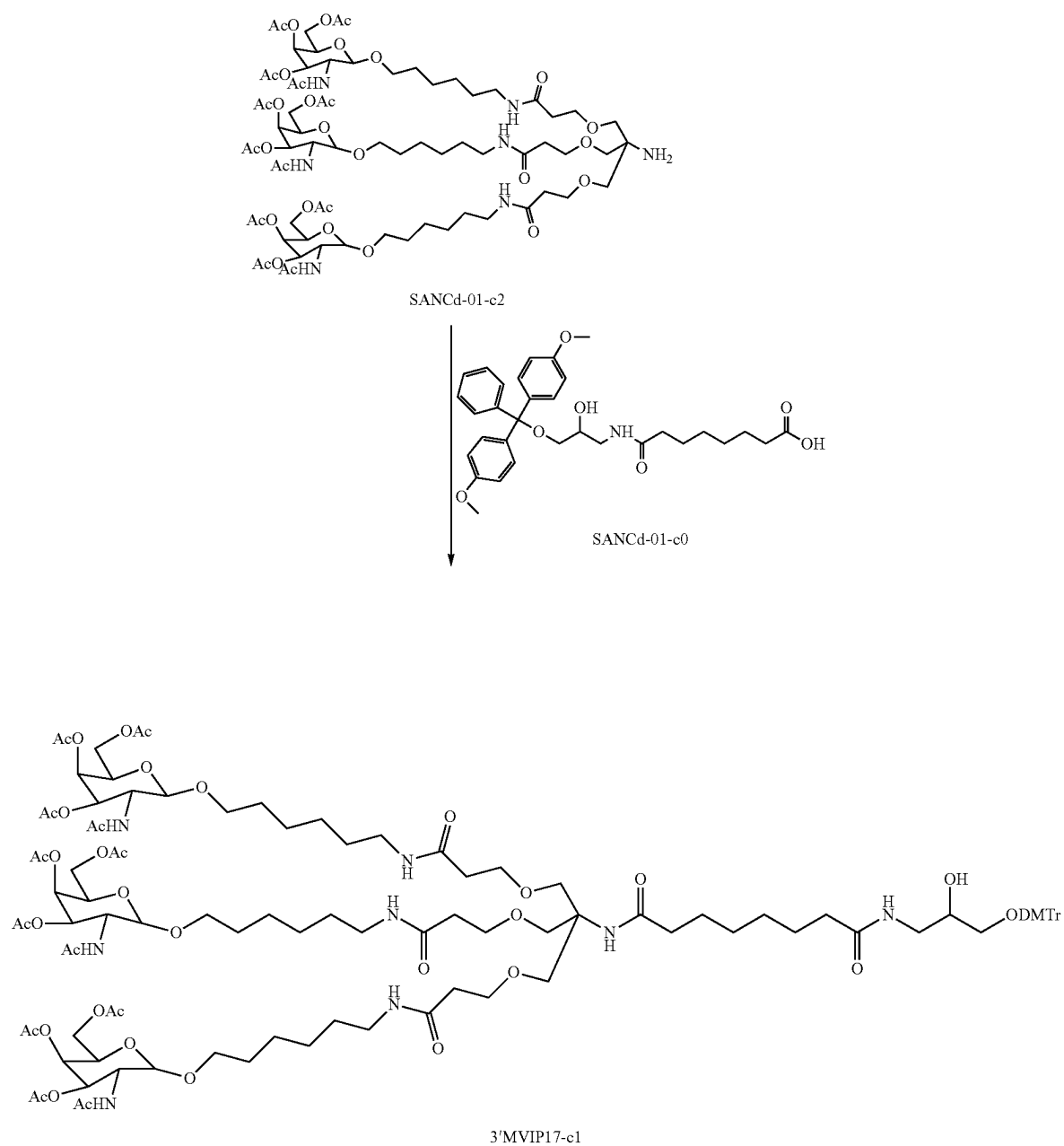
[0373] The synthesis steps are referred to 1.1.4 synthesis of ERCd-01-c1.

1.2.5 Synthesis of SANCd-01-c2



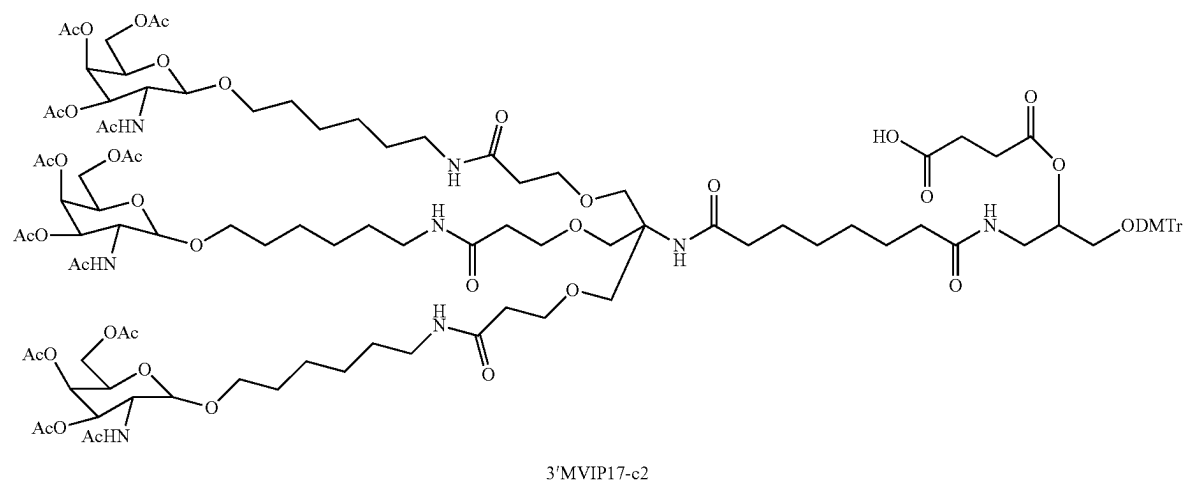
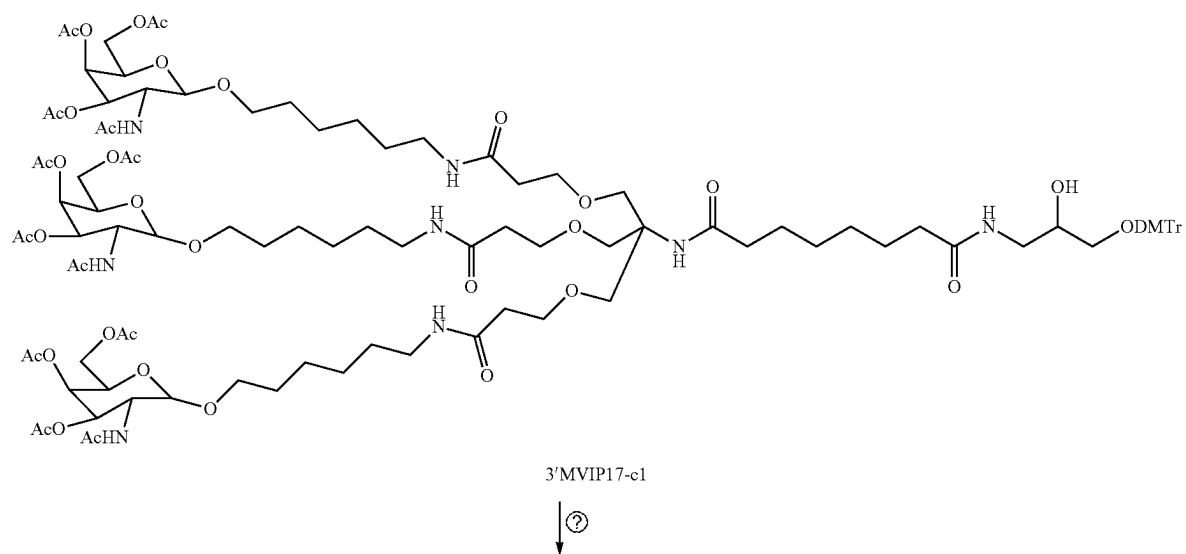
[0374] The synthesis steps are referred to 1.1.5 synthesis of ERCd-01-c2.

1.2.6 Synthesis of 3'MVIP17-c1



[0375] The synthesis steps are referred to 1.1.6 synthesis of 3'MVIP09-c1. The high-resolution mass spectrum of the synthesized 3'MVIP17-c1 is shown in FIG. 4.

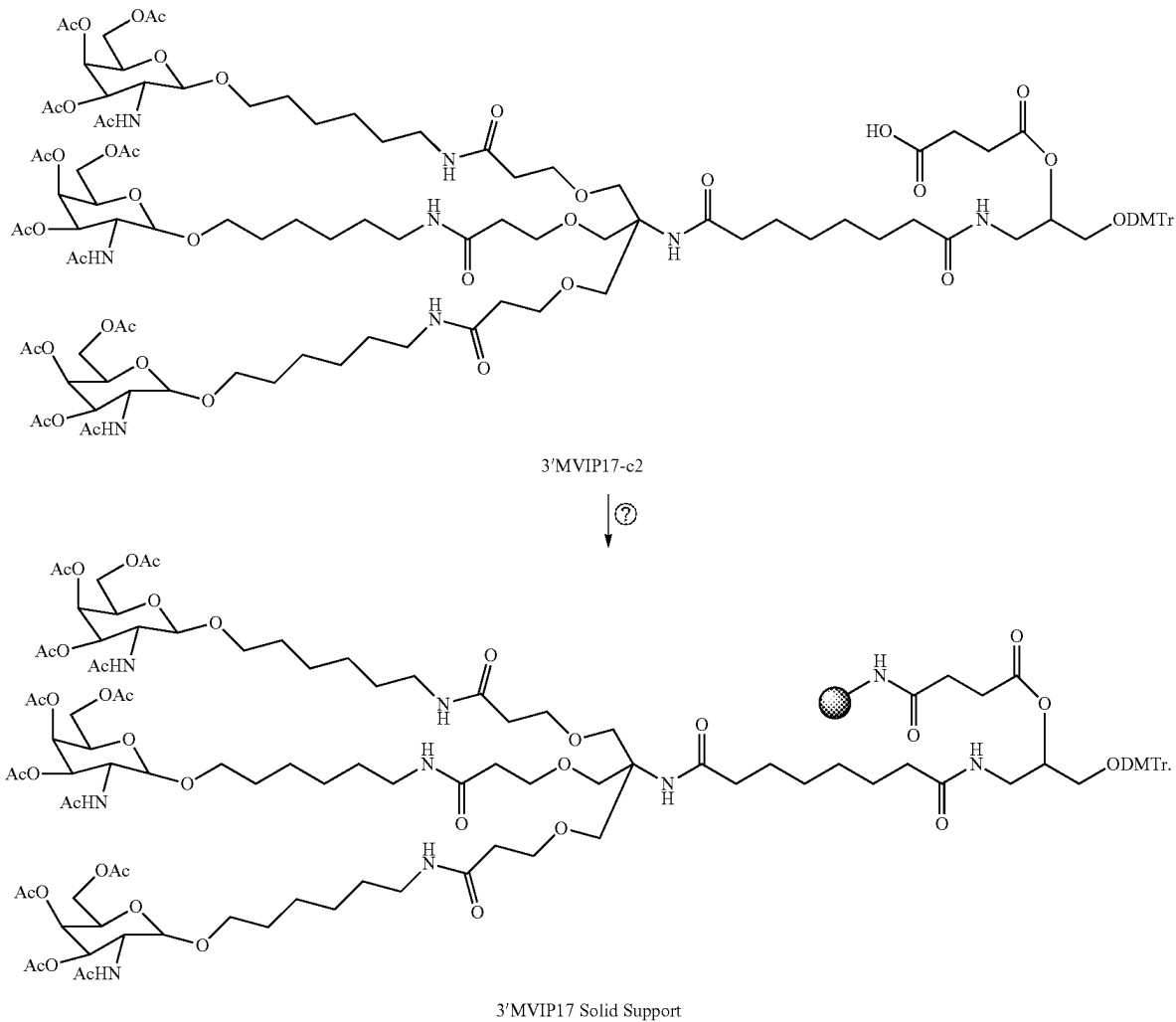
1.2.7 Synthesis of 3'MVIP17-c2



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[0376] The synthesis steps are referred to 1.1.7 synthesis of 3'MVIP09-c2.

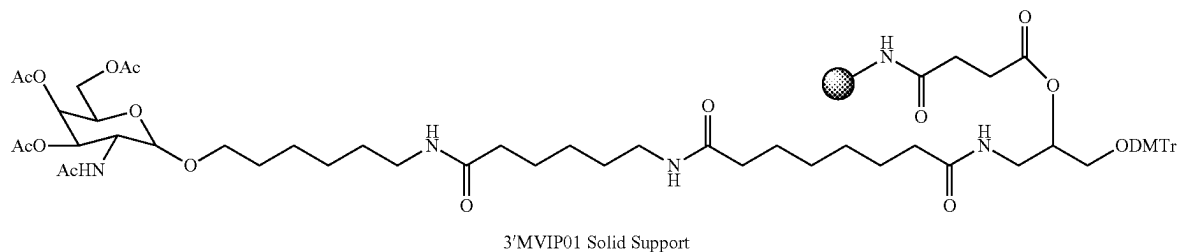
1.2.8 Synthesis of Solid Support of 3'MVIP17



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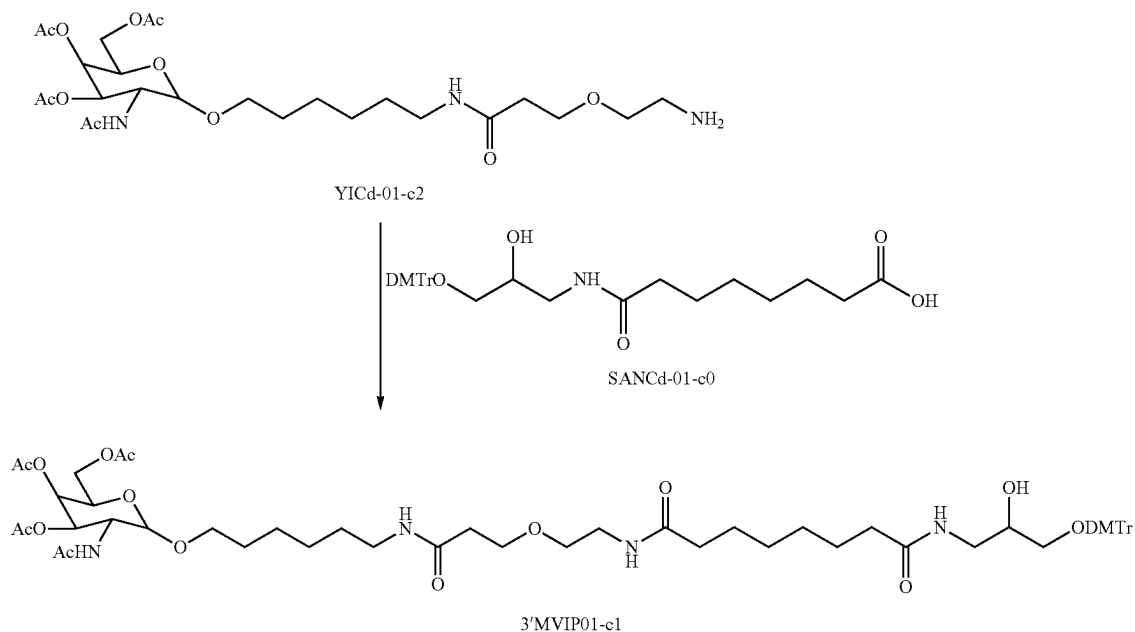
[0377] The synthesis steps are referred to 1.1.8 synthesis of Solid Support of 3'MVIP09.

1.3 Synthesis of Solid Support of 3'MVIP01 for the Antisense Strands of RNA Inhibitors Ky11-DS7106, Ky11-DS7107, and the Sense Strands of Ky11-DS7102, Ky11-DS7103



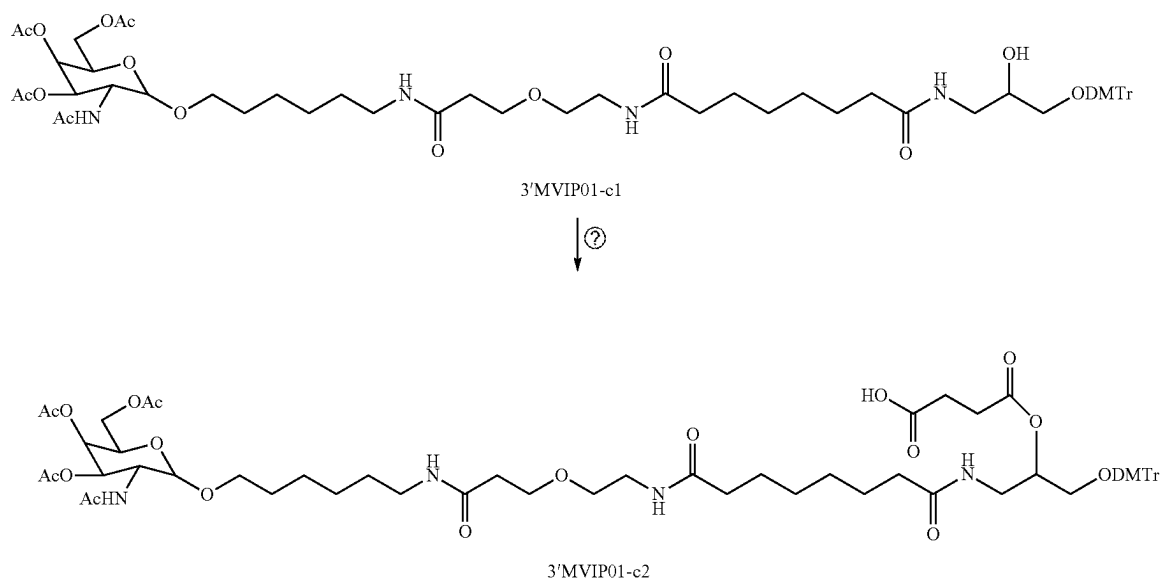
Description of Synthesis Process:

1.3.1 Synthesis of 3'MVIP01-c1



[0378] The synthesis steps are referred to 1.1.6 synthesis of 3'MVIP09-c1.

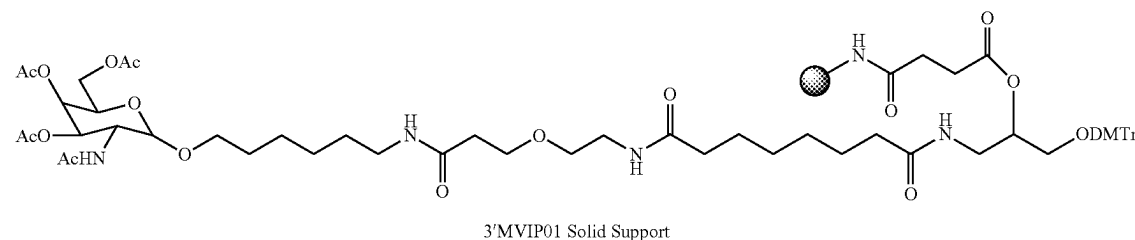
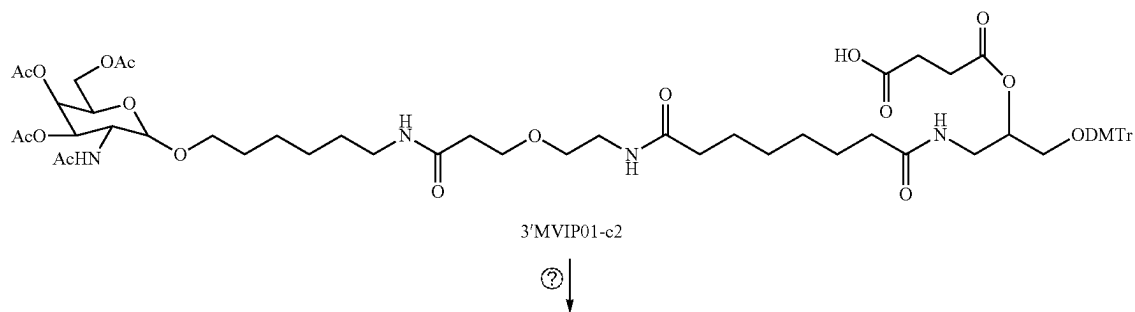
1.3.2 Synthesis of 3'MVIP01-c2



② indicates text missing or illegible when filed

[0379] The synthesis steps are referred to 1.1.7 synthesis of 3'MVIP09-c2.

1.3.3 Synthesis of Solid Support of 3'MVIP01

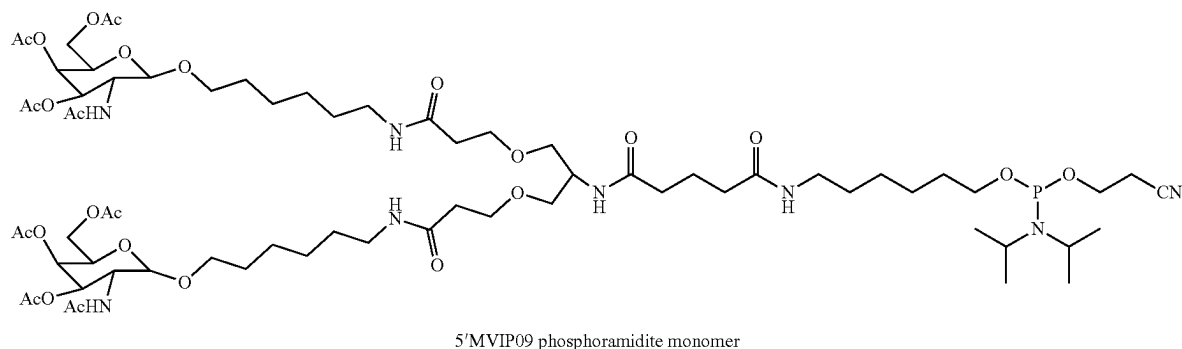


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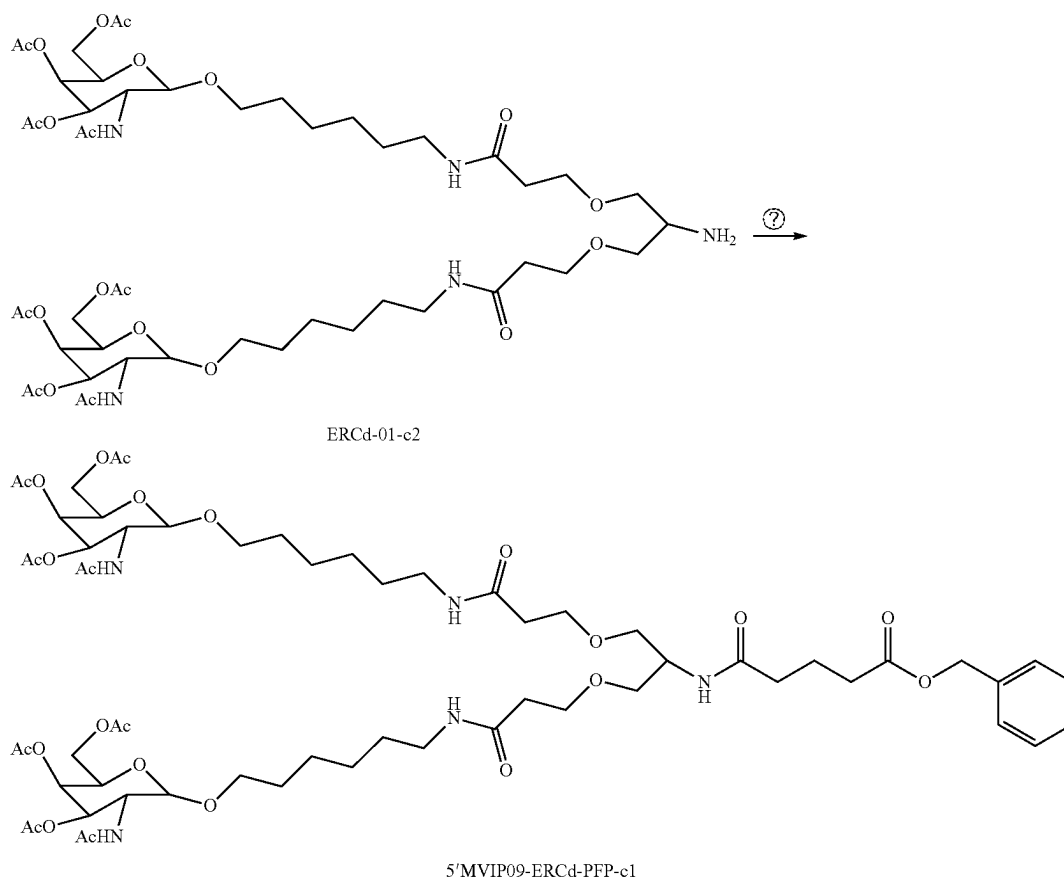
[0380] The synthesis steps are referred to 1.1.8 synthesis of Solid Support of 3'MVIP09.

2.5 Synthesis of 5' MVIP Phosphoramidite Monomer

2.1 the Synthesis of the 5'MVIP09 Phosphoramidite Monomer, which is the Last 5'MVIP Phosphoramidite Monomer Synthesized in the Solid-Phase Synthesis of the Sense Strands of the Ky11-DS71, Ky11-DS7101, Ky11-DS7102, Ky11-DS7103 when n is 2



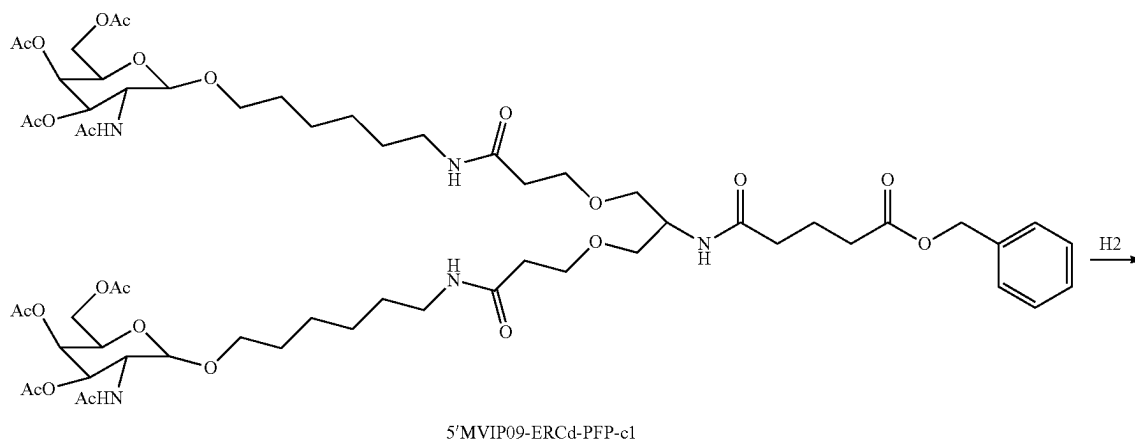
2.1.1 Synthesis of 5'MVIP09-ERCd-PFP-c1



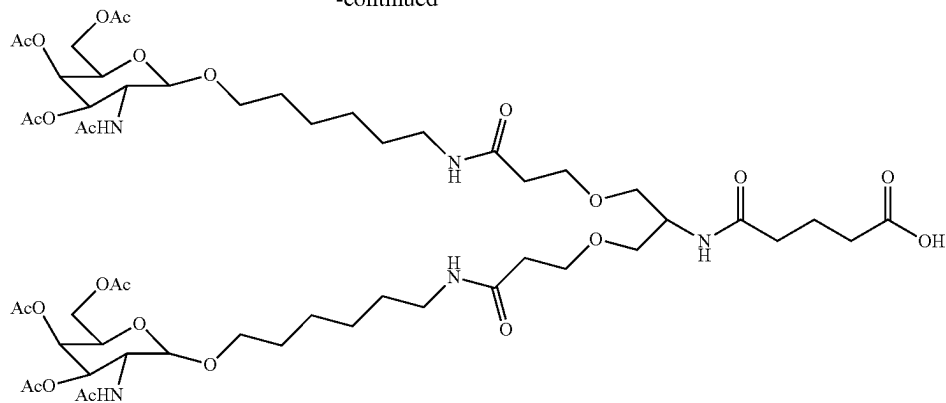
[0381] Weighed ERCd-01-c2 (2.18 g, 2.0 mmol) to dissolve in DMF (50 ml), added monobenzyl glutarate (0.53 g, 2.4 mmol), DIPEA (0.78 g) and TBTU (0.84 g), stirred at room temperature overnight, added water to quench (50 ml), extracted with DCM (30 ml*3), washed with 10% citric acid (50 ml*3), 50 ml saturated sodium bicarbonate and 100 ml

pyridine, dried with anhydrous sodium sulfate, filtered, conducted rotary evaporation, and purified by a column to obtain the product 5'MVIP09-ERCd-PFP-c1 (2.15 g).

2.1.2 Synthesis of 5'MVIP09-ERCd-PFP-c2



-continued

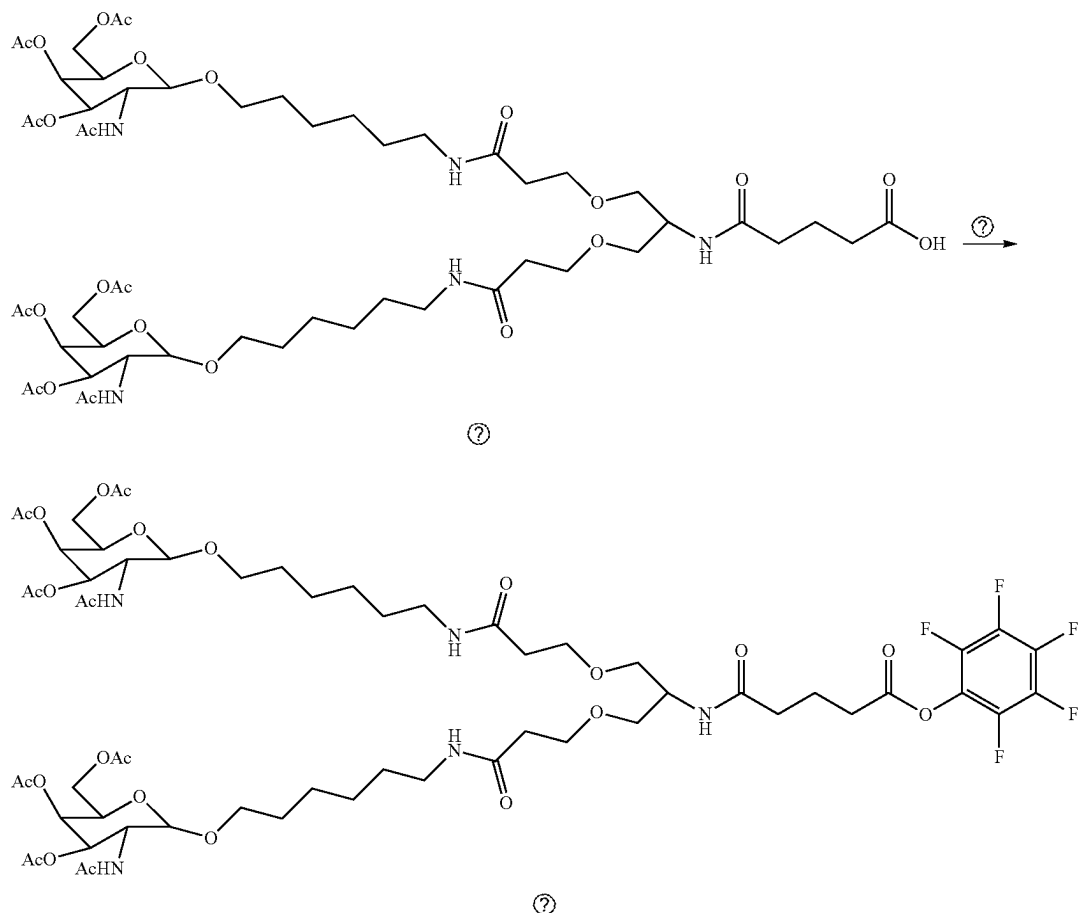


5'MVIP09-ERCd-PFP-c2

[0382] Weighed 5'MVIP09-ERCd-PFP-c1 (2.15 g, 1.66 mmol) and 10% palladium carbon (0.21 g), added methanol (50 ml), and hydrogenated overnight with stirring at room temperature, filtered palladium carbon with diatomite after the reaction, conducted rotary evaporation to obtain

5'MVIP09-ERCd-PFP-c2 crude product (1.9 g), and its high-resolution mass spectrum is shown in FIG. 5.

2.1.3 Synthesis of 5'MVIP09-ERCd-PFP

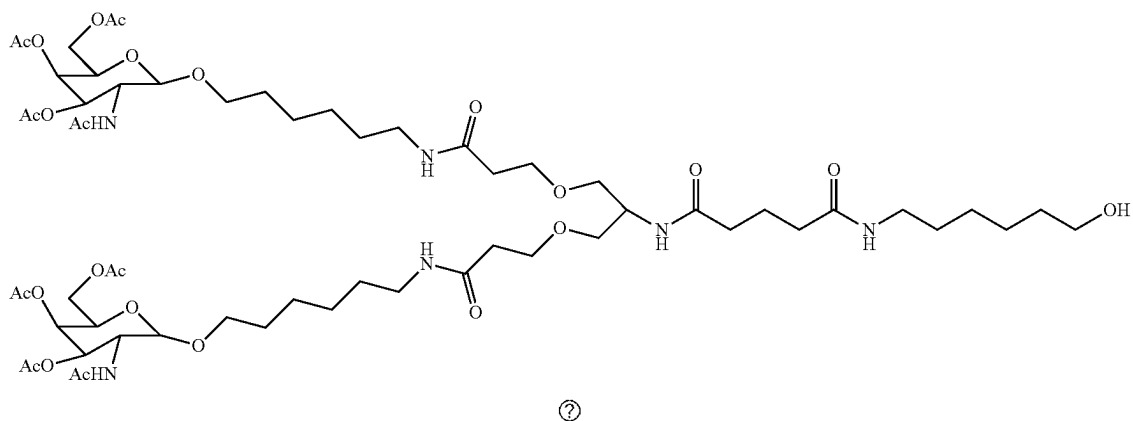
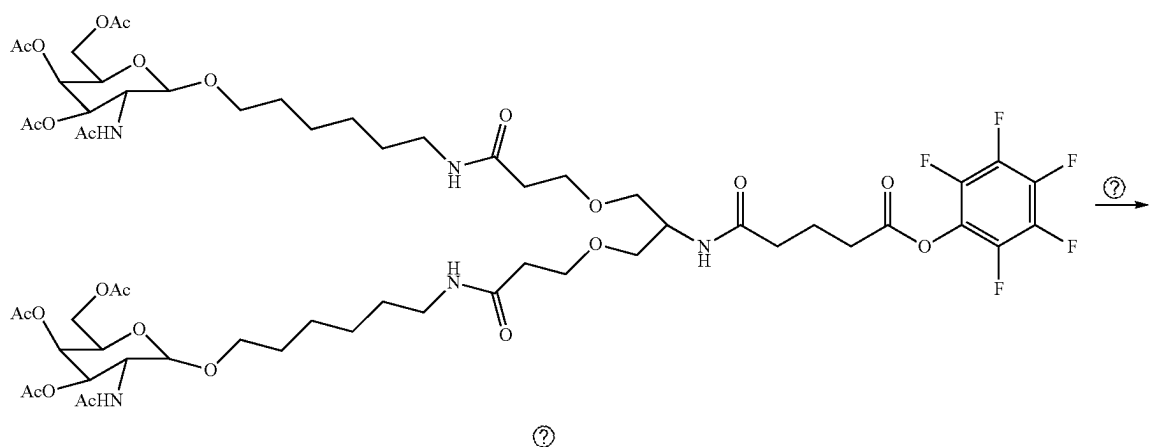


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[0383] Weighed 5'MVIP09-ERCd-PFP-c2 crude product (1.9 g, 1.58 mmol) to dissolve in DCM (60 ml), added DIPEA (1.33 g), cooled, added pentafluorophenol trifluoroacetate (2.21 g, 7.9 mmol), let it react for 2 h with stirring at room temperature, then conducted rotary evaporation, redissolved in DCM (60 ml), washed with saturated sodium bicarbonate (30 ml*3), 10% citric acid (30 ml*1), saturated salt water (50 ml*1), dried with anhydrous sodium sulfate,

filtered, conducted rotary evaporation to obtain 5'MVIP09-ERCd-PFP crude product (2.35 g). After vacuum drying by pump, it was directly used for the next reaction without purification.

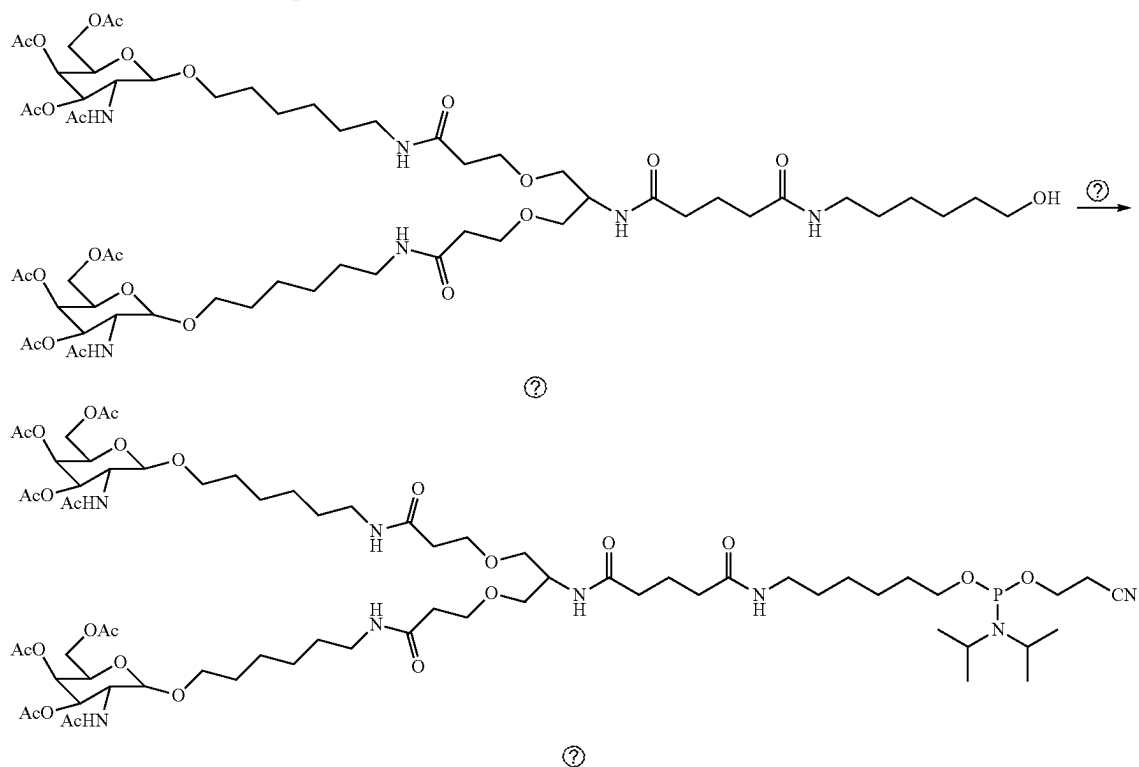
2.1.4 Synthesis of 5'MVIP09 Phosphoramidite Monomer-c1



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[0384] Dissolved 5'MVIP09-ERCd-PFP crude product (2.35 g, 1.58 mmol) in DCM (60 ml), added DIPEA (0.82 g, 6.32 mmol), 6-amino-1-hexanol (0.37 g, 3.16 mmol), and let it react overnight with stirring at room temperature. Added 10% citric acid (30 ml), extracted with DCM (30 ml*3), washed with saturated salt water (50 ml), dried with anhydrous sodium sulfate, filtered, conducted rotary evaporation, and purified by a column to obtain the product 5'MVIP09 monomer-c1 (1.73 g).

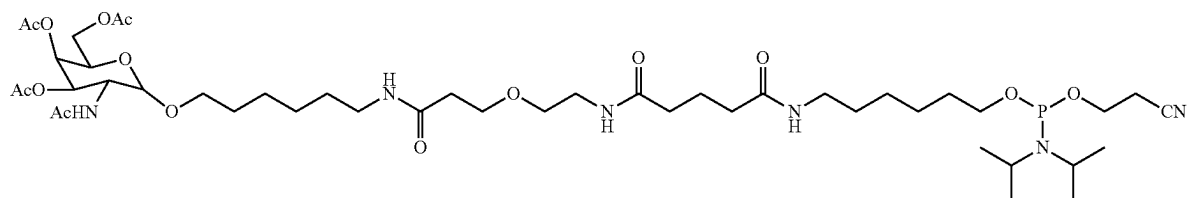
2.1.5 5'MVIP09 Phosphoramidite Monomer



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[0385] Weighed 5'MVIP09 phosphoramidite monomer-c1 (1.3 g, 1.0 mmol) to dissolve in acetonitrile (30 ml), added diisopropylamine triazole (0.22 g), added bis-(diisopropylamino) (2-cyanoethoxy) phosphine (0.36 g, 1.2 mmol) dropwise in an ice bath, let it react for 4 h at room temperature, after the reaction is qualified by centralized control of HPLC, concentrated and purified by a column to obtain the product 5'MVIP09 monomer (1.2 g).

2.2 the Synthesis of the 5'MVIP01 Phosphoramidite Monomer, which is Used as the Last 5'MVIP Phosphoramidite Monomer Synthesized in the Solid-Phase Synthesis of the Sense Strands of Ky11-DS7104, Ky11-DS7105, Ky11-DS7106, Ky11-DS7107, Ky11-DS7108, Ky11-DS7109 when n is 1



5'MVIP01 phosphoramidite monomer

[0386] Except for weighing YICd-01-c2 (1.12 g, 2.0 mmol) for the phosphoramidite monomer of 5'MVIP01, refer to 2.1.1~2.1.5 for the remaining operations.

Example 5 Synthesis of RNAi Agents with
Different siRNAs Conjugated to 5' MVIP09/3'
MVIP09

[0387] Description of synthesis of antisense strands conjugated to ligands in Table 13: Purged a reagent bottle with argon for at least 2 min. Added phosphoramidite monomer and acetonitrile into the reagent bottle in turn, tightened the bottle cap and shook until the solid is completely dissolved by visual inspection. Then added 3 Å molecular sieve and let it stand for more than 8 h for later use. Purged a reagent bottle with argon for at least 2 min. Added xanthane hydride and dried pyridine into the reagent bottle in turn, tightened the bottle cap, shook until the solid is completely dissolved by visual inspection, and stored temporarily for later use. Ensure to carry out the following operations at room temperature of 20~30° C.: weighed the 3'MVIP ligand, added it to a reagent bottle, then added acetonitrile, shook until mixed evenly. Transferred the ligand to a synthesis column, and eluted the residual ligand in the reagent bottle with acetonitrile and transferred to the synthesis column. After elution, added acetonitrile to fill the synthetic column, and recorded the amount of acetonitrile used. Installed and fixed the synthetic column according to the instrument instructions.

[0388] Connected the monomer solution, CAP A, CAP B, oxidant, thioreagent, activator, decapping agent and acetonitrile prepared above to the pipeline corresponding to AKTA PILOT100, and ensure that the pipeline is inserted into the bottom of the reagent bottle.

[0389] After the synthesis method is set, the instrument is ready for all work, clicked Run to start the synthesis. Recorded each detritylation peak area by online observation. During the synthesis process, added additional amount according to the actual amount of deprotection reagent used.

[0390] After the synthesis, purged the synthetic column with argon for >2 h, and unloaded the synthetic column according to the operating procedures. Transferred the solid supporter in the synthesis column to a reaction bottle, added methylamine aqueous solution and ammonia water, and put the reaction bottle into a shaker at 35° C. for 2-3 hours. Filtered the solution into a round bottom flask, washed the residual solid phase with 50% ethanol aqueous solution, filtered again and mixed with the previous filtrate, connected the round bottom flask with a rotary evaporator, set the water temperature at 50° C. and evaporated until there is no distillate, added ethanol into the round bottom flask, mixed well and evaporated again until there is no distillate. Repeated the operation until white powder appeared at the bottom of the flask. Prepared the obtained white powder into a solution, purified by a reverse chromatography column, and sampled to detect OD260 and purity. Divided the purified antisense strand solution into vials and lyophilized for future use, and stored the product sealed in a -20° C. refrigerator.

[0391] The synthesis process of sense strand conjugated with the ligand in Table 13 is the same as that of antisense strand, in which the carrier loaded in the column is Universal carrier. Added DIPEA to the obtained intermediate to pre-

pare a solution, added 5'MVIP phosphoramidite monomer, mixed well, and put the reaction bottle into a shaker at 35° C. for 2-3 hours.

[0392] Description of synthetic annealing process of RNA inhibitors in Table 13:

[0393] Took an antisense strand and a sense strand in Table 13, mixed them in the reaction bottle in an equimolar ratio of 1:1. Turned off the power of the water bath after 5 minutes of water bath at 95° C., let it naturally cool down to below 40° C. Added 3M sodium acetate aqueous solution to the solution containing double strands, mixed well, then added an appropriate volume of absolute ethanol, mixed well, and put the reaction solution into a -20° C. refrigerator for 45 min. Set the high-speed freezing centrifuge to pre-cooling at 4° C., put in the solution containing double strands after the temperature is reached, and started the centrifuge. Took out the centrifuged solution containing double strands, removed the supernatant, added ultrapure water to completely dissolve the solid, and took samples to detect OD260 and purity and obtained the RNAi agent in Table 14. Divided the purified finished solution into vials and lyophilized for future use, and stored the product sealed in a -20° C. refrigerator.

Example 6 Study on Evaluating the Activity of
RNA Inhibitors Containing 5' MVIP09/3' MVIP09
Structure by PCH

[0394] The primary cynomolgus monkey hepatocytes (PCH) were used to evaluate the effect of the combined ligand structure 5'MVIP09/3'MVIP09 conjugated at the ends of the sense and/or antisense strands of RNA inhibitors on the in vitro activity of the modified sequences.

[0395] On Day 0, PCH were seeded into a 48-well plate. As the cells were seeded, RNAi agents were freely uptaken up by them; On Day 1, added PCH; On Day 2, replaced with fresh culture medium. Every two days, replaced the fresh medium containing RNA inhibitor. The tested compounds were tested at three concentrations (500 nM, 100 nM, and 10 nM) and in triplicate wells. On day 8, collected the supernatant and detected the LPA mRNA level in the collected cell supernatant using qPCR. The results are shown in FIG. 6. Ky11-DS67-Ky11-DS72 and Ky11-DS74-Ky11-DS76 have significant effects on the inhibition of LPA mRNA expression levels in PCH cells.

Example 7 Exploratory Study on the In Vivo
Efficacy of RNA Inhibitors in Cynomolgus Monkey
Models

[0396] The effect of the combined ligand structure 5'MVIP09/3'MVIP09 conjugated at the ends of the sense strand and/or antisense strand of RNA inhibitor on the in vivo activity of the modified sequences. Prepared the corresponding RNA inhibitors Ky11-DS67, Ky11-DS68, Ky11-DS71, Ky11-DS73 and Ky11-DS74 according to the method described in Example 5.

[0397] Selected 12 4~7-year-old male cynomolgus monkeys, and randomly divided into control group and administration group according to body weight after the adaptive feeding, with two animals in each group. The dose was 3 mg/kg. The day of grouping and administration was defined as day0, and the time of administration was the morning of

that day. Added physiological saline to each RNA inhibitor to prepare solutions. Collected blood on Day 7, 14, 21, 28, and 35 after the first administration. After separating the plasma, determined the LP (a) levels and LDL-C.

[0398] Normalized to do and the control group, the changes in LP (a) levels and LDL-C rates in cynomolgus monkey blood after RNA inhibitor intervention are shown in FIGS. 7 and 8.

[0399] The experimental results showed that, Ky11-DS67, Ky11-DS68, Ky11-DS71, Ky11-DS73 and Ky11-DS74 all showed significant effects on the inhibition of LP (a) and LDL-C levels in cynomolgus monkeys with good sustainability.

Example 8 Exploratory Study on the In Vivo
Efficacy of RNA Inhibitors Formed by Different
Combinations of 5'MVIP/3'MVIP in Cynomolgus
Monkeys

[0400] This example examines the effect of different 5'MVIP/3'MVIP combinations conjugated at the ends of sense and/or antisense strands on the in vivo activity of RNA inhibitors. Specific implementation: The 5' end and/or 3' end of the sense strand GsUSAAf-UGfGfAfCAGAGUUAUCGsASU (SEQ ID NO. 460) is conjugated with different structures of 5'MVIP and/or 3'MVIP, and the 5' end and/or 3' end of the antisense strand CsGsAUAAfCUCUGfUCfCAUUAACSCSA (SEQ ID NO. 496) is conjugated with different structures of 5'MVIP

and/or 3'MVIP. Prepared the representative RNA inhibitors Ky11-DS7101, Ky11-DS7102, Ky11-DS7104, Ky11-DS7106, and Ky11-DS7108 in Table 17 according to the method described in Example 5, and selected 12 4-7-year-old male cynomolgus monkeys, and randomly divided into control group (n=2) and administration group (n=2) according to body weight after the adaptive feeding. The dosage was 3 mg/kg. The day of grouping and administration was defined as day0, and the time of administration was the morning of that day. Added physiological saline to each RNA inhibitor to prepare solutions. Collected blood on Day 7, 14, 21, 28, and 35 after the first administration. After separating the plasma, determined the LP (a) levels.

[0401] Normalized to do and the control group, the LP (a) levels in cynomolgus monkey blood after RNA inhibitor intervention are shown in FIG. 9.

[0402] The experimental results: the RNA inhibitors Ky11-DS7101, Ky11-DS7102, Ky11-DS7104, Ky11-DS7106 and Ky11-DS7108 all showed a reduction in LP (a) levels of greater than 90% on day 7 after administration. The ligand combination shown in Table 17 can effectively deliver modified RNA inhibitors.

[0403] The basic principles, main features and advantages of the present invention have been shown and described above. Those skilled in the art should understand that the above examples do not limit the present application in any way, and all technical solutions obtained by equivalent replacement or equivalent transformation shall fall within the scope of protection of the invention.

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SEQ ID NO: 41	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 41		
agagttgcaa ggacacttga t		21
SEQ ID NO: 42	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 42		
tggatctggg accaccgtga g		21
SEQ ID NO: 43	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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SEQUENCE: 43
tcttcagaag aagcctctgt g 21

SEQ ID NO: 44 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 44
tgcttcttca gaagaagcct c 21

SEQ ID NO: 45 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 45
agcaatcctg gaccccgagg c 21

SEQ ID NO: 46 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 46
tagcaatcct ggaccccgagg g 21

SEQ ID NO: 47 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 47
tgttgtcttc tgatgccagt g 21

SEQ ID NO: 48 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 48
tctgttgtcc tctgatgcc a 21

SEQ ID NO: 49 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 49
ttctgttgtc ctctgatgcc a 21

SEQ ID NO: 50 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 50
attctgttgt cctctgatgc c 21

SEQ ID NO: 51 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 51
aatattctgt tgcctctga t 21

SEQ ID NO: 52 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 52
ataatattct gttgtcctct g 21

SEQ ID NO: 53 moltype = RNA length = 21

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 53		
tggataatat tctgtgtgcc t		21
SEQ ID NO: 54	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 54		
tttgataat attctgttgt c		21
SEQ ID NO: 55	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 55		
aaggactaat ctcagcatct g		21
SEQ ID NO: 56	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 56		
aacaccaagg actaatctca g		21
SEQ ID NO: 57	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 57		
ataacaccaa ggactaatct c		21
SEQ ID NO: 58	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 58		
atggtataac accaaggact a		21
SEQ ID NO: 59	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 59		
aaacagccgt ggacgtcgca a		21
SEQ ID NO: 60	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 60		
agaaacagcc gtggacgtcg c		21
SEQ ID NO: 61	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 61		
tcagaaacag ccgtggacgt c		21
SEQ ID NO: 62	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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SEQUENCE: 62		
ttcagaaaca gccgtggacg t		21
SEQ ID NO: 63	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 63		
tgttcagaaa cagccgtgga c		21
SEQ ID NO: 64	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 64		
ttgttcagaa acagccgtgg a		21
SEQ ID NO: 65	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 65		
tgcttggtca gaaacagccg t		21
SEQ ID NO: 66	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 66		
tagcagtcct ggactgtggg g		21
SEQ ID NO: 67	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 67		
tgtggttctc tgatgccagt g		21
SEQ ID NO: 68	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 68		
attctgtggt tctctgatgc c		21
SEQ ID NO: 69	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 69		
agtattctgt gggtctctga t		21
SEQ ID NO: 70	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 70		
tccatcactg gacattgcgt c		21
SEQ ID NO: 71	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 71		
ttccatcact ggacattgcg t		21
SEQ ID NO: 72	moltype = RNA length = 21	

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 72		
attccatcac tggacattgc g		21
SEQ ID NO: 73	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 73		
agagttgatt ccatcactgg a		21
SEQ ID NO: 74	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 74		
ttgtgaggag agttgattcc a		21
SEQ ID NO: 75	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 75		
aaggaagctc tgtgcttgga a		21
SEQ ID NO: 76	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 76		
tcagaaggaa gctctgtgct t		21
SEQ ID NO: 77	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 77		
ttcagaagga agctctgtgc t		21
SEQ ID NO: 78	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 78		
tcttcagaag gaagctctgt g		21
SEQ ID NO: 79	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 79		
tgcttcttca gaaggaagct c		21
SEQ ID NO: 80	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 80		
tttcagttgg tgcttcttca g		21
SEQ ID NO: 81	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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SEQUENCE: 81
tgttttcagt tgggtgttct t 21

SEQ ID NO: 82 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 82
tgctgttttc agttggtgct t 21

SEQ ID NO: 83 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 83
tagcagtcct ggaccccagt g 21

SEQ ID NO: 84 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 84
atcacctcgg tagcagtcct g 21

SEQ ID NO: 85 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 85
tccatcacct cggtagcagt c 21

SEQ ID NO: 86 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 86
tgtccatcac ctcggtagca g 21

SEQ ID NO: 87 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 87
actctgtcca tcacctcggt a 21

SEQ ID NO: 88 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 88
aactctgtcc atcacctcgg t 21

SEQ ID NO: 89 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 89
taactctgtc catcacctcg g 21

SEQ ID NO: 90 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 90
atagtgggtg agagtgtgcc t 21

SEQ ID NO: 91 moltype = RNA length = 21

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 91		
aatgggatcc tccgatgcc a		21
SEQ ID NO: 92	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 92		
taatgggatc ctccgatgcc a		21
SEQ ID NO: 93	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 93		
ataatgggat cctccgatgc c		21
SEQ ID NO: 94	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 94		
agtataatgg gatcctccga t		21
SEQ ID NO: 95	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 95		
tagtataatg ggatcctccg a		21
SEQ ID NO: 96	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 96		
atagtataat gggatcctcc g		21
SEQ ID NO: 97	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 97		
tggatagtat aatgggatcc t		21
SEQ ID NO: 98	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 98		
atttggatag tataatggga t		21
SEQ ID NO: 99	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 99		
ttgtgaggac actcgattct g		21
SEQ ID NO: 100	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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SEQUENCE: 100
tgtgcttgga accggggcca c 21

SEQ ID NO: 101 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 101
ttttctcagg tgggtgcttgt t 21

SEQ ID NO: 102 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 102
atatgcctcg ataactccgt c 21

SEQ ID NO: 103 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 103
agttctcggt caggccagca t 21

SEQ ID NO: 104 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 104
tagttctcgg tcaggccagc a 21

SEQ ID NO: 105 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 105
agtagttctc ggtcaggcca g 21

SEQ ID NO: 106 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 106
tgcagtagtt ctcggtcagg c 21

SEQ ID NO: 107 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 107
attcctgcag tagttctcgg t 21

SEQ ID NO: 108 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 108
tctaggacac ctgattctgt t 21

SEQ ID NO: 109 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 109
tctctaggac acctgattct g 21

SEQ ID NO: 110 moltype = RNA length = 21

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 110		
tcagaaggag gccctaggct t		21
SEQ ID NO: 111	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 111		
ttcagaagga ggcctaggc t		21
SEQ ID NO: 112	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 112		
tcttgttcag aaggaggccc t		21
SEQ ID NO: 113	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 113		
agtcttgttc agaaggaggc c		21
SEQ ID NO: 114	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 114		
tacagtcttg ttcagaagga g		21
SEQ ID NO: 115	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 115		
atacagtctt gttcagaagg a		21
SEQ ID NO: 116	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 116		
aacatacagt cttgttcaga a		21
SEQ ID NO: 117	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 117		
tttgtccctg gaatgaacgt g		21
SEQ ID NO: 118	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 118		
atttgtccct ggaatgaacg t		21
SEQ ID NO: 119	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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SEQUENCE: 119
tttatttgtc cctggaatga a 21

SEQ ID NO: 120 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 120
atgtcaccat cagggttacg g 21

SEQ ID NO: 121 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 121
attgatgtca ccatcagggt t 21

SEQ ID NO: 122 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 122
attcattgtg tagcaccagg g 21

SEQ ID NO: 123 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 123
tggattcatt gtgtagcacc a 21

SEQ ID NO: 124 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 124
tcttggattc attgtgtagc a 21

SEQ ID NO: 125 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 125
ttttcttgga ttcattgtgt a 21

SEQ ID NO: 126 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 126
aaaaagtttt cttggattca t 21

SEQ ID NO: 127 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 127
agtcaaaaag ttttcttgga t 21

SEQ ID NO: 128 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 128
tagtcaaaaa gtttcttgga a 21

SEQ ID NO: 129 moltype = RNA length = 21

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 129		
tatcacagta gtcaaaaagt t		21
SEQ ID NO: 130	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 130		
acagagaggg atatcacagt a		21
SEQ ID NO: 131	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 131		
aaatgaagag gatgcacaga g		21
SEQ ID NO: 132	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 132		
aatgcttcca ggacatttct t		21
SEQ ID NO: 133	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 133		
ttcaagcagt gagcagcagt c		21
SEQ ID NO: 134	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 134		
tcttcaagca gtgagcagca g		21
SEQ ID NO: 135	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 135		
ttcttcaagc agtgagcagc a		21
SEQ ID NO: 136	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 136		
acttcttcaa gcagtgagca g		21
SEQ ID NO: 137	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 137		
aggacttctt caagcagtgga g		21
SEQ ID NO: 138	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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SEQUENCE: 138
aacatgagat tcgagggtca c 21

SEQ ID NO: 139 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 139
ttcctgaaca tgagattcga g 21

SEQ ID NO: 140 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 140
tttctgaac atgagattcg a 21

SEQ ID NO: 141 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 141
tctatttcct gaacatgaga t 21

SEQ ID NO: 142 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 142
tgcttggtg gggtccaaga a 21

SEQ ID NO: 143 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 143
atctgcttgt gtgggctcca a 21

SEQ ID NO: 144 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 144
tatctgcttg tgtgggctcc a 21

SEQ ID NO: 145 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 145
aatatctgct tgtgtgggct c 21

SEQ ID NO: 146 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 146
aaggcaatat ctgcttgtgt g 21

SEQ ID NO: 147 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 147
ttagcaaggc aatatctgct t 21

SEQ ID NO: 148 moltype = RNA length = 21

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 148		
tagtctgggg atggcagaca a		21
SEQ ID NO: 149	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 149		
tgtagtctgg ggatggcaga c		21
SEQ ID NO: 150	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 150		
accatgtagt ctggggatgg c		21
SEQ ID NO: 151	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 151		
ttcagtcctg gcggtgacca t		21
SEQ ID NO: 152	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 152		
attcagtcct ggcggtgacc a		21
SEQ ID NO: 153	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 153		
acattcagtc ctggcggtga c		21
SEQ ID NO: 154	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 154		
taacattcag tcctggcggt g		21
SEQ ID NO: 155	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 155		
tgtaacattc agtcctggcg g		21
SEQ ID NO: 156	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 156		
atgtaacatt cagtcctggc g		21
SEQ ID NO: 157	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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SEQUENCE: 157		
taacaaggag ctgggcttcc t		21
SEQ ID NO: 158	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 158		
tcaataacaa ggagctgggc t		21
SEQ ID NO: 159	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 159		
tctcaataac aaggagctgg g		21
SEQ ID NO: 160	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 160		
atttcaata acaaggagct g		21
SEQ ID NO: 161	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 161		
ttcatttca ataacaagga g		21
SEQ ID NO: 162	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 162		
acacttcatt ctcaataaca a		21
SEQ ID NO: 163	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 163		
tatagtgatt gcacacttca t		21
SEQ ID NO: 164	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 164		
acttatagtg attgcacact t		21
SEQ ID NO: 165	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 165		
atacttatag tgattgcaca c		21
SEQ ID NO: 166	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 166		
atacttat agtgattgca c		21
SEQ ID NO: 167	moltype = RNA length = 21	

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 167		
aataactta tagtgattgc a		21
SEQ ID NO: 168	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 168		
agcacaata tacttatagt g		21
SEQ ID NO: 169	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 169		
atgctcagca caaatatact t		21
SEQ ID NO: 170	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 170		
tgatgcttca ctctgtctcc c		21
SEQ ID NO: 171	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 171		
aggttgatgc ttcactctgt c		21
SEQ ID NO: 172	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 172		
aagtaggttg atgcttcact c		21
SEQ ID NO: 173	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 173		
tcagcttcta agtaggttga t		21
SEQ ID NO: 174	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 174		
ttaccacagt ttcagcttct a		21
SEQ ID NO: 175	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 175		
tccttaccga cgtttcagct t		21
SEQ ID NO: 176	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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SEQUENCE: 176
atccttacc acgtttcagc t 21

SEQ ID NO: 177 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 177
aaatccttac ccacgtttca g 21

SEQ ID NO: 178 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 178
attgctgtct attatttcca g 21

SEQ ID NO: 179 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 179
tcttcggttg attgctgtct a 21

SEQ ID NO: 180 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 180
atgccaaggt ttggcatagc t 21

SEQ ID NO: 181 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 181
aaaaatgcc aggtttggca t 21

SEQ ID NO: 182 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 182
aatacaaaa atgccaaggt t 21

SEQ ID NO: 183 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 183
aaatacaaaa aatgccaagg t 21

SEQ ID NO: 184 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 184
acaaaaatac caaaaatgcc a 21

SEQ ID NO: 185 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 185
agcttataca caaaaatacc a 21

SEQ ID NO: 186 moltype = RNA length = 21

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 186		
accttaaaag cttatacaca a		21
SEQ ID NO: 187	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 187		
agaccttaaa agcttataca c		21
SEQ ID NO: 188	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 188		
tcagtcagac cttaaaagct t		21
SEQ ID NO: 189	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 189		
agaatttgtc agtcagacct t		21
SEQ ID NO: 190	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 190		
tacagaattt gtcagtcaga c		21
SEQ ID NO: 191	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 191		
aatacagaat ttgtcagtc a g		21
SEQ ID NO: 192	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 192		
ttaatacaga attgtcagtc c		21
SEQ ID NO: 193	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 193		
agctatgaca ccttaataca g		21
SEQ ID NO: 194	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 194		
aatcaaaata agtgcagagt t		21
SEQ ID NO: 195	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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misc_feature      21
                  note = n is a,g,c or u
SEQUENCE: 195
ctgagcaaaag ccatgtggtc n                               21

SEQ ID NO: 196      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature      21
                  note = n is a,g,c or u
SEQUENCE: 196
gacaccacat caacataata n                               21

SEQ ID NO: 197      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature      21
                  note = n is a,g,c or u
SEQUENCE: 197
ccacatcaac ataataggac n                               21

SEQ ID NO: 198      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature      21
                  note = n is a,g,c or u
SEQUENCE: 198
cataatagga ccacagaaaa n                               21

SEQ ID NO: 199      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature      21
                  note = n is a,g,c or u
SEQUENCE: 199
cctgtggcag ccccttattg n                               21

SEQ ID NO: 200      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature      21
                  note = n is a,g,c or u
SEQUENCE: 200
cgtcgcgcgc tccaactatt n                               21

SEQ ID NO: 201      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature      21
                  note = n is a,g,c or u
SEQUENCE: 201
cgtcgcgcct ccaactatta n                               21

SEQ ID NO: 202      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature      21
                  note = n is a,g,c or u
SEQUENCE: 202
gtcgcgcctc caactattac n                               21

SEQ ID NO: 203      moltype = RNA length = 21

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 203		
caaggcacat acttcattac n		21
SEQ ID NO: 204	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 204		
gcacatactt cattactgtc n		21
SEQ ID NO: 205	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 205		
cattactgtc acaggaagaa n		21
SEQ ID NO: 206	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 206		
ctgccgaaat ccagatcctg n		21
SEQ ID NO: 207	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 207		
gtggcagccc cttggtgtta n		21
SEQ ID NO: 208	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 208		
ggcagcccct tgggtgtata n		21
SEQ ID NO: 209	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 209		
gacacgatgc tcagatgcag n		21
SEQ ID NO: 210	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	

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SEQUENCE: 210
cacgatgctc agatgcagaa n 21

SEQ ID NO: 211 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 21
note = n is a,g,c or u

SEQUENCE: 211
cgatgctcag atgcagaatg n 21

SEQ ID NO: 212 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 21
note = n is a,g,c or u

SEQUENCE: 212
ctcagatgca gaatggactg n 21

SEQ ID NO: 213 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 21
note = n is a,g,c or u

SEQUENCE: 213
ccttcgtccc tccgaatgtt n 21

SEQ ID NO: 214 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 21
note = n is a,g,c or u

SEQUENCE: 214
cttcgtccct ccgaatgtta n 21

SEQ ID NO: 215 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 21
note = n is a,g,c or u

SEQUENCE: 215
cgtccctccg aatgttattc n 21

SEQ ID NO: 216 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 21
note = n is a,g,c or u

SEQUENCE: 216
ctccaagcct agaggctttt n 21

SEQ ID NO: 217 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 21
note = n is a,g,c or u

SEQUENCE: 217
ccaagcctag aggctttttt n 21

SEQ ID NO: 218 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21

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misc_feature	mol_type = other RNA organism = synthetic construct 21 note = n is a,g,c or u	
SEQUENCE: 218		
caagcctaga ggcttttttt n		21
SEQ ID NO: 219	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 219		
gcctagaggc tttttttgaa n		21
SEQ ID NO: 220	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 220		
cctagaggct ttttttgaac n		21
SEQ ID NO: 221	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 221		
ggcttttttt gaacaagcac n		21
SEQ ID NO: 222	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 222		
ctttttttga acaagcactg n		21
SEQ ID NO: 223	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 223		
caagcactga ctgaggaaac n		21
SEQ ID NO: 224	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 224		
cgggggtaca ggactgctac n		21
SEQ ID NO: 225	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 225		
ggggtacagg actgctacta n		21

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SEQ ID NO: 226      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u
SEQUENCE: 226
gggtacagga ctgctactac n                               21

SEQ ID NO: 227      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u
SEQUENCE: 227
caggactgct actaccatta n                               21

SEQ ID NO: 228      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u
SEQUENCE: 228
ggactgctac taccattatg n                               21

SEQ ID NO: 229      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u
SEQUENCE: 229
ctgctactac cattatggac n                               21

SEQ ID NO: 230      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u
SEQUENCE: 230
gctactacca ttatggacag n                               21

SEQ ID NO: 231      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u
SEQUENCE: 231
ctaccattat ggacagagtt n                               21

SEQ ID NO: 232      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u
SEQUENCE: 232
cacaatgcct ggtgacagaa n                               21

SEQ ID NO: 233      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct

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misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 233		
caatgctgg tgacagaatc n		21
SEQ ID NO: 234	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 234		
cctggtgaca gaatcaagtg n		21
SEQ ID NO: 235	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 235		
caagtgtcct tgcaactctc n		21
SEQ ID NO: 236	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 236		
cacggtggtc ccagatccaa n		21
SEQ ID NO: 237	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 237		
cagaggcttc ttctgaagaa n		21
SEQ ID NO: 238	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 238		
ggcttcttct gaagaagcac n		21
SEQ ID NO: 239	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 239		
cccgggtcc aggattgcta n		21
SEQ ID NO: 240	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 240		
ccgggtcca ggattgctac n		21
SEQ ID NO: 241	moltype = RNA length = 21	

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 241		
ctggcatcag aggacaacag n		21
SEQ ID NO: 242	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 242		
ggcatcagag gacaacagaa n		21
SEQ ID NO: 243	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 243		
gcatcagagg acaacagaat n		21
SEQ ID NO: 244	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 244		
catcagagga caacagaata n		21
SEQ ID NO: 245	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 245		
cagaggacaa cagaatatta n		21
SEQ ID NO: 246	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 246		
gaggacaaca gaatattatc n		21
SEQ ID NO: 247	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 247		
gacaacagaa tattatccaa n		21
SEQ ID NO: 248	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	

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SEQUENCE: 248
caacagaata ttatccaaat n                                     21

SEQ ID NO: 249      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = other RNA
                   organism = synthetic construct
misc_feature       21
                   note = n is a,g,c or u

SEQUENCE: 249
gatgctgaga ttagtccttg n                                     21

SEQ ID NO: 250      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = other RNA
                   organism = synthetic construct
misc_feature       21
                   note = n is a,g,c or u

SEQUENCE: 250
gagattagtc cttggtgtta n                                     21

SEQ ID NO: 251      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = other RNA
                   organism = synthetic construct
misc_feature       21
                   note = n is a,g,c or u

SEQUENCE: 251
gattagtcct tgggtttata n                                     21

SEQ ID NO: 252      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = other RNA
                   organism = synthetic construct
misc_feature       21
                   note = n is a,g,c or u

SEQUENCE: 252
gtccttggtg ttataccatg n                                     21

SEQ ID NO: 253      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = other RNA
                   organism = synthetic construct
misc_feature       21
                   note = n is a,g,c or u

SEQUENCE: 253
gcgacgtcca cggctgtttc n                                     21

SEQ ID NO: 254      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = other RNA
                   organism = synthetic construct
misc_feature       21
                   note = n is a,g,c or u

SEQUENCE: 254
gacgtccacg gctgtttctg n                                     21

SEQ ID NO: 255      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = other RNA
                   organism = synthetic construct
misc_feature       21
                   note = n is a,g,c or u

SEQUENCE: 255
cgtccacggc tgtttctgaa n                                     21

SEQ ID NO: 256      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21

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	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 256		
gtccacggct gtttctgaac n		21
SEQ ID NO: 257	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 257		
ccacggctgt ttctgaacaa n		21
SEQ ID NO: 258	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 258		
cacggctggt tctgaacaag n		21
SEQ ID NO: 259	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 259		
ggctgtttct gaacaagcac n		21
SEQ ID NO: 260	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 260		
ccacagtcca ggactgctac n		21
SEQ ID NO: 261	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 261		
ctggcatcag agaaccacag n		21
SEQ ID NO: 262	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 262		
catcagagaa ccacagaata n		21
SEQ ID NO: 263	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 263		
cagagaacca cagaatacta n		21

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SEQ ID NO: 264	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 264		
cgcaatgtcc agtgatggaa n		21
SEQ ID NO: 265	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 265		
gcaatgtcca gtgatggaat n		21
SEQ ID NO: 266	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 266		
caatgtccag tgatggaatc n		21
SEQ ID NO: 267	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 267		
cagtgatgga atcaactctc n		21
SEQ ID NO: 268	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 268		
gaatcaactc tcctcacaaac n		21
SEQ ID NO: 269	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 269		
ccaagcacag agcttccttc n		21
SEQ ID NO: 270	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 270		
gcacagagct tccttctgaa n		21
SEQ ID NO: 271	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 271		
cacagagctt cttctgaag n		21
SEQ ID NO: 272	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 272		
cagagcttcc ttctgaagaa n		21
SEQ ID NO: 273	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 273		
gcttccttct gaagaagcac n		21
SEQ ID NO: 274	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 274		
gaagaagcac caactgaaaa n		21
SEQ ID NO: 275	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 275		
gaagcaccaa ctgaaaacag n		21
SEQ ID NO: 276	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 276		
gcaccaactg aaaacagcac n		21
SEQ ID NO: 277	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 277		
ctggggtcca ggactgctac n		21
SEQ ID NO: 278	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 278		
ggactgctac cgaggtgatg n		21
SEQ ID NO: 279	moltype = RNA length = 21	

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 279		
ctgctaccga ggtgatggac n		21
SEQ ID NO: 280	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 280		
gctaccgagg tgatggacag n		21
SEQ ID NO: 281	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 281		
ccgaggtgat ggacagagtt n		21
SEQ ID NO: 282	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 282		
cgaggtgatg gacagagtta n		21
SEQ ID NO: 283	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 283		
gaggtgatgg acagagttat n		21
SEQ ID NO: 284	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 284		
gcacactctc caccactatc n		21
SEQ ID NO: 285	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 285		
ggcatcggag gatcccatta n		21
SEQ ID NO: 286	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	

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SEQUENCE: 286
gcacggagg atcccattat n 21

SEQ ID NO: 287 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 21
 note = n is a,g,c or u

SEQUENCE: 287
catcggagga tcccattata n 21

SEQ ID NO: 288 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 21
 note = n is a,g,c or u

SEQUENCE: 288
cggaggatcc cattatacta n 21

SEQ ID NO: 289 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 21
 note = n is a,g,c or u

SEQUENCE: 289
ggaggatccc attatactat n 21

SEQ ID NO: 290 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 21
 note = n is a,g,c or u

SEQUENCE: 290
gaggatccca ttatactatc n 21

SEQ ID NO: 291 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 21
 note = n is a,g,c or u

SEQUENCE: 291
gatcccatta tactatccaa n 21

SEQ ID NO: 292 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 21
 note = n is a,g,c or u

SEQUENCE: 292
cccattatac tatccaaatg n 21

SEQ ID NO: 293 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 21
 note = n is a,g,c or u

SEQUENCE: 293
gaatcgagtg tcctcacaac n 21

SEQ ID NO: 294 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21

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	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 294		
ggccccggtt ccaagcacag n		21
SEQ ID NO: 295	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 295		
caagcaccac ctgagaaaag n		21
SEQ ID NO: 296	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 296		
cggagttatc gaggcataac n		21
SEQ ID NO: 297	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 297		
gctggcctga ccgagaacta n		21
SEQ ID NO: 298	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 298		
ctggcctgac cgagaactac n		21
SEQ ID NO: 299	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 299		
ggcctgaccg agaactactg n		21
SEQ ID NO: 300	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 300		
ctgaccgaga actactgcag n		21
SEQ ID NO: 301	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 301		
cgagaactac tgcaggaatc n		21

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SEQ ID NO: 302	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 302		
cagaatcagg tgcctagag n		21
SEQ ID NO: 303	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 303		
gaatcaggtg tcctagagac n		21
SEQ ID NO: 304	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 304		
gcctagggcc tccttctgaa n		21
SEQ ID NO: 305	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 305		
cctagggcct ccttctgaac n		21
SEQ ID NO: 306	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 306		
ggcctccttc tgaacaagac n		21
SEQ ID NO: 307	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 307		
cctccttctg aacaagactg n		21
SEQ ID NO: 308	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 308		
ccttctgaac aagactgtat n		21
SEQ ID NO: 309	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 309		
cttctgaaca agactgtatg n		21
SEQ ID NO: 310	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 310		
ctgaacaaga ctgtatgttt n		21
SEQ ID NO: 311	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 311		
cgttcattcc agggacaaat n		21
SEQ ID NO: 312	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 312		
gttcattcca gggacaaata n		21
SEQ ID NO: 313	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 313		
cattccaggg acaaataaat n		21
SEQ ID NO: 314	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 314		
gtaaccctga tggtgacatc n		21
SEQ ID NO: 315	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 315		
ccctgatggt gacatcaatg n		21
SEQ ID NO: 316	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 316		
ctggtgctac acaatgaatc n		21
SEQ ID NO: 317	moltype = RNA length = 21	

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 317		
gtgtacaca atgaatccaa n		21
SEQ ID NO: 318	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 318		
ctacacaatg aatccaagaa n		21
SEQ ID NO: 319	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 319		
cacaatgaat ccaagaaaac n		21
SEQ ID NO: 320	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 320		
gaatccaaga aaactttttg n		21
SEQ ID NO: 321	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 321		
ccaagaaaac tttttgacta n		21
SEQ ID NO: 322	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 322		
caagaaaact ttttgactac n		21
SEQ ID NO: 323	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 323		
ctttttgact actgtgatat n		21
SEQ ID NO: 324	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	

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SEQUENCE: 324
ctgtgatatc cctctctgtg n                                     21

SEQ ID NO: 325          moltype = RNA length = 21
FEATURE                Location/Qualifiers
source                 1..21
                      mol_type = other RNA
                      organism = synthetic construct
misc_feature           21
                      note = n is a,g,c or u

SEQUENCE: 325
ctgtgcatcc tcttcatttg n                                     21

SEQ ID NO: 326          moltype = RNA length = 21
FEATURE                Location/Qualifiers
source                 1..21
                      mol_type = other RNA
                      organism = synthetic construct
misc_feature           21
                      note = n is a,g,c or u

SEQUENCE: 326
gaaatgtcct ggaagcattg n                                     21

SEQ ID NO: 327          moltype = RNA length = 21
FEATURE                Location/Qualifiers
source                 1..21
                      mol_type = other RNA
                      organism = synthetic construct
misc_feature           21
                      note = n is a,g,c or u

SEQUENCE: 327
ctgctgctca ctgcttgaag n                                     21

SEQ ID NO: 328          moltype = RNA length = 21
FEATURE                Location/Qualifiers
source                 1..21
                      mol_type = other RNA
                      organism = synthetic construct
misc_feature           21
                      note = n is a,g,c or u

SEQUENCE: 328
gctgctcact gcttgaagaa n                                     21

SEQ ID NO: 329          moltype = RNA length = 21
FEATURE                Location/Qualifiers
source                 1..21
                      mol_type = other RNA
                      organism = synthetic construct
misc_feature           21
                      note = n is a,g,c or u

SEQUENCE: 329
ctgctcactg cttgaagaag n                                     21

SEQ ID NO: 330          moltype = RNA length = 21
FEATURE                Location/Qualifiers
source                 1..21
                      mol_type = other RNA
                      organism = synthetic construct
misc_feature           21
                      note = n is a,g,c or u

SEQUENCE: 330
gctcactgct tgaagaagtc n                                     21

SEQ ID NO: 331          moltype = RNA length = 21
FEATURE                Location/Qualifiers
source                 1..21
                      mol_type = other RNA
                      organism = synthetic construct
misc_feature           21
                      note = n is a,g,c or u

SEQUENCE: 331
cactgcttga agaagtcctc n                                     21

SEQ ID NO: 332          moltype = RNA length = 21
FEATURE                Location/Qualifiers
source                 1..21

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misc_feature	mol_type = other RNA organism = synthetic construct 21 note = n is a,g,c or u	
SEQUENCE: 332		
gaacctcgaa tctcatgttc n		21
SEQ ID NO: 333	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 333		
cgaatctcat gttcaggaaa n		21
SEQ ID NO: 334	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 334		
gaatctcatg ttcaggaaat n		21
SEQ ID NO: 335	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 335		
ctcatgttca ggaaatagaa n		21
SEQ ID NO: 336	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 336		
cttgagagccc acacaagcag n		21
SEQ ID NO: 337	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 337		
ggagcccaca caagcagata n		21
SEQ ID NO: 338	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 338		
gagcccacac aagcagatat n		21
SEQ ID NO: 339	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21 note = n is a,g,c or u	
SEQUENCE: 339		
gcccacacaa gcagatattg n		21

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SEQ ID NO: 340	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 340		
cacaagcaga tattgccttg n		21
SEQ ID NO: 341	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 341		
gcagatattg ccttgctaaa n		21
SEQ ID NO: 342	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 342		
gtctgccatc cccagactac n		21
SEQ ID NO: 343	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 343		
ctgccatccc cagactacat n		21
SEQ ID NO: 344	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 344		
catccccaga ctacatggtc n		21
SEQ ID NO: 345	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 345		
ggtcaccgcc aggactgaat n		21
SEQ ID NO: 346	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 346		
gtcaccgccca ggactgaatg n		21
SEQ ID NO: 347	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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misc_feature          21
                        note = n is a,g,c or u
SEQUENCE: 347
caccgccagg actgaatggt n                               21

SEQ ID NO: 348        moltype = RNA length = 21
FEATURE              Location/Qualifiers
source               1..21
                        mol_type = other RNA
                        organism = synthetic construct
misc_feature         21
                        note = n is a,g,c or u
SEQUENCE: 348
ccgccaggac tgaatgttac n                               21

SEQ ID NO: 349        moltype = RNA length = 21
FEATURE              Location/Qualifiers
source               1..21
                        mol_type = other RNA
                        organism = synthetic construct
misc_feature         21
                        note = n is a,g,c or u
SEQUENCE: 349
gccaggactg aatgtttacat n                             21

SEQ ID NO: 350        moltype = RNA length = 21
FEATURE              Location/Qualifiers
source               1..21
                        mol_type = other RNA
                        organism = synthetic construct
misc_feature         21
                        note = n is a,g,c or u
SEQUENCE: 350
ccaggactga atgttacatc n                               21

SEQ ID NO: 351        moltype = RNA length = 21
FEATURE              Location/Qualifiers
source               1..21
                        mol_type = other RNA
                        organism = synthetic construct
misc_feature         21
                        note = n is a,g,c or u
SEQUENCE: 351
gaagcccagc tccttggttat n                             21

SEQ ID NO: 352        moltype = RNA length = 21
FEATURE              Location/Qualifiers
source               1..21
                        mol_type = other RNA
                        organism = synthetic construct
misc_feature         21
                        note = n is a,g,c or u
SEQUENCE: 352
cccagctcct tgttattgag n                             21

SEQ ID NO: 353        moltype = RNA length = 21
FEATURE              Location/Qualifiers
source               1..21
                        mol_type = other RNA
                        organism = synthetic construct
misc_feature         21
                        note = n is a,g,c or u
SEQUENCE: 353
cagctccttg ttattgagaa n                             21

SEQ ID NO: 354        moltype = RNA length = 21
FEATURE              Location/Qualifiers
source               1..21
                        mol_type = other RNA
                        organism = synthetic construct
misc_feature         21
                        note = n is a,g,c or u
SEQUENCE: 354
gctccttggt attgagaatg n                             21

SEQ ID NO: 355        moltype = RNA length = 21

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 355		
ccttggtatt gagaatgaag n		21
SEQ ID NO: 356	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 356		
gttattgaga atgaagtgtg n		21
SEQ ID NO: 357	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 357		
gaagtgtgca atcactataa n		21
SEQ ID NO: 358	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 358		
gtgtgcaatc actataagta n		21
SEQ ID NO: 359	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 359		
gtgcaatcac tataagtata n		21
SEQ ID NO: 360	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 360		
gcaatcacta taagtatatt n		21
SEQ ID NO: 361	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 361		
caatcactat aagtatattt n		21
SEQ ID NO: 362	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	

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SEQUENCE: 362
ctataagtat atttgctg n                                     21

SEQ ID NO: 363      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u

SEQUENCE: 363
gtataattgt gctgagcatt n                                   21

SEQ ID NO: 364      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u

SEQUENCE: 364
gagacagagt gaagcatcaa n                                   21

SEQ ID NO: 365      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u

SEQUENCE: 365
cagagtgaag catcaaccta n                                   21

SEQ ID NO: 366      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u

SEQUENCE: 366
gtgaagcatc aacctactta n                                   21

SEQ ID NO: 367      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u

SEQUENCE: 367
caacctactt agaagctgaa n                                   21

SEQ ID NO: 368      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u

SEQUENCE: 368
gaagctgaaa cgtgggtaag n                                   21

SEQ ID NO: 369      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       21
                  note = n is a,g,c or u

SEQUENCE: 369
gctgaaacgt gggtaaggat n                                   21

SEQ ID NO: 370      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21

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	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 370		
ctgaaacgtg ggtaaggatt n		21
SEQ ID NO: 371	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 371		
gaaacgtggg taaggattta n		21
SEQ ID NO: 372	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 372		
ggaaataata gacagcaatc n		21
SEQ ID NO: 373	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 373		
gacagcaatc aaacgaagac n		21
SEQ ID NO: 374	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 374		
ctatgccaaa ccttggcatt n		21
SEQ ID NO: 375	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 375		
gccaaacctt ggcatttttg n		21
SEQ ID NO: 376	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 376		
ccttggcatt tttgtattt n		21
SEQ ID NO: 377	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 377		
cttggcattt ttgtatttt n		21

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SEQ ID NO: 378      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source              1..21
                    mol_type = other RNA
                    organism = synthetic construct
misc_feature        21
                    note = n is a,g,c or u
SEQUENCE: 378
gcatttttggtg tattttgtg n                               21

SEQ ID NO: 379      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source              1..21
                    mol_type = other RNA
                    organism = synthetic construct
misc_feature        21
                    note = n is a,g,c or u
SEQUENCE: 379
gtattttgtgt gtataagctt n                               21

SEQ ID NO: 380      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source              1..21
                    mol_type = other RNA
                    organism = synthetic construct
misc_feature        21
                    note = n is a,g,c or u
SEQUENCE: 380
gtgtataagc ttttaaggctc n                               21

SEQ ID NO: 381      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source              1..21
                    mol_type = other RNA
                    organism = synthetic construct
misc_feature        21
                    note = n is a,g,c or u
SEQUENCE: 381
gtataagctt ttaaggtctg n                               21

SEQ ID NO: 382      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source              1..21
                    mol_type = other RNA
                    organism = synthetic construct
misc_feature        21
                    note = n is a,g,c or u
SEQUENCE: 382
gcttttaagg tctgactgac n                               21

SEQ ID NO: 383      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source              1..21
                    mol_type = other RNA
                    organism = synthetic construct
misc_feature        21
                    note = n is a,g,c or u
SEQUENCE: 383
ggtctgactg acaaattctg n                               21

SEQ ID NO: 384      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source              1..21
                    mol_type = other RNA
                    organism = synthetic construct
misc_feature        21
                    note = n is a,g,c or u
SEQUENCE: 384
ctgactgaca aattctgtat n                               21

SEQ ID NO: 385      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source              1..21
                    mol_type = other RNA
                    organism = synthetic construct

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misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 385		
gactgacaaa ttctgtatta n		21
SEQ ID NO: 386	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 386		
ctgacaaatt ctgtattaag n		21
SEQ ID NO: 387	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 387		
gtattaaggt gtcataagta n		21
SEQ ID NO: 388	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	21	
	note = n is a,g,c or u	
SEQUENCE: 388		
ctctgcactt attttgattt n		21
SEQ ID NO: 389	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 389		
gacatttggt aaaaaataa		19
SEQ ID NO: 390	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 390		
gaatgatgag aaataatta		19
SEQ ID NO: 391	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 391		
gggcaggctc ggaaaaaaa		19
SEQ ID NO: 392	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 392		
ggttcttcta cttctttta		19
SEQ ID NO: 393	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 393		
ggacaaatac attttacaa		19
SEQ ID NO: 394	moltype = RNA length = 19	

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FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 394		
ggagatccc attatacta		19
SEQ ID NO: 395	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 395		
ccttggtatt gagaatgaa		19
SEQ ID NO: 396	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 396		
tggtaatgga cagagttatt t		21
SEQ ID NO: 397	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 397		
acagagttat cgaggctcat t		21
SEQ ID NO: 398	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 398		
gtaatggaca gagttatcga t		21
SEQ ID NO: 399	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 399		
gccccttggt gttatacaa		19
SEQ ID NO: 400	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 400		
gctccttatt gttatacga		19
SEQ ID NO: 401	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 401		
catggtaatg gacagagtta t		21
SEQ ID NO: 402	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 402		
ggacagagtt atcgaggctc a		21
SEQ ID NO: 403	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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SEQUENCE: 403		
agagttatcg aggctcatta t		21
SEQ ID NO: 404	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 404		
gaaggacaaa tacatttta		19
SEQ ID NO: 405	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 405		
cgttcattcc agggacaaa		19
SEQ ID NO: 406	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 406		
cgtgtttcaa ggtttgta		19
SEQ ID NO: 407	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 407		
tttattttta acaaagtca t		21
SEQ ID NO: 408	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 408		
taattatttc tcatcattcc c		21
SEQ ID NO: 409	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 409		
ttttttcca gacctgccca t		21
SEQ ID NO: 410	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 410		
taaaagaagt agaagaacca c		21
SEQ ID NO: 411	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 411		
ttgtaaaatg tatttgctct t		21
SEQ ID NO: 412	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 412		
tagtataatg ggatcctccg a		21
SEQ ID NO: 413	moltype = RNA length = 21	

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FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 413		
ttcattctca ataacaagga g		21
SEQ ID NO: 414	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 414		
ataactctgt ccattaccat t		21
SEQ ID NO: 415	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 415		
tgagcctcga taactctggt t		21
SEQ ID NO: 416	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 416		
cgataactct gtccattacc a		21
SEQ ID NO: 417	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 417		
ttgtataaca ccaaggggct g		21
SEQ ID NO: 418	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 418		
tcgtataaca ataaggagct g		21
SEQ ID NO: 419	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 419		
aactctgtcc attaccatgt t		21
SEQ ID NO: 420	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 420		
agcctcgata actctgtcct t		21
SEQ ID NO: 421	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 421		
aatgagcctc gataactctg t		21
SEQ ID NO: 422	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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SEQUENCE: 422
 taaaatgtat ttgtccttct c 21

SEQ ID NO: 423 moltype = RNA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 423
 ttgtccctg gaatgaacgt g 21

SEQ ID NO: 424 moltype = RNA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 424
 taacaaacct tgaaacacga g 21

SEQ ID NO: 425 moltype = RNA length = 19
 FEATURE Location/Qualifiers
 source 1..19
 mol_type = other RNA
 organism = synthetic construct
 misc_feature 1..19
 note = 2' -OH of the nucleotides at positions 1, 2, 4, 8,
 11-15, 17-19 from the 5' end of the sense strand is
 replaced by methoxy group

SEQUENCE: 425
 gacatttggt aaaaaataa 19

SEQ ID NO: 426 moltype = RNA length = 19
 FEATURE Location/Qualifiers
 source 1..19
 mol_type = other RNA
 organism = synthetic construct
 misc_feature 1..19
 note = 2' -OH of the nucleotides at positions 1-3, 5-6,
 8-13, 15-16, 19 from the 5' end of the sense strand is
 replaced by methoxy group

SEQUENCE: 426
 gaatgatgag aaataatta 19

SEQ ID NO: 427 moltype = RNA length = 19
 FEATURE Location/Qualifiers
 source 1..19
 mol_type = other RNA
 organism = synthetic construct
 misc_feature 1..19
 note = 2' -OH of the nucleotides at positions 1-3, 5-7,
 11-19 from the 5' end of the sense strand is replaced by
 methoxy group

SEQUENCE: 427
 gggcagggtct ggaaaaaaa 19

SEQ ID NO: 428 moltype = RNA length = 19
 FEATURE Location/Qualifiers
 source 1..19
 mol_type = other RNA
 organism = synthetic construct
 misc_feature 1..19
 note = 2' -OH of the nucleotides at positions 1-2, 10, 19
 from the 5' end of the sense strand is replaced by methoxy
 group

SEQUENCE: 428
 ggttcttcta cttctttta 19

SEQ ID NO: 429 moltype = RNA length = 19
 FEATURE Location/Qualifiers
 source 1..19
 mol_type = other RNA
 organism = synthetic construct
 misc_feature 1..19
 note = 2' -OH of the nucleotides at positions 1-3, 5-7, 9,
 11, 16, 18-19 from the 5' end of the sense strand is
 replaced by methoxy group

SEQUENCE: 429		
ggacaaatac attttacaa		19
SEQ ID NO: 430	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..19	
	note = 2' -OH of the nucleotides at positions 1-6, 11, 14, 16, 19 from the 5' end of the sense strand is replaced by methoxy group	
SEQUENCE: 430		
ggaggatccc attatacta		19
SEQ ID NO: 431	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..19	
	note = 2' -OH of the nucleotides at positions 5, 8, 11-15, 17-19 from the 5' end of the sense strand is replaced by methoxy group	
SEQUENCE: 431		
ccttggtatt gagaatgaa		19
SEQ ID NO: 432	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..19	
	note = 2' -OH of the nucleotides at positions 1-6, 8-10, 12, 14, 19 from the 5' end of the sense strand is replaced by methoxy group	
SEQUENCE: 432		
gaaggacaaa tacatttta		19
SEQ ID NO: 433	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..19	
	note = 2' -OH of the nucleotides at positions 1, 8-9, 11, 14, 16, 18-19 from the 5' end of the sense strand is replaced by methoxy group	
SEQUENCE: 433		
gcccccttggt gttatacaa		19
SEQ ID NO: 434	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..19	
	note = 2' -OH of the nucleotides at positions 2, 6, 11-15, 17-19 from the 5' end of the sense strand is replaced by methoxy group	
SEQUENCE: 434		
cgttcattcc agggacaaa		19
SEQ ID NO: 435	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..19	
	note = 2' -OH of the nucleotides at positions 1, 8, 11, 14, 16, 18-19 from the 5' end of the sense strand is replaced by methoxy group	
SEQUENCE: 435		
gctccttatt gttatacga		19
SEQ ID NO: 436	moltype = RNA length = 19	

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FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..19	
	note = 2' -OH of the nucleotides at positions 2, 4, 9-12, 16, 19 from the 5' end of the sense strand is replaced by methoxy group	
SEQUENCE: 436		
cgtgtttcaa ggtttgta		19
SEQ ID NO: 437	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..21	
	note = 2' -OH of the nucleotides at positions 1, 2, 4-8, 10-11, 13-15, 17-18, 20-21 from the 5' end of the antisense strand is replaced by methoxy group	
SEQUENCE: 437		
tttattttta acaaatgtca t		21
SEQ ID NO: 438	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..21	
	note = 2' -OH of the nucleotides at positions 2-4, 6-11, 13-14, 16-21 from the 5' end of the antisense strand is replaced by methoxy group	
SEQUENCE: 438		
taattatttc tcatcattcc c		21
SEQ ID NO: 439	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..21	
	note = 2' -OH of the nucleotides at positions 1-8, 10-14, 16-18, 20-21 from the 5' end of the antisense strand is replaced by methoxy group	
SEQUENCE: 439		
tttttttcca gacctgccca t		21
SEQ ID NO: 440	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..21	
	note = 2' -OH of the nucleotides at positions 2-9, 11-18, 20-21 from the 5' end of the antisense strand is replaced by methoxy group	
SEQUENCE: 440		
taaaagaagt agaagaacca c		21
SEQ ID NO: 441	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	1..21	
	note = 2' -OH of the nucleotides at positions 1, 3, 5-8, 10, 12-14, 16-21 from the 5' end of the antisense strand is replaced by methoxy group	
SEQUENCE: 441		
ttgtaaaatg tatttgctct t		21
SEQ ID NO: 442	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

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misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 2-3, 5, 7-8,
                  10-21 from the 5' end of the antisense strand is replaced
                  by methoxy group

SEQUENCE: 442
tagtataatg ggatcctccg a                               21

SEQ ID NO: 443      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 1-2, 4-8,
                  10-11, 13-21 from the 5' end of the antisense strand is
                  replaced by methoxy group

SEQUENCE: 443
ttcattctca ataacaagga g                               21

SEQ ID NO: 444      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 2-5, 7, 9-11,
                  13-21 from the 5' end of the antisense strand is replaced
                  by methoxy group

SEQUENCE: 444
taaaatgtat ttgtccttct c                               21

SEQ ID NO: 445      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 1, 3, 5, 7-8,
                  10-11, 13-19, 21 from the 5' end of the antisense strand
                  is replaced by methoxy group

SEQUENCE: 445
ttgtataaca ccaaggggct g                               21

SEQ ID NO: 446      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 1-2, 4-8,
                  10-19, 21 from the 5' end of the antisense strand is
                  replaced by methoxy group

SEQUENCE: 446
ttgtccctg gaatgaacgt g                               21

SEQ ID NO: 447      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 1-2, 4-8,
                  10-19, 21 from the 5' end of the antisense strand is
                  replaced by methoxy group

SEQUENCE: 447
tcgtataaca ataaggagct g                               21

SEQ ID NO: 448      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 2-3, 5-10,
                  12-15, 17-21 from the 5' end of the antisense strand is
                  replaced by methoxy group

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SEQUENCE: 448
taacaaacct tgaacacga g 21

SEQ ID NO: 449 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 5, 7, 8, 9, 12, 14 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 449
tggtaatgga cagagttatt t 21

SEQ ID NO: 450 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 5, 7, 8, 9, 11 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, the nucleotides at positions 1, 2 from the 3' end of the sense strand is thymidine, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 450
acagagttat cgaggctcat t 21

SEQ ID NO: 451 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 3, 5, 7, 8, 9, 11, 13, 15 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 451
tggtaatgga cagagttatt t 21

SEQ ID NO: 452 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 3, 5, 7, 8, 9, 13, 16 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, the nucleotides at positions 1, 2 from the 3' end of the sense strand is thymidine, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 452
acagagttat cgaggctcat t 21

SEQ ID NO: 453 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 5, 7, 8, 9, 11, 17 from the 5' end of the sense strand is replaced by

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fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 453
tggtaatgga cagagttatt t 21

SEQ ID NO: 454 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 5, 7, 8, 9, 16 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, the nucleotides at positions 1, 2 from the 3' end of the sense strand is thymidine, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 454
acagagttat cgaggctcat t 21

SEQ ID NO: 455 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 5, 7, 8, 9, 15 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 455
tggtaatgga cagagttatt t 21

SEQ ID NO: 456 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 5, 7, 8, 9 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 456
tggtaatgga cagagttatt t 21

SEQ ID NO: 457 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 5, 7, 8, 9 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, the nucleotides at positions 1, 2 from the 3' end of the sense strand is thymidine, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 457
acagagttat cgaggctcat t 21

SEQ ID NO: 458 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

-continued

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 3, 5, 7, 8, 9, 11, 13 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 458

gtaatggaca gagttatcga t 21

SEQ ID NO: 459 moltype = RNA length = 21

FEATURE Location/Qualifiers

source

1..21

mol_type = other RNA

organism = synthetic construct

misc_feature

1..21

note = 2' -OH of the nucleotides at positions 7, 8, 9, 14 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 459

gtaatggaca gagttatcga t 21

SEQ ID NO: 460 moltype = RNA length = 21

FEATURE Location/Qualifiers

source

1..21

mol_type = other RNA

organism = synthetic construct

misc_feature

1..21

note = 2' -OH of the nucleotides at positions 5, 7, 8, 9 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 460

gtaatggaca gagttatcga t 21

SEQ ID NO: 461 moltype = RNA length = 19

FEATURE Location/Qualifiers

source

1..19

mol_type = other RNA

organism = synthetic construct

misc_feature

1..19

note = 2' -OH of the nucleotides at positions 5, 7, 8, 9 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 461

gccccttggt gttatacaa 19

SEQ ID NO: 462 moltype = RNA length = 19

FEATURE Location/Qualifiers

source

1..19

mol_type = other RNA

organism = synthetic construct

misc_feature

1..19

note = 2' -OH of the nucleotides at positions 5, 7, 9, 15 from the 5' end of the sense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the sense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 462

gccccttggt gttatacaa 19

SEQ ID NO: 463 moltype = RNA length = 19

FEATURE Location/Qualifiers

source

1..19

mol_type = other RNA

organism = synthetic construct

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misc_feature      1..19
                  note = 2' -OH of the nucleotides at positions 3, 5, 7, 8,
                  9, 11, 13, 15 from the 5' end of the sense strand is
                  replaced by fluorine, 2' -OH of the remaining nucleotides
                  in the sense strand is replaced by methoxy group, and
                  there are two consecutive phosphorothioate linkages
                  between 3 adjacent nucleotides at the 5' and 3' ends of
                  the sense strand

SEQUENCE: 463
gcccttggt gttatacaa                                19

SEQ ID NO: 464      moltype = RNA length = 19
FEATURE            Location/Qualifiers
source             1..19
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..19
                  note = 2' -OH of the nucleotides at positions 5, 7, 8, 9
                  from the 5' end of the sense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the sense
                  strand is replaced by methoxy group, and there are two
                  consecutive phosphorothioate linkages between 3 adjacent
                  nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 464
gctccttatt gttatacga                                19

SEQ ID NO: 465      moltype = RNA length = 19
FEATURE            Location/Qualifiers
source             1..19
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..19
                  note = 2' -OH of the nucleotides at positions 3, 5, 7, 8,
                  9, 11, 13, 15 from the 5' end of the sense strand is
                  replaced by fluorine, 2' -OH of the remaining nucleotides
                  in the sense strand is replaced by methoxy group, and
                  there are two consecutive phosphorothioate linkages
                  between 3 adjacent nucleotides at the 5' and 3' ends of
                  the sense strand

SEQUENCE: 465
gctccttatt gttatacga                                19

SEQ ID NO: 466      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 5, 7, 8, 9,
                  15 from the 5' end of the sense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the sense
                  strand is replaced by methoxy group, and there are two
                  consecutive phosphorothioate linkages between 3 adjacent
                  nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 466
gtaatggaca gagttatcga t                             21

SEQ ID NO: 467      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 7, 8, 9 from
                  the 5' end of the sense strand is replaced by fluorine, 2'
                  -OH of the remaining nucleotides in the sense strand is
                  replaced by methoxy group, and there are two consecutive
                  phosphorothioate linkages between 3 adjacent nucleotides
                  at the 5' and 3' ends of the sense strand

SEQUENCE: 467
gtaatggaca gagttatcga t                             21

SEQ ID NO: 468      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA

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misc_feature      organism = synthetic construct
                  1..21
                  note = 2' -OH of the nucleotides at positions 5, 7, 8, 9,
                  19 from the 5' end of the sense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the sense
                  strand is replaced by methoxy group, and there are two
                  consecutive phosphorothioate linkages between 3 adjacent
                  nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 468
gtaatggaca gagttatcga t                               21

SEQ ID NO: 469    moltype = RNA length = 21
FEATURE          Location/Qualifiers
source           1..21
                 mol_type = other RNA
                 organism = synthetic construct

misc_feature      1..21
                 note = 2' -OH of the nucleotides at positions 3, 5, 7, 8, 9
                 from the 5' end of the sense strand is replaced by
                 fluorine, 2' -OH of the remaining nucleotides in the sense
                 strand is replaced by methoxy group, and there are two
                 consecutive phosphorothioate linkages between 3 adjacent
                 nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 469
gtaatggaca gagttatcga t                               21

SEQ ID NO: 470    moltype = RNA length = 21
FEATURE          Location/Qualifiers
source           1..21
                 mol_type = other RNA
                 organism = synthetic construct

misc_feature      1..21
                 note = 2' -OH of the nucleotides at positions 2, 5, 7, 8, 9
                 from the 5' end of the sense strand is replaced by
                 fluorine, 2' -OH of the remaining nucleotides in the sense
                 strand is replaced by methoxy group, and there are two
                 consecutive phosphorothioate linkages between 3 adjacent
                 nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 470
gtaatggaca gagttatcga t                               21

SEQ ID NO: 471    moltype = RNA length = 21
FEATURE          Location/Qualifiers
source           1..21
                 mol_type = other RNA
                 organism = synthetic construct

misc_feature      1..21
                 note = 2' -OH of the nucleotides at positions 2, 5, 7, 9
                 from the 5' end of the sense strand is replaced by
                 fluorine, 2' -OH of the remaining nucleotides in the sense
                 strand is replaced by methoxy group, and there are two
                 consecutive phosphorothioate linkages between 3 adjacent
                 nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 471
gtaatggaca gagttatcga t                               21

SEQ ID NO: 472    moltype = RNA length = 21
FEATURE          Location/Qualifiers
source           1..21
                 mol_type = other RNA
                 organism = synthetic construct

misc_feature      1..21
                 note = 2' -OH of the nucleotides at positions 5, 7, 8, 9,
                 10 from the 5' end of the sense strand is replaced by
                 fluorine, 2' -OH of the remaining nucleotides in the sense
                 strand is replaced by methoxy group, and there are two
                 consecutive phosphorothioate linkages between 3 adjacent
                 nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 472
gtaatggaca gagttatcga t                               21

SEQ ID NO: 473    moltype = RNA length = 21
FEATURE          Location/Qualifiers
source           1..21
                 mol_type = other RNA
                 organism = synthetic construct

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misc_feature 1..21
note = 2' -OH of the nucleotides at positions 7, 9, 10, 11
from the 5' end of the sense strand is replaced by
fluorine, 2' -OH of the remaining nucleotides in the sense
strand is replaced by methoxy group, and there are two
consecutive phosphorothioate linkages between 3 adjacent
nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 473
gtaatggaca gagttatcga t 21

SEQ ID NO: 474 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 7, 10, 11, 12
from the 5' end of the sense strand is replaced by
fluorine, 2' -OH of the remaining nucleotides in the sense
strand is replaced by methoxy group, and there are two
consecutive phosphorothioate linkages between 3 adjacent
nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 474
gtaatggaca gagttatcga t 21

SEQ ID NO: 475 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 5, 7, 9, 10
from the 5' end of the sense strand is replaced by
fluorine, 2' -OH of the remaining nucleotides in the sense
strand is replaced by methoxy group, and there are two
consecutive phosphorothioate linkages between 3 adjacent
nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 475
gtaatggaca gagttatcga t 21

SEQ ID NO: 476 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 7, 8, 9, 15
from the 5' end of the sense strand is replaced by
fluorine, 2' -OH of the remaining nucleotides in the sense
strand is replaced by methoxy group, and there are two
consecutive phosphorothioate linkages between 3 adjacent
nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 476
gtaatggaca gagttatcga t 21

SEQ ID NO: 477 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 1, 3, 5, 7,
8, 9, 12, 14, 16 from the 5' end of the sense strand is
replaced by fluorine, 2' -OH of the remaining nucleotides
in the sense strand is replaced by methoxy group, and
there are two consecutive phosphorothioate linkages
between 3 adjacent nucleotides at the 5' and 3' ends of
the sense strand

SEQUENCE: 477
gtaatggaca gagttatcga t 21

SEQ ID NO: 478 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

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misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 5, 7, 8, 9,
                  14 from the 5' end of the sense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the sense
                  strand is replaced by methoxy group, and there are two
                  consecutive phosphorothioate linkages between 3 adjacent
                  nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 478
gtaatggaca gagttatcga t                                     21

SEQ ID NO: 479      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 8, 9 from the
                  5' end of the sense strand is replaced by fluorine, 2' -OH
                  of the remaining nucleotides in the sense strand is
                  replaced by methoxy group, and there are two consecutive
                  phosphorothioate linkages between 3 adjacent nucleotides
                  at the 5' and 3' ends of the sense strand

SEQUENCE: 479
gtaatggaca gagttatcga t                                     21

SEQ ID NO: 480      moltype = RNA length = 19
FEATURE            Location/Qualifiers
source             1..19
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..19
                  note = 2' -OH of the nucleotides at positions 8, 9, 14 from
                  the 5' end of the sense strand is replaced by fluorine, 2'
                  -OH of the remaining nucleotides in the sense strand is
                  replaced by methoxy group, and there are two consecutive
                  phosphorothioate linkages between 3 adjacent nucleotides
                  at the 5' and 3' ends of the sense strand

SEQUENCE: 480
gctccttatt gttatacga                                       19

SEQ ID NO: 481      moltype = RNA length = 19
FEATURE            Location/Qualifiers
source             1..19
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..19
                  note = 2' -OH of the nucleotides at positions 7, 10, 11, 12
                  from the 5' end of the sense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the sense
                  strand is replaced by methoxy group, and there are two
                  consecutive phosphorothioate linkages between 3 adjacent
                  nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 481
gctccttatt gttatacga                                       19

SEQ ID NO: 482      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct

misc_feature      1..21
                  note = 2' -OH of the nucleotides at positions 3, 5, 7, 8,
                  9, 11, 13, 15 from the 5' end of the sense strand is
                  replaced by fluorine, 2' -OH of the remaining nucleotides
                  in the sense strand is replaced by methoxy group, the
                  nucleotides at positions 1, 2 from the 3' end of the sense
                  strand is thymidine, and there are two consecutive
                  phosphorothioate linkages between 3 adjacent nucleotides
                  at the 5' and 3' ends of the sense strand

SEQUENCE: 482
acagagttat cgaggctcat t                                     21

SEQ ID NO: 483      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA

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misc_feature      organism = synthetic construct
                  1..21
                  note = 2' -OH of the nucleotides at positions 5, 7, 8, 9,
                  16 from the 5' end of the sense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the sense
                  strand is replaced by methoxy group, and there are two
                  consecutive phosphorothioate linkages between 3 adjacent
                  nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 483
gtaatggaca gagttatcga t                               21

SEQ ID NO: 484    moltype = RNA length = 19
FEATURE          Location/Qualifiers
source           1..19
                 mol_type = other RNA
                 organism = synthetic construct

misc_feature      1..19
                 note = 2' -OH of the nucleotides at positions 5, 7, 8, 9,
                 10 from the 5' end of the sense strand is replaced by
                 fluorine, 2' -OH of the remaining nucleotides in the sense
                 strand is replaced by methoxy group, and there are two
                 consecutive phosphorothioate linkages between 3 adjacent
                 nucleotides at the 5' and 3' ends of the sense strand

SEQUENCE: 484
gctccttatt gttatacga                                   19

SEQ ID NO: 485    moltype = RNA length = 21
FEATURE          Location/Qualifiers
source           1..21
                 mol_type = other RNA
                 organism = synthetic construct

misc_feature      1..21
                 note = 2' -OH of the nucleotides at positions 2, 6, 8, 9,
                 10, 14-16 from the 5' end of the antisense strand is
                 replaced by fluorine, 2' -OH of the remaining nucleotides
                 in the antisense strand is replaced by methoxy group, and
                 there are two consecutive phosphorothioate linkages
                 between 3 adjacent nucleotides at the 5' and 3' ends of
                 the antisense strand

SEQUENCE: 485
ataactctgt ccattaccat t                               21

SEQ ID NO: 486    moltype = RNA length = 21
FEATURE          Location/Qualifiers
source           1..21
                 mol_type = other RNA
                 organism = synthetic construct

misc_feature      1..21
                 note = 2' -OH of the nucleotides at positions 2, 6, 8, 9,
                 10, 14, 15, 16 from the 5' end of the antisense strand is
                 replaced by fluorine, 2' -OH of the remaining nucleotides
                 in the antisense strand is replaced by methoxy group, and
                 there are two consecutive phosphorothioate linkages
                 between 3 adjacent nucleotides at the 5' and 3' ends of
                 the antisense strand

SEQUENCE: 486
tgagcctcga taactctggt t                               21

SEQ ID NO: 487    moltype = RNA length = 21
FEATURE          Location/Qualifiers
source           1..21
                 mol_type = other RNA
                 organism = synthetic construct

misc_feature      1..21
                 note = 2' -OH of the nucleotides at positions 2, 4, 6, 8,
                 14, 16 from the 5' end of the antisense strand is replaced
                 by fluorine, 2' -OH of the remaining nucleotides in the
                 antisense strand is replaced by methoxy group, and there
                 are two consecutive phosphorothioate linkages between 3
                 adjacent nucleotides at the 5' and 3' ends of the
                 antisense strand

SEQUENCE: 487
ataactctgt ccattaccat t                               21

SEQ ID NO: 488    moltype = RNA length = 21
FEATURE          Location/Qualifiers

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source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 4, 7, 12, 14 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 488
tgagcctcga taactctgtt t 21

SEQ ID NO: 489 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 4, 7, 12, 14 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 489
ataactctgt ccattaccat t 21

SEQ ID NO: 490 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 5, 7, 8, 9, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 490
tgagcctcga taactctgtt t 21

SEQ ID NO: 491 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 5, 7, 8, 9, 15 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 491
ataactctgt ccattaccat t 21

SEQ ID NO: 492 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 7, 12, 14 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 492

-continued

ataactctgt ccattaccat t 21

SEQ ID NO: 493 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 7, 12, 14
from the 5' end of the antisense strand is replaced by
fluorine, 2' -OH of the remaining nucleotides in the
antisense strand is replaced by methoxy group, and there
are two consecutive phosphorothioate linkages between 3
adjacent nucleotides at the 5' and 3' ends of the
antisense strand

SEQUENCE: 493
tgagcctcga taactctgtt t 21

SEQ ID NO: 494 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 4, 6, 8,
14, 16 from the 5' end of the antisense strand is replaced
by fluorine, 2' -OH of the remaining nucleotides in the
antisense strand is replaced by methoxy group, and there
are two consecutive phosphorothioate linkages between 3
adjacent nucleotides at the 5' and 3' ends of the
antisense strand

SEQUENCE: 494
cgataactct gtccattacc a 21

SEQ ID NO: 495 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 7, 8, 9, 14
from the 5' end of the antisense strand is replaced by
fluorine, 2' -OH of the remaining nucleotides in the
antisense strand is replaced by methoxy group, and there
are two consecutive phosphorothioate linkages between 3
adjacent nucleotides at the 5' and 3' ends of the
antisense strand

SEQUENCE: 495
cgataactct gtccattacc a 21

SEQ ID NO: 496 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 7, 12, 14
from the 5' end of the antisense strand is replaced by
fluorine, 2' -OH of the remaining nucleotides in the
antisense strand is replaced by methoxy group, and there
are two consecutive phosphorothioate linkages between 3
adjacent nucleotides at the 5' and 3' ends of the
antisense strand

SEQUENCE: 496
cgataactct gtccattacc a 21

SEQ ID NO: 497 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 7, 12, 14
from the 5' end of the antisense strand is replaced by
fluorine, 2' -OH of the remaining nucleotides in the
antisense strand is replaced by methoxy group, and there

-continued

are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 497
 ttgtataaca ccaaggggct g 21

SEQ ID NO: 498 moltype = RNA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = other RNA
 organism = synthetic construct
 misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 5, 7, 9, 14 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 498
 ttgtataaca ccaaggggct g 21

SEQ ID NO: 499 moltype = RNA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = other RNA
 organism = synthetic construct
 misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 2, 4, 6, 8, 14, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 499
 ttgtataaca ccaaggggct g 21

SEQ ID NO: 500 moltype = RNA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = other RNA
 organism = synthetic construct
 misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 7, 12, 14 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 500
 tcgtataaca ataaggagct g 21

SEQ ID NO: 501 moltype = RNA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = other RNA
 organism = synthetic construct
 misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 2, 4, 6, 8, 14, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 501
 tcgtataaca ataaggagct g 21

SEQ ID NO: 502 moltype = RNA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = other RNA
 organism = synthetic construct
 misc_feature 1..21

-continued

note = 2' -OH of the nucleotides at positions 5, 7, 8, 9, 15 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 502
cgataactct gtccattacc a 21

SEQ ID NO: 503 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 7, 8, 9 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 503
cgataactct gtccattacc a 21

SEQ ID NO: 504 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 5, 7, 8, 9 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 504
cgataactct gtccattacc a 21

SEQ ID NO: 505 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 4, 5, 7, 8, 9 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 505
cgataactct gtccattacc a 21

SEQ ID NO: 506 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct
misc_feature 1..21
 note = 2' -OH of the nucleotides at positions 2, 4, 5, 7, 8, 9 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 506
cgataactct gtccattacc a 21

SEQ ID NO: 507 moltype = RNA length = 21
FEATURE Location/Qualifiers

-continued

source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 4, 5, 7, 9 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 507
cgataactct gtccattacc a 21

SEQ ID NO: 508 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 6, 8, 9, 10, 14 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 508
cgataactct gtccattacc a 21

SEQ ID NO: 509 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 4, 12, 14 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 509
cgataactct gtccattacc a 21

SEQ ID NO: 510 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 6, 8, 10, 14, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 510
cgataactct gtccattacc a 21

SEQ ID NO: 511 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 10, 14, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 511

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cgataactct gtccattacc a                               21

SEQ ID NO: 512      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       1..21
                  note = 2' -OH of the nucleotides at positions 2, 6, 10, 14,
                  16 from the 5' end of the antisense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the
                  antisense strand is replaced by methoxy group, and there
                  are two consecutive phosphorothioate linkages between 3
                  adjacent nucleotides at the 5' and 3' ends of the
                  antisense strand

SEQUENCE: 512
cgataactct gtccattacc a                               21

SEQ ID NO: 513      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       1..21
                  note = 2' -OH of the nucleotides at positions 2, 4, 8, 14,
                  16, 18 from the 5' end of the antisense strand is replaced
                  by fluorine, 2' -OH of the remaining nucleotides in the
                  antisense strand is replaced by methoxy group, and there
                  are two consecutive phosphorothioate linkages between 3
                  adjacent nucleotides at the 5' and 3' ends of the
                  antisense strand

SEQUENCE: 513
cgataactct gtccattacc a                               21

SEQ ID NO: 514      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       1..21
                  note = 2' -OH of the nucleotides at positions 2, 6, 8, 14,
                  16 from the 5' end of the antisense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the
                  antisense strand is replaced by methoxy group, and there
                  are two consecutive phosphorothioate linkages between 3
                  adjacent nucleotides at the 5' and 3' ends of the
                  antisense strand

SEQUENCE: 514
cgataactct gtccattacc a                               21

SEQ ID NO: 515      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       1..21
                  note = 2' -OH of the nucleotides at positions 2, 6, 8, 9,
                  10, 14, 15, 16 from the 5' end of the antisense strand is
                  replaced by fluorine, 2' -OH of the remaining nucleotides
                  in the antisense strand is replaced by methoxy group, and
                  there are two consecutive phosphorothioate linkages
                  between 3 adjacent nucleotides at the 5' and 3' ends of
                  the antisense strand

SEQUENCE: 515
cgataactct gtccattacc a                               21

SEQ ID NO: 516      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       1..21
                  note = 2' -OH of the nucleotides at positions 8, 9, 14 from
                  the 5' end of the antisense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the
                  antisense strand is replaced by methoxy group, and there

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-continued

are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 516
tcgtataaca ataaggagct g 21

SEQ ID NO: 517 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 14, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 517
tcgtataaca ataaggagct g 21

SEQ ID NO: 518 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 4, 6, 8, 14, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 518
tgagcctcga taactctgtt t 21

SEQ ID NO: 519 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 5, 7, 8, 9, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 519
cgataactct gtccattacc a 21

SEQ ID NO: 520 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 6, 8, 9, 10, 14 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 520
tcgtataaca ataaggagct g 21

SEQ ID NO: 521 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21

-continued

note = 2' -OH of the nucleotides at positions 2, 6, 8, 9, 10, 14 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 521
ttgtataaca ccaaggggct g 21

SEQ ID NO: 522 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 4, 12, 14 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 522
ttgtataaca ccaaggggct g 21

SEQ ID NO: 523 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 6, 8, 9, 10, 14, 15, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 523
ttgtataaca ccaaggggct g 21

SEQ ID NO: 524 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 4, 12, 14 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 524
tcgtataaca ataaggagct g 21

SEQ ID NO: 525 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct
misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 6, 8, 9, 10, 14, 15, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 525
tcgtataaca ataaggagct g 21

SEQ ID NO: 526 moltype = RNA length = 21
FEATURE Location/Qualifiers

-continued

source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 6, 8, 10, 14, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 526
tcgtataaca ataaggagct g 21

SEQ ID NO: 527 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 14, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 527
tcgtataaca ataaggagct g 21

SEQ ID NO: 528 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 6, 14, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 528
tcgtataaca ataaggagct g 21

SEQ ID NO: 529 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 4, 8, 14, 16, 18 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 529
tcgtataaca ataaggagct g 21

SEQ ID NO: 530 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

misc_feature 1..21
note = 2' -OH of the nucleotides at positions 2, 6, 8, 14, 16 from the 5' end of the antisense strand is replaced by fluorine, 2' -OH of the remaining nucleotides in the antisense strand is replaced by methoxy group, and there are two consecutive phosphorothioate linkages between 3 adjacent nucleotides at the 5' and 3' ends of the antisense strand

SEQUENCE: 530

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tcgtataaca ataaggagct g                               21

SEQ ID NO: 531      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       1..21
                  note = 2' -OH of the nucleotides at positions 2, 6, 8, 10,
                  14, 16 from the 5' end of the antisense strand is replaced
                  by fluorine, 2' -OH of the remaining nucleotides in the
                  antisense strand is replaced by methoxy group, and there
                  are two consecutive phosphorothioate linkages between 3
                  adjacent nucleotides at the 5' and 3' ends of the
                  antisense strand

SEQUENCE: 531
ataactctgt ccattaccat t                               21

SEQ ID NO: 532      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       1..21
                  note = 2' -OH of the nucleotides at positions 2, 14, 16
                  from the 5' end of the antisense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the
                  antisense strand is replaced by methoxy group, and there
                  are two consecutive phosphorothioate linkages between 3
                  adjacent nucleotides at the 5' and 3' ends of the
                  antisense strand

SEQUENCE: 532
ataactctgt ccattaccat t                               21

SEQ ID NO: 533      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       1..21
                  note = 2' -OH of the nucleotides at positions 2, 6, 14, 16
                  from the 5' end of the antisense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the
                  antisense strand is replaced by methoxy group, and there
                  are two consecutive phosphorothioate linkages between 3
                  adjacent nucleotides at the 5' and 3' ends of the
                  antisense strand

SEQUENCE: 533
ataactctgt ccattaccat t                               21

SEQ ID NO: 534      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       1..21
                  note = 2' -OH of the nucleotides at positions 2, 4, 8, 14,
                  16, 18 from the 5' end of the antisense strand is replaced
                  by fluorine, 2' -OH of the remaining nucleotides in the
                  antisense strand is replaced by methoxy group, and there
                  are two consecutive phosphorothioate linkages between 3
                  adjacent nucleotides at the 5' and 3' ends of the
                  antisense strand

SEQUENCE: 534
ataactctgt ccattaccat t                               21

SEQ ID NO: 535      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                  mol_type = other RNA
                  organism = synthetic construct
misc_feature       1..21
                  note = 2' -OH of the nucleotides at positions 2, 6, 8, 14,
                  16 from the 5' end of the antisense strand is replaced by
                  fluorine, 2' -OH of the remaining nucleotides in the
                  antisense strand is replaced by methoxy group, and there

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                                are two consecutive phosphorothioate linkages between 3
                                adjacent nucleotides at the 5' and 3' ends of the
                                antisense strand
SEQUENCE: 535
ataactctgt ccattaccat t                                     21

SEQ ID NO: 536          moltype = RNA length = 21
FEATURE                Location/Qualifiers
source                 1..21
                        mol_type = other RNA
                        organism = synthetic construct
misc_feature           1..21
                        note = 2' -OH of the nucleotides at positions 2, 6, 14, 16
                        from the 5' end of the antisense strand is replaced by
                        fluorine, 2' -OH of the remaining nucleotides in the
                        antisense strand is replaced by methoxy group, and there
                        are two consecutive phosphorothioate linkages between 3
                        adjacent nucleotides at the 5' and 3' ends of the
                        antisense strand
SEQUENCE: 536
tgagcctcga taactctggt t                                     21

SEQ ID NO: 537          moltype = RNA length = 21
FEATURE                Location/Qualifiers
source                 1..21
                        mol_type = other RNA
                        organism = synthetic construct
misc_feature           1..21
                        note = 2' -OH of the nucleotides at positions 2, 4, 8, 14,
                        16, 18 from the 5' end of the antisense strand is replaced
                        by fluorine, 2' -OH of the remaining nucleotides in the
                        antisense strand is replaced by methoxy group, and there
                        are two consecutive phosphorothioate linkages between 3
                        adjacent nucleotides at the 5' and 3' ends of the
                        antisense strand
SEQUENCE: 537
tgagcctcga taactctggt t                                     21

SEQ ID NO: 538          moltype = RNA length = 21
FEATURE                Location/Qualifiers
source                 1..21
                        mol_type = other RNA
                        organism = synthetic construct
misc_feature           1..21
                        note = 2' -OH of the nucleotides at positions 2, 6, 8, 10,
                        14, 16 from the 5' end of the antisense strand is replaced
                        by fluorine, 2' -OH of the remaining nucleotides in the
                        antisense strand is replaced by methoxy group, and there
                        are two consecutive phosphorothioate linkages between 3
                        adjacent nucleotides at the 5' and 3' ends of the
                        antisense strand
SEQUENCE: 538
tgagcctcga taactctggt t                                     21

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1. An RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting the expression of an LPA gene comprising a complementary region of an antisense strand, the complementary region comprises at least 15 consecutive nucleotides, wherein the antisense strand comprises the following nucleotide sequence: any sequence of SEQ ID NO. 414-418, a sequence having at least 15 consecutive nucleotides identical to that in SEQ ID NO. 414-418, or a sequence having no more than three different nucleotides from the above sequences.

2. The RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene according to claim 1, wherein the sense strand comprises the following nucleotide sequence: any sequence of SEQ ID NO. 396-400, a sequence having at least 15 consecutive nucleotides identical to that in SEQ ID NO. 396-400, or a sequence having no more than three different nucleotides from the above sequences.

3. The RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene according to claim 2, wherein the sense and antisense strands are selected from one or more of the following combinations:

1) The sense strand:
(SEQ ID NO. 396)
ugguauggacagaguauuuu,

the antisense strand:
(SEQ ID NO. 414)
auaacucuguccauuaccuu;

2) The sense strand:
(SEQ ID NO. 397)
acagaguauaucgaggcucatt,

5. The RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene according to claim 4, wherein the sense and antisense strands are selected from one or more of the following combinations:

1) The sense strand: (SEQ ID NO. 456)
UsGsGUfAAfUfGfGACAGAGUUAUsUsU,
the antisense strand: (SEQ ID NO. 492)
AsUsAACUfCUGUCfCAfUUACCAsUsU;
2) The sense strand: (SEQ ID NO. 457)
AsCsAGfAGfUfUfAUCGAGGCUCAsTsT,
the antisense strand: (SEQ ID NO. 493)
UsGsAGCCfUCGAUfAAfCUCUGUsUsU;

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3) The sense strand:

(SEQ ID NO. 460)

GsUsAAfUGfGfAfCAGAGUUAUCGsAsU,

the antisense strand:

(SEQ ID NO. 496)

CsGsAUAAfCUCUGfUCfCAUUACsCsA;

4) The sense strand:

(SEQ ID NO. 463)

GsCsfcCCfCUfUfGfGUfGUfUAFUACsAsA,

the antisense strand:

(SEQ ID NO. 499)

UsfUsGfUAFUAFACACCAfAGGGGCsUsG;
and

5) The sense strand:

(SEQ ID NO. 465)

GsCsfcUCfCUfUfAfUUfGUfUAFUACsGsA,

the antisense strand:

(SEQ ID NO. 501)

UsfCsGfUAFUAFACAUAfAGfGAGCsUsG;

wherein, G = 2'-O-methylguanylate, A = 2'-O-methyladenyate, U
= 2'-O-methyluridyate, C = 2'-O-methylcytidyate, Gs = 2'-O-
methyl-3'-thioguanlyate, As = 2'-O-methyl-3'-thioadenyate,
Us = 2'-O-methyl-3'-thiouridyate, Cs = 2'-O-methyl-3'-thio-
cytidyate, fG = 2'-fluoroguanlyate, fA = 2'-fluoroadenyate,
fU = 2'-fluorouridyate, fC = 2'-fluorocytidyate, fGs = 2'-
fluoro-3'-thioguanlyate, fAs = 2'-fluoro-3'-thioadenyate,
fUs = 2'-fluoro-3'-thiouridyate, fCs = 2'-fluoro-3'-thiogua-
nylate.

6. The RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene according to claim 1, wherein the RNA inhibitor is selected from one or more of Ky 11-DS38, Ky 11-DS39, Ky 11-DS42, Ky 11-DS45, and Ky 11-DS47.

7. The RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene according to claim 1, further comprising a ligand, wherein the ligand is conjugated to the sense strand and/or antisense strand.

8. The RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene according to claim 7, wherein the RNA inhibitor is selected from one or more of Ky 11-DS71 and Ky 11-DS7101 to Ky 11-DS7109.

9. A pharmaceutical composition containing the RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene according to claim 1, further comprising a delivery vehicle, and/or a physiologically acceptable excipient and/or carrier and/or diluent.

10. Use of the RNA inhibitor or a pharmaceutically acceptable salt thereof for inhibiting expression of the LPA gene according to claim 1 in the preparation of drugs for the prevention or treatment of diseases or conditions or for reducing the risk of diseases or conditions.

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